

# Transcendence of a planar reflection–group growth series and its relaxed companion

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## Abstract

Let  $W_{\text{univ}} = \langle R_0, R_1, R_2 \rangle$  be the universal right-triangle reflection group, the group generated by the reflections in the sides of a right triangle with generic (transcendental) angle parameter, to which every right triangle group agrees up to a finite, effectively computable depth [6]. Let  $U(x) = \sum_{n \geq 0} u_n x^n$  be its geodesic word-length growth series (the integer sequence OEIS A396406 [52]), with relaxed companion  $V$ . We prove:

- (i) each catalytic building-block  $q$ -series  $\Sigma_0, \Sigma_1, S_0, S_1$  is, by the Pólya–Carlson theorem, either a rational function whose poles are roots of unity or has  $|q| = 1$  as a natural boundary and is transcendental over  $\mathbb{Q}(q)$ . For the two denominators  $\Sigma_1, S_1$  the first alternative is excluded with no growth hypothesis, by the infinitude of their level sets  $\Sigma_1 = 1, S_1 = 1$  in  $(0, 1)$  against the finiteness of the zero set of a nonzero rational function (Corollary 3); that route consumes the same single analytic input as everything else here, Lemma 10. For the two numerators  $\Sigma_0, S_0$  the exclusion is instead under a coefficient-growth hypothesis, verified exactly for the first 34 square indices for  $\Sigma_1$  and for all four blocks in the uniform form of Hypothesis 5 by exact integer arithmetic out to degree 1156, and not proved (Theorem 2, Remark 6); what the pole route reaches for the numerators is the joint statement that  $\Sigma_0$  and  $S_0$  are not both rational;
- (ii) the common growth rate is  $\beta_2 = 1.4916177871\dots$ , and  $\frac{3}{2}$  is excluded;
- (iii) the travel resolvent  $\Sigma_0/(1 - \Sigma_1)$  is transcendental over  $\mathbb{Q}(x)$ , unconditionally (Theorem 75), on three inputs all proved here: the pole set, from the rectangle bound of Lemma 10; the non-vanishing of the denominator block at the travel poles,  $|S_e(q_m)| \geq 0.35\sqrt{\tau_m}$ , by an elementary explicit chain with all constants numerical and no cited asymptotic theorem (Appendix A); and the non-vanishing of both numerators, from the exact identity  $\Sigma_0(q_m)S_0(q_m) = 2q_m/(1 - q_m)$  (Proposition 25), a consequence of the  $q$ -Pythagorean identity at a zero of the  $q$ -cosine. The travel poles are the zeros of the Hahn–Exton  $q$ -Bessel function  $J_{-1/2}^{(3)}(Z(q); q^2)$ ,  $Z = \sqrt{2(1 - q)/q}$ , which up to a positive factor is the Fredholm determinant  $\det(I - \mathcal{T})$  of the travel transfer operator (Theorem 19). The block assembly itself is developed in §5: the travel and the bulk block are one resolvent of the semiseparable kernel  $q^{\max(a,b)}$  seen twice, one three-term recursion covers both, and this proves the travel telescoping  $G_0^T = \Sigma_0/(1 - \Sigma_1)$ , the bulk dressing  $b_0 = S_0/(1 - S_1)$ , the reciprocity  $t_0 = b_1$  (equivalently  $\det P = 1$ ), the identity  $t_1 = P_{12}/S_e$ , the Möbius factorisation of the gap-marked bulk resolvent, and the *shape* of the local form at a travel pole: its leading numerator is the square of one junction pairing  $\langle \lambda, R \rangle$ , which has an exact closed form in  $t_1$  and is positive at every travel pole (Theorem 72; analytic for  $\tau \leq 5 \cdot 10^{-3}$ , a certificate at the first twelve poles). Hence  $V$  and  $U$  are transcendental over  $\mathbb{Q}(x)$  (Theorems 76, 81). More holds: both are meromorphic on  $|x| < 1$  with infinitely many poles there, so they are not  $D$ -finite ( $u_n$

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and  $v_n$  are not P-recursive), satisfy no linear  $q$ -difference equation with  $Q$  not a root of unity and no Mahler equation, and, by the Pólya–Bertrandias theorem, have  $|x| = 1$  as a natural boundary (Theorems 88, 92); for the travel resolvent the same holds unconditionally, with natural boundary  $|q| = 1$ . The invariance of the travel block (T) is Proposition 1, and the former hypothesis (L) is a theorem. All of this rests on the strand-walk transfer model (M). Its local cost law is now a theorem inside the crossing optimisation (Lemma 51, Corollary 52), and its marker clause is corrected: with  $d_L$  the last bulk deposit and  $d_R$  the first travel deposit, the junction cost is  $\max(|d_L - 1|, |d_R|)$ , which depends on the sign of  $d_L$  and not on its magnitude alone; the form  $\max(|d_L| - 1, |d_R|)$  carried by earlier versions is false on every cell with  $d_L \leq -2$  and  $|d_R| \leq |d_L|$  (Remark 55). The transfer model (M) is a theorem (Theorem 66): the metric identity and the faithfulness of the model to  $W_{\text{univ}}$  are formalised in Lean, and the block decomposition (M3), corrected here, has a hand proof (Appendix D) checked against exact enumeration to  $x^{26}$ . The amplitude estimate  $(\star)$  ( $|P_{12}(q_m)|w_m^3 < 2$  at every travel pole) is proved: the gate block is the symmetric second difference of the  $q$ -cosine at its own zero, a two-interval uniform certificate bounds it by  $0.619\tau^{3/2} < \tau^{3/2}/\sqrt{2}$  for every pole with  $\tau \leq 5 \cdot 10^{-3}$ , and the six poles above that threshold are checked directly at 120-digit precision. An exact identity,  $S_e = \frac{qZ}{2}\mathfrak{s}(Z) - P_{12}$ , turns the same certificate into the denominator bound  $|S_e| \geq 0.63\sqrt{\tau}$ , so the gate closes at every travel pole. The turning-point amplitude atom on the  $U$ -side, long the special-function obstacle, is an explicit rational inequality *formally verified in Lean 4 / Mathlib [35]* (Lemma 79), as is the whole site cost chain of the model, Lemma 51 with Corollaries 52, 53 and 54 (`SiteCost.lean`, `MarkedSite.lean`, `PairingMatrix.lean`, `Realisation.lean`, `CutCross.lean`);

- (iv) an orthogonal, analysis-free route via Christol’s theorem would settle  $U$  without any of the above: the  $p$ -kernel of  $(u_n \bmod p)$  is computed to be maximally non-automatic at  $p = 3, 5$  and machine-checked at  $p = 3$ , though a proof for dense-support series is open.

Every analytic input consumed by  $V$  and by  $U$  is proved here, and so is the algebra of the block assembly; the combinatorial assembly (M3) is a hand proof, not formalised, and two finite ranges rest on computer-assisted interval certificates. On  $U$  every coefficient is derived elementarily and machine-checked, the remainders are at the standard simple-saddle level, and the gate  $(\star)$  and the denominator bound) is proved, and with it the junction pairing at every travel pole (analytically for  $\tau \leq 5 \cdot 10^{-3}$ , by certificate at the first twelve poles). The precise rigor standing, including the arithmetic model of the committed certificates, is stated in the text. In sharp contrast with the rational growth of hyperbolic groups and the algebraic growth series of Parry’s wreath products, the geometric word metric places this virtually-lamplighter group outside every algebraic class, and outside the  $D$ -finite one.

## 1 Introduction

A right triangle  $T$  has three sides; the reflections across them generate a group  $W_T = \langle R_0, R_1, R_2 \rangle \leq \text{Aff}(\mathbb{R}^2)$ . For a right triangle whose angle parameter  $\zeta_T$  is generic,  $W_T$  is the *universal* group  $W_{\text{univ}}$  of [6], and every other right triangle group agrees with  $W_{\text{univ}}$  in the ball of some finite radius and deviates beyond it; the radius is effective [6, Theorem “Effective universality”, thm:effective-universality] (for the  $(1, 2)$  triangle the first deviation is at depth 33, [6, Theorem “Sharp uniform universality at depth 32”, thm:universality-sharp]). Irrationality of the angles alone (Niven [40]) does not give genericity: a triangle in  $\mathbb{Q}^2$  with unequal legs has irrational angles, yet it is not universal. Throughout,  $U(x) = \sum_n u_n x^n$  is the growth series of  $W_{\text{univ}}$ , which counts elements by minimal word length in  $\{R_0, R_1, R_2\}$ .  $W_{\text{univ}}$  is not a Coxeter group [26], and, unlike in the rational Coxeter case governed by Steinberg’s formula [54], its growth is not a priori rational. The sequence  $u_n = 1, 3, 5, 8, 13, 21, 34, 55, 89, \dots$  is OEIS A396406; it coincides with the Fibonacci numbers  $F_{n+3}$  for  $1 \leq n \leq 9$  and first deviates at  $n = 10$ . This paper settles the arithmetic nature of the travel resolvent  $\Sigma_0/(1 - \Sigma_1)$ , which is transcendental over  $\mathbb{Q}(x)$  with every input proved self-contained

(Theorem 75), and proves that both growth series are transcendental over  $\mathbb{Q}(x)$  (Theorems 76, 81): the junction non-vanishing (R-J) is Theorem 72, the travel-block invariance (T) is Proposition 1, and the transfer model (M) is Theorem 66, whose assembly step is a hand proof (Appendix D) and is not formalised. The algebraic layer between the blocks and those two series, which earlier versions of this paper carried as the hypotheses (R) and (L), is proved in §5, and so is the local cost law of (M), together with the correction of its marker clause (§5.6). The analytic side is closed on both: the amplitude estimate ( $\star$ ) and the gate (numerator bound and denominator lower bound) are proved at every travel pole in Appendix B (Theorem 109, Lemma 110, Corollary 111), and the two numerator non-vanishing statements the resolvents need are proved in §4.3. Section 8 goes past algebraicity.  $U$  and  $V$  continue meromorphically to  $|x| < 1$  (Lemma 67), and their infinitely many poles there make them non- $D$ -finite, so that  $u_n$  and  $v_n$  are not P-recursive (Theorem 88); they exclude linear  $q$ -difference equations with  $Q$  not a root of unity and Mahler equations (Proposition 90); and, with the Pólya–Bertrandias rationality theorem, they make  $|x| = 1$  a natural boundary (Theorem 92). For the travel resolvent these statements are unconditional, with natural boundary  $|q| = 1$ .  $U$  also has a pole at  $-x_m$  for every travel pole (Proposition 85).

**Context.** For word-growth series of groups, rationality is the norm in the classical families: hyperbolic groups have rational growth series with respect to every finite generating set (Cannon [10]), Coxeter systems are governed by Steinberg’s rational formula [24, §5.12], and for wreath products (including lamplighter-type groups) Parry computed the growth series for natural generating sets and found them *algebraic* [43]. Provably transcendental growth series are rare. The classical example is Stoll’s [55]: the higher Heisenberg group  $H_5$  has a transcendental growth series with respect to its standard generating set, and a rational one with respect to a specially chosen generating set; the three-dimensional Heisenberg group, by contrast, has rational growth for every finite generating set (Duchin–Shapiro [17]). Non- $D$ -finite growth series are known as well. For groups of polynomial growth, and for groups of intermediate growth such as Grigorchuk’s [22], the growth series has radius of convergence 1 and integer coefficients, so by Pólya–Carlson [12] it is rational or has  $|x| = 1$  as a natural boundary; in the second case, which includes Stoll’s series for  $H_5$  and every group of intermediate growth, it is not  $D$ -finite. And a finitely presented group with unsolvable word problem has a non-computable growth function, while a  $D$ -finite series with rational coefficients has computable coefficients, so its growth series is not  $D$ -finite either. The group studied here is virtually the lamplighter  $\mathbb{Z} \wr \mathbb{Z}$  [6], and its generating set is not constructed but imposed by Euclidean geometry; the three side-reflections of a generic right triangle. The results below show that for this natural geometric system the growth series is transcendental and not  $D$ -finite, by an explicit mechanism: infinitely many poles inside the disc  $|x| < 1$  of meromorphy, beyond the radius of convergence  $1/\beta_2$ , together with a natural boundary on  $|x| = 1$  (§8). For a group generated by reflections this is the opposite of the Coxeter case, where Steinberg’s formula makes the growth series rational. It also places the example in sharp contrast with the algebraicity of Parry’s wreath-product series: the geometric word metric moves a virtually-lamplighter group outside every algebraic class. The cogrowth series is settled by general theory:  $W_{\text{univ}}$  is amenable, being virtually metabelian, and not virtually nilpotent, so its cogrowth series is not  $D$ -finite by Bell–Mishna [4]. The higher-dimensional picture is subtler than a clean dimensional dichotomy [7]: the abstract right-angled Coxeter envelope of an  $n$ -orthoscheme has rational growth (rate  $r_n = 1 + 2 \cos \frac{2\pi}{n+3}$ ), but the geometric group is an amenable proper quotient that matches this rational envelope only to a finite depth before deviating below it. The plane is the one dimension in which the deviation has been computed past its onset; there it is the transcendental series treated here; and whether every higher dimension is likewise eventually transcendental is an open problem.

**The relaxed/true dichotomy.** The word length  $\ell_T(g)$  is realised by strand walks in a finite transition system; permitting free isolated cycles yields a smaller, analytically tamer *relaxed* length  $\ell_R(g)$ , with  $\ell_T = \ell_R + 2c(g)$ ,  $c(g) \in \mathbb{Z}_{\geq 0}$  a cycle count. The inequality  $\ell_T \leq \ell_R + 2c$  is proved in [6]. With  $\ell_R$  and  $c$  the site-local closed forms of that paper, the identity  $\ell_T = \ell_R + 2c$  holds for every element and is formalised in Lean (`metricAll`; §7.1); that the closed form  $\ell_R$  is also the minimum over relaxed realisations is verified there, and bears only on reading  $V$  as a relaxed count. The relaxed counting function is the companion  $V$ ; one has  $v_n \geq u_n$ , equality for  $n \leq 7$ , first strict at  $n = 8$ . Both are governed by a linear  $q$ -difference (dilation  $t \mapsto q^2 t$ ) *catalytic functional equation* (§2) that is  $q$ -holonomic and not of polynomial-kernel type.

**Method.** The catalytic transfer is assembled from four integer  $q$ -series  $\Sigma_0, \Sigma_1, S_0, S_1$  with  $j^2$ -gapped coefficients. Pólya–Carlson gives a dichotomy for them (rational with roots-of-unity poles, or a natural boundary and transcendence over  $\mathbb{Q}(q)$ ). For the two denominators the rational alternative is excluded outright: their level sets  $\Sigma_1 = 1$  and  $S_1 = 1$  in  $(0, 1)$  are infinite, and a rational function that is not identically zero has finitely many zeros in  $\mathbb{C}$  (Corollary 3). For the two numerators it is excluded by super-polynomial coefficient growth along a sparse set of indices, verified for the first 34 square indices for  $\Sigma_1$  and out to degree 1156 for all four blocks, and not proved (§3). Both  $V$  and  $U$  acquire poles at the *travel poles*  $q_m$ , the roots of  $\Sigma_1^T(q) = 1$ ; transcendence follows from accumulation of these poles at  $q = 1$  once a certain auxiliary integral  $T_2$  is shown to be small (§4). The travel poles are the zeros of the Hahn–Exton  $q$ -Bessel function  $J_{-1/2}^{(3)}(Z(q); q^2)$ ,  $Z = \sqrt{2(1-q)}/q$ , because that function is, up to a positive factor, the Fredholm determinant  $\det(I - \mathcal{T})$  of the travel transfer operator (Theorem 19, Corollary 21). What the theorems consume is only  $|T_2| \rightarrow 0$ , and that is proved self-contained in Lemma 10 by an absolute bound on a rectangle: no saddle, no path through a saddle, no cited asymptotic theorem. Its quantitative part is a verified enclosure of the contour majorant, in interval arithmetic and quantified over parameter boxes, on  $\tau \in [2/101^2, 0.02]$ , together with an explicit-constant estimate below that; nothing there is a scan over sample values of  $\tau$ . It is the single analytic input of this kind, shared by  $V$  (§6) and  $U$  (§7). The sharp size of  $T_2$ , with the constant  $\sqrt{2}/36$ , is recorded separately in Remark 9 as a numerically certain statement whose proof is incomplete (Remark 11); nothing here depends on it.

The  $U$ -side carries one extra ingredient. Writing the cocycle off-diagonal  $P_{12}$  in exact closed form as a Hahn–Exton  $q$ -Bessel function  $J_{3/2}^{(3)}$ , the transcendence gate reduces to one amplitude bound on that series at the travel poles. This bound is *proved* (Appendix B):  $P_{12}$  is the symmetric second difference of the  $q$ -cosine at its own zero (Proposition 102), so the gate is a second-derivative estimate on a window of width  $\tau$ , delivered by a two-interval uniform certificate; an exact identity then converts the same certificate into the denominator lower bound, closing the gate (Corollary 111). The assembled residue at a travel pole is the square of a junction pairing  $\langle \lambda, R \rangle$  (Proposition 68), which has an exact closed form in  $t_1$ ; the gate makes it positive at every pole with  $\tau \leq 5 \cdot 10^{-3}$ , and a certificate covers the first twelve (Theorem 72). The invariance of the travel block under the cycle marker, (T), is proved (Proposition 1). The sharp constants behind the picture are *derived* in Remark 16: the pole–zero offset  $\delta w = \frac{\sqrt{2}}{8}\tau^{3/2}$  and  $|P_{12}|/\tau^{3/2} \rightarrow 1/(4\sqrt{2})$ , a fourfold margin below the gate threshold. (The turning-point atom feeding that analysis, the rational envelope  $h(X) = (1 + \frac{3}{2X^2})/(1 + \frac{1}{X^2}) \in [1, \frac{3}{2}]$  of Lemma 79, is formally verified in Lean 4.)  $U$  and  $V$  therefore rest on the same self-contained *analysis*; no special-function input separates them, and after §5 they share the junction pairing (R-J), now Theorem 72, and the transfer model (M), now Theorem 66.

Section 9 records an independent arithmetic route: by Christol’s theorem [13, 14], if  $(u_n \bmod p)$  is not  $p$ -automatic for one prime  $p$  then  $U$  is transcendental over  $\mathbb{Q}(x)$ ; the  $p$ -kernel is maximally non-automatic at  $p = 3, 5$ .

Appendix C treats a question separate from all of the above: whether the *number*  $\beta_2$  is transcendental. That is open. What the appendix contributes is a reduction of it to a value statement for a Hahn–Exton  $q$ -Bessel function at an algebraic base, and an exact location of the obstruction, at the arithmetic ratio  $\rho = \frac{1}{2}$ , one notch beyond the range in which  $q$ -irrationality criteria are proved. No statement in the body depends on it.

## 2 The catalytic equation and its building blocks

Write  $q = x^2 = e^{-\tau}$ ,  $w = \sqrt{2/\tau}$ . The relaxed length statistic satisfies a linear  $q$ -difference equation with dilation  $t \mapsto q^2 t$ ; telescoping it produces four integer power series in  $q$ ,

$$\Sigma_1(q) = \sum_{i \geq 1} (-1)^{i-1} \frac{w^{2i}}{(2i)!} \eta_i, \quad S_e(q) = \sum_{j \geq 0} \frac{(-2(1-q))^j q^{j(j+1)}}{(q; q)_{2j}}, \quad (1)$$

and their companions  $\Sigma_0, S_0$ , where  $\eta_j = e^{-B_j}$  is the Gaussian-decaying form factor of §4 (equivalently  $\eta_j = q^{j^2+j} [2(1-q)]^j (2j)! / [(q; q)_{2j} W^{2j}]$ ; the identification of this display with  $1 - \Sigma_1 = \cos(Z; q^2)$  of Proposition 17 is the collapse used in Lemma 12). We write  $\Sigma_1^T = \Sigma_1$  when the travel rôle is to be emphasised. The relaxed travel resolvent is  $G_0^T = \Sigma_0 / (1 - \Sigma_1)$  (Proposition 37), and  $U$  inherits its denominator  $1 - \Sigma_1$  (Proposition 1); how the two blocks assemble into  $V$  and  $U$ , and how far that assembly is proved, is §5. The *travel poles*

$$q_1 < q_2 < \cdots \uparrow 1, \quad \Sigma_1^T(q_m) = 1,$$

form an infinite set accumulating at  $q = 1$  (Lemma 12, proved in §4 once the collapse machinery is in place).

**Proposition 1** (Invariance of the travel block). *The denominator  $1 - \Sigma_1$  is the identical, cycle-marker-free object in the generating functions of  $U$  and of  $V$ ; hence  $U$  and  $V$  share the travel poles  $q_m$ .*

*Proof.* A pure-travel element has no gap edge, since its span is  $I_k$  and  $f_j = \pm 1$  there, so Theorem 62 gives  $c = 0$ : some relaxed-optimal realisation has no isolated cycle. That realisation is a single open walk, hence admissible for  $\ell_T$ , so  $\ell_T \leq \ell_R$ ; and  $\ell_T \geq \ell_R$  because the relaxed minimum ranges over a superset of the realisations. Therefore  $\ell_T = \ell_R$  on pure-travel elements and the two gradings of  $T$  agree termwise. The travel recursion is assembled from pure-travel runs alone, appending one  $f = \pm 1$  edge at a time, so it is computed in the sector  $c \equiv 0$  and the travel block carries  $y^0$  only.  $\square$

Earlier versions of this paper recorded the proposition as a verified conjecture resting on two unproved statements, the localisation of cycles and the metric identity  $\ell_T = \ell_R + 2c$ . The first is now Corollary 63. The second is not needed: only the inequality  $\ell_T \leq \ell_R$  is used, and it follows from the realisation exhibited by Theorem 62 together with the trivial reverse inequality, so the open lower bound of the metric identity does not enter. Statements below that invoke this proposition no longer inherit a conditional standing from it; see Theorem 81 and Remark 83.

## 3 Transcendence of the blocks

The four blocks are the  $k$ -recursion sums

$$\Sigma_k = \sum_{j \geq 0} A_{k+2j} \prod_{i < j} C_{k+2i}, \quad S_k = \sum_{j \geq 0} \alpha_{k+2j} \prod_{i < j} \gamma_{k+2i}, \quad (2)$$

$$A_k = \frac{2q}{1 - q^{k+1}}, \quad C_k = \frac{2q^{k+3}}{1 - q^{k+2}} - \frac{2q^{k+2}}{1 - q^{k+1}}, \quad \alpha_k = \frac{2q^{k+1}}{1 - q^{k+1}}, \quad \gamma_k = \alpha_{k+1} - \alpha_k,$$

$\Sigma_0, \Sigma_1$  carrying the travel rôle and  $S_0, S_1$  the bulk rôle;  $\Sigma_1$  and  $S_1$  are the denominators of (1) and  $\Sigma_0, S_0$  their numerators. Every factor is a Lambert series with integer coefficients. The factor  $C_{1+2i}$  starts at order  $2i + 3$ , so  $\prod_{i < j} C_{1+2i}$  starts at order  $\sum_{i < j} (2i + 3) = j^2 + 2j$  and, with  $A_{1+2j}$  starting at order 1, the  $j$ -th summand of  $\Sigma_1$  starts at order exactly  $(j + 1)^2$ ; only the finitely many  $j \leq \sqrt{n} - 1$  reach  $[q^n]$ . Each coefficient is therefore a finite integer sum, and  $\Sigma_1 = 2q + 2q^3 - 4q^4 + 6q^5 - 4q^6 + \dots$ . That integrality, with convergence on  $|q| < 1$ , is the whole input to Pólya–Carlson.

**Theorem 2.** *Let  $\Sigma$  be any of  $\Sigma_0, \Sigma_1, S_0, S_1$ . Then  $\Sigma$  has integer coefficients and converges in  $|q| < 1$ , and exactly one of the following holds:*

- (a)  $\Sigma$  is a rational function whose poles are roots of unity; or
- (b)  $|q| = 1$  is a natural boundary of  $\Sigma$ , and  $\Sigma$  is transcendental over  $\mathbb{Q}(q)$ .

Alternative (a) is excluded, and hence (b) holds, for any block whose coefficients grow super-polynomially along a subsequence. For  $\Sigma_1$  that growth is the hypothesis

$$|[q^{(j+1)^2}] \Sigma_1| \geq 2^{j+1} \quad \text{for infinitely many } j, \quad (3)$$

which is not proved here: see Remark 6.

*Proof.* The coefficients are integers by the order count above: each is a finite sum of integers. Convergence in  $|q| < 1$  is read off the closed forms: each block is a series whose  $j$ -th term carries the Gaussian factor  $q^{j^2 + O(j)}$  against a denominator  $(q; q)_{2j}$  or  $(q; q)_{2j+1}$  bounded below in modulus by  $|(q; q)_\infty| > 0$  on compact subsets of the punctured disc (for  $\Sigma_1$  use  $1 - \Sigma_1 = \cos(Z; q^2) = \sum_{j \geq 0} (-1)^j q^{j^2} (2(1 - q))^j / (q; q)_{2j}$  of Proposition 17, in which the pole of  $Z^2 = 2(1 - q)/q$  at the origin is cancelled), so each series converges absolutely and locally uniformly there. Hence Pólya–Carlson [12] applies and gives the dichotomy (a)/(b): a power series with integer coefficients convergent in the unit disc is either a rational function whose poles are roots of unity, or has the unit circle as a natural boundary; a function with a natural boundary is transcendental over  $\mathbb{Q}(q)$ . In case (a) the coefficient sequence satisfies a linear recurrence all of whose characteristic roots are roots of unity, hence  $|[q^n] \Sigma| \leq Cn^d$  for some  $C, d$ ; a subsequence growing faster than every polynomial in  $n$  is therefore incompatible with (a). Under (3),  $|[q^n] \Sigma_1| \geq 2^{\sqrt{n}}$  along  $n = (j + 1)^2$ , which is such a subsequence.  $\square$

For the two denominators the growth hypothesis is not needed at all: alternative (a) is excluded by the pole set instead.

**Corollary 3** (the pole route to alternative (a)). *Alternative (a) of Theorem 2 fails for  $\Sigma_1$  and fails for  $S_1$ ; hence for each of the two denominators  $|q| = 1$  is a natural boundary and the block is transcendental over  $\mathbb{Q}(q)$ . It also fails for at least one of the two numerators  $\Sigma_0, S_0$ . Neither statement uses (3) or Hypothesis 5.*

*Proof.* A rational function that is not identically zero has finitely many zeros in  $\mathbb{C}$ . The constraint is global: unlike the identity theorem it requires no accumulation point inside a domain, so a zero set that is merely infinite already contradicts it, wherever in  $\mathbb{C}$  the zeros sit and wherever they accumulate.

Suppose  $\Sigma_1$  satisfies (a), so that on  $|q| < 1$  it agrees with a rational function whose poles are roots of unity. Then  $1 - \Sigma_1$  is rational, and it is not identically zero, because  $\Sigma_1(0) = 0$ : every factor  $A_k, C_k$  of (2) vanishes at  $q = 0$ , equivalently  $1 - \Sigma_1 = \cos(Z; q^2)$  has constant term 1

(Proposition 17). A nonzero rational function has finitely many zeros; but  $\{q \in (0, 1) : \Sigma_1(q) = 1\}$  is infinite by Lemma 12. Alternative (a) is therefore impossible and Theorem 2 leaves (b). The same argument applies to  $S_1$ , with  $1 - S_1 = S_e$  and  $S_1(0) = 0$  (Proposition 17) and with Lemma 13 in place of Lemma 12.

For the numerators, suppose  $\Sigma_0$  and  $S_0$  both satisfy (a). Then  $\Sigma_0 S_0$  is rational and, by Proposition 25, agrees with the rational function  $2q/(1 - q)$  at every travel pole, hence at infinitely many points of  $(0, 1)$  (Lemma 12); two rational functions agreeing at infinitely many points are equal. They are not:  $\Sigma_0 S_0 = 4q^2 + 8q^3 + 4q^4 + \dots$  vanishes to order 2 at  $q = 0$  and  $2q/(1 - q)$  to order 1. This says nothing about either numerator separately, (14) constraining only the product.  $\square$

The two lemmas quoted are proved in §4 and the two identities in §4.1; of Theorem 2 they use only its first clause, that the blocks converge on  $|q| < 1$ , so the dependency graph is acyclic. The series data  $\Sigma_1 = 2q + 2q^3 - 4q^4 + \dots$ ,  $S_1 = 2q^2 + 2q^4 - 2q^6 + \dots$ ,  $\Sigma_0 S_0 = 4q^2 + 8q^3 + 4q^4 + \dots$  and the two collapses  $1 - \Sigma_1 = \cos(Z; q^2)$ ,  $1 - S_1 = S_e$  are recomputed in exact integer arithmetic at two truncation degrees in `f2_pole_route.py`.

*Remark 4* (what the pole route costs). Corollary 3 makes Theorem 2 unconditional for  $\Sigma_1$  and  $S_1$ , and the two routes to it are of different kinds. (3) is checked in exact integer arithmetic, with no precision to state, at 34 indices, and remains a finite verification of a statement with infinitely many instances. The pole route runs on Lemma 12, hence on Lemma 10, and that lemma is proved on all of  $\tau \in (0, 0.02]$ : a verified enclosure of the contour majorant, in interval arithmetic and quantified over parameter boxes rather than sampled, on  $\tau \geq 2/101^2$ , and an explicit-constant estimate below that. The two infinitude lemmas consume it along the infinite sequences  $\tau^{(m)} = 2/(m\pi)^2$  and  $\tau^{[m]}$ ,  $m \geq 4$ , all of which lie in that range. So the pole route carries no unproved arithmetic input, while (3) does; and for  $\Sigma_0$  and for  $S_0$  individually, (4) remains the only route. What the corollary buys besides the label is a shorter dependency list: the exclusion of (a) for the two denominators consumes the same single analytic input as Theorems 75, 76, 81 and Corollary 78, instead of a second input of its own. Proposition 22 gives both infinitude statements a second proof with no  $T_2$  input, so for the two denominators, and for the joint statement on the numerators, the exclusion of (a) does not need Lemma 10 at all.

The hypothesis (3) concerns  $\Sigma_1$  alone. Its counterpart for the other three blocks is the following uniform statement, which is what alternative (a) has to be weighed against for  $\Sigma_0$  and  $S_0$ , the two blocks Corollary 3 does not reach individually.

**Hypothesis 5** (coefficient growth). Let  $\Sigma$  be any of  $\Sigma_0, \Sigma_1, S_0, S_1$ . Then

$$|[q^n]\Sigma| \geq 2^{\sqrt{n}-1} \quad \text{for infinitely many } n. \quad (4)$$

Statement (3) is the  $\Sigma_1$  instance of (4) in the sharper form that names the witnessing indices  $n = (j + 1)^2$ . Either form excludes alternative (a) for the block concerned,  $2^{\sqrt{n}-1}$  outgrowing every polynomial in  $n$ .

*Remark 6* (why (3) is expected, and how far it is verified). (3) is verified, not proved, and the reason it is expected is exact rather than heuristic. By the order count above, the  $j$ -th summand of the telescoping  $\Sigma_1 = \sum_{j \geq 0} (\prod_{i < j} C_{1+2i}) A_{1+2j}$  has lowest degree exactly  $(j + 1)^2$ , with coefficient exactly  $(-1)^j 2^{j+1}$  (the leading 2 of  $A_{1+2j}$  against the leading  $-2$  of each of the  $j$  factors  $C_{1+2i}$ ), and summands of index  $> j$  start above that degree. Hence

$$[q^{(j+1)^2}]\Sigma_1 = (-1)^j 2^{j+1} + \left( \text{the degree-}(j+1)^2 \text{ tails of the } j \text{ summands of index } < j \right),$$

a *finite* correction to an exactly known leading term: (3) says precisely that those  $j$  tails do not cancel it. They do not merely fail to cancel; they reinforce. Exact integer evaluation of (2) truncated at degree 1156 gives

$$[q^{(j+1)^2}]_{\Sigma_1} = 2, -4, 14, -52, 178, -856, 4626, -27524, 150214, -816268, \dots$$

and confirms (3) for every  $j \leq 33$ , with the sign  $(-1)^j$  of the leading term at every one of them and a margin that grows without bound:  $|[q^{(j+1)^2}]_{\Sigma_1}|/2^{j+1}$  is 797 at  $j = 9$ ,  $1.2 \cdot 10^8$  at  $j = 20$  and  $2.7 \cdot 10^{14}$  at  $j = 33$ , where the value is  $-4649405079364022238518260$  against the bound  $2^{34}$ ; the successive ratios  $|[q^{(j+2)^2}]_{\Sigma_1}|/|[q^{(j+1)^2}]_{\Sigma_1}|$  rise from 2 to 6.22 over that range. Hypothesis 5 holds for all four blocks with a witness in every window  $[(j+1)^2, (j+2)^2)$  for every  $j \leq 32$ , the range being cut off by the truncation degree and not by a failure. The witnessing diagonals are  $n = (j+1)(j+2)$  for  $S_1$  and  $n = (j+1)^2$  for  $S_0$ , each carrying  $|[q^n]\Sigma| \geq 2^{j+1}$  throughout  $4 \leq j \leq 32$  and  $3 \leq j \leq 32$  respectively; for  $\Sigma_0$  the leading diagonal cancels: its values at  $n = (j+1)^2$  are 2, 2, 2, 10, 18, 34, 266, 306, 8170, 330, which drop below  $2^{j+1}$  at  $j = 1, \dots, 5$  and again at  $j = 9$ , so its witnesses are off-diagonal and (4) is the only form available for it.

The computation is exact integer arithmetic in  $\mathbb{Z}[[q]]/(q^{1161})$  (`blocks_growth.py`); no floating point enters, so no precision is quoted. It is checked against a second and independent derivation of the same coefficients from the  $q$ -trigonometric closed forms of Proposition 17 and Lemma 101 (agreement to degree 1160 for each of  $\Sigma_1, S_1, S_0$ ), and against the bulk block  $S_0/(1 - S_1) = 2q + 2q^2 + 6q^3 + 2q^4 + 18q^5 + 6q^6 + 42q^7 + \dots$ . For  $j \leq 9$  the bound on  $\Sigma_1$  is also machine-checked in Lean 4 (`PolyaCarlson.lean`, theorem `coeff_bound`); that proof uses `native_decide` and therefore depends, besides `propext`, on the compiler-trust axiom `PolyaCarlson.coeff_bound._native.native_decide.ax_1` which is outside the three standard axioms.

No finite verification can establish a statement quantified over infinitely many indices. (3) is numerically certain and unproved, and the transcendence conclusion (b) of Theorem 2 inherits that standing wherever it is the route used, which after Corollary 3 is  $\Sigma_0$  and  $S_0$ . What would close it is a lower bound on the degree- $(j+1)^2$  coefficient of the  $j$  preceding Lambert products, uniform in  $j$ : the statement is finite for each  $j$  and the obstruction is uniformity, not any single case.

The dichotomy of Theorem 2 uses no analytic gate. The exclusion of alternative (a) rests on (3) for  $\Sigma_0$  and  $S_0$ , and on the pole route of Corollary 3, hence on the analytic input of §4, for  $\Sigma_1$  and  $S_1$ . Neither transfers directly to  $V$  or  $U$ , which are *ratios* of blocks in which the boundary singularities can cancel; those require the pole-accumulation argument below, which is independent of (3).

## 4 The steepest-descent estimate

The one analytic input for both  $V$  and  $U$  is the smallness of an auxiliary integral. Let  $\varphi(y) = \log \frac{\sinh(y/2)}{y/2} = \sum_{n \geq 1} \varphi_n y^{2n}$  with  $\varphi_n = \frac{(-1)^{n+1} \zeta(2n)}{n(2\pi)^{2n}}$ . The form factor  $B_s$  is defined by the closed form (6) of Lemma 7; the familiar expansion  $B_s \sim \sum_{n \geq 1} \varphi_n \tau^{2n} (P_n(2s) - s)$ ,  $P_n(M) = \sum_{m=1}^M m^{2n}$  the Faulhaber sum, is its Euler–Maclaurin *asymptotic* series (equality at integer  $2s$ , divergent otherwise; Lemma 7(v)) and is used only truncated. The block  $S_e$  decomposes as  $S_e = \cos W - T_2$  with  $W = we^{-\tau/2}$  and

$$T_2 = \sum_{n \geq 1} (-1)^n g_n \frac{W^{2n}}{\Gamma(2n+1)}, \quad g_s = 1 - e^{-B_s}. \quad (5)$$

What the theorems consume is only  $T_2 \rightarrow 0$ ; its sharp size is recorded separately in Remark 9 and is not used.

**Lemma 7** (amplitude regularity). *For  $\operatorname{Re} s > -\frac{1}{2}$  set  $M = 2s$  and*

$$B_s = \frac{\tau}{4}M(M+1) - M \log \tau - \log \Gamma(M+1) + \log(q; q)_\infty - \log(q^{M+1}; q)_\infty - s \varphi(\tau), \quad (6)$$

$\varphi(y) = \log \frac{\sinh(y/2)}{y/2}$  as above, each  $q$ -Pochhammer logarithm meaning the sum of principal logarithms  $\sum_{j \geq 0} \operatorname{Log}(1 - q^{M+1+j})$ : every factor has  $|q^{M+1+j}| < 1$ , hence  $\operatorname{Re}(1 - q^{M+1+j}) > 0$ , so this is the analytic branch on the half-plane (the logarithm of the product would differ by multiples of  $2\pi i$ ). Then  $B_s$  is analytic on  $\operatorname{Re} s > -\frac{1}{2}$ , and at every integer  $n \geq 0$

$$B_n = \sum_{i=1}^{2n} \varphi(i\tau) - n \varphi(\tau), \quad (7)$$

so (6) is the analytic interpolation of the collapse coefficients  $\eta_n = e^{-B_n}$  of (1). On the strip

$$\mathcal{S} = \{-\frac{1}{2} < \operatorname{Re} s \leq 2w, |\Im s| \leq 2w\}, \quad \tau \leq 2/\pi^2,$$

one has the explicit bounds

$$|B_s| \leq 30.3 \sqrt{\tau}, \quad |B'_s| \leq 7.6 \tau, \quad (8)$$

and, on the smaller region  $|M| \leq 2w$  which contains the saddles, the truncation bound that Remark 9 consumes:

$$|B_s - \varphi_1 \tau^2 (P_1(M) - s)| \leq 0.02 \tau^{3/2}, \quad \varphi_1 = \frac{1}{24}. \quad (9)$$

Consequently  $|e^{-B_s}| \leq e^{30.3\sqrt{\tau}}$ , the amplitude  $A := 1 - e^{-B_s}$  satisfies  $|A| \leq 1 + e^{30.3\sqrt{\tau}} \rightarrow 2$ , and along any rectifiable  $\Gamma \subset \mathcal{S}$  of length  $\ell(\Gamma) \leq cw$ ,

$$V_\Gamma(A) \leq \sup_\Gamma |e^{-B_s}| \int_\Gamma |B'_s| |ds| \leq e^{30.3\sqrt{\tau}} \cdot 7.6 \tau \cdot cw = O(\sqrt{\tau}) \rightarrow 0,$$

using  $\tau w = \sqrt{2\tau}$ . The strip contains both saddles  $s^* = -\frac{1}{4} \pm i(\frac{w}{2} + O(\sqrt{\tau}))$  of Remark 9 and the  $O(w)$ -length portion of its steepest-descent contour through them (which is all the  $\ell(\Gamma) \leq cw$  clause above requires, the Gaussian width being  $O(\sqrt{w})$ ) and, being built from the larger argument  $w = We^{\tau/2}$ , it contains the corresponding  $W$ -strip as well.

*Proof.* (i) Closed form, and (7). Unwinding  $\eta_n = q^{n^2+n} [2(1-q)]^n (2n)! / [(q; q)_{2n} W^{2n}]$  with  $W^2 = (2/\tau) e^{-\tau}$  gives  $\eta_n = e^{-\tau n^2} [(1-q)\tau]^n (2n)! / (q; q)_{2n}$ . Now  $1 - q^i = i\tau e^{-i\tau/2} \frac{\sinh(i\tau/2)}{i\tau/2}$ , i.e.  $\log(1 - q^i) = \log(i\tau) - \frac{i\tau}{2} + \varphi(i\tau)$ ; summing over  $1 \leq i \leq M$ ,

$$\log(q; q)_M = \log \Gamma(M+1) + M \log \tau - \frac{\tau}{4} M(M+1) + \sum_{i=1}^M \varphi(i\tau). \quad (10)$$

Since  $\log[(1-q)\tau] = -\frac{\tau}{2} + 2 \log \tau + \varphi(\tau)$ , substituting (10) into  $-\log \eta_n$  collapses every elementary term and leaves exactly (7); and rewriting  $\log(q; q)_M = \log(q; q)_\infty - \log(q^{M+1}; q)_\infty$  in (10) turns (7) into (6) for all  $s$ . Every ingredient of (6) is analytic in  $s$  for  $\operatorname{Re} s > -\frac{1}{2}$ : there  $|q^{M+1}| < 1$  keeps  $(q^{M+1}; q)_\infty$  zero-free and the branch above canonical, and  $\operatorname{Re}(M+1) > 0$  keeps  $M+1$  off the poles of  $\Gamma$ . (Both singular factors do in fact cancel at  $M = -1$ , so  $B_s$  continues past  $\operatorname{Re} s = -\frac{1}{2}$ ; we never need that.)

(ii) *Splitting into Gamma blocks with exact cancellation.* By the Weierstrass product  $\frac{\sinh(y/2)}{y/2} = \prod_{k \geq 1} (1 + \frac{y^2}{4\pi^2 k^2})$  we have  $\varphi(y) = \sum_{k \geq 1} \log(1 + y^2/(4\pi^2 k^2))$ , so with  $R_k := 2\pi k/\tau$  and  $\prod_{n=1}^M (1 + n^2 a^2) = a^{2M} \Gamma(M+1-\frac{i}{a})\Gamma(M+1+\frac{i}{a})/[\Gamma(1-\frac{i}{a})\Gamma(1+\frac{i}{a})]$  at  $a = 1/R_k$ , (7) becomes

$$B_s = \sum_{k \geq 1} G(M; R_k) - s \varphi(\tau), \quad G(M; R) := \int_0^M [\psi(1+u+iR) + \psi(1+u-iR) - 2 \log R] du, \quad (11)$$

the integral along the segment  $[0, M]$  fixing the branch (legitimate since  $|M| < R_1$  puts the segment far from the poles of  $\psi(1+u \pm iR)$  at  $u = -1 - j \mp iR$ ). *The  $-2 \log R$  is not a normalisation: it is the exact cancellation of the two leading Stirling terms*, and it is what makes the  $k$ -sum converge; without it each block would grow like  $2M \log R$ .

Identity (11) is proved at integers, where the  $i$ -sum is finite and the interchange of  $\sum_i$  and  $\sum_k$  is trivial; it extends to all  $s$  as follows.

*Domain.* Both sides are analytic on the half-plane  $\operatorname{Re} M > -1$ : the left by (i), the right because the segment  $[0, M]$  then stays in  $\operatorname{Re} u > -1$  while the poles of  $\psi(1+u \pm iR)$  lie in  $\operatorname{Re} u \leq -1$ , and because  $\sum_k G$  converges locally uniformly there, all but finitely many  $k$  having  $R_k > 2|M|$  and hence  $|G(M; R_k)| = O(|M|^3/R_k^2)$ . (The disc  $|M| < R_1$  of step (iii) is where the *quantitative* majorant lives; it is not the domain of analyticity, and the recursion below would walk out of it.)

*Recursion.* Write  $\tilde{B} = B_s + s\varphi(\tau)$  for either side with the elementary term restored. Both obey the *same* unit recursion in  $M$ : on the left  $\frac{\tau}{4}[(M+1)(M+2) - M(M+1)] - \log \tau - \log(M+1) + \log(1 - q^{M+1}) = \varphi((M+1)\tau)$  term by term, and on the right  $G(M+1; R) - G(M; R) = \log(1 + (M+1)^2/R^2)$ , which sums over  $k$  to the same  $\varphi((M+1)\tau)$  by the Weierstrass product. (For  $B_s$  itself both increments are  $\varphi((M+1)\tau) - \frac{1}{2}\varphi(\tau)$ , the extra term being common to the two sides.)

*Conclusion.* Hence the difference  $D$  is 1-periodic on  $\operatorname{Re} M > -1$ , so it continues to a 1-periodic entire function. Its Fourier coefficients satisfy  $|c_m| \leq e^{2\pi m y} \sup_{0 \leq x \leq 1} |D(x+iy)|$  for every real  $y$ , and  $D$  grows at most polynomially: on the left by Stirling, on the right by splitting the  $k$ -sum at  $R_k = 2|M|$ : the  $O(|M|\tau)$  low terms are each  $O(|M| \log |M|)$  by Stirling and the remainder is  $O(|M|^3 \sum_{R_k > 2|M|} R_k^{-2}) = O(|M|^2 \tau)$ . Letting  $y \rightarrow -\infty$  for  $m > 0$  and  $y \rightarrow +\infty$  for  $m < 0$  gives  $c_m = 0$  for  $m \neq 0$ , so  $D$  is constant; and  $D(0) = 0$  by (7) at  $n = 0$ , so  $D \equiv 0$ .

(iii) *The integrand.* For  $|z| \geq 1$  with  $|\arg z| \leq \frac{3\pi}{4}$  one has  $|\psi(1+z) - \log z - \frac{1}{2z}| \leq |z|^{-2}$  (the constant is 0.13 on that sector; the bound fails without limit as  $\arg z \rightarrow \pi$ , which is why the sector, not a distance to the cut, is the right hypothesis). On  $\mathcal{S}$ ,  $|M| \leq 4\sqrt{2}w$  and  $|M|/R_1 = \frac{4\sqrt{2}w\tau}{2\pi} = \frac{4}{\pi}\sqrt{\tau} \leq 0.58$ , so every  $R_k > |M|$ ; and along the segment  $|\arg(u \pm iR_k)| \leq 120^\circ < 135^\circ$  with  $|u \pm iR_k| \geq R_1 - |M| > 13$ , so the estimate applies throughout. Adding the two conjugate terms, the logarithms combine with  $-2 \log R$  to  $\log(1 + u^2/R^2)$  and the  $\frac{1}{2z}$  terms to  $u/(u^2 + R^2)$ , whence for  $|u| \leq |M|$

$$|\psi(1+u+iR) + \psi(1+u-iR) - 2 \log R| \leq \frac{|M|^2 + |M|}{R^2 - |M|^2} + \frac{2}{(R - |M|)^2} =: \gamma(R).$$

(iv) *Summation.* Hence  $|G(M; R_k)| \leq |M| \gamma(R_k)$  and, from (11),

$$|B_s| \leq |M| \sum_{k \geq 1} \gamma(R_k) + |s| \varphi(\tau), \quad |B'_s| \leq 2 \sum_{k \geq 1} \gamma(R_k) + \varphi(\tau),$$

the second because  $\partial_M G = \psi(1+M+iR) + \psi(1+M-iR) - 2 \log R$  and  $M = 2s$ . Both sums are dominated by  $\sum_k R_k^{-2} = \tau^2/24$ ; carrying the finite corrections explicitly at  $|M| = 4\sqrt{2}w$ ,  $\tau \leq 2/\pi^2$  gives  $|M| \sum_k \gamma(R_k) + |s| \varphi(\tau) \leq 30.3\sqrt{\tau}$  and  $2 \sum_k \gamma(R_k) + \varphi(\tau) \leq 7.6 \tau$ , which is (8). Both ratios are *increasing* in  $\tau$  on the range (30.20, 25.14, 23.26, 22.21, 21.64 and 7.55, 6.28, 5.81, 5.55, 5.41

at  $\tau = 2/\pi^2, 0.1, 0.05, 0.02, 0.005$ , decreasing to the limits  $\frac{64}{3}$  and  $\frac{16}{3}$  as  $\tau \rightarrow 0$ ), so each attains its maximum at the right endpoint  $\tau = 2/\pi^2$ , and the stated constants hold throughout.

For (9): expand the integrand of  $G$  one order further, now *keeping the third Stirling term*  $-\frac{1}{12z^2}$ , which is what supplies the linear part of  $P_1$ : its integral is  $\frac{M}{6(M^2+R^2)} = \frac{M}{6R^2} + O(|M|^3/R^4)$ . With  $\int_0^M \log(1+u^2/R^2)du = \frac{M^3}{3R^2} + O(|M|^5/R^4)$  and  $\int_0^M u/(u^2+R^2)du = \frac{M^2}{2R^2} + O(|M|^4/R^4)$  this gives  $R^2G(M; R) \rightarrow P_1(M) = \frac{M^3}{3} + \frac{M^2}{2} + \frac{M}{6}$ , so with  $\sum_k R_k^{-2} = \tau^2/24$  exactly,

$$B_s = \varphi_1 \tau^2 (P_1(M) - s) + O(|M|^5 \tau^4) + O(|s| \tau^4).$$

On  $|M| \leq 2w$  every remainder is  $O(\tau^{3/2})$ , and the constant is 0.02 (worst 0.0177 at  $\tau = 2/\pi^2$ , decreasing).

(v) *What is not used.* The  $\tau$ -expansion  $B_s = \sum_{n \geq 1} \varphi_n \tau^{2n} (P_n(2s) - s)$ ,  $P_n$  the Faulhaber sum, is the Euler–Maclaurin *asymptotic* expansion of (6): it is *divergent* at non-integer  $2s$ , since  $P_n(2s)$  carries a Bernoulli part of size  $(2n)! e^{2\pi i \Im 2s} / (2\pi)^{2n+1}$ . Nothing above sums it; it is used only in the truncated form “leading term plus remainder”, as in Remark 9, where the  $n = 1$  term at the saddle supplies the constant  $\sqrt{2}/36$ .  $\square$

The algebraic skeleton of this proof is machine-checked in Lean 4 (`BboundedSkeleton.lean`, ten theorems, no `sorry`, axioms `propext`, `Classical.choice`, `Quot.sound`). Five carry the arithmetic content: the Gauss sum, the summation (10) built on it, the telescoping cancellation (7), the scaling identity  $(\tau^2 w^3)^2 = 8\tau$  (exact for  $w$ , only  $e^{-3\tau/2}$ -approximate for  $W$ ), and the strip condition  $|M|/R_1 < 1$ . The other five (the per-factor rewriting, the collapse of  $\log[(1-q)\tau]$ , the cancellation  $2M \log a + 2M \log R = 0$  at  $aR = 1$ , the  $w$ -definition, and the bounded-variation arithmetic) are *rearrangements of their hypotheses*, verified as such: they certify that the paper’s algebra is what it claims, not the transcendental facts fed into them. Those facts (the Weierstrass product, the digamma remainder, the convergence of the  $k$ -sum, and the continuation argument of step (ii)) are not formalised.

*Remark 8* (sharpness). The majorant of step (iv) is lossy by about a factor 6: the true extremes on  $\mathcal{S}$  are  $\operatorname{Re} B_s \geq -5.03\sqrt{\tau}$  and  $|B'_s| \leq 5.46\tau$ , and  $\sup_{\mathcal{S}} |A| = 9.02$  at  $\tau = 2/\pi^2$  (`star_gate_certificate.py`, part (7)). Only the proved bounds (8) are used anywhere.

The sketch exhibits the saddles explicitly and derives the constants by Cauchy’s theorem, Stirling’s formula and elementary Gaussian integration, and it would end by invoking a second-order remainder estimate for a simple non-degenerate saddle, [42, Ch. 4, Thm. 7.1]. Its hypotheses are *not* verified here (the saddle is flat rather than non-degenerate, and the amplitude’s bounded variation is available only on a bounded piece of the path), which is one of the reasons the sketch is not a proof (Remark 11). No theorem of this paper rests on it.

*Remark 9* (the sharp size of  $T_2$ ; HEURISTIC). Numerically,  $T_2 = \frac{\sqrt{2}}{36} \sqrt{\tau} \sin X - \frac{161}{1296} \tau \cos X + O(\tau^{3/2})$  uniformly in the phase, for any argument  $X$  with  $\tau X^2 \in [2e^{-\tau}, 2]$ ; the constant and the exponent of the remainder are certain to eight digits over  $m \leq 1024$ . The sketch below indicates why, by steepest descent, but it does *not* constitute a proof, for the reasons set out in Remark 11. Accordingly this statement carries no weight anywhere in this paper: the only property of  $T_2$  that the theorems consume is the crude bound  $|T_2| \rightarrow 0$  of Lemma 10, which is proved, and which is what Lemma 12 cites.

*Sketch, not a proof; see Remark 11.* Since  $\cos W = \sum_{n \geq 0} (-1)^n W^{2n} / (2n)!$ , (5) gives the exact alternating sum  $S_e = \cos W - T_2 = \sum_{n \geq 0} (-1)^n e^{-B_n} W^{2n} / (2n)!$  (with  $B_0 = 0$ ). Let  $b(z) = e^{-B_z} W^{2z} / \Gamma(2z + 1)$ , analytic for  $\operatorname{Re} z > -\frac{1}{2}$  (by (6) and Lemma 7; *not* by termwise polynomiality, the expansion being divergent off  $z$ ). As  $\pi \csc \pi z$  has residue  $(-1)^n$  at each integer, Cauchy’s

theorem gives  $S_e = \frac{1}{2\pi i} \oint b(z) \pi \csc(\pi z) dz$  over a contour enclosing  $\{0, 1, 2, \dots\}$ . Write the integrand as  $e^{h(z)} \pi \csc \pi z$  with  $h(z) = 2z \log W - \log \Gamma(2z + 1)$ ; including the  $\csc$  phase, the exponent is stationary where  $2 \log W - 2\psi(2z + 1) \pm i\pi = 0$ , so by  $\psi(w) = \log w + O(1/w)$  there are two simple, non-degenerate saddles

$$z_{\pm}^* = \pm \frac{i}{2} W (1 + O(1/W)), \quad h''(z_{\pm}^*) = \pm \frac{4i}{W} + O(W^{-2}) \neq 0.$$

Deform the contour to the steepest-descent paths through  $z_{\pm}^*$ ; this is legitimate because  $b$  decays super-exponentially as  $\operatorname{Re} z \rightarrow +\infty$  and  $\csc$  suppresses  $|\Im z| \rightarrow \infty$ . A direct Stirling evaluation shows that at the modulation-free level  $B \equiv 0$  the two saddles sum to  $\cos W$  *exactly*: the divergent factors  $e^{\pm \pi W/2}$  from  $\csc(\pi z_{\pm}^*)$  and from  $1/\Gamma(\pm iW + 1)$  cancel identically, each saddle contributing  $\frac{1}{2} e^{\pm iW}$ .

Restoring the modulation scales each saddle by  $e^{-B_{z_{\pm}^*}}$ ; since  $P_1(2z) = \frac{2z(2z+1)(4z+1)}{6}$  and  $\varphi_1 = \frac{1}{24}$ ,

$$B_{z_+^*} = \varphi_1 \tau^2 (P_1(iW) - \frac{iW}{2}) + O(\tau) \quad (\text{the } \tau^{3/2} \text{ part of the remainder is (9); the } O(\tau) \text{ also absorbs } B_{z_+^*} - B_{iW/2} \text{ and})$$

(using  $\tau^2 W^3 = 2\sqrt{2} \sqrt{\tau} e^{-3\tau/2}$ , the factor  $e^{-3\tau/2}$  being absorbed in the  $O(\tau)$ ), and  $B_{z_-^*} = \overline{B_{z_+^*}}$ . Hence

$$S_e = \frac{1}{2} e^{-B_{z_+^*}} e^{iW} + \frac{1}{2} e^{-B_{z_-^*}} e^{-iW} + E = \cos W - \frac{\sqrt{2}}{36} \sqrt{\tau} \sin W + E.$$

The remainder  $E$  is  $O(\tau)$  by the classical second-order estimate for a simple saddle [42, Ch. 4, Thm. 7.1]: parametrising the descent path by arclength  $s$ , the integrand equals  $e^{h(z^*)} e^{-s^2} (1 + O(sW^{-1/2}) + O(\tau))$ , the next saddle coefficient is  $\propto h'''(z^*) = O(W^{-2}) = O(\tau)$ , and the bounded-variation hypothesis on the amplitude  $e^{-B_z}$  along the path is supplied by Lemma 7. This citation is the proof's one non-elementary step. The saddle prediction matches  $S_e$  to  $O(\tau)$  over  $\tau = 0.02, 0.01, 0.005$  numerically (`saddle_verify.py`). Thus  $T_2 = \frac{\sqrt{2}}{36} \sqrt{\tau} \sin W + O(\tau)$ , with  $\sin W = \sin w + O(\sqrt{\tau})$ .  $\square$

**Lemma 10** (absolute bound on  $T_2$ ). *For  $\tau \leq 0.02$ ,  $|T_2| \leq 0.17 \tau^{1/4}$ ; in particular  $|T_2| \leq 0.064 < 1$  there, and  $|T_2| \rightarrow 0$  as  $\tau \rightarrow 0$ . The argument uses no saddle, no path through a saddle, and no cited asymptotic theorem: only the Mellin–Barnes residues, a rectangle contained in the strip  $\mathcal{S}$  of Lemma 7, and the amplitude bounds proved there. The same holds at any argument  $X$  with  $\tau X^2 \in [2e^{-\tau}, 2]$ ; the constant above is the one for  $X = w$ , which is the case Lemma 12 uses and the worse of the two.*

*Proof.* Since  $\operatorname{Res}_{s=n} \pi / \sin \pi s = (-1)^n$  and  $h(s) = X^{2s} g_s / \Gamma(2s + 1)$  is analytic on  $\operatorname{Re} s > -\frac{1}{2}$  (Lemma 7(i)), the terms of (5) are the residues of  $h(s) \pi / \sin \pi s$  at  $s = 1, 2, \dots$ . Split the series at  $n = \lfloor 2w \rfloor$ .

*The tail.* At integers,  $B_n = \sum_{i \leq 2n} \varphi(i\tau) - n\varphi(\tau) \geq n\varphi(\tau) \geq 0$  by (7) and the monotonicity of  $\varphi$ , so  $0 \leq g_n = 1 - e^{-B_n} \leq 1$  and  $|\sum_{n > \sigma_*} (-1)^n g_n X^{2n} / (2n)!| \leq \sum_{n > \sigma_*} X^{2n} / (2n)!$ . This is a sawtooth in  $\tau$ , jumping each time  $2w$  crosses an integer, with a superexponentially decreasing envelope:  $7.5 \cdot 10^{-10}$  at  $\tau = 0.02$  and  $1.9 \cdot 10^{-11}$  at  $\tau = 0.0127$ . It is bounded together with the contour integral below and is nowhere the binding term.

*The head.* The remaining residues are enclosed by the rectangle  $R = \{\frac{1}{2} \leq \operatorname{Re} s \leq \sigma_*, |\Im s| \leq X/2\}$ , where  $\sigma_*$  is the largest half-odd-integer below  $2w$  (so  $|\sin \pi s| = \cosh \pi t \geq 1$  on the right edge; taking the edge at  $2w$  itself would put it through a pole whenever  $2w \in \mathbb{Z}$ , e.g. at  $\tau = 0.02$ ), which lies *wholly inside*  $\mathcal{S}$ , so the amplitude bounds of Lemma 7 apply at every point of  $\partial R$ ; no bound on  $B_s$  outside  $\mathcal{S}$  is needed anywhere. On  $\operatorname{Re} s = \frac{1}{2}$  we have  $|\sin \pi s| = \cosh \pi t \geq 1$ , so no pole is grazed. On the horizontal edges the two exponentials cancel:  $|\pi / \sin \pi s| \leq \pi / \sinh(\pi X/2) \leq 2.01 \pi e^{-\pi X/2}$  (valid once  $X \geq 1.69$ , hence throughout  $\tau \leq 0.02$ , where  $X \geq 9.9$ ), while  $1/|\Gamma(2\sigma + 1 + iX)|$  carries

the compensating  $e^{+\pi X/2}$ ; the resulting kernel is  $O(\sqrt{2\pi/X})$ , attaining 0.7887 at  $\sigma = \frac{1}{2}$  against  $\sqrt{2\pi/X} = 0.7927$  when  $\tau = 0.02$ .

For the amplitude: where  $|2s| \leq 2w$ , (9) gives  $|B_s| \leq \frac{\tau^2}{24} \left| \frac{M^3}{3} + \frac{M^2}{2} - \frac{M}{3} \right| + 0.02 \tau^{3/2}$  with  $M = 2s$ , hence  $|g_s| = |1 - e^{-B_s}| \leq |B_s| e^{|B_s|}$ ; elsewhere on  $\partial R$ ,  $|g_s| \leq 1 + e^{30.3\sqrt{\tau}}$  by (8). Write  $\mathcal{B}(\tau, X)$  for the resulting majorant: the boundary integral of those bounds against  $|X^{2s}/\Gamma(2s+1)| |\pi/\sin \pi s|$ , divided by  $2\pi$ , plus the tail. The prose above fixes the geometry; what remains is to bound  $\mathcal{B}$  on the whole region  $\tau \in (0, 0.02]$ ,  $\tau X^2 \in [2e^{-\tau}, 2]$ , and that is done in two ranges.

(1)  $\tau \geq 2/101^2$ : *verified enclosure*. On  $w = \sqrt{2/\tau} \in [10, 101]$ ,  $X \in [we^{-\tau/2}, w]$ , the majorant is enclosed in MPFR interval arithmetic with outward directed rounding at every operation, by a branch and bound on the parameter box: a box is accepted when the enclosed supremum of  $\mathcal{B}$  over the whole box does not exceed the infimum of  $0.17 \tau^{1/4}$  on it, and is bisected otherwise. The contour integrals are enclosed cellwise, each cell contributing its integrand enclosure times its width, so what is returned bounds the integral rather than estimating it; the cells tile each edge exactly, and the initial boxes are cut at the half-integer steps of  $\sigma_*$ , so no box straddles a jump of the contour. The scan terminates with 232 accepted boxes covering the range and no failures, and returns  $\mathcal{B} \leq 0.16978 \tau^{1/4}$ , a box-level figure comparing the largest bound on a box with the smallest  $\tau^{1/4}$  on it. Run at 96 and again at 160 bits it produces the same box decomposition and the same figure. Pointwise at  $\tau = 0.02$ ,  $X = w = 10$ , where the cross-check below locates the maximum of the ratio, the enclosure is  $\mathcal{B} \leq 0.061685 = 0.164031 \tau^{1/4}$ , and the scan repeated against the constant 0.166 in place of 0.17 also terminates without failures. (`tools/t2abs_iv`; the earlier adaptive-quadrature certificate `t2abs_certificate.py` shares no code with it and is kept as a cross-check, its seven tabulated ratios 0.1623, 0.0428, 0.0308, 0.0226, 0.0173, 0.0136, 0.0115 at  $\tau = 2 \cdot 10^{-2}, \dots, 10^{-5}$  being reproduced from above.)

(2)  $\tau < 2/101^2$ : *explicit constants*. No finite scan reaches  $\tau \rightarrow 0$ . Here  $X \geq 101e^{-1/101^2} > 100$ ,  $w \leq X e^{1/X^2} \leq 1.0001X$  and  $\tau \leq 2/X^2$ , so  $0.17 \tau^{1/4} = 0.17 \cdot 2^{1/4} w^{-1/2} \geq 0.2021 X^{-1/2}$ , and it suffices to bound  $\mathcal{B}$  by that.

On the left edge  $2s + 1 = 2 + 2it$ , and  $|\Gamma(1 + iy)|^2 = \pi y / \sinh \pi y$  makes the kernel exact:  $|X^{2s}/\Gamma(2s+1)| |\pi/\sin \pi s| = X\pi \sqrt{g(\pi t)}/\sqrt{1+4t^2}$  with  $g(u) = \tanh u/u \leq \min(1, 1/u)$ , hence at most  $X\pi$  for  $t \leq 1/\pi$  and at most  $\frac{1}{2}X\sqrt{\pi} t^{-3/2}$  beyond. There  $|M| = \sqrt{1+4t^2} \leq 1+2t \leq 1+X < 2w$ , so (9) applies along the whole edge; with  $\frac{(1+2t)^3}{3} + \frac{(1+2t)^2}{2} + \frac{1+2t}{3} = \frac{8}{3}t^3 + 6t^2 + \frac{14}{3}t + \frac{7}{6}$  integrated against  $t^{-3/2}$ , and collecting the range  $t \leq 1/\pi$  and the  $0.02 \tau^{3/2}$  remainder on the same power of  $X$ , the edge contributes  $L \leq 0.03033 X^{-1/2}$  for  $X \geq 100$ .

On a horizontal edge put  $a = 2\sigma + 1$ ,  $u = a/X$  and  $\psi(u) = u - \arctan u - \frac{u}{2} \log(1+u^2)$ , so  $\psi'(u) = -\frac{1}{2} \log(1+u^2)$  and  $\psi$  is negative, decreasing and concave on  $u > 0$ . Stirling with Binet's remainder,  $\log \Gamma(z) = (z - \frac{1}{2}) \log z - z + \frac{1}{2} \log 2\pi + \mu(z)$  with  $|\mu(z)| \leq \mu(\operatorname{Re} z) \leq 1/(12 \operatorname{Re} z)$  (the kernel of Binet's first integral is positive on  $(0, \infty)$ ), cancels every term of size  $a \log X$  and leaves

$$\frac{X^{2\sigma}}{|\Gamma(a + iX)|} \cdot \frac{\pi}{\sinh(\pi X/2)} = \frac{\sqrt{2\pi}}{1 - e^{-\pi X}} X^{-1/2} (1+u^2)^{1/4} e^{X\psi(u)} e^{-\operatorname{Re} \mu}, \quad |\operatorname{Re} \mu| \leq \frac{1}{12a} \leq \frac{1}{24}. \quad (12)$$

With  $d\sigma = \frac{X}{2} du$  and  $|M| \leq X\sqrt{1+u^2}$ , (9) applies for  $u \leq \sqrt{3}$ , where  $|M| \leq 2X \leq 2w$ , and gives  $|g_s| \leq 1.00449[(1+u^2)^{3/2}/(18X) + 0.33502 X^{-2}]$ ; for  $u > \sqrt{3}$  only  $|g_s| \leq 1 + e^{30.3\sqrt{\tau}} \leq 2.5352$  is used. Two elementary inequalities close the  $u$ -integral:  $\psi(u) \leq -u^3/12$  on  $[0, 1]$ , since  $\log(1+x) \geq x/2$  there; and  $\psi(u) \leq \psi(1) + \psi'(1)(u-1)$  for  $u \geq 1$ , by concavity. With  $\int_0^\infty e^{-Xu^{3/12}} du = \Gamma(\frac{4}{3})(12/X)^{1/3}$  and  $(1+u^2)^{7/4} \leq 2^{7/4}$  on  $[0, 1]$ , one edge contributes  $H \leq 0.50142 X^{-5/6} + 1.06902 X^{-11/6} + 4.5 \cdot 10^{-9} e^{-0.131(X-100)}$ .

The right edge and the tail are exponentially small. On  $\operatorname{Re} s = \sigma_*$  one has  $|\pi/\sin \pi s| \leq \pi$  and  $a = 2\sigma_* + 1 > 4X - 1$ , while  $\log |\Gamma(a + iY)| \geq (a - \frac{1}{2}) \log a - Y^2/a - a + \frac{1}{2} \log 2\pi - \frac{1}{12a}$  for  $|Y| \leq X$ ;

since  $c \mapsto c(1 - \log c)$  decreases and  $a/X \in [3.99, 4.011]$ , the kernel there is at most  $\pi e^{-1.2805X}$ . And  $n_0 = \lfloor \sigma_* \rfloor + 1 > 2X - 1$  with  $(2n)! \geq (2n/e)^{2n}$  makes the tail at most  $1.07 e^{-1.51X}$ . Both are below  $10^{-50}$  throughout. Collecting, for  $X \geq 100$ ,

$$\mathcal{B} \leq 0.00966 X^{-1/2} + 0.15961 X^{-5/6} + 0.34028 X^{-11/6},$$

which is at most  $0.2021 X^{-1/2}$  as soon as  $0.15961 X^{-1/3} + 0.34028 X^{-4/3} \leq 0.2021 - 0.00966 = 0.19244$ ; the left side decreases and equals  $0.03512$  at  $X = 100$ . The constants of this range are regenerated, and the two  $\psi$  inequalities and the rearrangement (12) independently checked against `mpmath`'s complex  $\Gamma$ , in `t2abs_smalltau.py` at 40 and again at 60 digits.

At  $\tau = \tau_4 = 0.012665$ , the largest value Lemma 12 uses, the bound is  $0.0144$ , and what Lemma 12 consumes there is only  $|T_2| < 1$ .  $\square$

*Remark 11* (why Remark 9 is not used). The steepest-descent sketch is recorded because it produces the sharp constant, but it is not a proof, on three counts. (i) It asserts that the two saddles sum to  $\cos W$  *exactly*; they do not: the leading saddle term carries a phase  $e^{-i/(24W)}$ , so the two sum to  $\cos W + \frac{1}{24W} \sin W + O(W^{-2})$ . (ii) It sizes the first saddle correction by  $h'''(z^*) = O(W^{-2})$ ; but the saddle is *flat* ( $h'' = 4i/W \rightarrow 0$ ), so the expansion parameter is  $W^{-1}$  and the correction is  $c_1 = i/(24W) + O(W^{-2})$ . Errors (i) and (ii) are each about three quarters of the quantity being computed and cancel exactly, which is why the stated constant is nevertheless right; taken literally the chain yields  $\sqrt{2}/144$ , four times too small. (iii) The deformation is justified by the claim that  $\text{csc}$  suppresses  $|\Im z| \rightarrow \infty$ , which is false for the exact  $B$ , as the numbers in the previous proof show; and the uniform remainder is attributed to a steepest-descent error estimate whose hypotheses (a parameter-independent phase and amplitude, an explicit path, an explicit truncation order) are not verified here. A complete proof needs the two-term flat-saddle expansion together with an error estimate on the curved path; that is not written, and nothing in this paper waits on it.

**Lemma 12** (infinitude and accumulation of the travel poles). *The set  $\{q \in (0, 1) : \Sigma_1^T(q) = 1\}$  is infinite, and together with Lemma 103 it accumulates exactly at  $q = 1$ . In particular there is a travel pole with  $w \in (m\pi, (m+1)\pi)$  for every large  $m$ .*

*Proof.* By Proposition 17,  $1 - \Sigma_1^T = \cos(Z; q^2)$  with  $Z^2 = z_0^2 e^\tau$ . The collapse above defines the radius-free coefficients  $e^{-B_n} = q^{n^2+n} [2(1-q)]^n (2n)! / [(q; q)_{2n} W^{2n}]$  through  $S_e = \cos(z_0; q^2) = \sum_{n \geq 0} (-1)^n e^{-B_n} W^{2n} / (2n)!$ ; replacing  $z_0^2$  by  $Z^2 = z_0^2 e^\tau$  multiplies the  $n$ -th coefficient by  $e^{n\tau}$ , i.e. replaces  $W$  by  $W e^{\tau/2} = w$  exactly:

$$1 - \Sigma_1^T = \sum_{n \geq 0} (-1)^n e^{-B_n} \frac{w^{2n}}{(2n)!} = \cos w - T_2(w),$$

with  $T_2(\cdot)$  the series (5) evaluated at argument  $w$  in place of  $W$ ; the coefficients  $e^{-B_n}$  are the *same* ones, so no re-derivation of the amplitude is involved. Only the argument moves, and it moves inside the region where the two estimates are already proved: Lemma 7 is stated on a strip built from the larger argument  $w$  (and containing the  $W$ -strip), so it applies verbatim, and Lemma 10 is stated for any argument with  $\tau X^2 \in [2e^{-\tau}, 2]$ , hence at  $X = w$ , where  $\tau w^2 = 2$  identically. Thus  $|T_2(w)| \leq 0.064$  once  $\tau \leq 0.02$ , which along the extreme-phase sequence  $w = m\pi$ ,  $\tau^{(m)} = 2/(m\pi)^2$ , holds for every  $m \geq 4$  (there  $\tau \leq \tau_4 = 0.012665$  and the bound is  $0.0144$ ). Along that sequence

$$1 - \Sigma_1^T|_{w=m\pi} = (-1)^m - T_2(m\pi),$$

which has the sign of  $(-1)^m$  for all  $m \geq m_0$ , with  $m_0 = 4$  from Lemma 10 (only  $|T_2(m\pi)| < 1$  is used, a far weaker input than the sharp size recorded in Remark 9). By continuity of  $q \mapsto \Sigma_1^T(q)$

the intermediate value theorem yields a zero of  $1 - \Sigma_1^T$  with  $w \in (m\pi, (m+1)\pi)$  for every  $m \geq m_0$ : infinitely many travel poles, with  $\tau \downarrow 0$ , i.e.  $q \uparrow 1$ . Since every travel pole has  $q > \frac{1}{10}$  and the pole set is locally finite in  $(0, 1)$  (Lemma 103), its only accumulation point is  $q = 1$ .  $\square$

The alternation is in fact immediate:  $|T_2(\pi)| \leq 0.027$  by direct evaluation and  $|T_2(m\pi)| = O(\tau_m)$  thereafter, so  $m_0 = 1$  and there is a travel pole in every window  $(m\pi, (m+1)\pi)$ ,  $m \geq 1$ . The dominant pole  $q_1$  is the one below this ladder, in  $w \in (0, \pi)$ . The transport identity and the sign alternation (checked for  $m \leq 40$ , worst  $|T_2(m\pi)| = 0.0267$ ) are verified in `star_gate_certificate.py`.

*Alternative proof of the first sentence, without  $T_2$ .* Proposition 22 proves that the travel poles are infinite from the Fredholm determinant of Theorem 19 alone:  $\lambda_1(q) > (1 - q^2)/(2q)$ , the uniform domination  $|t_k| \leq (2|y|)^k/(2k-1)!!$  with Hurwitz's theorem, which gives  $\lambda_k(q) \sim (1-q)(2k-1)^2\pi^2/8$  at fixed  $k$ , and the fact that distinct branches cross the level 1 at distinct  $q$ ; no single-crossing property is needed. With Lemma 103 this also gives the accumulation at  $q = 1$ . It does not localise the poles: the window statement above, a travel pole with  $w \in (m\pi, (m+1)\pi)$  for every large  $m$ , still uses Lemma 10.

**Lemma 13** (infinitude of the zeros of the bulk block). *The zero set of  $S_e$  in  $(0, 1)$ , equivalently  $\{q \in (0, 1) : S_1(q) = 1\}$ , is infinite.*

*Proof.* Here no transport is needed:  $S_e = \cos W - T_2$  at  $W = we^{-\tau/2}$  is (5) itself, the series for which both estimates were set up. The map  $\tau \mapsto W(\tau) = \sqrt{2/\tau} e^{-\tau/2}$  is continuous and strictly decreasing on  $(0, \infty)$  with  $W \rightarrow \infty$  as  $\tau \downarrow 0$ , so each integer  $m \geq 4$  gives a unique  $\tau^{[m]}$  with  $W(\tau^{[m]}) = m\pi$ ; since  $W < w$ ,  $\tau^{[m]} < 2/(m\pi)^2 \leq \tau_4 = 0.012665$ , and  $\tau W^2 = 2e^{-\tau}$  identically, so Lemma 10 applies at every one of them and only  $|T_2| < 1$  is used. Then  $S_e = (-1)^m - T_2$  has the sign of  $(-1)^m$  along the sequence, and the  $\tau^{[m]}$  decrease in  $m$ , so the intermediate value theorem applied to the continuous function  $q \mapsto S_e(q)$  (Theorem 2, first clause) gives a zero in each of the disjoint intervals  $(e^{-\tau^{[m]}}, e^{-\tau^{[m+1]}})$ ,  $m \geq 4$ . Finally  $S_e = 1 - S_1$  by Proposition 17.  $\square$

The zeros of  $S_e$  are not the travel poles: those are the zeros of  $\cos(Z; q^2)$ , which is the same function a half  $q$ -step away (Proposition 17), and  $S_e$  is bounded away from 0 at every travel pole by Proposition 94. Lemma 13 is used only in Corollary 3. The alternation it asserts is checked directly for  $4 \leq m \leq 40$  in `f2_pole_route.py`, at 60 and again at 100 digits: the series for  $S_e$  cancels heavily against intermediate terms far larger than its value, and 30 digits already returns wrong signs there.

*Remark 14* (the Hubbard–Stratonovich integral representation). Writing  $q^{j(j+1)} = q^j e^{-j^2\tau}$  and linearising the Gaussian through  $e^{-j^2\tau} = (4\pi\tau)^{-1/2} \int_{\mathbb{R}} e^{-u^2/(4\tau)} e^{iju} du$ , the denominator block admits the *exact* real integral

$$S_e = \frac{1}{\sqrt{4\pi\tau}} \int_{-\infty}^{\infty} e^{-u^2/(4\tau)} \Psi(u, q) du, \quad \Psi(u, q) = \sum_{j \geq 0} \frac{(-2(1-q)q e^{iu})^j}{(q; q)_{2j}}, \quad (13)$$

the interchange of sum and integral being justified by absolute convergence ( $\sum_j |Z|^j/(q; q)_{2j} < \infty$ ,  $Z = -2(1-q)q e^{iu}$ , against the integrable Gaussian). Because the coefficients of  $\Psi$  in  $\zeta = e^{iu}$  are real,  $\Psi(-u, q) = \overline{\Psi(u, q)}$ , so (13) is real and may be written over a single variable,  $S_e = (4\pi\tau)^{-1/2} \int_{\mathbb{R}} e^{-u^2/(4\tau)} \operatorname{Re} \Psi du$ ; the exact Gaussian moment underlying it,  $\int_{\mathbb{R}} e^{-u^2/(4\tau)} e^{iju} du = \sqrt{4\pi\tau} e^{-\tau j^2}$ , is machine-checked in Lean 4 (`GaussHS.lean`, theorem `gaussian_moment`; axioms `propext`, `Classical.choice`, `Quot.sound`, no sorry).

This representation, though exact and convenient, is *not* an analytic shortcut. The amplitude  $\Psi(\cdot, q)$  is entire in  $\zeta$ , but it is *not* uniformly bounded as  $q \rightarrow 1^-$ : on any fixed window it grows like  $\cosh(w \sin(u/2))$  (numerically  $|\Psi|$  already exceeds 300 at  $u = 1$ ,  $\tau = 10^{-2}$ ), and its termwise limit

$\cos(w e^{iu/2})$  differs from  $\Psi$  by an  $O(1)$  factor at the dominant scale  $j \sim \tau^{-1/2}$  (where the subleading part of  $(q; q)_{2j}$  is itself  $O(1)$ ; numerically  $\Psi/\cos(w e^{iu/2}) \approx 0.62$ ). The size of  $S_e$  is therefore governed by the balance between this exponential growth and the Gaussian; a simple *complex* saddle at  $u^* \approx -\sqrt{2}\tau$ , of the same order of difficulty as the direct sum, *not* a bounded-amplitude stationary-phase integral with trivial hypotheses [57, Ch. II]. In particular the bounded-variation input of Lemma 7 is *not* eliminated by passing to (13); it, together with the rectangle bound of Lemma 10, remains the analytic core. The value of (13) is representational (one real variable, an entire integrand, an exact Lean-checked moment), not a reduction of hypotheses; the numerical agreement of its saddle value with  $S_e$  is recorded in `gaussint_verify.py`.

*Remark 15* (an elementary derivation of the coefficients of Remark 9). The leading term and the constant  $\sqrt{2}/36$  of Remark 9 can be obtained without the steepest-descent theorem, by an exact algebraic reduction. Writing  $1 - e^{-k\tau} = 2e^{-k\tau/2} \sinh(k\tau/2)$  throughout, the  $j$ -th term of  $S_e = \sum_{j \geq 0} (-2(1-q))^j q^{j(j+1)} / (q; q)_{2j}$  collapses to

$$S_e = \sum_{j \geq 0} (-1)^j e^{-j\tau} \frac{\sinh^j(\tau/2)}{\prod_{k=1}^{2j} \sinh(k\tau/2)} = \sum_{j \geq 0} (-1)^j \frac{(2/\tau)^j}{(2j)!} e^{-j\tau + E_j}, \quad E_j = \frac{j\tau^2}{24} - \frac{\tau^2}{144} (2j)(2j+1)(4j+1) + O(\tau^4 j^5),$$

$E_j$  being the exact  $\sinh$ -curvature defect. Two elementary facts finish the extraction. First, the factor  $e^{-j\tau}$  resums *in closed form*:  $\sum_j (-1)^j (2e^{-\tau}/\tau)^j / (2j)! = \cos W$  with  $W = \sqrt{2/\tau} e^{-\tau/2}$ , which is exactly the frequency of Remark 9; so the phase correction needs no estimate at all. Second, the entire-function moment identity  $\sum_j (-1)^j y^j j^m / (2j)! = (y d/dy)^m \cos \sqrt{y}$  evaluates the curvature term through its leading part  $-\frac{\tau^2}{9} (y d/dy)^3 \cos \sqrt{y} = -\frac{\tau^2}{9} \cdot \frac{W^3}{8} \sin W = -\frac{\sqrt{2}}{36} \sqrt{\tau} \sin W$  (using  $\tau^2 W^3 = 2\sqrt{2} \sqrt{\tau} e^{-3\tau/2}$ , the factor  $e^{-3\tau/2}$  being absorbed in the  $O(\tau)$ ), giving

$$S_e = \cos W - \frac{\sqrt{2}}{36} \sqrt{\tau} \sin W + O(\tau), \quad W = \sqrt{2/\tau} e^{-\tau/2},$$

all constants explicit and verified to ten digits (`elemcos_verify.py`). Carried to higher moments the same reduction produces every subsequent coefficient as a finite combination of  $W^k \sin W$ ,  $W^k \cos W$ , so it serves the  $O(\tau^2)$ -relative demand of the  $U$ -numerator in the same elementary terms. The one input it does *not* render elementary is the remainder estimate; that the alternating expansion is asymptotic with  $O(\tau)$  error. That step is a *cancellation* statement: not only does the triangle inequality on  $\sum_j (2/\tau)^j / (2j)! |\cdot|$  return  $\cosh W \asymp e^{\sqrt{2/\tau}}$ , but the partial sums of the  $\cos W$  series themselves swing up to  $\asymp e^{c\sqrt{2/\tau}}$  before cancelling to  $O(1)$  (numerically  $\max_J |S_J| = 9, 48, 1459, 7.6 \times 10^4$  at  $\tau = 0.1, 0.05, 0.02, 0.01$ ). Hence *every* partial-summation method (Abel, summation-by-parts, Euler transformation) fails, since each is controlled by these exponentially large partial sums; only a global resummation captures the cancellation. This global resummation is exactly the two-saddle steepest-descent computation carried out self-contained in the proof of Remark 9: the saddles  $z_{\pm}^* = \pm iW/2$  supply the cancellation the partial sums cannot. Together, the elementary extraction here and that saddle argument reduce Remark 9 to Cauchy's theorem, Stirling's formula, elementary Gaussian integration, the bounded-variation bound of Lemma 7, and *one* cited estimate: the classical remainder bound for a simple saddle [42, Ch. 4, Thm. 7.1], used once in the proof of Remark 9. It is not independent of that citation, and we do not claim otherwise; Appendix A provides the route that is.

*Remark 16* (derivation of the  $U$ -numerator amplitude). The sharp picture behind Theorem 109: the asymptotic value  $|\sum_k d_k| = \frac{3}{8\sqrt{2}} \tau^{5/2} (1 + O(\sqrt{\tau}))$  at the travel poles, on the one explicit series  $P_{12} = \frac{2q^3}{1-q^3} \sum_k d_k$ ,  $d_k = (-2)^k (1-q)^k q^{k^2+3k} / [(q^2; q^2)_k (q^5; q^2)_k]$ , which submits to the identical

reduction. Writing  $1 - q^m = 2e^{-m\tau/2} \sinh(m\tau/2)$  throughout collapses the terms *exactly* to

$$d_k = (-1)^k e^{-k\tau} \frac{\sinh^{k+1}(\frac{\tau}{2}) \sinh(\frac{3\tau}{2}) \sinh((k+1)\tau)}{\prod_{m=1}^{2k+3} \sinh(\frac{m\tau}{2})},$$

and the elementary expansion (exact linear absorption into the argument, Faulhaber sums for the sinh-curvature, the moment identities  $k^m \mapsto (y d/dy)^m$ ) yields, with  $\Phi(y) = (\sin W - W \cos W)/(2W^3)$ ,  $W = \sqrt{y}$ ,  $D = y d/dy$ ,

$$\sum_k d_k = 6 \left[ \Phi + c_3 D^3 \Phi + c_2 D^2 \Phi + \frac{c_3^2}{2} D^6 \Phi + (c_5 + c_2 c_3) D^5 \Phi + \frac{c_3^3}{6} D^9 \Phi \right] (y_*) + O(\tau^3),$$

$y_* = \frac{2}{\tau} e^{-\tau-23\tau^2/36}$ ,  $c_2 = -\frac{5}{12}\tau^2$ ,  $c_3 = -\frac{1}{9}\tau^2$ ,  $c_5 = \frac{1}{450}\tau^4$ ; *every constant an exact rational, derived, not fitted* (`elemY3_verify.py`; the error is verified  $O(\tau^3)$  over  $\tau \in [0.005, 0.08]$  and the formula reproduces the exact series at the poles  $m = 7, \dots, 33$  to the displayed accuracy). At the travel poles the expansion gives  $|\sum_k d_k|/\tau^{5/2} \rightarrow 3/(8\sqrt{2}) = 0.2652$ , i.e. exactly the observed gate value  $|P_{12}|/\tau^{3/2} \rightarrow 1/(4\sqrt{2})$  with its fourfold margin: structurally, the pole sits at offset  $\delta w$  from the zero of the displayed expansion, and the constant  $3/(8\sqrt{2})$  arises as  $(3\tau/2) \delta w / \tau^{5/2}$ .

The *phase-matching step* that fixes  $\delta w$  is carried out by the same machinery. The pole equation  $\Sigma_1(q_m) = 1$  collapses exactly to  $1 - \Sigma_1 = \sum_{j \geq 0} (-1)^j \sinh^j(\frac{\tau}{2}) / \prod_{m=1}^{2j} \sinh(\frac{m\tau}{2})$ ; the  $S_e$ -series stripped of its  $e^{-j\tau}$  phase; so the identical expansion applies with  $y_* = \frac{2}{\tau} e^{\tau^2/36}$ ,  $c_2 = -\frac{1}{12}\tau^2$  and the *universal*  $c_3 = -\frac{1}{9}\tau^2$ ,  $c_5 = \frac{1}{450}\tau^4$ . Solving the two phase conditions perturbatively in the common variable  $w = \sqrt{2/\tau}$  near each grid point  $(M + \frac{1}{2})\pi$  gives

$$\chi_{\text{pole}} = -\frac{\sqrt{2}}{36} \sqrt{\tau} + \frac{41\sqrt{2}}{2400} \tau^{3/2} + O(\tau^2), \quad \chi_{\text{zero}} = -\frac{\sqrt{2}}{36} \sqrt{\tau} - \frac{259\sqrt{2}}{2400} \tau^{3/2} + O(\tau^2).$$

The  $\sqrt{\tau}$  phases agree *identically*; both are produced by the universal cubic curvature  $c_3$ , and the constant  $\sqrt{2}/36$  is that of Remark 9; this is precisely why the travel poles track the zeros of  $Y_3(1)$ . The  $\tau^{3/2}$  terms differ by exactly

$$\delta w = \chi_{\text{pole}} - \chi_{\text{zero}} = \frac{\sqrt{2}}{8} \tau^{3/2} + O(\tau^2), \quad \text{whence} \quad \left| \sum_k d_k \right| = \frac{3\tau}{2} \delta w (1 + O(\sqrt{\tau})) = \frac{3}{8\sqrt{2}} \tau^{5/2} (1 + O(\sqrt{\tau})),$$

and therefore  $|P_{12}|/\tau^{3/2} \rightarrow 1/(4\sqrt{2}) = 0.17678 < 1/\sqrt{2}$ : *the numerator gate, derived, with its fourfold margin*. Verification: the two phase series match the exact pole and zero locations to seven digits at  $m = 11, 17, 23, 29$ , and  $\delta w / \tau^{3/2} = 0.17699, 0.17687, 0.17683, 0.17681 \rightarrow \sqrt{2}/8 = 0.176777$  (`phase_match_verify.py`); the pole-side series independently reproduces the McMahon-type coefficients of the travel-pole expansion. The remainder discipline throughout is that of Remark 9: every coefficient extracted elementarily, the  $O(\cdot)$  terms controlled by the same two-saddle argument.

#### 4.1 The travel-pole equation as a $q$ -trigonometric identity

The two series that govern the poles admit an exact identification with the  $q$ -cosine of Koornwinder and Swarttouw [30],  $\cos(z; q^2) := {}_1\phi_1(0; q; q^2, q^2 z^2) = \sum_{j \geq 0} (-1)^j q^{j^2+j} z^{2j} / (q; q)_{2j}$ .

**Proposition 17.** *Let  $z_0 = \sqrt{2(1-q)}$  and  $Z = z_0/\sqrt{q}$ . Then*

$$1 - S_1 = S_e = \cos(z_0; q^2), \quad 1 - \Sigma_1 = \cos(Z; q^2).$$

*In particular the travel poles  $q_m$  are exactly the solutions of  $\cos(Z; q^2) = 0$ , and  $S_e$  is the same function evaluated one half  $q$ -step below its own zero.*

*Proof.* Collapse the  $k$ -recursion (2) on its two odd ladders, the travel one carrying  $\Sigma_1$  and the bulk one carrying  $S_1$ . Write  $1 - q^m = 2e^{-m\tau/2} \sinh(m\tau/2)$  in the sinh-forms of the two series (§4): the products collapse to  $1 - \Sigma_1 = \sum_j (-1)^j (2(1 - q))^j q^{j^2} / (q; q)_{2j}$  and  $1 - S_1 = \sum_j (-1)^j (2(1 - q))^j q^{j^2+j} / (q; q)_{2j}$ , the second being  $S_e$  as defined in (1). Comparing with the series for  $\cos(z; q^2)$  term by term forces  $q^2 z^2 = 2(1 - q)q$ , i.e.  $z = Z$ , in the first case and  $q^2 z^2 = 2(1 - q)q^2$ , i.e.  $z = z_0$ , in the second; both series converge for  $|q| < 1$ , the factor  $q^{j^2}$  dominating. Both collapses are recomputed as identities of integer power series, to degree 1160 in `blocks_growth.py` and at two truncation degrees in `f2_pole_route.py`. The per-term identity is machine-checked in Lean 4 (`GramCasoratian.lean`, theorem `sigma1_term`; axioms `propext`, `Classical.choice`, `Quot.sound`, no sorry).  $\square$

*The Hahn–Exton normalisation.* Write  $J_\nu^{(3)}(x; q^2) = x^\nu \frac{(q^{2\nu+2}; q^2)_\infty}{(q^2; q^2)_\infty} {}_1\phi_1(0; q^{2\nu+2}; q^2, q^2 x^2)$  for the Hahn–Exton  $q$ -Bessel function, in the normalisation of [29, eq. (2.4)] with base  $q^2$ , and put

$$c_q := \frac{(q^2; q^2)_\infty}{(q; q^2)_\infty} > 0.$$

Since  $(q^2; q^2)_k (q; q^2)_k = (q; q)_{2k}$  and  $(q^2; q^2)_k (q^3; q^2)_k = (q; q)_{2k+1} / (1 - q)$ , comparing coefficients gives, for  $x > 0$ ,

$$\cos(x; q^2) = c_q x^{1/2} J_{-1/2}^{(3)}(x; q^2), \quad \sin(x; q^2) = c_q x^{1/2} J_{1/2}^{(3)}(x; q^2).$$

So the travel poles are the zeros in  $(0, 1)$  of  $q \mapsto J_{-1/2}^{(3)}(Z(q); q^2)$  (Corollary 21); Proposition 118(i) records the same identity in the variable of Appendix C.

*Remark 18.* The identification is consistent with the pole data: at all 58 tabulated poles the computed value of  $\cos(z_0; q^2)$  agrees with the stored  $S_e$  to at least thirty digits, and  $|S_e|/\sqrt{\tau}$  lies in  $[0.698, 0.708]$ , approaching  $1/\sqrt{2}$  (`qtrig_verify.py`). The half-step structure accounts for the size of the gate quantity:  $Z - z_0 = z_0(1 - \sqrt{q})/\sqrt{q} = \tau^{3/2}/\sqrt{2} + O(\tau^{5/2})$ , the pole condition annihilates  $\cos(\cdot; q^2)$  at  $Z$ , and the  $q$ -derivative rule  $(1 - q)D_{q,z} \sin(z; q^2) = \cos(z; q^2)$  of [30] gives the function slope  $1/(1 - q) \sim 1/\tau$  near the zero, so the sampled value is of size  $(\tau^{3/2}/\sqrt{2})(1/\tau) = \sqrt{\tau}/2$ , the observed constant. Turning this observation into a proved lower bound on  $|S_e(q_m)|$  independent of Remark 9 requires uniform control of  $\cos'(\cdot; q^2)$  between  $z_0$  and  $Z$  as  $q \rightarrow 1^-$ . That control is supplied by Appendix A (Lemma 98 bounds  $\max |f''|$  on  $[qZ, Z]$ , and Proposition 94 converts it into the lower bound), independently of Remark 9.

## 4.2 The pole equation as a Fredholm determinant

The  $q$ -cosine of Proposition 17 is the Fredholm determinant of the travel operator. Throughout this subsection  $0 < q < 1$ ,  $D_e = \text{diag}(2q^{1+s})_{s \geq 0}$  and  $K = [q^{\max(s, s')}]_{s, s' \geq 0}$ , so that  $\mathcal{T} = D_e K$  is the travel operator of Definition 33. Put

$$\mathbf{S} := D_e^{1/2} K D_e^{1/2}, \quad \mathbf{S}[s, s'] = 2q^{1+(s+s')/2+\max(s, s')}, \quad F(\lambda, q) := \sum_{k \geq 0} \frac{(-2\lambda(1 - q))^k q^{k^2}}{(q; q)_{2k}} \quad (\lambda \in \mathbb{C}).$$

The spectral parameter  $\lambda$  is not to be confused with the junction vectors  $\lambda^{(y)}$  of §5.8.

**Theorem 19** (the travel determinant). *Let  $0 < q < 1$  and  $Z = \sqrt{2(1 - q)/q}$ .*

(i)  *$\mathbf{S}$  is a positive definite trace-class operator on  $\ell^2(\mathbb{Z}_{\geq 0})$ , with  $\text{tr } \mathbf{S} = 2q/(1 - q^2)$ .*

(ii)  $\mathcal{T}$  and  $\mathbf{S}$  have the same principal minors,  $\det \mathcal{T}[I, I] = \det \mathbf{S}[I, I]$  for every finite  $I \subset \mathbb{Z}_{\geq 0}$ , and the same non-zero eigenvalues, the eigenvectors corresponding by  $R = D_e^{1/2} \phi$ : if  $\mathbf{S} \phi = \kappa \phi$  with  $\phi \in \ell^2$  and  $\kappa \neq 0$ , then  $R \in \ell^1$  and  $\mathcal{T}R = \kappa R$ ; conversely, if  $R \in \ell^1$  and  $\mathcal{T}R = \kappa R$  with  $\kappa \neq 0$ , then  $\phi = D_e^{-1/2} R \in \ell^2$  and  $\mathbf{S} \phi = \kappa \phi$ .

(iii) For every  $\lambda \in \mathbb{C}$ , with absolutely convergent series,

$$\det(I - \lambda \mathbf{S}) = \sum_{k \geq 0} (-\lambda)^k \sum_{|I|=k} \det \mathbf{S}[I, I] = F(\lambda, q),$$

$$F(\lambda, q) = \cos(\sqrt{\lambda} Z; q^2) = c_q(\sqrt{\lambda} Z)^{1/2} J_{-1/2}^{(3)}(\sqrt{\lambda} Z; q^2),$$

the last two expressions being even in  $\sqrt{\lambda}$ . We write  $\det(I - \lambda \mathcal{T})$  for the same Fredholm series formed with the minors of  $\mathcal{T}$ ; by (ii) it is the same function.

(iv) The eigenvalues of  $\mathbf{S}$  are  $1/\lambda_k(q)$ ,  $k \geq 1$ , with  $0 < \lambda_1(q) < \lambda_2(q) < \dots \rightarrow \infty$ , all simple;  $\sum_k 1/\lambda_k(q) = 2q/(1 - q^2)$  and  $F(\lambda, q) = \prod_{k \geq 1} (1 - \lambda/\lambda_k(q))$ .

(v)  $1 - \Sigma_1 = F(1, q) = \det(I - \mathcal{T})$ , and  $S_e = F(q, q) = \det(I - q\mathcal{T}) = \det(I - M_0)$ : the bulk operator  $M_0$  of Definition 33 is  $q\mathcal{T}$  after the index shift  $b = s + 1$ .

*Proof.* (i) For  $x \in \ell^2$  put  $X = D_e^{1/2} x$ , which lies in  $\ell^1$  by Cauchy–Schwarz. Since  $q^{\max(s, s')} = (1 - q) \sum_{t \geq \max(s, s')} q^t$ ,

$$\langle x, \mathbf{S}x \rangle = \langle X, KX \rangle = (1 - q) \sum_{t \geq 0} q^t \left| \sum_{s \leq t} X_s \right|^2 \geq 0,$$

the double sums converging absolutely, with equality only if every partial sum of  $X$  vanishes, that is  $X = 0$ , that is  $x = 0$ .  $\mathbf{S}$  is Hilbert–Schmidt, hence bounded, and it is positive definite; a positive operator with summable diagonal is trace class, with trace  $\sum_s 2q^{1+s}q^s = 2q/(1 - q^2)$ .

(ii)  $\mathcal{T}[I, I] = D_e[I, I] K[I, I]$  and  $\mathbf{S}[I, I] = D_e[I, I]^{1/2} K[I, I] D_e[I, I]^{1/2}$  with  $D_e[I, I]$  diagonal, so the two determinants agree. If  $\mathbf{S} \phi = \kappa \phi$ , then  $R = D_e^{1/2} \phi \in \ell^1$  and  $\mathcal{T}R = D_e^{1/2} \mathbf{S} \phi = \kappa R$ . Conversely,  $\mathcal{T}R = \kappa R$  gives  $R = \kappa^{-1} D_e(KR)$  with  $|(KR)_s| \leq \|R\|_{\ell^1}$ , so  $\phi = D_e^{-1/2} R = \kappa^{-1} D_e^{1/2} KR \in \ell^2$  and  $\mathbf{S} \phi = D_e^{1/2} KR = \kappa \phi$ . The operators are not boundedly similar,  $D_e^{-1/2}$  being unbounded; only (ii) is used.

(iii) For trace-class  $A$ ,  $\det(I - \lambda A) = \sum_k (-\lambda)^k \text{tr } \Lambda^k(A)$  [51, Ch. 3] (the von Koch form; compare Fredholm’s formula for continuous kernels, [51, Thm. 3.10]), and computing the trace of  $\Lambda^k(\mathbf{S})$  in the orthonormal basis  $e_{s_1} \wedge \dots \wedge e_{s_k}$ ,  $s_1 < \dots < s_k$ , whose diagonal entries are the principal minors, gives the first equality. The principal minors of the positive  $\mathbf{S}$  are  $\geq 0$ , so the inner sums converge absolutely, to the values computed next, and the outer series then converges absolutely for every  $\lambda$ . For  $I = \{s_1 < \dots < s_k\}$  the matrix  $K[I, I]$  is  $(a_{\max(i, j)})$  with  $a_i = q^{s_i}$  decreasing; subtracting row  $i + 1$  from row  $i$  for  $i < k$  leaves a lower triangular matrix with diagonal  $a_1 - a_2, \dots, a_{k-1} - a_k, a_k$ , so  $\det K[I, I] = q^{s_k} \prod_{i < k} (q^{s_i} - q^{s_{i+1}})$  and, with the gaps  $g_i = s_{i+1} - s_i \geq 1$ ,

$$\det \mathbf{S}[I, I] = 2^k q^k q^{2(s_1 + \dots + s_k)} \prod_{i < k} (1 - q^{g_i}).$$

Since  $s_1 + \dots + s_k = ks_1 + \sum_{j < k} (k - j)g_j$ , the sum over  $s_1 \geq 0$  and  $g_1, \dots, g_{k-1} \geq 1$  factorises; with  $\sum_{g \geq 1} q^{2rg} (1 - q^g) = q^{2r} (1 - q) / ((1 - q^{2r})(1 - q^{2r+1}))$ ,

$$\sum_{|I|=k} \det \mathbf{S}[I, I] = \frac{(2q)^k}{1 - q^{2k}} \prod_{r=1}^{k-1} \frac{q^{2r} (1 - q)}{(1 - q^{2r})(1 - q^{2r+1})} = \frac{(2(1 - q))^k q^{k^2}}{(q; q)_{2k}}.$$

Hence  $\det(I - \lambda \mathbf{S}) = F(\lambda, q)$ ; the case  $k = 1$  is the trace. Finally  $(2\lambda(1 - q))^k q^{k^2} = q^{k^2+k}(\sqrt{\lambda} Z)^{2k}$ , since  $Z^2 = 2(1 - q)/q$ , so  $F(\lambda, q) = \cos(\sqrt{\lambda} Z; q^2)$ , and the Hahn–Exton form is the normalisation after Proposition 17.

(iv) For positive trace-class  $\mathbf{S}$  the determinant is the product  $\prod_k (1 - \lambda \kappa_k)$  over its eigenvalues  $\kappa_k > 0$  counted with multiplicity [51, Ch. 3], so the  $1/\kappa_k$  are the zeros of  $F(\cdot, q)$ , the multiplicity of  $\kappa_k$  being the order of the zero. By (iii) these are the  $\lambda = x^2/Z^2$  with  $x > 0$  a zero of  $\cos(x; q^2) = c_q x^{1/2} J_{-1/2}^{(3)}(x; q^2)$  ( $\cos(\cdot; q^2)$  is even with  $\cos(0; q^2) = 1$ ). For  $\nu > -1$  the zeros of  $J_\nu^{(3)}(\cdot; q^2)$  are real and simple and form an infinite sequence [29, Cor. 3.2, Lemma 3.3, Thm. 3.4]; here  $\nu = -\frac{1}{2}$  with base  $q^2$ . So the eigenvalues are simple and infinitely many, and the sum of the  $1/\lambda_k$  is the trace.

(v) At  $\lambda = 1$ ,  $F(1, q) = \sum_k (-1)^k (2(1 - q))^k q^{k^2} / (q; q)_{2k} = 1 - \Sigma_1$  (Proposition 17); at  $\lambda = q$ ,  $F(q, q) = \sum_k (-2(1 - q))^k q^{k^2+k} / (q; q)_{2k} = S_e$  by (1). And  $M_0[s + 1, s' + 1] = 2q^{s+1} q^{\max(s, s')+1} = q \mathcal{T}[s, s']$ , so  $M_0$  has the principal minors of  $q \mathcal{T}$ .  $\square$

The identity  $\det(I - \lambda \mathbf{S}) = F(\lambda, q)$  was checked in double precision against truncated matrices at  $q = 0.3, 0.6, 0.8$  and four values of  $\lambda$  (one of them non-real), to within  $3.2 \cdot 10^{-15}$ , and the gap sum against a brute-force sum of minors for  $k \leq 3$  at  $q = 0.3$ .

*Remark 20* (a  $q$ -string). With the Jackson integral  $\int_0^1 f d_q t = (1 - q) \sum_{s \geq 0} q^s f(q^s)$ , let  $K_q f(x) = \int_0^1 \min(x, t) f(t) d_q t$  on the lattice  $x \in \{q^s\}$ . At  $x = q^s$ ,  $(K D_e u)_s = \sum_{s'} q^{\max(s, s')} 2q^{1+s'} u_{s'} = \frac{2q}{1-q} (K_q u)(q^s)$ , that is,

$$\mathcal{T}^\top = K D_e = \frac{2q}{1-q} K_q,$$

so up to the transposition, which changes neither the principal minors nor the eigenvalues,  $\mathcal{T} = \frac{2q}{1-q} K_q$ . The kernel  $\min(x, t)$  is the Green function of  $-d^2/dx^2$  on  $[0, 1]$  with a Dirichlet end at 0 and a Neumann end at 1:  $\mathcal{T}$  is the compliance operator of a  $q$ -string with uniform Jackson mass, fixed at 0 and free at 1, and  $u = KR$  is its displacement ( $u_s \rightarrow 0$  as  $x = q^s \rightarrow 0$ ). As  $q \rightarrow 1$  the kernel tends to the classical one, with eigenvalues  $4/((2k - 1)^2 \pi^2)$ ; this is the heuristic behind  $\lambda_k(q) \sim (1 - q)(2k - 1)^2 \pi^2 / 8$ , proved for fixed  $k$  in Proposition 22.

**Corollary 21** (the travel poles are Hahn–Exton zeros). *A point  $q \in (0, 1)$  is a travel pole if and only if*

$$J_{-1/2}^{(3)}(Z(q); q^2) = 0, \quad Z(q) = \sqrt{2(1 - q)/q},$$

*if and only if 1 is an eigenvalue of  $\mathcal{T}(q)$ , that is  $\lambda_k(q) = 1$  for some  $k$ . That  $k$  is unique, the eigenvalue is simple, its eigenvector is the travel null vector  $R$  of Proposition 68, and  $\phi = D_e^{-1/2} R$  is the corresponding eigenvector of  $\mathbf{S}$ .*

*Proof.* Theorem 19(iii)–(v), with  $c_q > 0$  and  $Z > 0$ , and Proposition 17; uniqueness of  $k$  is the simplicity in Theorem 19(iv), and the eigenvector is Theorem 19(ii) with Lemma 43.  $\square$

The zero theory used is that of Koelink–Swarttouw [29], at  $\nu = -\frac{1}{2}$  and base  $q^2$ , where its hypothesis  $\nu > -1$  holds. The eigenvalues also agree with the Jacobi-matrix description of the zeros of  $J_\nu^{(3)}$  by Štampach and Šťovíček [53], in its Friedrichs extension; the positive form of  $\mathbf{S}$  is the Friedrichs realisation, on which  $u_s = (KR)_s \rightarrow 0$ . At  $q_1, \dots, q_{10}$ , located at 90-digit precision,  $|J_{-1/2}^{(3)}(Z; q^2)| \leq 2 \cdot 10^{-79}$ , and exactly  $m - 1$  zeros of  $\cos(\cdot; q_m^2)$  lie below  $Z(q_m)$  (a numerical check, not a certificate).

**Proposition 22** (infinitely many travel poles, without  $T_2$ ). (a)  $\lambda_1(q) > (1 - q^2)/(2q)$  on  $(0, 1)$ .

In particular every  $\lambda_k(q) > 1$  for  $q \leq \sqrt{2} - 1$ , and there is no travel pole in  $(0, \sqrt{2} - 1]$ .

(b) For each fixed  $k$ ,  $\lambda_k(q)/(1 - q) \rightarrow (2k - 1)^2 \pi^2/8$  as  $q \rightarrow 1^-$ .

(c) Each  $\lambda_k$  is continuous on  $(0, 1)$ , and there is  $c_k \in (\sqrt{2} - 1, 1)$  with  $\lambda_k(c_k) = 1$ . The  $c_k$  are pairwise distinct travel poles; in particular there are infinitely many travel poles.

None of this uses  $T_2$ , Lemma 10, or a single-crossing property.

*Proof.* (a) The largest eigenvalue of the positive trace-class  $\mathbf{S}$  is strictly less than its trace, since  $\mathbf{S}$  has infinitely many positive eigenvalues:  $1/\lambda_1 < 2q/(1 - q^2)$ . And  $(1 - q^2)/(2q) \geq 1$  exactly for  $q \leq \sqrt{2} - 1$ . At such  $q$  no  $\lambda_k$  equals 1, so  $1 - \Sigma_1 = F(1, q) \neq 0$  by Theorem 19(iv).

(b) Put  $\varepsilon = 1 - q$  and  $\lambda = \varepsilon y$ . The  $k$ -th term of  $F(\varepsilon y, q)$  is  $t_k(y, \varepsilon) = (-1)^k (2y\varepsilon^2)^k q^{k^2} / (q; q)_{2k}$ . By the arithmetic–geometric mean inequality,  $1 - q^n = \varepsilon \sum_{i < n} q^i \geq n\varepsilon q^{(n-1)/2}$ , so  $(q; q)_{2k} \geq \varepsilon^{2k} (2k)! q^{k(2k-1)/2}$  and

$$|t_k(y, \varepsilon)| \leq \frac{(2|y|)^k q^{k/2}}{(2k)!} \leq \frac{(2|y|)^k}{(2k - 1)!!},$$

uniformly in  $\varepsilon \in (0, 1)$ . Each  $t_k(y, \varepsilon)$  tends to  $(-1)^k (2y)^k / (2k)!$  as  $\varepsilon \rightarrow 0$ , because  $q^{k^2} \rightarrow 1$  and  $(q; q)_{2k} / \varepsilon^{2k} \rightarrow (2k)!$ . The domination is summable locally uniformly in  $y \in \mathbb{C}$ , so  $F(\varepsilon y, 1 - \varepsilon) \rightarrow \cos \sqrt{2y}$  uniformly on compact sets. The limit has the simple zeros  $y_j = (2j - 1)^2 \pi^2/8$ , and the zeros of  $y \mapsto F(\varepsilon y, 1 - \varepsilon)$  are the  $\lambda_j(q)/\varepsilon$ , positive and ordered. By Hurwitz’s theorem, for  $\varepsilon$  small each disc  $|y - y_j| < \delta$ ,  $j \leq k$ , contains exactly one of them and  $|y| \leq y_k + \delta$  contains no other; hence  $\lambda_j(q)/\varepsilon \rightarrow y_j$  for  $j \leq k$ .

(c)  $q \mapsto \mathbf{S}(q)$  is continuous in trace norm, hence in operator norm, on  $(0, 1)$ : its entries are continuous and  $\sum_{s, s'} |\mathbf{S}[s, s']|$  converges locally uniformly. The ordered eigenvalues of positive compact operators are 1-Lipschitz in operator norm (min–max), so each  $1/\lambda_k$ , and so each  $\lambda_k$ , is continuous. By (a)  $\lambda_k > 1$  on  $(0, \sqrt{2} - 1]$ , and by (b)  $\lambda_k(q) \rightarrow 0$  as  $q \rightarrow 1$ , so the intermediate value theorem gives  $c_k$ ; it is a travel pole by Corollary 21. If  $c_j = c_k$  with  $j \neq k$ , then  $\lambda_j(c_k) = \lambda_k(c_k) = 1$ , against simplicity (Theorem 19(iv)).

The same argument with  $\lambda_k(q)/q$  in place of  $\lambda_k(q)$  gives Lemma 13 without  $T_2$  too:  $S_e = F(q, q) = \prod_k (1 - q/\lambda_k(q))$  vanishes exactly where some  $\lambda_k(q) = q$ ;  $\lambda_k(q)/q > (1 - q^2)/(2q^2) > 1$  for  $q < 1/\sqrt{3}$  by (a);  $\lambda_k(q)/q \rightarrow 0$  by (b); and two branches cannot both equal  $q$  at the same point.  $\square$

### 4.3 The numerators, and why they do not vanish at the poles

Proposition 17 identifies the two *denominators*. The two numerators are the companion  $q$ -sine at the same two arguments, and that is what settles their non-vanishing at the travel poles. Throughout,  $\mathfrak{s}(u) = q^{-1/2} \sin(q^{1/2}u; q^2)$  is the half-step  $q$ -sine of Lemma 101,  $\sin(z; q^2) = \sum_{j \geq 0} (-1)^j q^{j^2+j} z^{2j+1} / (q; q)_{2j+1}$ .

**Lemma 23** (the numerators in closed form). *Identically in  $0 < q < 1$ ,*

$$\Sigma_0 = \frac{2q}{Z} \mathfrak{s}(Z), \quad S_0 = \frac{2q}{z_0} \mathfrak{s}(z_0).$$

With Proposition 17 this identifies all four blocks as one  $q$ -trigonometric pair sampled at two arguments a half  $q$ -step apart:  $(\Sigma_0, 1 - \Sigma_1)$  is  $(\frac{2q}{u} \mathfrak{s}(u), \cos(u; q^2))$  at  $u = Z$ , and  $(S_0, S_e)$  is the same pair at  $u = z_0 = \sqrt{q} Z$ . In particular  $S_0 = \frac{2q}{1-q} S_o$  with  $S_o$  the odd bulk block of Lemma 101, which is the second identity of the dictionary (33) read as a statement about the explicit series, and is therefore independent of the transfer model.

*Proof.* Collapse the  $k$ -recursion (2) on the even ladders, exactly as Proposition 17 does on the odd ones. From  $\alpha_k = 2q^{k+1}/(1 - q^{k+1})$ ,

$$|\gamma_k| = \alpha_k - \alpha_{k+1} = \frac{2(1-q)q^{k+1}}{(1-q^{k+1})(1-q^{k+2})}, \quad |C_k| = \frac{2(1-q)q^{k+2}}{(1-q^{k+1})(1-q^{k+2})},$$

$\gamma_k$  and  $C_k$  themselves being negative on  $(0, 1)$ , so both even ladders alternate. For the bulk one,  $\sum_{i < j} (2i+1) = j^2$  and the denominators telescope to  $(q; q)_{2j}$ , giving  $\prod_{i < j} |\gamma_{2i}| = [2(1-q)]^j q^{j^2} / (q; q)_{2j}$  and hence

$$S_0 = \sum_{j \geq 0} (-1)^j \alpha_{2j} \prod_{i < j} |\gamma_{2i}| = 2 \sum_{j \geq 0} (-1)^j \frac{[2(1-q)]^j q^{(j+1)^2}}{(q; q)_{2j+1}}.$$

Write  $q^{(j+1)^2} = q q^{j^2+j} q^j$  and  $[2(1-q)]^j q^j = (q^{1/2} z_0)^{2j}$ : the sum is  $(2q/(q^{1/2} z_0)) \sin(q^{1/2} z_0; q^2) = (2q/z_0) \mathfrak{s}(z_0)$ . For the travel ladder,  $\sum_{i < j} (2i+2) = j^2 + j$  and the denominators telescope to  $(q; q)_{2j}$  again, so  $\prod_{i < j} |C_{2i}| = [2(1-q)]^j q^{j^2+j} / (q; q)_{2j}$  and, with  $A_{2j} = 2q/(1 - q^{2j+1})$ ,

$$\Sigma_0 = 2q \sum_{j \geq 0} (-1)^j \frac{[2(1-q)]^j q^{j^2+j}}{(q; q)_{2j+1}} = \frac{2q}{z_0} \sin(z_0; q^2) = \frac{2q^{3/2}}{z_0} \mathfrak{s}(Z) = \frac{2q}{Z} \mathfrak{s}(Z),$$

using  $[2(1-q)]^j = z_0^{2j}$  and  $z_0 = q^{1/2} Z$ . The  $S_o$  statement is Lemma 101's  $S_o = \frac{z_0}{2} \mathfrak{s}(z_0)$  combined with  $z_0^2 = 2(1-q)$ . Both collapses are the exponent bookkeeping machine-checked in `QTrigIdentities.lean` and are verified numerically off the poles, hence as identities, in `qnumerator_certificate.py`.  $\square$

**Lemma 24** (half-step rules and the pole invariant). *Identically in  $0 < q < 1$ , with  $c(u) = \cos(u; q^2)$ :*

- (i)  $c(qu) = c(u) + q^{3/2} u \mathfrak{s}(q^{1/2} u)$ ;
- (ii)  $\mathfrak{s}(qu) = \mathfrak{s}(u) - u c(q^{1/2} u)$ ;
- (iii)  $c(u) c(q^{1/2} u) + q^{3/2} \mathfrak{s}(u) \mathfrak{s}(q^{1/2} u) = 1$ .

*Consequently, at every travel pole, where  $c(Z) = 0$ :*

$$q^{3/2} \mathfrak{s}(z_0) \mathfrak{s}(Z) = 1,$$

*so neither  $\mathfrak{s}(Z)$  nor  $\mathfrak{s}(z_0)$  vanishes there.*

*Proof.* (i) and (ii) are termwise. From  $(q; q)_{2j} = (q; q)_{2j-1} (1 - q^{2j})$ ,  $c(u) - c(qu) = \sum_{j \geq 1} (-1)^j q^{j^2+j} u^{2j} / (q; q)_{2j-1}$ ; shifting  $j \mapsto j+1$  and factoring out  $-qu$  leaves  $-qu \sin(qu; q^2) = -q^{3/2} u \mathfrak{s}(q^{1/2} u)$ , which is (i). From  $(q; q)_{2j+1} = (q; q)_{2j} (1 - q^{2j+1})$ ,  $\mathfrak{s}(u) - \mathfrak{s}(qu) = q^{-1/2} \sum_{j \geq 0} (-1)^j q^{j^2+j} (q^{1/2} u)^{2j+1} / (q; q)_{2j}$ ; factoring out  $q^{1/2} u$  leaves  $u c(q^{1/2} u)$ , which is (ii). This is the exponent matching machine-checked as `cshift_exponent` and `sshift_exponent`. For (iii), substitute (i) and (ii) into the Hahn–Exton  $q$ -Wronskian  $\mathfrak{s}(u) c(qu) - c(u) \mathfrak{s}(qu) = u$  ([56, eq. (4.3.8)]; [41, eq. (4.21)]): the  $\mathfrak{s}(u) c(u)$  terms cancel and what is left is  $u [c(u) c(q^{1/2} u) + q^{3/2} \mathfrak{s}(u) \mathfrak{s}(q^{1/2} u)]$ , so the  $q$ -Wronskian and (iii) are equivalent for  $u \neq 0$ . Identity (iii) is the  $q$ -Pythagorean identity of [30]. Taking  $u = Z$  and  $c(Z) = 0$  gives the pole invariant, which is the step already used inside the proof of Proposition 102.  $\square$

**Proposition 25** (the numerators do not vanish at the travel poles). *At every travel pole  $q_m$ ,*

$$\Sigma_0(q_m) S_0(q_m) = \frac{2q_m}{1 - q_m} \neq 0. \quad (14)$$

*In particular  $\Sigma_0(q_m) \neq 0$  and  $S_0(q_m) \neq 0$ , at every travel pole, with no asymptotic input and no exceptional set. Moreover  $|\Sigma_0(q_m)| |S_0(q_m)| \geq q_m w_m^2$ , so at least one of the two numerators exceeds  $\sqrt{q_m} w_m$  in modulus.*

*Proof.* By Lemma 23,  $\Sigma_0 S_0 = (4q^2/(z_0 Z))\mathfrak{s}(Z)\mathfrak{s}(z_0)$ , and by Lemma 24 the product  $\mathfrak{s}(Z)\mathfrak{s}(z_0)$  equals  $q^{-3/2}$  at a travel pole. Since  $z_0 Z = z_0^2/\sqrt{q} = 2(1-q)/\sqrt{q}$ , the two powers of  $q$  combine to  $4q^2 \cdot q^{-3/2} \cdot \sqrt{q}/(2(1-q)) = 2q/(1-q)$ . The size statement follows from  $1-q \leq \tau = 2/w^2$ .  $\square$

*Remark 26.* The two numerators are individually of size  $w_m$ :  $|\Sigma_0(q_m)|/w_m$  and  $|S_0(q_m)|/w_m$  rise monotonically to 1 over the first 32 travel poles, from 0.731 and 0.893 at the dominant pole  $q_1 = 0.449453\dots$  to 0.99992 and 0.99997 at the last, with  $\Sigma_0$  strictly alternating in sign (`qnumerator_certificate.py`, `mpmath` at two precisions, worst deviation in (14)  $4.5 \cdot 10^{-28}$ ; the dominant pole is included, it lying below the extreme-phase ladder and being bracketed separately). This matches the asymptotic  $\Sigma_0 \sim w \sin w$ , the amplitude companion of  $1 - \Sigma_1 \sim 1 - \cos w$ ; but (14) is an identity and the asymptotic is not used anywhere.

#### 4.4 Spectral consequences at the travel poles

Theorem 19 turns questions about the travel poles into questions about the eigenvalue branches  $\lambda_k(q)$ . This subsection records what that gives, and at what standing. Write  $\mathbf{B}(q) := 1 - \Sigma_1(q) = \det(I - \mathcal{T}(q))$ .

**Lemma 27** (Hellmann–Feynman). *The map  $q \mapsto \mathbf{S}(q)$  is real-analytic from  $(0, 1)$  into the trace-class operators, and each  $\lambda_k$  is real-analytic on  $(0, 1)$ . Let  $\mathcal{T}R = \lambda_k(q)^{-1}R$  with  $R \neq 0$  real (Theorem 19(ii)), and put  $u = KR$ . Then*

$$\frac{\lambda'_k(q)}{\lambda_k(q)} = -\frac{\mathfrak{e}_1 + \mathfrak{e}_2}{q} - \frac{1 - 2q}{q(1 - q)}, \quad \mathfrak{e}_1 = \frac{\sum_s s q^{-s} (u_{s+1} - u_s)^2}{\sum_s q^{-s} (u_{s+1} - u_s)^2}, \quad \mathfrak{e}_2 = \frac{\sum_s s q^s u_s^2}{\sum_s q^s u_s^2}.$$

In particular  $\lambda'_k(q) < 0$  if and only if  $\mathfrak{e}_1 + \mathfrak{e}_2 > (2q - 1)/(1 - q)$ .

*Proof.* With the principal branch of  $q^{1/2}$ , the entries  $2q^{1+(s+s')/2+\max(s,s')}$  are holomorphic on  $\{|q| < 1, \operatorname{Re} q > 0\}$  and the sum of their moduli converges locally uniformly there, so  $\mathbf{S}$  is holomorphic in trace norm, and self-adjoint for real  $q$ . Its non-zero eigenvalues are isolated and simple (Theorem 19(iv)), so by analytic perturbation theory for a bounded holomorphic self-adjoint family [28, Ch. II and Ch. VII] each eigenvalue  $\kappa(q) = 1/\lambda_k(q)$  and its eigenvector  $\phi(q)$  are real-analytic, and  $\kappa' = \langle \phi, \mathbf{S}'\phi \rangle / \langle \phi, \phi \rangle$ . No  $q$ -dependent domain enters:  $\mathbf{S}(q)$  is a bounded family.

It remains to evaluate  $\langle \phi, \mathbf{S}'\phi \rangle$ . Put  $R = D_e^{1/2}\phi$  and  $u = KR$ , so that  $D_e u = \mathcal{T}R = \kappa R$  and  $\phi_s^2 = \kappa^{-2} 2q^{1+s} u_s^2$ ; put  $N = \sum_s q^s u_s^2$  and  $E = \sum_s q^{-s} (u_{s+1} - u_s)^2$ . Since  $q \partial_q \mathbf{S}[s, s'] = (1 + \frac{s+s'}{2} + \max(s, s')) \mathbf{S}[s, s']$ ,

$$q \langle \phi, \mathbf{S}'\phi \rangle = \kappa \langle \phi, \phi \rangle + \kappa \sum_s s \phi_s^2 + \sum_{s, s'} R_s R_{s'} \max(s, s') q^{\max(s, s')},$$

and the first two terms are  $(2qN/\kappa)(1 + \mathfrak{e}_2)$ . For the third apply  $q \partial_q$  to  $q^{\max(s, s')} = (1 - q) \sum_{t \geq \max(s, s')} q^t$ : with  $\rho_t = \sum_{s \leq t} R_s$ , bounded by  $\|R\|_{\ell^1}$ , it equals  $\sum_t (t(1 - q) - q) q^t \rho_t^2$ , all sums converging absolutely. Since  $u_{t+1} - u_t = \sum_{s'} (q^{\max(t+1, s')} - q^{\max(t, s')}) R_{s'} = -(1 - q) q^t \rho_t$ , one has  $q^t \rho_t^2 = q^{-t} (u_{t+1} - u_t)^2 / (1 - q)^2$ , so the third term is  $E((1 - q)\mathfrak{e}_1 - q)/(1 - q)^2$ . Finally  $\langle R, KR \rangle = (1 - q) \sum_t q^t \rho_t^2 = E/(1 - q)$  and  $\langle R, KR \rangle = \sum_s R_s u_s = 2qN/\kappa$ , so  $E = 2q(1 - q)N/\kappa$ . Collecting,  $q \langle \phi, \mathbf{S}'\phi \rangle = (2qN/\kappa)(1 + \mathfrak{e}_1 + \mathfrak{e}_2 - q/(1 - q))$  and  $\langle \phi, \phi \rangle = 2qN/\kappa^2$ , so  $\kappa'/\kappa = (\mathfrak{e}_1 + \mathfrak{e}_2)/q + (1 - 2q)/(q(1 - q))$ , and  $\lambda'_k/\lambda_k = -\kappa'/\kappa$ .  $\square$

At  $q_1, \dots, q_6$  the right-hand side agrees to 12 digits with  $\lambda'_m(q_m) = -\partial_q F / \partial_\lambda F$  computed from the explicit  $F$  of Theorem 19. At a travel pole the eigenvalue 1 has eigenvector the travel null vector (Corollary 21), so the following is a statement about  $R$  alone.

**Hypothesis 28** (( $T_{\text{poles}}^*$ )). At every travel pole  $q_m > \frac{1}{2}$ , the travel null vector  $R$ , with  $u = KR$  and  $\mathfrak{e}_1, \mathfrak{e}_2$  as in Lemma 27, satisfies

$$\mathfrak{e}_1 + \mathfrak{e}_2 > \frac{2q_m - 1}{1 - q_m}.$$

At a pole  $q_m \leq \frac{1}{2}$  (only  $q_1 = 0.4494\dots$ ) the inequality holds automatically: the right-hand side is  $\leq 0$  and  $\mathfrak{e}_2 > 0$ , since  $u$  is not supported at  $s = 0$  (if it were,  $R = \kappa^{-1}D_e u$  would be a multiple of  $\delta_0$  and  $(KR)_1 = qR_0 \neq 0$ ).

**Proposition 29** (simple poles and the sign of  $\mathbf{B}'$ ; conditional on ( $T_{\text{poles}}^*$ )). Assume Hypothesis 28. Then:

- (i) each  $\lambda_k$  takes the value 1 at exactly one point  $c_k \in (0, 1)$ , and  $c_1 < c_2 < \dots$ ; the  $m$ -th travel pole is  $q_m = c_m$ , and  $\lambda_j(q_m) < 1$  for  $j < m$ ,  $\lambda_m(q_m) = 1$ ,  $\lambda_j(q_m) > 1$  for  $j > m$ ;
- (ii)  $\lambda'_m(q_m) < 0$ , and  $\mathbf{B}$  has a simple zero at  $q_m$ :  $\mathbf{B}'(q_m) = \lambda'_m(q_m) \prod_{j \neq m} (1 - 1/\lambda_j(q_m)) \neq 0$ . Hence in Proposition 68  $p = \pi = 1$ : every travel pole is a simple pole of  $(I - \mathcal{T})^{-1}$ ;
- (iii)  $\text{sign } \mathbf{B}'(q_m) = (-1)^m$ .

*Proof.* (i) At a point  $c$  with  $\lambda_k(c) = 1$ ,  $c$  is a travel pole with eigenvector  $R$  (Corollary 21), so Hypothesis 28 and the note after it, with Lemma 27, give  $\lambda'_k(c) < 0$ . A real-analytic function that is strictly decreasing at every point where it equals 1 takes the value 1 at most once: between two consecutive such points it would be  $< 1$  just after the first and  $> 1$  just before the second, and the intermediate value theorem would give a third point between them. With Proposition 22,  $\lambda_k$  takes the value 1 exactly once, at  $c_k$ , and  $\lambda_k < 1$  exactly on  $(c_k, 1)$ . Since  $\lambda_k(c_{k+1}) < \lambda_{k+1}(c_{k+1}) = 1$ ,  $c_k < c_{k+1}$ . The travel poles are the points where some  $\lambda_k$  equals 1 (Corollary 21), so their increasing enumeration is  $q_m = c_m$ , and  $\lambda_j(q_m) < 1$  exactly for  $j < m$ .

(ii) Near  $q_m$  let  $P(q)$  be the analytic rank-one eigenprojection of  $\kappa_m = 1/\lambda_m$  [28, Ch. VII]. Since  $\mathbf{S}$  commutes with  $P$ ,  $\mathbf{B} = \det(I - \mathbf{S}) = (1 - \kappa_m) \det(I - \mathbf{S}(I - P))$ , and the second factor is real-analytic near  $q_m$ , equal to  $\prod_{j \neq m} (1 - 1/\lambda_j(q_m)) \neq 0$  at  $q_m$ . So  $\mathbf{B}'(q_m) = -\kappa'_m(q_m) \prod_{j \neq m} (1 - 1/\lambda_j(q_m))$  with  $-\kappa'_m(q_m) = \lambda'_m(q_m) < 0$ , and  $p = 1$ . For  $\pi$ , on  $\ell^1$

$$(I - \mathcal{T})^{-1} = I + D_e^{1/2}(I - \mathbf{S})^{-1}D_e^{1/2}K,$$

as follows from  $(I - D_e K)D_e^{1/2} = D_e^{1/2}(I - \mathbf{S})$  and  $D_e^{1/2}KD_e K = \mathbf{S}D_e^{1/2}K$ ; here  $D_e^{1/2}K : \ell^1 \rightarrow \ell^2$  and  $D_e^{1/2} : \ell^2 \rightarrow \ell^1$  are bounded and analytic in  $q$ . Near  $q_m$ ,  $(I - \mathbf{S})^{-1} = P/(1 - \kappa_m)$  plus the reduced resolvent, which is holomorphic there because the other eigenvalues stay away from 1; and  $1 - \kappa_m$  has a simple zero. So  $(I - \mathcal{T})^{-1}$  has a pole of order at most 1, and  $\pi \geq p \geq 1$  (Proposition 68) gives  $\pi = p = 1$ .

(iii) By (i) the product in (ii) has exactly  $m - 1$  negative factors, and  $\lambda'_m(q_m) < 0$ .  $\square$

The alternation  $\text{sign } \mathbf{B}'(q_m) = (-1)^m$  is checked numerically for  $m \leq 10$  ( $\mathbf{B}'(q_1) = -2.630$ ,  $\mathbf{B}'(q_2) = 28.71$ ,  $\dots$ ,  $\mathbf{B}'(q_{10}) = 6661$ ).

**Proposition 30** (the sign of  $S_e$ , and half-interlacing). Put  $\mathbf{v} = D_e^{1/2}\mathbf{1} = (\sqrt{2q^{1+s}})_{s \geq 0} \in \ell^2$ .

- (a) Off the travel poles in  $(0, 1)$ ,  $\Sigma_0/\mathbf{B} = \langle \mathbf{v}, (I - \mathbf{S})^{-1}\mathbf{v} \rangle$ . At a travel pole  $q_m$ , with  $\lambda_k(q_m) = 1$  and  $\phi$  the normalised eigenvector,

$$\Sigma_0(q_m) = w_m \prod_{j \neq k} (1 - 1/\lambda_j(q_m)), \quad w_m = \langle \mathbf{v}, \phi \rangle^2 > 0.$$

- (b) Under Hypothesis 28,  $\text{sign } \Sigma_0(q_m) = (-1)^{m-1}$ ; hence  $\Sigma_0$  has at least one zero in each interval  $(q_m, q_{m+1})$ .
- (c) Under Hypothesis 28, if moreover  $\lambda_{m-1}(q_m) < q_m$ , then  $\text{sign } S_e(q_m) = (-1)^{m-1}$ .

*Proof.* (a) By Proposition 37,  $\Sigma_0/\mathbf{B} = \mathbf{1}^\top (I - \mathcal{T})^{-1} D_e \mathbf{1}$  off the poles, and by the resolvent identity in the proof of Proposition 29,  $(I - \mathcal{T})^{-1} D_e = D_e + D_e^{1/2} (I - \mathbf{S})^{-1} \mathbf{S} D_e^{1/2} = D_e^{1/2} (I - \mathbf{S})^{-1} D_e^{1/2}$ ; and  $\mathbf{1}^\top D_e^{1/2} g = \langle \mathbf{v}, g \rangle$  for  $g \in \ell^2$ . For the value at a pole fix  $q$  and put  $N_q(\lambda) = \det(I - \lambda \mathbf{S} + \mathbf{v} \mathbf{v}^\top) - \det(I - \lambda \mathbf{S})$ , entire in  $\lambda$ . Where  $I - \lambda \mathbf{S}$  is invertible, the rank-one identity  $\det(A + \mathbf{v} \mathbf{v}^\top) = \det A (1 + \langle \mathbf{v}, A^{-1} \mathbf{v} \rangle)$  and the expansion of  $\mathbf{S}$  in its orthonormal eigenbasis  $(\phi_j)$  (complete,  $\mathbf{S}$  being compact, self-adjoint and injective) give

$$N_q(\lambda) = \det(I - \lambda \mathbf{S}) \langle \mathbf{v}, (I - \lambda \mathbf{S})^{-1} \mathbf{v} \rangle = \sum_j w_j \prod_{i \neq j} (1 - \lambda / \lambda_i(q)), \quad w_j = \langle \mathbf{v}, \phi_j \rangle^2,$$

and since  $|\prod_{i \neq j} (1 - \lambda / \lambda_i)| \leq e^{|\lambda| \text{tr} \mathbf{S}}$  and  $\sum_j w_j = \|\mathbf{v}\|^2$ , the series converges locally uniformly and the identity holds for all  $\lambda$ . At  $\lambda = 1$  and  $q$  off the poles,  $N_q(1) = \Sigma_0(q)$ ; both sides are continuous in  $q$ ,  $N_q(1)$  as a difference of Fredholm determinants of trace-norm continuous families, so  $N_q(1) = \Sigma_0(q)$  on all of  $(0, 1)$ . At  $q = q_m$  every term  $j \neq k$  of the series contains the factor  $1 - 1/\lambda_k(q_m) = 0$ . This gives the formula, and  $w_m > 0$  because  $\Sigma_0(q_m) \neq 0$  (Proposition 25).

(b) By Proposition 29(i),  $k = m$  and exactly  $m - 1$  factors are negative. So  $\Sigma_0$  alternates in sign along the poles, and, being continuous, vanishes between consecutive ones.

(c) By Theorem 19(iv),(v) at  $\lambda = q$ ,  $S_e(q_m) = \prod_j (1 - q_m / \lambda_j(q_m))$ . For  $j > m$ ,  $\lambda_j > 1 > q_m$ ; for  $j = m$  the factor is  $1 - q_m > 0$ ; for  $j < m$ ,  $\lambda_j \leq \lambda_{m-1} < q_m$  and the factor is negative.  $\square$

The extra condition in (c) would follow from a shift inequality  $\lambda_n(q) \leq q^2 \lambda_{n+1}(q)$ , which gives  $\lambda_{m-1}(q_m) \leq q_m^2 < q_m$ ; that inequality is *not* proved in this paper, so (c) carries it as a hypothesis. Numerically, at  $m = 2, \dots, 6$ ,  $\lambda_{m-1}(q_m) = 0.112, 0.360, 0.510, 0.605, 0.670$ , well below  $q_m^2$ , and  $S_e(q_m) = 0.520, -0.210, 0.127, -0.091, 0.071, -0.058$  for  $m = 1, \dots, 6$  (50 digits). The sign alternation of  $\Sigma_0$  is the one recorded for the first 32 poles in Remark 26.

*Remark 31* (standing of Propositions 29 and 30). Earlier versions of this paper stated neither Hypothesis 28 nor the three inputs that our working notes carried beside it for the same purpose: (I1) a Sturm oscillation theorem for the Friedrichs realisation of the eigenvalue problem; (I2) Hellmann–Feynman on a  $q$ -dependent form domain; (I3) Proposition 17 with Lemma 12. Here (I2) is Lemma 27, proved because  $\mathbf{S}(q)$  is a bounded norm-analytic family, so Kato’s theory applies with no domain question; (I3) is covered by Proposition 17, Theorem 19 and Proposition 22, and Lemma 12 is not used; (I1) is not used at all: it would turn the index statement of Proposition 29(i) into the count of  $m - 1$  sign changes of  $R_s$ , which nothing here needs. What remains is Hypothesis 28 itself, which is *not* proved here. Numerically, the ratio of its two sides exceeds 2 at every pole  $2 \leq m \leq 50$  and decreases towards 2 (2.145 at  $m = 2$ , 2.014 at  $m = 5$ ). A proof with the constant 1.25 in front of the right-hand side, from a flux identity expressing  $\mathbf{e}_1 + \mathbf{e}_2$  as twice a mean position, an amplitude bound, and an interval certificate on  $q \in [\frac{1}{2}, 1 - 10^{-6}]$ , has been written and independently reviewed, but it is not part of this paper. So Proposition 29 and Proposition 30(b) are proved given Hypothesis 28, Proposition 30(c) given also the shift inequality, and Proposition 30(a) unconditionally. No theorem of this paper depends on them: Proposition 68 needs only  $\pi \geq p \geq 1$ , and Theorem 72 gives the non-vanishing of the residue directly.

*Remark 32* (what the known theory does not give). Two statements about the travel poles lie outside what Theorem 19 and the literature on Hahn–Exton zeros deliver.

*The accumulation law.* The zero theory of [29, 53] concerns the zeros in  $x$  at fixed  $q$ ; the travel poles sit on the diagonal  $x = Z(q)$ . With Hurwitz's theorem it gives each branch at fixed  $k$ ,  $\lambda_k(q) \sim \varepsilon(2k-1)^2\pi^2/8$  with  $\varepsilon = 1 - q$  (Proposition 22(b)). The  $m$ -th pole is where  $\lambda_m = 1$ , that is in the regime  $k = m \asymp \varepsilon^{-1/2}$ , and no fixed- $k$  statement reaches it. So the law  $\varepsilon_m = 1 - q_m \sim 8/(\pi^2(2m-1)^2)$ , equivalently  $w_m \sim (m - \frac{1}{2})\pi$ , recorded numerically in Remark 113, is not a consequence. It needs a count lemma: an asymptotic for  $\#\{k : \lambda_k(q) < 1\}$ , the number of zeros of  $\cos(\cdot; q^2)$  in  $(0, Z(q))$ , uniform as  $q \rightarrow 1$ . Lemma 10 puts at least one travel pole in each window of  $w$  (the proof of Lemma 12), and Proposition 29(i) identifies the index under Hypothesis 28, but neither bounds the number of poles in a window. The count lemma is open.

*Complex  $q$ .* The identity  $\det(I - \mathcal{T}) = \cos(Z; q^2)$  continues analytically to complex  $q$ , but positivity and self-adjointness do not, and the zero theory cited is for  $0 < q < 1$ . We know of no result on zeros of  $q$ -Bessel functions at complex base that would locate the non-real zeros of Remark 93; their density on the circle remains a conjecture.

## 5 The transfer model and the block assembly

The travel block and the bulk block are one object seen twice. Both are the resolvent of the semiseparable kernel  $q^{\max(a,b)}$  against a geometric diagonal weight, and a single three-term recursion in the section index  $k$  covers both: its coefficients are  $\alpha_k, \gamma_k$  in the bulk case and  $A_k, C_k$  in the travel case, which is why the four series of (2) are four ladders on one recursion and not four separate constructions. This section develops that into the block assembly the rest of the paper consumes.

Sections 5.1–5.5 are unconditional statements about explicitly given operators, and §5.6 is an unconditional statement about the crossing optimisation that those operators encode. Section 5.8 states the strand-walk input (M) that identifies the one with the other and proves it (Theorem 66, with the assembly step proved in Appendix D). Section 5.9 evaluates the junction pairing at the travel poles in closed form.

Throughout,  $|q| < 1$  and  $x = q^{1/2}$ . Besides the four blocks of (2) we use the third ladder on the same recursion,

$$T_k = \sum_{j \geq 0} \alpha_{k+2j+1} \prod_{i < j} \gamma_{k+2i} \quad (k \geq 0). \quad (15)$$

Since  $\gamma_k = -2q^{k+1}(1-q)/[(1-q^{k+1})(1-q^{k+2})]$  and  $C_k = q\gamma_k$ , on every compact subset of  $|q| < 1$  there is a  $\Xi$  with  $|\gamma_k|, |C_k| \leq \Xi|q|^{k+1}$ , so  $|\prod_{i < j} \gamma_{k+2i}| \leq \Xi^j|q|^{j^2+jk}$  and likewise for  $C$ ; as  $\alpha_k$  and  $A_k$  are bounded there,  $S_k, \Sigma_k, T_k$  converge absolutely and locally uniformly and are holomorphic on  $|q| < 1$  for every  $k \geq 0$ .

### 5.1 The section identity

**Definition 33** (section identity). Fix  $s_0 \in \{0, 1\}$  and  $\vartheta \in \{0, 1\}$  with  $s_0 + \vartheta = 1$ , and put  $e_s = 2q^{\vartheta+s}$  for  $s \geq s_0$ . Given a bounded *source*  $\mathbf{c} = (c_s)_{s \geq s_0}$ , a family  $\Phi = (\Phi_s)_{s \geq s_0}$  satisfies the *section identity* with data  $(s_0, \vartheta, \mathbf{c})$  if

$$\Phi_s = e_s \left( c_s + \sum_{s' \geq s_0} q^{\max(s, s')} \Phi_{s'} \right), \quad s \geq s_0. \quad (16)$$

Its *sections* are  $G_k = \sum_{s \geq s_0} q^{ks} \Phi_s$ ,  $k \geq 0$ . In operator form  $(I - M)\Phi = E$  with

$$M[s, s'] = e_s q^{\max(s, s')}, \quad E_s = e_s c_s. \quad (17)$$

The *bulk* instance is  $(s_0, \vartheta) = (1, 0)$  and the *travel* instance is  $(s_0, \vartheta) = (0, 1)$ ; we write  $M_0$  for the bulk operator and  $\mathcal{T}$  for the travel one.

**Lemma 34** (existence, uniqueness, analyticity). *For  $|q| < \frac{1}{3}$  and any bounded  $\mathbf{c}$ , equation (16) has a unique solution in  $\ell^1$ , namely  $\Phi = \sum_{n \geq 0} M^n E$ , the series converging in  $\ell^1$  and coefficientwise in  $q$ , one extra power of  $q$  per term. One has  $|\Phi_s| \leq C|q|^{\vartheta+s}$  with  $C$  independent of  $s$ , and each section  $G_k$  is holomorphic at  $q = 0$ .*

*Proof.*  $\|M\|_{\ell^1 \rightarrow \ell^1} = \sup_{s'} \sum_s |M[s, s']| \leq \sup_{s'} \sum_{s \geq s_0} 2|q|^{\vartheta+s} = 2|q|/(1-|q|) < 1$  for  $|q| < \frac{1}{3}$ , using  $\vartheta + s_0 = 1$  and  $|q|^{\max(s, s')} \leq 1$ . Hence  $I - M$  is invertible on  $\ell^1$  and the Neumann series converges. Every entry of  $M$  is divisible by  $q^{\vartheta+s} q^{\max(s, s')}$ , hence by  $q$ , so  $M^n E$  is divisible by  $q^{n+1}$  and the series converges coefficientwise. Reading the  $s$ -th row of  $\Phi = E + M\Phi$  gives  $|\Phi_s| \leq 2|q|^{\vartheta+s} (\|\mathbf{c}\|_\infty + \|\Phi\|_{\ell^1})$ . Since  $|\Phi_s| = O(|q|^s)$ , each  $G_k$  is a locally uniformly convergent sum of holomorphic functions.  $\square$

## 5.2 One telescoping lemma for both blocks

**Lemma 35** (catalytic telescoping). *Let  $0 < |q| < 1$  and let  $\Phi \in \ell^1$  satisfy (16) with data  $(s_0, \vartheta, \mathbf{c})$ . Put*

$$a_k = \frac{2q^{\vartheta+(k+1)s_0}}{1-q^{k+1}}, \quad b_k = \frac{2q^{\vartheta+k+2}}{1-q^{k+2}} - \frac{2q^{\vartheta+k+1}}{1-q^{k+1}}, \quad C_k = \sum_{s \geq s_0} e_s c_s q^{ks},$$

*and  $\mathbf{a}_k = \sum_{j \geq 0} a_{k+2j} \prod_{i < j} b_{k+2i}$ ,  $\mathbf{c}_k = \sum_{j \geq 0} C_{k+2j} \prod_{i < j} b_{k+2i}$ . Then for every  $k \geq 0$*

$$G_k = C_k + a_k G_1 + b_k G_{k+2}, \quad (18)$$

*the series  $\mathbf{a}_k, \mathbf{c}_k$  converge,  $G_k = \mathbf{c}_k + \mathbf{a}_k G_1$ , and if  $\mathbf{a}_1 \neq 1$  then*

$$G_1 = \frac{\mathbf{c}_1}{1-\mathbf{a}_1}, \quad G_k = \mathbf{c}_k + \mathbf{a}_k \frac{\mathbf{c}_1}{1-\mathbf{a}_1} \quad (k \geq 0). \quad (19)$$

*In the bulk instance  $a_k = \alpha_k$ ,  $b_k = \gamma_k$  and  $\mathbf{a}_k = S_k$ ; in the travel instance  $a_k = A_k$ ,  $b_k = C_k$  and  $\mathbf{a}_k = \Sigma_k$ .*

*Proof.* Split the inner sum of (16) at  $s' = s$ , where  $\max(s, s')$  switches:

$$\Phi_s = e_s c_s + e_s \sum_{s' \geq s} q^{s'} \Phi_{s'} + e_s q^s \sum_{s_0 \leq s' < s} \Phi_{s'}.$$

Multiply by  $q^{ks}$  and sum over  $s \geq s_0$ . All three double sums converge absolutely, since  $\Phi \in \ell^1$  and  $|q| < 1$ , so the orders of summation may be exchanged. The first gives  $C_k$ . In the second, with  $e_s = 2q^{\vartheta+s}$ ,

$$\sum_{s \geq s_0} 2q^{\vartheta+(k+1)s} \sum_{s' \geq s} q^{s'} \Phi_{s'} = 2q^{\vartheta} \sum_{s' \geq s_0} q^{s'} \Phi_{s'} \sum_{s=s_0}^{s'} q^{(k+1)s} = \frac{2q^{\vartheta}}{1-q^{k+1}} \sum_{s' \geq s_0} q^{s'} \Phi_{s'} (q^{(k+1)s_0} - q^{(k+1)(s'+1)}),$$

which is  $a_k G_1 - \frac{2q^{\vartheta+k+1}}{1-q^{k+1}} G_{k+2}$ , because  $\sum_{s'} q^{s'} \Phi_{s'} = G_1$  and  $\sum_{s'} q^{(k+2)s'} \Phi_{s'} = G_{k+2}$ . In the third,

$$\sum_{s \geq s_0} 2q^{\vartheta+(k+2)s} \sum_{s_0 \leq s' < s} \Phi_{s'} = 2q^{\vartheta} \sum_{s' \geq s_0} \Phi_{s'} \sum_{s > s'} q^{(k+2)s} = \frac{2q^{\vartheta+k+2}}{1-q^{k+2}} G_{k+2}.$$

Adding the three gives (18). Iterating it  $J$  times,

$$G_k = \sum_{j < J} (C_{k+2j} + a_{k+2j} G_1) \prod_{i < j} b_{k+2i} + G_{k+2J} \prod_{i < J} b_{k+2i}.$$

For fixed  $q$  one has  $|b_m| \leq \Xi|q|^{m+1}$  with  $\Xi = 4/(1 - |q|)$ , so  $|\prod_{i < J} b_{k+2i}| \leq \Xi^J |q|^{J^2 + Jk} \rightarrow 0$ ; and  $|G_m| \leq \|\Phi\|_{\ell^1}$  uniformly in  $m$ , so the remainder vanishes as  $J \rightarrow \infty$ . The same bound, against  $\sup_k |a_k| < \infty$  and  $\sup_k |\mathcal{C}_k| \leq 2\|\mathbf{c}\|_\infty/(1 - |q|)$ , makes  $\mathbf{a}_k$  and  $\mathbf{c}_k$  converge. This gives  $G_k = \mathbf{c}_k + \mathbf{a}_k G_1$ ; taking  $k = 1$  gives  $G_1(1 - \mathbf{a}_1) = \mathbf{c}_1$ , whence (19). The identifications of  $a_k, b_k$  are read off the displays:  $(s_0, \vartheta) = (1, 0)$  gives  $a_k = 2q^{k+1}/(1 - q^{k+1}) = \alpha_k$  and  $b_k = \alpha_{k+1} - \alpha_k = \gamma_k$ , while  $(s_0, \vartheta) = (0, 1)$  gives  $a_k = 2q/(1 - q^{k+1}) = A_k$  and  $b_k = 2q^{k+3}/(1 - q^{k+2}) - 2q^{k+2}/(1 - q^{k+1}) = C_k$ .  $\square$

*Remark 36* (where each feature of the model is used). Three features enter (18), each exactly once. That the coupling is  $q^{\max(s, s')}$  is what makes the split at  $s' = s$  produce two *geometric* sums; a symmetric kernel not of the form  $\mu^{\max}$  destroys the three-term relation. That  $e_s$  is geometric in  $s$  is what makes those sums summable in closed form; a non-geometric edge weight leaves  $G_k$  coupled to a continuum of sections. That the source is geometric is used only to evaluate  $\mathcal{C}_k$ ; both (18) and (19) hold for any bounded source. The step  $k \mapsto k + 2$ , and with it the two-ladder structure of  $S_k$  and  $\Sigma_k$ , comes from the two occurrences of  $q^{(k+2)s'}$  above.

**Proposition 37** (the travel  $k$ -sum telescopes). *Let  $\Phi$  solve the travel section identity,  $(s_0, \vartheta, \mathbf{c}) = (0, 1, \mathbf{1})$ . Then, as meromorphic functions on  $|q| < 1$ ,*

$$G_k^T = \frac{\Sigma_k}{1 - \Sigma_1} \quad (k \geq 0), \quad \text{in particular} \quad G_0^T = \sum_{s \geq 0} \Phi_s = \frac{\Sigma_0}{1 - \Sigma_1}.$$

*Proof.* Lemma 35 with  $(s_0, \vartheta) = (0, 1)$  and  $c_s \equiv 1$  gives  $\mathcal{C}_k = \sum_{s \geq 0} 2q^{1+s} q^{ks} = 2q/(1 - q^{k+1}) = A_k = a_k$ , so  $\mathbf{c}_k = \mathbf{a}_k = \Sigma_k$  and (19) reads  $G_k^T = \Sigma_k + \Sigma_k \Sigma_1/(1 - \Sigma_1) = \Sigma_k/(1 - \Sigma_1)$ . Both sides are holomorphic near  $q = 0$  and the right-hand side is meromorphic on  $|q| < 1$ , so the identity extends by analytic continuation. Combinatorially this is the Neumann series of Lemma 34:  $\sum_s \Phi_s = \sum_{n \geq 0} \mathbf{1}^\top \mathcal{T}^n E$ , whose  $n$ -th term is the generating function of the travel runs with exactly  $n + 1$  edges.  $\square$

**Proposition 38** (the bulk dressing). *Let  $\Phi$  solve the bulk section identity with  $\mathbf{c} = \mathbf{1}$  and let  $\Psi$  solve it with  $c_s = q^s$ . Set*

$$b_0 = \sum_{s \geq 1} \Phi_s, \quad t_0 = \sum_{s \geq 1} q^s \Phi_s, \quad b_1 = \sum_{s \geq 1} \Psi_s, \quad t_1 = \sum_{s \geq 1} q^s \Psi_s.$$

*Then, as meromorphic functions on  $|q| < 1$ ,*

$$b_0 = \frac{S_0}{1 - S_1}, \quad t_0 = \frac{S_1}{1 - S_1}, \quad t_1 = \frac{T_1}{1 - S_1}, \quad b_1 = T_0 + \frac{S_0 T_1}{1 - S_1}.$$

*Proof.* Lemma 35 with  $(s_0, \vartheta) = (1, 0)$ . For  $\mathbf{c} = \mathbf{1}$ ,  $\mathcal{C}_k = \sum_{s \geq 1} 2q^{s(k+1)} = \alpha_k = a_k$ , so  $\mathbf{c}_k = \mathbf{a}_k = S_k$  and  $G_k = S_k/(1 - S_1)$ ; take  $k = 0, 1$ . For  $c_s = q^s$ ,  $\mathcal{C}_k = \sum_{s \geq 1} 2q^{s(k+2)} = \alpha_{k+1}$ , so  $\mathbf{c}_k = \sum_j \alpha_{k+2j+1} \prod_{i < j} \gamma_{k+2i} = T_k$  by (15), and (19) gives  $t_1 = G_1 = T_1/(1 - S_1)$  and  $b_1 = G_0 = T_0 + S_0 T_1/(1 - S_1)$ . Analytic continuation as before.  $\square$

With  $1 - S_1 = S_e$  (Proposition 17) the first of these is  $b_0 = S_0/S_e$ , the third line of the dictionary (33); and  $b_0$  is holomorphic and non-zero at every travel pole, by Propositions 94 and 25 respectively.

### 5.3 The cocycle, the reciprocity, and the dictionary

The bulk section identity has a second, equivalent, description as a two-point problem for a first-order system. That system is the cocycle of §7.4, and comparing the two descriptions identifies  $t_1$  with  $P_{12}/S_e$ .

**Corollary 39** (the cocycle is the same object). *Let  $\Phi$  solve the bulk section identity with source  $\mathbf{c}$ , and put  $L_b = \sum_{a \leq b} \Phi_a$  and  $\mathcal{E}_b = -\sum_{a \geq b} q^a \Phi_a$ . Then  $\Phi_b = 2q^b c_b + 2q^{2b} L_{b-1} - 2q^b \mathcal{E}_b$  and*

$$\begin{pmatrix} L_b \\ \mathcal{E}_{b+1} \end{pmatrix} = \mathcal{M}_b \begin{pmatrix} L_{b-1} \\ \mathcal{E}_b \end{pmatrix} + \begin{pmatrix} 2q^b c_b \\ 2q^{2b} c_b \end{pmatrix}, \quad \mathcal{M}_b = \begin{pmatrix} 1 + 2q^{2b} & -2q^b \\ 2q^{3b} & 1 - 2q^{2b} \end{pmatrix}, \quad (20)$$

with  $L_0 = 0$ ,  $\mathcal{E}_1 = -t_0$  or  $-t_1$  according to the source, and  $\mathcal{E}_b \rightarrow 0$ . Let  $(\ell, r)$  and  $(\ell', r')$  be the solutions of the homogeneous recursion ( $\mathbf{c} = 0$ ) with data  $(\ell_0, r_1) = (1, 0)$  and  $(\ell'_0, r'_1) = (0, 1)$ , and put  $(P_{11}, P_{12}) = \lim_b (\ell_b, r_{b+1})$  and  $(P_{21}, P_{22}) = \lim_b (\ell'_b, r'_{b+1})$ , the limits existing because  $\sum_b \|\mathcal{M}_b - I\| < \infty$ . Eliminating  $L$  between the two lines of (20) at  $\mathbf{c} = 0$  gives

$$\mathcal{E}_{b+2} = (1 + q^3 - 2(1 - q)q^{2b+2})\mathcal{E}_{b+1} - q^3 \mathcal{E}_b,$$

which is the scalar three-term recursion of §7.4; so  $r$  and  $r'$  are its two standard solutions and  $P_{12}, P_{22}$  their limits,  $P_{12}$  being the entry whose closed form is (35). Then, identically in  $0 < q < 1$ ,

$$P_{11} = 1 + T_0, \quad P_{12} = T_1, \quad P_{21} = -S_0, \quad P_{22} = S_e = 1 - S_1.$$

In particular  $t_1 = P_{12}/S_e$  is an identity between independently defined objects, not a definition.

*Proof.* Substituting  $L_b = L_{b-1} + \Phi_b$  and  $\mathcal{E}_{b+1} = \mathcal{E}_b + q^b \Phi_b$  into  $\Phi_b = 2q^b(c_b + q^b L_b - \mathcal{E}_{b+1})$ , which is (16) after the split at  $s' = b$ , the two terms  $\pm 2q^{2b} \Phi_b$  cancel and leave  $\Phi_b = 2q^b c_b + 2q^{2b} L_{b-1} - 2q^b \mathcal{E}_b$ ; feeding this back gives (20). The homogeneous solutions form a two-dimensional space, since (20) determines  $(L_b, \mathcal{E}_{b+1})$  from  $(L_{b-1}, \mathcal{E}_b)$ .

Take  $\mathbf{c} = \mathbf{1}$ , so  $L_0 = 0$ ,  $L_b \rightarrow b_0$ ,  $\mathcal{E}_1 = -t_0$  and  $\mathcal{E}_{b+1} \rightarrow 0$ . Then  $(L, \mathcal{E} - 1)$  solves the homogeneous recursion: in the first line the source  $2q^b c_b = 2q^b$  is cancelled by  $-2q^b \cdot (-1)$ , and in the second the source  $2q^{2b} c_b = 2q^{2b}$  is cancelled by  $(1 - 2q^{2b})(-1) - (-1)$ . Its data are  $(0, -1 - t_0)$ , so  $(L, \mathcal{E} - 1) = -(1 + t_0)(\ell', r')$ , and letting  $b \rightarrow \infty$ ,

$$b_0 = -(1 + t_0)P_{21}, \quad -1 = -(1 + t_0)P_{22}.$$

By Proposition 38 and  $S_e = 1 - S_1$  one has  $1 + t_0 = 1 + S_1/S_e = 1/S_e$ , whence  $P_{22} = S_e$  and  $P_{21} = -b_0 S_e = -S_0$ .

Take now  $c_s = q^s$ , with data  $L'_0 = 0$ ,  $L'_b \rightarrow b_1$ ,  $\mathcal{E}'_1 = -t_1$ ,  $\mathcal{E}'_{b+1} \rightarrow 0$ . Here  $(L' + 1, \mathcal{E}')$  solves the homogeneous recursion: in the first line  $(1 + 2q^{2b}) \cdot 1$  exceeds 1 by  $2q^{2b}$ , which is the source  $2q^b c_b = 2q^{2b}$ , and in the second  $2q^{3b} \cdot 1$  is the source  $2q^{2b} c_b = 2q^{3b}$ . Its data are  $(1, -t_1)$ , so  $(L' + 1, \mathcal{E}') = (\ell, r) - t_1(\ell', r')$ , and letting  $b \rightarrow \infty$ ,

$$b_1 + 1 = P_{11} - t_1 P_{21} = P_{11} + t_1 S_0, \quad 0 = P_{12} - t_1 P_{22} = P_{12} - t_1 S_e.$$

The second gives  $P_{12} = t_1 S_e = T_1$  by Proposition 38, and the first then gives  $P_{11} = b_1 + 1 - t_1 S_0 = T_0 + S_0 T_1/S_e + 1 - (T_1/S_e)S_0 = 1 + T_0$ . All four limits are taken for  $|q| < \frac{1}{3}$ , where Lemma 34 applies, and extend to  $0 < q < 1$  by analytic continuation, both sides being meromorphic there.  $\square$

**Proposition 40** (reciprocity).  $t_0 = b_1$  identically in  $q$ . Equivalently  $T_0 S_e + T_1 S_0 = S_1$ , equivalently  $P_{11} P_{22} - P_{12} P_{21} = 1$ .

*Proof.* Two independent proofs.

*Kernel symmetry.* Write  $D = \text{diag}(2q^b)_{b \geq 1}$  and  $K[b, a] = q^{\max(a, b)}$ , so that  $M_0 = DK$ ,  $E = D\mathbf{1}$ , and  $u := (2q^{2b})_b = Dv$  with  $v = (q^a)_a$ . The two sources of Proposition 38 are  $E$  and  $u$ , so  $\Phi = (I - M_0)^{-1}E$ ,  $\Psi = (I - M_0)^{-1}u$ , and

$$t_0 = v^\top (I - M_0)^{-1}E, \quad b_1 = \mathbf{1}^\top (I - M_0)^{-1}u.$$

For  $|q| < \frac{1}{3}$ ,  $X := (I - M_0)^{-1}D = \sum_{n \geq 0} (DK)^n D$ , and the  $n$ -th summand is  $DKD \cdots KD$ , a palindromic product of the symmetric matrices  $D$  and  $K$ , hence symmetric; the series converges entrywise. Therefore  $t_0 = v^\top X\mathbf{1} = \sum_{a, b} v_a X[a, b] = \sum_{a, b} X[b, a]v_a = \mathbf{1}^\top Xv = b_1$ , every sum being absolutely convergent. Both sides are meromorphic on  $|q| < 1$  by Proposition 38, so the identity extends.

*Casoratian.*  $\det \mathcal{M}_b = (1 + 2q^{2b})(1 - 2q^{2b}) + 4q^{3b}q^b = 1$ , so  $\ell_b r'_{b+1} - \ell'_b r_{b+1}$  does not depend on  $b$ ; at  $b = 0$  it is 1, whence  $P_{11}P_{22} - P_{21}P_{12} = 1$ . Substituting the four entries of Corollary 39 gives  $(1 + T_0)S_e + T_1S_0 = 1$ , that is  $T_0S_e + T_1S_0 = 1 - S_e = S_1$ ; dividing by  $S_e$  and using Proposition 38 turns this into  $b_1 = T_0 + S_0T_1/S_e = S_1/S_e = t_0$ .  $\square$

The symmetry of  $K$  is used and is necessary: replacing  $q^{\max(a, b)}$  by the asymmetric  $q^{\max(a, b)} + q^{a+2b}$  makes  $t_0 - b_1 = 4q^8 + O(q^9)$  (`assembly_certificate.py`, Part 2). The determinant identity  $P_{11}P_{22} - P_{12}P_{21} = 1$  is the Casoratian of the cocycle and is automatic; what carries content is the identification of its four entries in Corollary 39, which turns it into the reciprocity.

**Corollary 41** (the dictionary). *Identically in  $0 < q < 1$ ,*

$$S_e = 1 - S_1, \quad S_o = \frac{1-q}{2q}S_0, \quad b_0 = \frac{S_0}{S_e}, \quad t_0 = b_1 = \frac{S_1}{S_e}, \quad t_1 = \frac{T_1}{S_e} = \frac{P_{12}}{S_e},$$

and consequently  $\kappa := t_0b_1 - b_0t_1 = (S_1^2 - S_0P_{12})/S_e^2$ . This is (33) together with its two remaining entries.

*Proof.* The first line is Proposition 17 and Lemma 23; the rest is Proposition 38 with Proposition 40 and Corollary 39, and  $\kappa$  is substitution.  $\square$

## 5.4 Compactness, the null space, and the singularities

**Lemma 42** (compactness and meromorphy). *For  $|q| < 1$  the operator  $M$  of (17), and its gap-marked version with kernel (21) below, are compact on  $\ell^1$ . Hence  $q \mapsto (I - M)^{-1}$  is meromorphic on  $|q| < 1$ , with poles of finite order, and at each pole  $\ker(I - M) \neq 0$ .*

*Proof.* Let  $M_N$  agree with  $M$  on rows  $s \leq N$  and vanish above. Then  $\|M - M_N\|_{\ell^1 \rightarrow \ell^1} \leq \sup_{s'} \sum_{s > N} |e_s| |q|^{\max(s, s')} \leq \sum_{s > N} 2|q|^{\vartheta+s} \rightarrow 0$ , so  $M$  is a norm limit of finite-rank operators and is compact; the rank-one gap term is compact a fortiori. The family is holomorphic in  $q$  on the connected domain  $|q| < 1$  and  $I - M$  is invertible near  $q = 0$  (Lemma 34), so the analytic Fredholm theorem applies.  $\square$

**Lemma 43** (the null space is at most one-dimensional). *Let  $0 < |q| < 1$  and let  $R \in \ell^1$  satisfy  $R_s = e_s \sum_{s'} q^{\max(s, s')} R_{s'}$  for all  $s \geq s_0$ . Then its sections satisfy  $G_k^R = G_1^R \mathbf{a}_k$  for all  $k \geq 0$ , and  $G_1^R = 0$  forces  $R = 0$ . In particular  $\dim \ker(I - M) \leq 1$ , and a non-zero kernel element exists only where  $\mathbf{a}_1 = 1$ .*

*Proof.* Lemma 35 with  $\mathbf{c} = 0$  gives  $\mathcal{C}_k = 0$ , hence  $\mathbf{c}_k = 0$  and  $G_k^R = \mathbf{a}_k G_1^R$ ; its derivation used only  $R \in \ell^1$  and  $|q| < 1$ . Taking  $k = 1$  gives  $G_1^R = \mathbf{a}_1 G_1^R$ , so  $\mathbf{a}_1 = 1$  unless  $G_1^R = 0$ . If  $G_1^R = 0$  then  $G_k^R = 0$  for every  $k$ . The function  $F(t) = \sum_{s \geq s_0} R_s t^s$  is holomorphic on  $|t| < 1$  because  $R \in \ell^1$ , and  $F(q^k) = G_k^R = 0$  for all  $k \geq 0$ ; the points  $q^k$  accumulate at the interior point  $t = 0$ , so  $F \equiv 0$  and  $R = 0$ . Hence the kernel is spanned by the class normalised by  $G_1^R = 1$ .  $\square$

**Corollary 44** (singularities of the two gapless resolvents). *On  $|q| < 1$ , the poles of  $(I - M_0)^{-1}$  lie in the zero set of  $S_e$  and the poles of  $(I - \mathcal{T})^{-1}$  lie in the zero set of  $1 - \Sigma_1$ ; conversely every travel pole is a pole of  $(I - \mathcal{T})^{-1}$ . In particular  $b_0, b_1, t_0, t_1$  are holomorphic at every travel pole  $q_m$ , where  $S_e(q_m) \neq 0$  by Proposition 94, and  $b_0(q_m) = S_0(q_m)/S_e(q_m) \neq 0$  by Proposition 25.*

*Proof.* By Lemma 42 a pole forces a non-zero kernel element, and by Lemma 43 that forces  $\mathbf{a}_1 = 1$ , which is  $S_1 = 1$ , i.e.  $S_e = 0$ , in the bulk instance and  $\Sigma_1 = 1$  in the travel one. Conversely, at a travel pole  $\Sigma_0/(1 - \Sigma_1)$  is unbounded because  $\Sigma_0(q_m) \neq 0$  (Proposition 25), so by Proposition 37 the travel resolvent is singular there.  $\square$

## 5.5 The gap term is rank one: the Möbius factorisation

Marking the gap bridges changes the bulk kernel by one term. With  $\mathbf{g} = \mathbf{g}(q, y) = \sum_{L \geq 1} q^L y^{L-1} = q/(1 - qy)$ , the gap-marked bulk kernel and system are

$$K_{\mathbf{g}}(a, b) = q^{\max(a, b)} + q^{a+b} \mathbf{g}, \quad P_b = 2q^b \left( q^b + \sum_{a \geq 1} K_{\mathbf{g}}(a, b) P_a \right), \quad B(q, y) = \sum_{b \geq 1} P_b, \quad (21)$$

the source  $q^b$  being the cost of the outer span-end site of a bulk run (Lemma 144). In operator form  $(I - M_0 - \mathbf{g}uv^\top)P = u$  with

$$M_0[b, a] = 2q^b q^{\max(a, b)}, \quad E_b = 2q^b, \quad u_b = 2q^{2b}, \quad v_a = q^a, \quad (22)$$

since  $2q^b \cdot q^{a+b} = (2q^{2b})(q^a)$ : the gap term is exactly rank one. With the four scalars of Proposition 38,

$$b_0 = \mathbf{1}^\top (I - M_0)^{-1} E, \quad b_1 = \mathbf{1}^\top (I - M_0)^{-1} u, \quad t_0 = v^\top (I - M_0)^{-1} E, \quad t_1 = v^\top (I - M_0)^{-1} u.$$

**Proposition 45** (Möbius factorisation). *Let  $q$  be such that  $I - M_0$  is invertible on  $\ell^1$  and  $1 - \mathbf{g}t_1 \neq 0$ . Then  $I - M_0 - \mathbf{g}uv^\top$  is invertible on  $\ell^1$ ,*

$$(I - M_0 - \mathbf{g}uv^\top)^{-1} = R_0 + \frac{\mathbf{g}}{1 - \mathbf{g}t_1} R_0 uv^\top R_0, \quad R_0 = (I - M_0)^{-1}, \quad (23)$$

and the  $E$ -sourced sum  $B^E(q, y) := \mathbf{1}^\top (I - M_0 - \mathbf{g}uv^\top)^{-1} E$  is  $(b_0 + \mathbf{g}\kappa)/(1 - \mathbf{g}t_1)$  with  $\kappa = t_0 b_1 - b_0 t_1$ , which is (31). (The bulk sum of the model has source  $u$ ; see Remark 46.) In particular  $B^E$  is a Möbius function of  $\mathbf{g}$ , with a single zero, at  $\mathbf{g}_0 = -b_0/\kappa$ , and a single pole, at  $\mathbf{g} = 1/t_1$ .

*Proof.* Sherman–Morrison. Put  $\delta = 1 - \mathbf{g}v^\top R_0 u = 1 - \mathbf{g}t_1 \neq 0$ . Then the right-hand side of (23) is a two-sided inverse of  $I - M_0 - \mathbf{g}uv^\top$ :

$$(I - M_0 - \mathbf{g}uv^\top)(R_0 + \mathbf{g}\delta^{-1}R_0 uv^\top R_0) = I + \mathbf{g}(\delta^{-1} - 1 - \mathbf{g}\delta^{-1}t_1)uv^\top R_0 = I,$$

since  $\delta^{-1}(1 - \delta - \mathbf{g}t_1) = 0$ ; the other side is identical. All pairings converge absolutely:  $u, E \in \ell^1$ ,  $v, \mathbf{1} \in \ell^\infty$ , and  $R_0$  is bounded on  $\ell^1$ . Applying  $\mathbf{1}^\top(\cdot)E$ ,

$$B^E = \mathbf{1}^\top R_0 E + \mathbf{g}\delta^{-1}(\mathbf{1}^\top R_0 u)(v^\top R_0 E) = b_0 + \frac{\mathbf{g}b_1 t_0}{1 - \mathbf{g}t_1} = \frac{b_0 + \mathbf{g}(t_0 b_1 - b_0 t_1)}{1 - \mathbf{g}t_1}. \quad \square$$

*Remark 46* (the bulk sum of the model). The system (21) has right-hand side  $u$ , not  $E$ . By (23),

$$B(q, y) = \mathbf{1}^\top (I - M_0 - \mathfrak{g}uv^\top)^{-1}u = \frac{b_1}{1 - \mathfrak{g}t_1}, \quad B_0(q, y) := 1 + \mathfrak{g} \sum_{b \geq 1} q^b P_b = \frac{1}{1 - \mathfrak{g}t_1},$$

where  $B_0$  is the weight of the junction-adjacent bulk chains that are empty or begin with a gap run. The Möbius form  $(b_0 + \mathfrak{g}\kappa)/(1 - \mathfrak{g}t_1)$  of Proposition 45 is the resolvent of  $E$ . It enters below only through  $b_0$ ,  $t_1$  and the gate, and not as the bulk sum.

*Remark 47* (the two hypotheses are the two the paper already carries). Invertibility of  $I - M_0$  is  $S_e \neq 0$  (Corollary 44), which at the travel poles is Proposition 94. And  $1 - \mathfrak{g}_V t_1 \neq 0$  is  $s \neq 1$ , the gate half  $|s| < 1$  of (32), which is Theorem 109 with Lemma 110;  $1 - \mathfrak{g}_U t_1 \neq 0$  follows, since  $\mathfrak{g}_U = \mathfrak{g}_V/(1 + q) < \mathfrak{g}_V$  and  $t_1 > 0$  at the poles. So Proposition 45 consumes no hypothesis that §7.3 was not already carrying.

## Machine verification of the telescoping, the factorisation and the reciprocity

Proposition 45 is formally verified in Lean 4 against Mathlib [35] in `Mobius.lean` and `MobiusL1.lean`. Its hypotheses are its Lean hypotheses: ‘ $I - M_0$  is invertible’ is an operator  $R_0$  with  $R_0(I - M_0) = (I - M_0)R_0 = I$ , and  $1 - \mathfrak{g}t_1 \neq 0$  is carried as stated. With those in hand nothing analytic remains. `Mobius.lean` proves the factorisation for an arbitrary module over an arbitrary field: `opN_opS` and `opS_opN` are the two halves of (23), `solution_unique` is the unique solvability of (21), `B_mobius` is (31), and `B_eq_zero_iff` and `B_pole` are the single zero at  $\mathfrak{g}_0 = -b_0/\kappa$  and the single pole at  $\mathfrak{g} = 1/t_1$ .

`MobiusL1.lean` instantiates that statement at the case the proposition is about,  $K = \mathbb{C}$  and  $E = \ell^1$  with the concrete data of (22). For  $|q| < 1$  a kernel with rows dominated by a summable sequence defines a  $\mathbb{C}$ -linear operator on  $\ell^1$  (`opOfKer`) and a bounded sequence defines a functional (`funcOfSeq`); `M0`, `uElt`, `EElt`, `vFun` and `oneFun` are  $M_0$ ,  $u$ ,  $E$ ,  $v^\top$  and  $\mathbf{1}^\top$  of (22), with the index  $n$  standing for site  $n + 1$ . `rank_one` is the identity  $2q^b q^{a+b} = (2q^{2b})(q^a)$ , `gapSystem_iff` identifies the operator equation  $(I - M_0 - \mathfrak{g}uv^\top)P = E$  with the display (21) coordinatewise, `l1_solution_unique` is the unique  $\ell^1$  solvability of (21), `l1_smresolvent_left` and `l1_smresolvent_right` are (23) for those operators, and `l1_B_of_solution` is the conclusion: an  $\ell^1$  solution of (21) has  $B = \sum_b P_b = (b_0 + \mathfrak{g}\kappa)/(1 - \mathfrak{g}t_1)$ .

The first sentence of Proposition 40,  $t_0 = b_1$ , is verified in `Reciprocity.lean` in the operator form its first proof uses, with  $E$  and  $F$  kept apart:  $D: F \rightarrow E$  and  $K: E \rightarrow F$  symmetric for the pairing,  $M_0 = DK$ , and the two sources  $D\mathbf{1}$  and  $Dv$ . `pushThrough` is  $(I - DK)^{-1}D = D(I - KD)^{-1}$ , which replaces the palindromic Neumann series of the paper’s proof by an algebraic identity; `X_symm` is the symmetry of  $X$  and `t0_eq_b1` the conclusion. The necessity of the symmetry of  $K$  is recorded as `K_symm_needed`.

Lemma 35 is verified in `Telescope.lean`. The site  $s = n + s_0$  is indexed by  $n \in \mathbb{Z}_{\geq 0}$ , so  $e_s$  is `eCo` and  $s_0 + \vartheta = 1$  is the hypothesis `hs`; the source is bounded and  $\Phi \in \ell^1$  is carried as summability of  $\|\Phi_n\|$ , which is what `MobiusL1.summable_norm` supplies for a  $\Phi$  given as an element of the  $\ell^1$  space of `MobiusL1.lean` (`l1_three_term`, `l1_G_eq`). `row_sum` is the split of the inner sum at  $s' = s$  together with the finite and the infinite geometric sum it produces; `three_term` is (18), the interchange of the two summations being Fubini against the majorant  $2|q|^{n+1}\|\Phi_m\|$ , which is summable on  $\mathbb{Z}_{\geq 0}^2$  as a product; `norm_bTel_le` is the bound  $|b_k| \leq \Xi|q|^{k+1}$  with  $\Xi = 4/(1 - |q|)$  and `summable_bProd_norm` is the paper’s  $\Xi^J|q|^{J^2+Jk} \rightarrow 0$ , obtained by the ratio test; `summable_aFrak` and `summable_cFrak` are the convergence of  $\mathfrak{a}_k$  and  $\mathfrak{c}_k$ ; `G_eq` is  $G_k = \mathfrak{c}_k + \mathfrak{a}_k G_1$ , proved by telescoping  $R_j = G_{k+2j} \prod_{i < j} b_{k+2i}$  and letting  $R_j \rightarrow 0$ ; `G_one_solved` and `G_solved` are (19). The last sentence of the lemma, the identification of the two instances, is `aTel_bulk`, `bTel_bulk`, `aTel_travel` and `bTel_travel`.

Proposition 37 and Proposition 38 are verified in `SectionLadders.lean`, which evaluates  $\mathcal{C}_k$  at the three geometric sources: `CC_travel` is  $\mathcal{C}_k = 2q/(1 - q^{k+1}) = A_k$ , `CC_bulk` is  $\mathcal{C}_k = \alpha_k$  and `CC_bulk_geo` is  $\mathcal{C}_k = \alpha_{k+1}$ , whence `cFrak_travel`, `cFrak_bulk` and `cFrak_bulk_geo` identify  $\mathbf{c}_k$  with  $\Sigma_k$ , with  $S_k$  and with  $T_k$ . `travelsum` is Proposition 37 and `bulkdress_b0_t0`, `bulkdress_t1_b1` are the four scalars of Proposition 38. `Sladder_eq`, `SigmaLadder_eq` and `Tladder_eq` are the three ladders written out as in (2) and (15). What is verified is the implication, not the existence of the solution: the  $\ell^1$  solution is a hypothesis, which is where the paper's  $|q| < \frac{1}{3}$  lives, and no analytic continuation step is needed, `G_eq` holding on the whole of  $|q| < 1$  for any  $\ell^1$  solution. Lemma 34 is not formalised.

The first of the two clauses of Proposition 40 that begin “equivalently” is verified as `SectionLadders.recip_ladder` given the two bulk solutions,  $t_0 = b_1$  and  $T_0 S_e + T_1 S_0 = S_1$  are the same statement, the equivalence being `recip_ladder_iff`. The reciprocity itself enters there as a hypothesis; it is proved separately in `Reciprocity.lean`, in the operator form recorded above, and the two are not joined, that joining being the instantiation of `Reciprocity.lean` at the concrete  $\ell^1$  resolvent.

Not formalised in this section: Corollary 39, and with it the second “equivalently” clause of Proposition 40, the one in terms of  $P_{ij}$ . The obstruction is the existence of the four limits  $\lim_b(\ell_b, r_{b+1})$  and  $\lim_b(\ell'_b, r'_{b+1})$ : Mathlib carries no convergence theorem for an infinite product of matrices  $\prod_b \mathcal{M}_b$  with  $\sum_b \|\mathcal{M}_b - I\| < \infty$ , and supplying one is a statement about products, not about this cocycle. The rest of the corollary, the passage from (16) to (20) and the identification of the four entries once the limits exist, needs only the split already carried out in `Telescope.row_sum`.

**Lemma 48** (a positivity segment). *For  $0 < q \leq 0.45$  one has  $S_e(q) \geq 0.32$  and  $|s(q)| = \frac{q}{1-q} |P_{12}|/S_e \leq 0.78$ . Since  $q_1 = 0.4494536 \dots < 0.45$ , both hold on  $(0, q_1]$ .*

*Proof.* By Proposition 17,  $S_e = \sum_{j \geq 0} (-1)^j [2(1-q)]^j q^{j^2+j} / (q; q)_{2j}$ , and by (38),  $P_{12} = \sum_{k \geq 1} (-1)^{k-1} [2(1-q)]^k q^{k^2+k+1} (1 - q^{2k}) / (q; q)_{2k+1}$ . For every  $n$ ,  $(q; q)_n \geq (q; q)_\infty \geq 1 - q - q^2$ , the last step by truncating Euler's pentagonal series  $1 - q - q^2 + q^5 + q^7 - \dots$ , whose tail is positive for  $q \leq \frac{1}{2}$ ; on  $q \leq 0.45$  this is  $\geq 0.3475$ . Each term  $[2(1-q)]^j q^{j^2+j}$  with  $j \geq 1$  increases on  $(0, 0.45]$ , since  $(j^2+j)/q - j/(1-q) > 0$  there, so it is at most its value at  $q = 0.45$ : successively 0.222750, 0.010048, 0.000092, the remaining terms summing to less than  $10^{-5}$ . Hence  $S_e \geq 1 - 0.23290/0.3475 \geq 0.32$ . The same estimate applied to  $[2(1-q)]^k q^{k^2+k+1}$ , likewise increasing, and to  $(1 - q^{2k}) \leq 1$ , gives 0.100238, 0.004521, 0.000042 and a tail below  $10^{-6}$ , so  $|P_{12}| \leq 0.104801/0.3475 \leq 0.3016$  and  $|s| \leq \frac{0.45}{0.55} \cdot \frac{0.3016}{0.32} \leq 0.78$ .  $\square$

## 5.6 The site cost

The subsections above concern the operators of Definition 33. This one concerns the combinatorial quantity those operators are meant to encode: the per-site cost of the crossing optimisation of [6]. Its value is computed here in closed form, which settles the first half of the model input (M) below and corrects the marker clause of the form in which that input was previously stated.

**Definition 49** (the pairing optimisation). Fix marker data  $(\varepsilon^*, \delta^*) \in \{\pm 1\} \times \{0, 1\}$ , a displacement  $k^* \in \mathbb{Z}$ , and deposits  $d_j \in \mathbb{Z}$  on the integer edges  $j$ , edge  $j$  joining site  $j$  to site  $j + 1$ . The travel indicator is  $f_j = 1$  for  $0 \leq j < k^*$ ,  $f_j = -1$  for  $k^* \leq j < 0$  and  $f_j = 0$  otherwise, and  $d_j \equiv f_j \pmod{2}$ . A *realisation* consists of

- crossing counts  $m_j \geq \max(|d_j|, |f_j|)$  with  $m_j \equiv d_j \pmod{2}$ , whose support is an interval containing 0; edge  $j$  is crossed  $u_j = (m_j + f_j)/2$  times upward and  $\text{dn}_j = (m_j - f_j)/2$  times downward;
- a sign in  $\{\pm\}$  on each crossing, with  $p_j^u$  of the up-crossings and  $p_j^d$  of the down-crossings carrying  $+$ , subject to  $d_j = 2p_j^d - \text{dn}_j + u_j - 2p_j^u$ ;

- at every site  $s$ , a bijection between the arrivals and the departures at  $s$ .

The arrivals at  $s$  are the up-crossings of edge  $s - 1$ , of class  $(L, \pm)$  by their sign, and the down-crossings of edge  $s$ , of class  $(R, \pm)$ ; the departures are the down-crossings of edge  $s - 1$ , of class  $(L, \pm)$ , and the up-crossings of edge  $s$ , of class  $(R, \pm)$ . In addition there is a virtual arrival of class  $(L, +)$  at  $s = 0$ , and a virtual departure at  $s = k^*$ , of class  $(R, \varepsilon^*)$  if  $\delta^* = 1$  and  $(L, \varepsilon^*)$  if  $\delta^* = 0$ . A matched pair costs 0 on the same side with equal signs, 2 on the same side with opposite signs, and 1 on opposite sides. The cost of a realisation is  $\sum_j m_j$  plus the sum of the pairing costs;  $\ell_R$  is the minimum cost, and  $c$  is the least number of components other than the one carrying the two virtual events, over the realisations of minimum cost.

*Remark 50* (standing of Definition 49). This is the optimisation of [6, Metric formula, as verified], restated. That it computes the relaxed word length is verified there and is not proved. Everything in this subsection is a statement about the optimisation itself and is proved outright. The growth-series statements of this paper do not consume it: they go through the site-local closed forms  $\ell_R, c$  and the Lean identity `metricAll` (Theorem 66), and the optimisation enters only the interpretation of  $V$  as a relaxed count.

**Lemma 51** (the transportation value). *Let  $A^\pm, B^\pm, C^\pm, D^\pm$  be non-negative integers with  $A^+ + A^- + B^+ + B^- = C^+ + C^- + D^+ + D^-$ , and consider the transportation problem with supplies  $A^+, A^-, B^+, B^-$  and demands  $C^+, C^-, D^+, D^-$  in the four classes  $(L, +), (L, -), (R, +), (R, -)$  under the cost of Definition 49. Put*

$$\alpha = (C^+ - C^-) - (A^+ - A^-), \quad \beta = (B^+ - B^-) - (D^+ - D^-), \quad \Phi = (A^+ + A^-) - (C^+ + C^-).$$

*Then the minimum cost equals  $\max(|\alpha|, |\beta|, |\Phi|)$ .*

*Proof.* Write  $N_L^a = A^+ + A^-$ ,  $N_L^d = C^+ + C^-$ ,  $N_R^a = B^+ + B^-$ ,  $N_R^d = D^+ + D^-$ , so  $\Phi = N_L^a - N_L^d = N_R^d - N_R^a$ , and  $\sigma_L^a = A^+ - A^-$ ,  $\sigma_L^d = C^+ - C^-$ , and similarly on the right, so that  $\alpha = \sigma_L^d - \sigma_L^a$  and  $\beta = \sigma_R^a - \sigma_R^d$ .

Let  $\pi$  be the mass sent from left supplies to right demands and  $\pi'$  that from right supplies to left demands. The mass staying on the left is  $g = N_L^a - \pi = N_L^d - \pi'$ , so  $\pi' = \pi - \Phi$ ; put  $P = \pi + \pi' = 2\pi - \Phi$ . Cross transports cost 1 each and their cost does not depend on the signs, so for fixed  $\pi, \pi'$  the cost is  $P + 2\text{mis}_L + 2\text{mis}_R$ , where  $\text{mis}_L$  is the least number of opposite-sign pairs among the  $g$  same-side pairs on the left, minimised over which supplies stay. If  $a^+$  of the staying supplies and  $c^+$  of the served demands carry  $+$ , that number is  $|a^+ - c^+|$ , since  $\min(a^+, c^+) + \min(g - a^+, g - c^+) = g - |a^+ - c^+|$  pairs can be matched with equal signs and no more. Here  $a^+$  ranges over the integer interval  $[\max(0, g - A^-), \min(A^+, g)]$ , which is non-empty because  $0 \leq g \leq A^+ + A^-$ , and  $c^+$  over  $[\max(0, g - C^-), \min(C^+, g)]$ , likewise non-empty. The least  $|a^+ - c^+|$  over two non-empty integer intervals  $[l_1, r_1], [l_2, r_2]$  is  $\max(0, l_1 - r_2, l_2 - r_1)$ . Here  $l_1 - r_2 = \max(0, g - A^-) - \min(C^+, g)$ , which is  $g - A^- - C^+$  when  $g \geq A^-$  and  $C^+ \leq g$ , and is at most 0 otherwise, so  $\max(0, l_1 - r_2) = \max(0, g - A^- - C^+)$ ; symmetrically  $\max(0, l_2 - r_1) = \max(0, g - g - C^- - A^+)$ , and

$$\text{mis}_L = \max(0, g - A^- - C^+, g - C^- - A^+).$$

Substituting  $A^\pm = (N_L^a \pm \sigma_L^a)/2$  and  $C^\pm = (N_L^d \pm \sigma_L^d)/2$  gives  $g - A^- - C^+ = \frac{\Phi - \alpha}{2} - \pi$  and  $g - C^- - A^+ = \frac{\Phi + \alpha}{2} - \pi$ , hence  $2\text{mis}_L = \max(0, |\alpha| + \Phi - 2\pi) = \max(0, |\alpha| - P)$ . The same computation on the right gives  $2\text{mis}_R = \max(0, |\beta| - \Phi - 2\pi') = \max(0, |\beta| - P)$ . Therefore the cost is

$$\phi(P) = P + (|\alpha| - P)^+ + (|\beta| - P)^+,$$

which for  $P \geq 0$  is at least  $\max(|\alpha|, |\beta|)$ , with equality exactly on  $\min(|\alpha|, |\beta|) \leq P \leq \max(|\alpha|, |\beta|)$ , and which equals  $P$  above that range.

It remains to identify the admissible  $P$ . From  $\pi \geq 0$  and  $\pi' \geq 0$  one gets  $P \geq |\Phi|$ ; from  $\pi \leq \min(N_L^a, N_R^d)$  and  $\pi' \leq \min(N_R^a, N_L^d)$  one gets, using  $\Phi = N_L^a - N_L^d = N_R^d - N_R^a$ , the single constraint  $P \leq \min(N_L^a + N_L^d, N_R^a + N_R^d)$ . Finally  $P \equiv \Phi \pmod{2}$ , and  $\alpha \equiv \beta \equiv \Phi \pmod{2}$  because  $\sigma \equiv N \pmod{2}$  on each side. Now  $|\alpha| \leq |\sigma_L^d| + |\sigma_L^a| \leq N_L^d + N_L^a$  and  $|\Phi| \leq N_L^a + N_L^d$ , and symmetrically  $|\beta|, |\Phi| \leq N_R^a + N_R^d$ . Hence  $P^* := \max(|\Phi|, \min(|\alpha|, |\beta|))$  is admissible: it has the right parity, it is at least  $|\Phi|$ , and it is at most  $\max(|\Phi|, |\alpha|) \leq N_L^a + N_L^d$  and at most  $\max(|\Phi|, |\beta|) \leq N_R^a + N_R^d$ . If  $|\Phi| \leq \max(|\alpha|, |\beta|)$  then  $P^*$  lies in the equality range and  $\phi(P^*) = \max(|\alpha|, |\beta|)$ , which is optimal since  $\phi \geq \max(|\alpha|, |\beta|)$  throughout. If  $|\Phi| > \max(|\alpha|, |\beta|)$  then every admissible  $P$  satisfies  $P \geq |\Phi|$  and  $\phi(P) = P$ , so the minimum is  $|\Phi|$ , attained at  $P^* = |\Phi|$ . In both cases the minimum is  $\max(|\alpha|, |\beta|, |\Phi|)$ .  $\square$

**Corollary 52** (the local cost law). *In Definition 49, let  $\alpha_s, \beta_s, \Phi_s$  be the three quantities of Lemma 51 formed from the arrivals and departures at site  $s$ . Then*

$$\alpha_s = d_{s-1} - [s = 0] + \varepsilon^*[s = k^*][\delta^* = 0], \quad \beta_s = d_s - \varepsilon^*[s = k^*][\delta^* = 1],$$

$$\Phi_s = f_{s-1} + [s = 0] - [s = k^*][\delta^* = 0],$$

one has  $|\Phi_s| \leq \min(|\alpha_s|, |\beta_s|)$  in every case, and the minimal pairing cost at  $s$  is

$$\text{Site}(s) = \max(|\alpha_s|, |\beta_s|),$$

independently of the crossing counts  $m_j$ , of the sign splits  $p_j^u, p_j^d$ , and of the travel indicators. In particular at a site carrying no virtual event  $\text{Site}(s) = \max(|d_{s-1}|, |d_s|)$ .

*Proof.* The displays are read off Definition 49: the up-crossings of edge  $s-1$  contribute  $\sigma_L^a$ , the down-crossings contribute  $\sigma_L^d$ , and  $d_{s-1} = 2p_{s-1}^d - \text{dn}_{s-1} + u_{s-1} - 2p_{s-1}^u = \sigma_L^d - \sigma_L^a$ , which is  $\alpha_s$  before the virtual events; the virtual arrival adds one to  $A^+$  and the virtual departure adds one to  $C^{\varepsilon^*}$  or to  $D^{\varepsilon^*}$ . Likewise  $d_s = \sigma_R^a - \sigma_R^d$  and  $\Phi_s = u_{s-1} - \text{dn}_{s-1} = f_{s-1}$  before the virtual events. Since the sign splits enter only through  $\sigma^a, \sigma^d$ , and those enter  $\alpha_s, \beta_s, \Phi_s$  only through the fixed combinations  $d_{s-1}, d_s, f_{s-1}$ , none of the three depends on  $m_j$  or on  $p_j^u, p_j^d$ .

For  $|\Phi_s| \leq \min(|\alpha_s|, |\beta_s|)$ : at a site with no virtual event  $\Phi_s = f_{s-1} = f_s$ , since arrivals and departures balance, and  $|d_j| \geq |f_j|$ , since a bulk deposit is even with  $f = 0$  and a travel deposit is odd with  $|f| = 1$ . The virtual events sit at  $s = 0$  and  $s = k^*$ , which for  $k^* \neq 0$  are exactly the two sites where  $f$  jumps, and there are eight remaining cases. For  $s = 0 < k^*$ :  $\Phi_0 = f_{-1} + 1 = 1$ , while  $\alpha_0 = d_{-1} - 1$  is odd and  $\beta_0 = d_0$  is odd. For  $s = 0$  and  $k^* < 0$ :  $\Phi_0 = -1 + 1 = 0$ . For  $s = k^* > 0$ :  $\Phi = 1 - 1 = 0$  if  $\delta^* = 0$ , and  $\Phi = 1$  with  $\alpha = d_{k^*-1}$  odd,  $\beta = d_{k^*} - \varepsilon^*$  odd, if  $\delta^* = 1$ . For  $s = k^* < 0$ :  $\Phi = -1$  with  $\alpha = d_{k^*-1} + \varepsilon^*$  odd and  $\beta = d_{k^*}$  odd if  $\delta^* = 0$ , and  $\Phi = 0$  if  $\delta^* = 1$ . For  $k^* = 0$ , where the two virtual events share the site:  $\Phi_0 = 0$  if  $\delta^* = 0$ , and  $\Phi_0 = 1$  with  $\alpha_0 = d_{-1} - 1$  and  $\beta_0 = d_0 - \varepsilon^*$  both odd if  $\delta^* = 1$ . In each case  $|\Phi_s| \leq \min(|\alpha_s|, |\beta_s|)$ , so Lemma 51 gives  $\text{Site}(s) = \max(|\alpha_s|, |\beta_s|)$ .  $\square$

**Corollary 53** (the closed form for  $\ell_R$ , and rigidity of the crossing counts). *Let the span be the least interval of edges containing 0, all  $j$  with  $d_j \neq 0$  and all  $j$  with  $f_j \neq 0$ . Every realisation of minimum cost has support exactly the span and crossing counts*

$$m_j = \max(|d_j|, |f_j|), \quad \text{forced to 2 when that value is 0,}$$

and

$$\ell_R = \sum_{j \in \text{span}} m_j + \sum_s \max(|\alpha_s|, |\beta_s|),$$

the second sum over the sites of the span.

*Proof.* By Corollary 52 the pairing cost at every site is independent of the  $m_j$ , so the total cost is  $\sum_j m_j$  plus a quantity that does not depend on the  $m_j$ ; hence a minimum-cost realisation minimises  $\sum_j m_j$  alone, subject to  $m_j \geq \max(|d_j|, |f_j|)$ ,  $m_j \equiv d_j$ , and the support being an interval containing 0. Since  $d_j \equiv f_j$ , the value  $\max(|d_j|, |f_j|)$  has the parity of  $d_j$  and is the least admissible value on the span except where it vanishes, where the support condition forces  $m_j \geq 2$ . Enlarging the support beyond the span adds an edge with  $m \geq 2$  and creates one further site, of cost  $\max(0, 0) = 0$ , without changing any other site cost, so it adds 2 and never helps.  $\square$

Corollary 52 is the first clause of (M1) below, proved. The marker clause needs the sign, and the form previously used does not carry it.

**Corollary 54** (the marker junctions). *Let  $k^* > 0$ , let  $d_L = d_{-1}$  be the last bulk deposit (even) and  $d_R = d_0$  the first travel deposit (odd). Then the cost of the junction at site 0 is*

$$\text{Site}_0(d_L, d_R) = \max(|d_L - 1|, |d_R|),$$

for all four marker data. At the far junction, site  $k^*$ , with  $d_L = d_{k^*-1}$  odd and  $d_R = d_{k^*}$  even, the cost is  $\max(|d_L + \varepsilon^*|, |d_R|)$  if  $\delta^* = 0$  and  $\max(|d_L|, |d_R - \varepsilon^*|)$  if  $\delta^* = 1$ ; it is not independent of  $(\varepsilon^*, \delta^*)$  and it is not the mirror image of  $\text{Site}_0$ .

*Proof.* Corollary 52. At site 0 neither  $\varepsilon^*$  nor  $\delta^*$  occurs.  $\square$

*Remark 55* (the earlier marker form is false). Earlier versions of this paper, and [8], carried  $\text{Site}_0(d_L, d_R) = \max(|d_L| - 1, |d_R|)$ . That agrees with Corollary 54 exactly on  $d_L \geq 0$  and fails on every cell with  $d_L \leq -2$  and  $|d_R| \leq |d_L|$ : the smallest is  $d_L = -2$ ,  $d_R = 1$ , where the true cost is 3 and the earlier form gives 1. The verification on which the earlier form rested covered  $0 \leq d_L \leq 12$  only. The junction cost therefore depends on the *sign* of the last bulk deposit, not on its magnitude alone.

The correction is local. Every bulk edge cost and every interior bulk site cost depends on the deposit only through its magnitude (Corollary 52), so reversing the sign of the junction-adjacent deposit of a bulk run is a weight-preserving involution on bulk runs; writing  $B_\sigma$  for the generating function of the bulk runs whose junction-adjacent deposit has magnitude  $\sigma$  (so  $B_{2b} = P_b$  of (21), and  $B_0 = 1/(1 - \mathfrak{g}t_1) \neq 0$  of Remark 46 collects the empty chain and the chains opening with a gap run), the two signs carry  $B_\sigma/2$  each, and the near-junction vector of (M3) is

$$\mu_s^{(y)} = B_0 x^{2s+1} + \sum_{b \geq 1} \frac{P_b}{2} \left( x^{\max(2b-1, 2s+1)} + x^{\max(2b+1, 2s+1)} \right), \quad (24)$$

in place of  $\sum_\sigma B_\sigma x^{\max(\sigma-1, 2s+1)}$ . The far-junction vector is

$$E_s^{(y)} = B_0^{[s]} x^{2s} + B_0 x^{2s+2} + \sum_{b \geq 1} P_b \left( x^{\max(2s, 2b)} + x^{\max(2s+2, 2b)} \right), \quad B_0^{[s]} = \begin{cases} B_{0z} := 1 + y \mathfrak{g} \sum_b q^b P_b, & s = 0, \\ B_0, & s \geq 1, \end{cases} \quad (25)$$

and  $\lambda^{(y)} := E^{(y)} + 2\mu^{(y)}$ . The slot  $B_{0z}$  carries the junction cut: at  $y = 1$  it equals  $B_0$ , and at  $y = q$  it equals  $B_0(1 - t_1 q / (1 + q))$ . This changes no rank statement below: Proposition 69 concludes

that the junction is not of rank one, and the symmetrised matrix (24) has rank 6 on the block  $\sigma \in \{0, 2, 4, 6, 8, 10\}$ ,  $2s + 1 \in \{1, 3, 5, 7, 9, 11\}$ , by exact elimination over  $\mathbb{Q}$  at  $x = \frac{1}{2}$ , against rank 5 for the earlier form on the same block.

**Proposition 56** (the cut criterion and the shield lower bound). *Call a site  $s$  of the span cut if  $\alpha_s = \beta_s = \Phi_s = 0$ , and let  $Z$  be the set of cut sites interior to the span. At a cut site every minimum-cost pairing matches each arrival with a departure on its own side, so no strand crosses  $s$ ; at any other interior site  $s$  some minimum-cost pairing lets a strand cross. Consequently every minimum-cost realisation has at least  $|Z| + 1$  components and*

$$c \geq |Z|.$$

*A maximal run of  $L$  gap edges of the bulk ( $d_j = f_j = 0$ ) flanked by bulk deposits and containing no virtual event contributes exactly its  $L - 1$  interior sites to  $Z$ .*

*Proof.* At a cut site  $\phi(P) = P$  in the notation of Lemma 51, so the unique minimiser is  $P = 0$ , which is the assertion. At any other interior site  $\max(|\alpha_s|, |\beta_s|) > 0$ , since  $|\Phi_s| \leq \min(|\alpha_s|, |\beta_s|)$ ; the equality range of  $\phi$  is  $[\min(|\alpha_s|, |\beta_s|), \max(|\alpha_s|, |\beta_s|)]$  and contains a value  $P > 0$  admissible for the caps of Lemma 51: namely  $\min(|\alpha_s|, |\beta_s|)$  when that is positive, which is the choice  $P^*$  of Lemma 51; and 2 otherwise, since  $\min(|\alpha_s|, |\beta_s|) = 0$  forces  $\Phi_s = 0$ , hence  $\alpha_s \equiv \beta_s \equiv 0 \pmod{2}$  and  $\max(|\alpha_s|, |\beta_s|) \geq 2$ . The two caps of Lemma 51 are  $m_{s-1}$  and  $m_s$ , each augmented by 1 for a virtual event of  $s$  on that side, and both are at least 2 when  $\Phi_s = 0$ : an edge with  $f = 0$  has  $m \geq 2$  on the span by Corollary 53, while  $\Phi_s = f_{s-1} + [s = 0] - [s = k^*][\delta^* = 0] = f_s + [s = k^*][\delta^* = 1]$  shows that an adjacent edge with  $|f| = 1$  forces a virtual event of  $s$  on its side, which supplies the missing unit. Removing the  $|Z|$  cut sites disconnects the crossings of the span into at least  $|Z| + 1$  classes, no two of which can lie in the same component; one component carries the virtual events, so at least  $|Z|$  are isolated cycles. For the last sentence, an interior site of such a run has  $d_{s-1} = d_s = 0$  and  $f_{s-1} = 0$  and no virtual event, hence is cut, while the two sites at the ends of the run have  $|\alpha|$  resp.  $|\beta|$  equal to the flanking bulk deposit, which is even and non-zero.  $\square$

*Remark 57* (what the shield law still owes). Proposition 56 gives  $c \geq |Z|$ . The reverse inequality  $c \leq |Z|$  asks for one minimum-cost realisation whose components are exactly the  $|Z| + 1$  classes cut out by  $Z$ , that is, for pairings that connect all crossings within each class. It is not proved here. It is verified by exhaustive enumeration, over every sign split and every site bijection, on all bulk configurations with  $k^* = 0$ , up to 8 edges, even deposits of magnitude at most 4 and both signs, and all four marker data: 1 048 544 configurations, gap runs up to length 6, 0 exceptions to both  $c = |Z|$  and the closed form of Corollary 53 (`code/zeta_probe/tools/sitecost`, mode `shield`). The closed form  $|Z|$  also recovers, by inspection, the boundary shield  $\text{sh}_R = \mathbf{1}[\varepsilon^* = -1 \text{ or } \delta^* = 1]$  that [6, the explicit transducer] records at  $k^* = 0$ : by Corollary 52 the marker site is cut exactly when  $d_{-1} = 0$ ,  $\delta^* = 0$  and  $\varepsilon^* = 1$ , which is exactly where that indicator vanishes.

*Remark 58* (the boundary term in the cut count, and the check at  $k^* \neq 0$ ). The set  $Z$  of Proposition 56 consists of cut sites *interior* to the span. That count is short by one whenever both markers sit at an endpoint of the span on the side opposite the edges, since cutting there isolates the marker component and leaves every crossing in closed walks. The configuration is exactly

$$k^* = 0, \quad \delta^* = 0, \quad \text{lo} = 0 < \text{hi},$$

and it is the boundary shield of Remark 57: the marker site is cut precisely where  $\text{sh}_R$  vanishes. The clause  $\text{lo} = 0 < \text{hi}$  is needed as well, since without it the identity is miscounted, having  $c = 0$  and one cut site.

Writing  $Z^+$  for  $Z$  together with that site when the displayed condition holds, the shield law  $c = |Z^+|$  has no exceptions on the 3336511 elements of word length at most 31, over all  $k^*$ , with  $c = (\ell_T - \ell_R)/2$  computed from breadth-first word length and the closed form of Corollary 53 (code/zeta\_probe/tools/nogap). This extends the check of Remark 57, which covers the bulk  $k^* = 0$  only. Enlarging  $Z$  by every marker site instead, rather than by the boundary configuration above, produces 8555 violations of Proposition 56 at word length 19, which is how the condition was located.

The count  $|Z^+|$  is the site-local cut count `cTrue` of `CorrectedSpan.lean`, and  $\ell_T = \ell_R + 2|Z^+|$ , with  $\ell_R$  the closed form of Corollary 53, is now the Lean theorem `metricAll` (Theorem 66). What remains unproved is only the reverse inequality  $c \leq |Z^+|$  for the component count  $c$  of the optimisation of Definition 49; the growth-series results do not use it.

## 5.7 The gap-free case: the reverse inequality holds when $Z$ is empty

Call an edge  $j$  of the span a *gap edge* when  $d_j = 0$  and  $f_j = 0$ ; by Corollary 53 such an edge is forced to  $m_j = 2$  by reachability. The reverse inequality is out of reach in general (Remark 57), but it holds, with a short proof, on the elements that carry no gap edge at all.

**Lemma 59** (the sign split is forced at minimum multiplicity). *Let  $j$  be a span edge with no gap, so  $d_j \neq 0$ . Then  $m_j^* = |d_j|$ , and the split  $p_j^d = p_j^u + (d_j - f_j)/2$  subject to  $0 \leq p_j^u \leq u_j$  and  $0 \leq p_j^d \leq dn_j$  has exactly one solution:  $p_j^u = 0$  when  $d_j > 0$  and  $p_j^u = u_j$  when  $d_j < 0$ . Consequently every up-crossing of edge  $j$  carries the sign  $-\text{sgn}(d_j)$  and every down-crossing the sign  $+\text{sgn}(d_j)$ .*

*Proof.*  $d_j \equiv f_j \pmod{2}$  and  $d_j \neq 0$  give  $|d_j| \geq |f_j|$  in both parities, so  $m_j^* = |d_j|$  and  $2dn_j = d_j - f_j$  when  $d_j > 0$ . Then  $p_j^d = p_j^u + dn_j \leq dn_j$  forces  $p_j^u \leq 0$ , hence  $p_j^u = 0$ . The case  $d_j < 0$  is the mirror image.  $\square$

**Lemma 60** (the pair cost sees only the side pattern). *At a site both of whose edges are gap-free, the left arrivals all carry one sign and the left departures the opposite sign, and likewise on the right. Hence  $\text{pc} = 2$  for a bounce and 1 for a pass, independently of which strands are matched.*

*Proof.* Left arrivals are the up-crossings of the left edge and left departures its down-crossings, so Lemma 59 gives them opposite signs; same on the right.  $\square$

**Lemma 61** (free swaps). *Under Lemma 60, the 2-swap exchanging the departures of two pairs changes the site cost by 0 exactly when the two arrivals lie on the same side or the two departures do; the only cost-lowering patterns are two bounces on opposite sides. In particular a pair that is a bounce never blocks.*

*Proof.* Sixteen side patterns, checked exhaustively; formalised as `NoGapMerge.swap_free_iff`, `swap_neg_iff` and `bounce_never_blocks`.  $\square$

**Theorem 62** (no gap edge forces  $c = 0$ ). *If  $g$  has no gap edge then  $c(g) = 0$ .*

*Proof.* Since the site cost of Lemma 51 depends on neither the crossing counts nor the sign splits, every relaxed-optimal realisation has  $m_j = m_j^*$ , so Lemmas 59–61 apply at every site. Take a relaxed-optimal realisation with the fewest components and suppose there are at least two. If two components hold strands of the leftmost span edge, that site is shared and both have  $(R \rightarrow R)$  bounces there, so by Lemma 61 they merge at no cost. Otherwise one component owns that edge; let  $B$  be another component whose leftmost site  $r$  is minimal, so  $r$  exceeds the left end of the span. Edge  $r - 1$  then lies in the span with  $m_{r-1} \geq 1$ , and its strands are not  $B$ 's, so site  $r$  is shared;  $B$ 's

ends at  $r$  come only from edge  $r$ , so  $B$  bounces there and again the merge is free. A 2-swap across two components merges them, so the component count drops at no cost, contradicting minimality. Hence there is one component and no isolated cycle.

One configuration escapes the last step: if  $r$  carries the virtual arrival,  $B$  may have a single  $(L \rightarrow R)$  pass and no bounce, and a neighbour with an opposing  $(R \rightarrow L)$  pass and  $d_{r-1} < 0$  would block. That neighbour then has no left arrival at  $r$ , so its strands on edge  $r - 1$  are all down-crossings, and any up-crossing there would belong to a third component which merges with  $B$  freely; so the blocker also needs  $u_{r-1} = 0$ , that is  $d_{r-1} = f_{r-1} = -1$  and a single strand. But  $f_{r-1} = -1$  forces  $k^* < 0$ , and  $r = 0$  since  $r$  carries the virtual arrival, so the marker component runs from site 0 to site  $k^* < 0$  and must traverse edge  $r - 1$ ; its one strand is therefore the marker component's, contradicting the assumption that it belongs to another. The configuration does not occur.  $\square$

**Corollary 63.** *If the lamp support of  $g$  lies inside its travel interval then  $c(g) = 0$ .*

*Proof.* Support inside  $I_k$  makes the span equal to  $I_k$ , on which  $f_j = \pm 1$ , so no edge is a gap edge; apply Theorem 62.  $\square$

Corollary 63 is the statement used in the lifting argument for the true series, where it had been verified but not proved. Verification of Theorem 62 independent of the proof: 2 505 271 elements of word length at most 31, no exceptions (`code/zeta_probe/tools/nogap`).

*Remark 64* (the argument does not extend past  $Z = \emptyset$ ). Theorem 62 rests on Lemma 59, which needs  $d_j \neq 0$ . A gap edge has  $m = 2$  and  $d = 0$ , so  $p^d = p^u$  with  $u = dn = 1$  admits two splits and the homogeneity of Lemma 60 fails. Enumerating all 256 sign-and-side configurations of a 2-swap, 52 block, of which 36 share an arrival or a departure side, and a cost-0 bounce is blocked in 28; both criteria of Lemma 61 are therefore false in general. The reverse shield inequality is not reachable by this route.

*Remark 65* (the relaxed length decomposes at every  $k^*$ ). Corollary 53 is verified by direct enumeration only for  $k^* = 0$  (Remark 57), but it holds at every  $k^*$ . The site cost depends on neither  $m_j$  nor the sign splits, so  $\sum_j m_j$  is minimised separately at  $m_j = m_j^*$ ; edge  $j$  presents a departure at site  $j$  and an arrival at site  $j + 1$ , so the matchings at distinct sites act on disjoint sets of ends and are chosen independently; and in the relaxed model every family of per-site matchings is admissible, isolated cycles being free. No term falls below its local minimum and all local minima are attained together. The true metric behaves differently precisely because the single-walk requirement couples the sites, which is what Theorem 62 has to handle.

## Machine verification of the site cost

Lemma 51, Corollary 52, Corollary 53, Corollary 54, the counterexample of Remark 55 and all of Proposition 56 except its component count are formally verified in Lean 4 against Mathlib [35], in six files: `SiteCost.lean` (the transportation problem at one site), `MarkedSite.lean` (the virtual events and the local cost law), `PairingMatrix.lean` (bijections against multiplicity matrices), `Realisation.lean` (the whole edge path), `CutCross.lean` (the middle sentence of Proposition 56) and `CutComponents.lean` (the counting step of the component bound, over an abstract graph).

A pairing at a site is the structure `Plan`: sixteen non-negative integers with the eight matching equations of Definition 49; `Plan.cost` is its cost and `Plan.cross` the number of pairs on opposite sides. The minimum of Lemma 51 is formalised as `IsLeast`, so both halves are proved separately: `cost_lower_bound` holds for every pairing and `exists_plan_cost_eq` exhibits one of exactly that

cost. Definition 49 asks for a *bijection* between the arrivals and the departures rather than for a matrix, and the two are reconciled: `planOfBij` sends a bijection to its multiplicity matrix without changing the cost, and `bijOfMatrix` realises every matrix with the right row and column sums by a bijection, so `bij_min` is Lemma 51 stated over bijections and `MarkedSite.site_cost_bij` is Corollary 52 stated over bijections. A site is the structure `MarkedSite`: the displacement  $k^*$ , the site  $s$ , the marker data, and the crossing data of the two adjacent edges, constrained to realise the travel indicator `travel`. A realisation of a whole path is the structure `Realisation` over a `PathData`, which carries  $k^*$ , the marker data, the deposits and the span  $[A, B]$ ; `PathData.emptyPath` is a witness that the hypotheses are consistent.

| Paper step                                                                                             | Lean declaration                                                       | axioms               |
|--------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|----------------------|
| $\text{cost} \geq \max( \alpha ,  \beta ,  \Phi )$ , six duals                                         | <code>cost_lower_bound</code>                                          | standard             |
| a pairing attaining it                                                                                 | <code>exists_plan_cost_eq</code>                                       | standard             |
| Lemma 51, the minimum                                                                                  | <code>transport_min</code>                                             | standard             |
| a bijection gives a matrix of equal cost                                                               | <code>planOfBij_cost</code>                                            | standard             |
| a matrix gives a bijection of equal cost                                                               | <code>bijOfMatrix_cost</code>                                          | standard             |
| Lemma 51 over bijections                                                                               | <code>bij_min</code>                                                   | standard             |
| collapse to $\max( \alpha ,  \beta )$                                                                  | <code>siteCost_eq</code>                                               | standard             |
| $\alpha_s = d_{s-1} - [s=0] + \varepsilon^*[s=k^*][\delta^*=0]$                                        | <code>MarkedSite.alpha_eq</code>                                       | standard             |
| $\beta_s = d_s - \varepsilon^*[s=k^*][\delta^*=1]$                                                     | <code>MarkedSite.beta_eq</code>                                        | standard             |
| $\Phi_s = f_{s-1} + [s=0] - [s=k^*][\delta^*=0]$                                                       | <code>MarkedSite.Phi_eq</code>                                         | standard             |
| $ \Phi_s  \leq \min( \alpha_s ,  \beta_s )$ , all site types                                           | <code>MarkedSite.Phi_le_min</code>                                     | standard             |
| Corollary 52, the minimum at any site                                                                  | <code>MarkedSite.site_cost</code>                                      | standard             |
| Corollary 52 over bijections                                                                           | <code>MarkedSite.site_cost_bij</code>                                  | standard             |
| independence of $m_j$ and of $p_j^u, p_j^d$                                                            | <code>MarkedSite.cost_indep</code>                                     | propept              |
| interior site cost $\max( d_{s-1} ,  d_s )$                                                            | <code>MarkedSite.cost_interior</code>                                  | propept              |
| Corollary 54, near junction                                                                            | <code>PathData.marker_near</code>                                      | propept              |
| Corollary 54, far junction, $\delta^*=0$                                                               | <code>PathData.marker_far_left</code>                                  | propept              |
| Corollary 54, far junction, $\delta^*=1$                                                               | <code>PathData.marker_far_right</code>                                 | propept              |
| and it is neither $(\varepsilon^*, \delta^*)$ -free<br>nor the mirror of <code>Site<sub>0</sub></code> | <code>PathData.marker_far_depends</code><br>(the same declaration)     | none                 |
| off the span every site is empty                                                                       | <code>Realisation.pair_cost_zero_outside</code>                        | standard             |
| Corollary 53, the minimum                                                                              | <code>lR_closed</code>                                                 | standard             |
| Corollary 53, rigidity of $m_j$                                                                        | <code>Realisation.rigidity</code>                                      | standard             |
| Proposition 56, cut site: no crossing                                                                  | <code>cut_forces_no_cross</code>                                       | standard             |
| the same over a realisation                                                                            | <code>Realisation.cut_no_cross</code>                                  | standard             |
| the transportation value at a prescribed cross mass                                                    | <code>exists_plan_of_cross</code>                                      | standard             |
| the two caps $m_{s-1}, m_s \geq 2$ where $\Phi_s = 0$                                                  | <code>Realisation.capL_two</code><br><code>Realisation.capR_two</code> | standard<br>standard |
| Proposition 56, a non-cut site: a crossing                                                             | <code>Realisation.exists_cross_of_not_cut</code>                       | standard             |
| Proposition 56, a gap run's cut sites                                                                  | <code>PathData.gap_run_cut</code>                                      | standard             |
| $Z$ interior to the span                                                                               | <code>PathData.cutSet_lt</code> , <code>cutSet_le</code>               | standard             |
| every edge of the span carries a crossing                                                              | <code>Realisation.m_pos_of_min</code>                                  | standard             |
| blk constant on components                                                                             | <code>CutComponents.blk_reachable</code>                               | propept              |
| blk attains $0, \dots,  Z $ on the span                                                                | <code>CutComponents.exists_pos_with_gz</code>                          | standard             |
| at least $ Z  + 1$ components, given the graph                                                         | <code>CutComponents.exists_injective_components</code>                 | standard             |
| and $c \geq  Z $ from it                                                                               | <code>..._avoiding</code>                                              | standard             |
| the two marker forms agree on $d_L \geq 0$                                                             | <code>marker_forms_agree</code>                                        | standard             |
| and differ at $(d_L, d_R) = (-2, 1)$                                                                   | <code>marker_forms_differ</code>                                       | none                 |

“standard” means `propept`, `Classical.choice`, `Quot.sound`, or a subset of them; several of the rows above in fact report only `propept` and `Quot.sound`. The last row is closed by `decide` and depends

on no axiom at all, as do `PathData.marker_far_depends` and `Realisation.site_cost_eq`. The cost of a realisation is summed over the sites of its support interval, which is legitimate because both virtual events lie there (`Realisation.virtual_in_window`) and every site outside it is empty.

The middle sentence of Proposition 56, that at an interior site which is not cut some minimum-cost pairing lets a strand cross, is verified in the form the proof uses. `exists_plan_cost_eq` settles every case except  $\min(|\alpha_s|, |\beta_s|) = 0$  with  $\Phi_s = 0$ ; there `exists_plan_of_cross` runs the construction of Lemma 51 at the prescribed cross mass  $P = 2$ , which is admissible because  $\Phi_s = 0$  forces both caps to be at least 2, and that is a path-level fact, proved as `Realisation.capL_two` and `Realisation.capR_two` from  $m_j \geq \max(|d_j|, |f_j|)$  on the span together with  $\Phi_s = f_{s-1} + [s=0] - [s=k^*][\delta^*=0] = f_s + [s=k^*][\delta^*=1]$ .

The counting step of the component bound is verified in `CutComponents.lean`, over an abstract graph. Write  $\text{blk}(v)$  for the number of cut sites at or below the edge carrying the crossing  $v$ . `blk_adj` and `blk_reachable` are that `blk` is constant on connected components, given the locality hypothesis `Local`: an edge of the graph joins crossings on the two edges adjacent to one site, and joins crossings on *different* edges only at a site that is not cut. `exists_pos_with_gz` is the discrete intermediate value argument, that `blk` takes every value in  $0, \dots, |Z|$  on the span, and `exists_injective_components` concludes with an injection of  $\{0, \dots, |Z|\}$  into the connected components, which is “at least  $|Z| + 1$  components” with no finiteness hypothesis. `exists_injective_components_avoiding` is the passage to  $c \geq |Z|$  once the two virtual events are known to share a component. Two of the three inputs are supplied against the structures of `Realisation.lean`: `PathData.cutSet` is  $Z$  with its two bounds (`cutSet_lt`, `cutSet_le`), and `Realisation.m_pos_of_min` is the occupancy, every edge of the span of a minimum-cost realisation carrying a crossing.

Two things in this subsection are *not* formalised. First, that the optimisation of Definition 49 computes the relaxed word length (Remark 50, an input of [6], verified there and not proved). Second, the strand graph itself, and with it the component count  $c \geq |Z|$ . `Realisation.pair` stores the multiplicity matrix of the site pairing rather than the bijection, so the vertex set and the incidence relation that `CutComponents.lean` quantifies over are not definable against the present structure and would need a further definition. `Realisation.cut_no_cross` already carries the mathematical content of the locality hypothesis at the level of the matrix; what is missing is the passage from a matrix with vanishing cross mass to an incidence relation on individual crossings.

## 5.8 The model (M), and the assembly

Everything above is a statement about the explicit operators of Definition 33 and (21), or about the optimisation of Definition 49. What connects those to the growth series is the strand-walk input of Route B, which we state once and name.

**Theorem 66** (the transfer model (M)). (M1) Local cost law. *The site cost between adjacent active edges carrying deposits  $d, d'$  is  $x^{\max(|d|, |d'|)}$ , independent of the crossing counts and of the travel indicator. Bulk deposits are even,  $|d| = 2s$ , and travel deposits odd,  $|d| = 2s + 1$ , so this cost is  $q^{\max(s, s')}$  in the bulk and  $x^{2\max(s, s') + 1}$  on the travel side, and the edge weights are  $2q^s$  and  $2x^{2s+1}$  respectively: Definition 33 in its two instances. At the near marker site the cost is  $x^{\text{Site}_0(d_L, d_R)}$  with  $\text{Site}_0(d_L, d_R) = \max(|d_L - 1|, |d_R|)$ ,  $d_L$  the last bulk deposit and  $d_R$  the first travel deposit, for all four marker data  $(\varepsilon^*, \delta^*)$ ; at the far marker site it is the  $(\varepsilon^*, \delta^*)$ -dependent cost of Corollary 54.* (M2) Shield law. *A maximal interior run of  $L$  forced gap edges costs  $q^L$  and carries  $L - 1$  isolated cycles; hence the gap-bridge kernel is  $\mathbf{g}(q, y) = q/(1 - qy)$  and the gap-marked bulk kernel is the  $K_{\mathbf{g}}$  of (21).* (M3) Decomposition. *The bivariate series of §7.1 is  $\mathcal{W} = \mathcal{W}_+ + \mathcal{W}_- + \mathcal{W}_0$ , the sectors*

$k^* > 0$ ,  $k^* < 0$ ,  $k^* = 0$ , with

$$\mathcal{W}_+ = x^{-1} \langle \mu^{(y)}, (I - \mathcal{T})^{-1} D_e \lambda^{(y)} \rangle, \quad \mathcal{W}_- = x^{-1} \langle \tfrac{1}{2} E^{(y)}, (I - \mathcal{T})^{-1} D_e \lambda^{(y)} \rangle, \quad \mathcal{W}_+ + \mathcal{W}_- = \frac{1}{2x} \langle \lambda^{(y)}, (I - \mathcal{T})^{-1} D_e \lambda^{(y)} \rangle, \quad (26)$$

where  $\mathcal{T} = D_e K$  is the travel operator of Definition 33,  $D_e = \text{diag}(e_s)$  with  $e_s = 2q^{1+s}$ , and  $\mu^{(y)}, E^{(y)}, \lambda^{(y)}$  are as in (24)–(25). The sum over the marker data  $(\varepsilon^*, \delta^*)$  is already carried out in these vectors:  $\delta^* = 1$  at the far site gives  $2\mu$  and  $\delta^* = 0$  gives  $E$  when  $k^* > 0$ , and the roles are exchanged when  $k^* < 0$ . The  $k^* = 0$  sector is  $\mathcal{W}_0 = A_0 + B_0 A_1 + B_0^2 A_2$ , with  $A_i$  built from bulk quantities alone; it contains no travel resolvent.

*Proof and standing.* Let  $(k^*, \varepsilon^*, \delta^*, d)$  parametrise the elements of  $W_{\text{univ}}$  as in [6, Theorem “Normal form and lamp model”, thm:normal-form]. Two facts are formalised in Lean 4 with the standard axioms only:

- the affine realisation intertwines the three generators and is injective, and every element of the model is reachable, so the word length of the model is the Cayley length in  $W_{\text{univ}}$  (RJP.hi.lean: `intertwine_s1, intertwine_s2, intertwine_s3, injective; reachability is reaches_of_phiZ` in `PhiLipschitz.lean`);
- `wordLength =  $\ell_R + 2c$`  for every element, with  $\ell_R$  and  $c$  the site-local closed forms `lRTrue, cTrue` of `CorrectedSpan.lean` (RJMetricAll.lean, theorem `metricAll`).

Given these,  $\ell_R$  and  $c$  are sums of site and edge costs whose only non-bulk sites are 0 and  $k^*$  (Corollaries 52, 54; a gap run of length  $L$  contributes  $q^L y^{L-1}$  and a travel-interior site is never a cut). Summing these site costs chain by chain gives (21) with source  $u$ , the vectors (24)–(25), and the three sectors of (26); this is Theorem 138, proved in Appendix D, and the regrouping of  $\mathcal{W}_0$  is Remark 148. That last step is a hand proof, with two independent adversarial reviews, and it is not formalised. Every identity holds in  $\mathbb{Z}[y][[x]]$ , and the series agree with enumeration as recorded in §7.1.  $\square$

Earlier versions carried (M) as a hypothesis, with three named weak links: the reverse shield inequality  $c \leq L - 1$  (Remark 57), the metric formula, and (M3). The first two are no longer separate inputs. The cut count  $c$  that enters  $\mathcal{W}$  is the site-local count `cTrue`, and `metricAll` proves `wordLength =  $\ell_R + 2c$`  for it directly, so neither the reverse inequality nor the optimisation of Definition 49 is consumed. One caveat:  $\ell_R$  in  $\mathcal{W}$  is the closed-form relaxed length of [6]. That it also equals the minimum over relaxed realisations ([6], `rem:metric-formula-status`) affects only the *interpretation* of  $V$  as a relaxed count, not  $U$  and not the series identities. The third, (M3), is proved in Appendix D, in the corrected form stated above. Nothing in §§2–4.3, §4, §§5.1–5.6 above, or Appendices A–B uses it: those speak only about the explicit series of (2), about  $P_{12}$ , about the operators just defined, and about the optimisation of Definition 49.

$V = \mathcal{W}(x, 1)$  and  $U = \mathcal{W}(x, q)$  (§7.1), the two evaluations differing only through  $\mathbf{g}_V = \frac{q}{1-q}$ ,  $\mathbf{g}_U = \frac{q}{1-q^2}$  and the slot  $B_{0z}$ ; and  $\lambda^{(y)}, \mu^{(y)}, \mathcal{W}_0$  are holomorphic at a travel pole  $q_m$ , by (23), as soon as  $S_e(q_m) \neq 0$  and  $1 - \mathbf{g}t_1 \neq 0$  there, which is Proposition 94 and Corollary 111 ( $|\mathbf{g}_V t_1| \leq 0.983$ , and  $|\mathbf{g}_U t_1| < |\mathbf{g}_V t_1|$ ). In particular  $\mathcal{W}_0$  has no travel pole: by Remark 148 it is a polynomial in  $B_0$  with coefficients built from the bulk vector  $P$  and monomials, so its only possible singularities are at  $S_e = 0$  and at  $1 - \mathbf{g}t_1 = 0$ .

The growth series are defined as power series, with radius of convergence  $x_1 = \sqrt{q_1} = 1/\beta_2 = 0.6704 \dots$  (Corollary 78), and the local forms below are statements about functions on a larger disc. The following lemma supplies that continuation and identifies it with the series.

**Lemma 67** (meromorphic continuation). *Let  $y$  be either a complex constant with  $|y| \leq 1$  or  $y = x^2$ . Then the specialisation  $x \mapsto \mathcal{W}(x, y)$  of the series of Theorem 138 converges near  $x = 0$  and is the Taylor series at 0 of a single-valued meromorphic function on the disc  $|x| < 1$ , given there by (26) and  $\mathcal{W}_0 = A_0 + B_0 A_1 + B_0^2 A_2$  read as analytic expressions. Its poles in  $0 < |x| < 1$  lie over points  $q = x^2$  at which  $1 - \Sigma_1(q) = 0$ ,  $S_e(q) = 0$  or  $1 - \mathfrak{g}(q, y) t_1(q) = 0$ . In particular  $V = \mathcal{W}(x, 1)$  and  $U = \mathcal{W}(x, x^2)$  are meromorphic on  $|x| < 1$ , and  $G_0^T = \Sigma_0/(1 - \Sigma_1)$  is meromorphic on  $|q| < 1$ . (The continuation concerns the explicit operators of §§5.1–5.5; its identification with  $U$  and  $V$  is Theorem 66.)*

*Proof.* Put  $q = x^2$ , so  $|q| < 1$ ;  $\mathfrak{g} = q/(1 - qy)$  is holomorphic on the disc, since  $|qy| < 1$  in both cases.

*Operators.*  $q \mapsto \mathcal{T}(q)$  and  $q \mapsto M_0(q)$  are holomorphic on  $|q| < 1$  as maps into the bounded operators on  $\ell^1$ , in operator norm: the row truncations  $M_N$  in the proof of Lemma 42 have finitely many non-zero rows, each a norm-convergent  $\ell^\infty$ -valued power series in  $q$  (the row  $s$  is  $e_s \sum_{s'} q^{\max(s, s')} \delta_{s'}$ ), hence are holomorphic in norm, and  $\|M - M_N\| \leq \sum_{s > N} 2|q|^{\vartheta+s} \rightarrow 0$  uniformly on  $|q| \leq r < 1$ ; a locally uniform norm limit of holomorphic operator-valued maps is holomorphic. With compactness (Lemma 42) and invertibility near  $q = 0$  (Lemma 34), the analytic Fredholm theorem makes  $(I - \mathcal{T})^{-1}$  and  $R_0 = (I - M_0)^{-1}$  meromorphic on  $|q| < 1$  in operator norm, with poles only where  $1 - \Sigma_1 = 0$ , resp.  $S_e = 0$  (Corollary 44).

*Bulk quantities.*  $u = (2q^{2b})_b$  is a norm-convergent, hence holomorphic,  $\ell^1$ -valued function, so  $\Psi = R_0 u$  is meromorphic  $\ell^1$ -valued and  $t_1 = v^\top \Psi$  is meromorphic.  $1 - \mathfrak{g} t_1$  tends to 1 as  $x \rightarrow 0$ , so it is not identically zero, and  $B_0 = 1/(1 - \mathfrak{g} t_1)$ ,  $P = B_0 \Psi$ ,  $\mathfrak{g} S = B_0 - 1$  and  $B_{0z} = 1 + y(B_0 - 1)$  are meromorphic (Remark 148).

*Boundary vectors and the sectors  $k^* \neq 0$ .* Every entry of (40)–(42) is a combination of  $B_0$ ,  $B_{0z}$  and sums  $\sum_b P_b x^{m_b}$  with  $m_b \geq 0$ , so  $|\mu_s^{(y)}|, |E_s^{(y)}|, |\lambda_s^{(y)}| \leq C(|B_0| + |B_{0z}| + \|P\|_{\ell^1})$  uniformly in  $s$ , locally uniformly in  $x$  away from the poles of  $B_0$  and  $\Psi$ . Hence  $D_e \lambda^{(y)}$  has  $\ell^1$ -tails  $\sum_{s > N} |e_s \lambda_s^{(y)}| = O(|q|^{N+2})$ , locally uniformly, and is meromorphic  $\ell^1$ -valued; so is  $(I - \mathcal{T})^{-1} D_e \lambda^{(y)}$ . Its pairing with the bounded vectors  $\mu^{(y)}$  and  $\frac{1}{2} E^{(y)}$  is an absolutely convergent sum, locally uniformly convergent away from the poles (an  $\ell^1$ -valued continuous map has uniformly small tails on compact sets), hence meromorphic. This gives  $\mathcal{W}_\pm$ ; the prefactor  $x^{-1}$  is at most a pole at  $x = 0$ , and is removed below.

*The sector  $k^* = 0$ .* By Remark 148,  $\mathcal{W}_0 = A_0 + B_0 A_1 + B_0^2 A_2$ . The  $A_i$  are not finite sums: each is an absolutely convergent double sum, over the left and the right chain kinds  $\mathbf{E}, \mathbf{Z}, \mathbf{N}_{\pm, b}$  of (49) ( $b \geq 1$ ), of monomials  $x^j y^\chi$  with  $j \geq 0$ ,  $\chi \in \{0, 1\}$ , times at most two entries of  $\Psi$ . Since  $|x^j y^\chi| \leq 1$ , the double sum is dominated by a constant times  $(1 + \|\Psi\|_{\ell^1})^2$ , converges locally uniformly where  $\Psi$  is holomorphic, and is meromorphic.

*Identification with the  $x$ -adic series.* Choose  $r_0 > 0$  so small that on  $|x| < r_0$  the operators  $\mathcal{T}$ ,  $M_0$  and  $M_0 + \mathfrak{g} u v^\top$  have  $\ell^1$ -norm below  $\frac{1}{2}$  (Lemma 34; the rank-one term has norm at most  $\|\mathfrak{g}\| \|u\|_{\ell^1}$ ) and  $|\mathfrak{g} t_1| < \frac{1}{2}$ . There every object above is the sum of a norm-convergent Neumann series  $((I - \mathcal{T})^{-1}, R_0, P = (I - M_0 - \mathfrak{g} u v^\top)^{-1} u$ , and  $B_0 = \sum_n (\mathfrak{g} t_1)^n$ ), whose terms are holomorphic on  $|x| < r_0$  and whose Taylor series at 0 are the terms of the  $x$ -adic Neumann series of Lemma 140, of  $x$ -adic order tending to infinity. If  $\sum_n h_n$  converges uniformly on a disc about 0, the Taylor coefficients of the sum are the sums of the Taylor coefficients of the  $h_n$  (Weierstrass); when  $\text{ord } h_n \rightarrow \infty$  each of these sums is finite and is the  $x$ -adic sum. Applying this to each Neumann series, to the absolutely convergent pairings and to the double sums  $A_i$ , the Taylor series at 0 of the analytic  $\mathcal{W}(x, y)$  is the  $x$ -adic series of Theorem 138, specialised at  $y$  (the specialisation  $\mathbb{Z}[y][[x]] \rightarrow \mathbb{C}[[x]]$  at a constant  $y$ , or at  $y = x^2$ , is  $x$ -adically continuous and commutes with the sums). In particular that function is holomorphic at  $x = 0$ . At  $y = 1$  the series is  $V$ , and at  $y = x^2$  it is  $U$  (Corollary 139). A power series that agrees near 0 with a function meromorphic on the disc is continued by it, uniquely. For

$G_0^T$ ,  $\Sigma_0$  and  $1 - \Sigma_1 \neq 0$  are holomorphic on  $|q| < 1$  (Theorem 2, first clause).  $\square$

**Proposition 68** (the shape of the local form). *Let  $q_m$  be a travel pole, let  $p \geq 1$  be the order of the zero of  $1 - \Sigma_1$  there (finite, Lemma 103), and let  $\pi$  be the order of the pole of  $(I - \mathcal{T})^{-1}$  there (finite, Lemma 42). Then  $\pi \geq p \geq 1$  and, for each  $y$ ,*

$$\mathcal{W}(x, y) = \frac{\mathcal{N}_y}{(q - q_m)^\pi} + (\text{poles of order } < \pi) + H_m, \quad H_m \text{ holomorphic at } q_m,$$

with the leading numerator a perfect square

$$\mathcal{N}_y = \frac{c_\star}{2x_m} \Pi_y(q_m)^2, \quad \Pi_y(q_m) := \langle \lambda^{(y)}, R \rangle, \quad c_\star \neq 0, \quad x_m = \sqrt{q_m}, \quad (27)$$

where  $R$  spans  $\ker(I - \mathcal{T}(q_m))$ , normalised by its sections,

$$\sum_{s \geq 0} q^{ks} R_s = \Sigma_k(q_m) \quad (k \geq 0), \quad \text{in particular } \sum_s R_s = \Sigma_0(q_m), \quad \sum_s q^s R_s = \Sigma_1(q_m) = 1,$$

and  $L$  spans  $\ker(I - \mathcal{T}(q_m)^\top)$ , explicitly  $L_s = R_s/e_s$ . There are no cross terms between the sectors. In particular  $\Pi_y(q_m) \neq 0$  implies that  $q_m$  is a pole of  $\mathcal{W}(\cdot, y)$ , of order exactly  $\pi$ .

*Proof.* Write  $\mathcal{R}(q) = (I - \mathcal{T}(q))^{-1} = \sum_{j=1}^\pi A_j(q - q_m)^{-j} + \text{hol}$ , with  $A_\pi \neq 0$ . Multiplying  $(I - \mathcal{T}(q))\mathcal{R}(q) = I$  by  $(q - q_m)^\pi$  and letting  $q \rightarrow q_m$  gives  $(I - \mathcal{T}(q_m))A_\pi = 0$ , so  $\text{ran } A_\pi \subseteq \ker(I - \mathcal{T}(q_m))$ , which is one-dimensional by Lemma 43; the same step on  $\mathcal{R}(q)(I - \mathcal{T}(q)) = I$  gives  $A_\pi(I - \mathcal{T}(q_m)) = 0$ . Hence  $A_\pi x = c_\star \langle L, x \rangle R$  with  $c_\star \neq 0$ ,  $R$  spanning the kernel and  $L$  the cokernel. The normalisation of  $R$  is Lemma 43:  $G_k^R = G_1^R \Sigma_k$  and  $\Sigma_1(q_m) = 1$ , so  $G_1^R = 1$  gives  $G_k^R = \Sigma_k(q_m)$ . That  $L_s = R_s/e_s$  spans the cokernel follows from  $\mathcal{T} = D_e K$  with  $D_e = \text{diag}(e_s)$  and  $K$  symmetric: from  $D_e K R = R$  one gets  $K D_e (D_e^{-1} R) = K R = D_e^{-1} R$ , that is  $\mathcal{T}^\top L = L$ , and  $L \in \ell^\infty$  because  $|R_s| \leq |e_s| \|R\|_{\ell^1}$ ; the cokernel is one-dimensional by Fredholm. Substituting the Laurent expansion into  $\mathcal{W}_+ + \mathcal{W}_-$  of (26), and using that  $\lambda^{(y)}, \mathcal{W}_0$  are holomorphic at  $q_m$ , the leading coefficient is  $(2x_m)^{-1} \langle \lambda, A_\pi D_e \lambda \rangle = (2x_m)^{-1} c_\star \langle L, D_e \lambda \rangle \langle \lambda, R \rangle = (2x_m)^{-1} c_\star \langle \lambda, R \rangle^2$ , since  $D_e L = R$ . It is non-zero exactly when  $\Pi_y(q_m) \neq 0$ , and then  $q_m$  is a pole of order  $\pi$ . Finally  $\langle \mathbf{1}, R \rangle = \Sigma_0/(1 - \Sigma_1)$  (Proposition 37) has a pole of order exactly  $p$  at  $q_m$ , since  $\Sigma_0(q_m) \neq 0$  (Proposition 25), while a matrix element of  $\mathcal{R}$  has pole order at most  $\pi$ ; hence  $\pi \geq p$ .  $\square$

**Proposition 69** (the marker junction is not rank one). *The junction matrix  $J[\sigma, s]$  of (24) has rank at least 3. Already the one-sign matrix  $[x^{\max(\sigma-1, 2s+1)}]$  has rank at least 3: on  $\sigma \in \{0, 4, 8\}$ ,  $2s+1 \in \{1, 5, 9\}$  it is*

$$\begin{pmatrix} x & x^5 & x^9 \\ x^3 & x^5 & x^9 \\ x^7 & x^7 & x^9 \end{pmatrix}, \quad \det = x^9 (x - x^3)(x^5 - x^7) \neq 0 \quad (0 < x < 1),$$

its rank on  $\sigma \in \{0, 2, 4, 6, 8\}$ ,  $2s+1 \in \{1, 3, 5, 7, 9\}$  is 4, and the symmetrised matrix (24) has rank 6 on  $\sigma \in \{0, 2, 4, 6, 8, 10\}$ ,  $2s+1 \in \{1, 3, 5, 7, 9, 11\}$ . By contrast the gap-bridge term  $q^{a+b}$  of (21) is rank one, which is why (31) factorises. Consequently  $\mu^{(y)}$  is not a function of the total bulk sum  $B(q, y)$  alone, and  $\Pi_y(q_m) \neq 0$  does not follow from  $\Sigma_0(q_m) \neq 0$  together with  $B(q_m, y) \neq 0$ . At a travel pole, however, the pairing against the null vector  $R$  collapses to a function of  $t_1$  alone (Theorem 72).

*Proof.* The displayed block is the case  $\sigma - 1 \in \{-1, 3, 7\}$ ,  $2s + 1 \in \{1, 5, 9\}$ , which interleaves as  $-1 < 1 < 3 < 5 < 7 < 9$ ; subtracting the second row from the first gives  $(x - x^3, 0, 0)$ , and expanding along it leaves  $(x - x^3)(x^5x^9 - x^9x^7) = x^9(x - x^3)(x^5 - x^7)$ . The two rank statements are exact Gaussian elimination over  $\mathbb{Q}$ , at  $q = \frac{1}{3}$  and at  $x = \frac{1}{2}$  respectively (`assembly_certificate.py`, Part 4). For the second-last sentence: by (M3),  $\mu^{(y)}$  is the image of  $(B_\sigma)_\sigma$  under the junction matrix, whose rank is at least 3.  $\square$

**Theorem 70** (the bulk dictionary at the travel poles). *At every travel pole  $q_m$  the factorisation (31) holds with  $\kappa = t_0b_1 - b_0t_1$  and  $t_0 = b_1$ , and the dictionary (33) holds identically in  $0 < q < 1$ , in the completed form of Corollary 41. Consequently the bulk quantities  $B_0 = 1/(1 - \mathfrak{g}t_1)$  and  $B = b_1/(1 - \mathfrak{g}t_1)$  of Remark 46 are finite at every travel pole, for  $y = 1$  and  $y = q$ .*

*Proof.* Proposition 45, whose two hypotheses are supplied by Corollary 44 with Proposition 94 and by Remark 47; Proposition 40; Corollary 41. Finiteness is  $|\mathfrak{g}_V t_1| \leq 0.983$  (Corollary 111) and  $\mathfrak{g}_U < \mathfrak{g}_V$ .  $\square$

*Remark 71* (what (L) consumed). That the bulk transfer is the operator of (22), that is, (M1) and (M2). Of these, (M1) is now Corollary 52, the exponent  $q^{\max(s, s')}$  of the bulk kernel being  $x^{\max(|d|, |d'|)}$  at even deposits; and of (M2) the cost  $q^L$  of a gap run and the inequality  $c \geq L - 1$  are Corollary 53 and Proposition 56, and  $c \leq L - 1$  follows from the metric identity `metricAll` with the site-local cut count (Theorem 66), which no longer consumes the optimisation of Definition 49. Everything the paper previously listed as unproved *downstream* of that, namely the factorisation (31), the reciprocity  $t_0 = b_1$ , the entry  $t_1 = P_{12}/S_e$ , and the resolvent lines of (33), is proved above. The residue at  $q_m$  is governed not by the bulk sum but by the pairing  $\Pi_y(q_m)$  (Proposition 68), which Theorem 72 evaluates.

## 5.9 The junction pairing (R-J)

(R-J) is the statement that  $\Pi_y(q_m) = \langle \lambda^{(y)}, R \rangle \neq 0$  at infinitely many travel poles, for  $y = 1$  and  $y = q$ . Earlier versions carried it as a hypothesis. It is now a theorem.

**Theorem 72** (closed form of the junction pairing). *Let  $q_m$  be a travel pole,  $x = \sqrt{q_m}$ , and  $R$  normalised as in Proposition 68. Then*

$$(1 - \mathfrak{g}_V t_1) \Pi_1(q_m) = \frac{2q(1+x)}{1-q} \left( t_1 + \frac{1+x}{2x} \right), \quad (28)$$

$$\Pi_q(q_m) = \frac{2q B_0(1+x)}{1-x} \left( t_1 \frac{1-x+q}{1+q} + \frac{1}{2x} \right), \quad B_0 = \frac{1}{1 - \mathfrak{g}_U t_1}. \quad (29)$$

*Consequently  $\Pi_1(q_m) > 0$  and  $\Pi_q(q_m) > 0$  at every travel pole  $q_m$ .*

*Proof.* Throughout  $q = q_m$ .

*Reduction to the ungapped bulk vector.* Put  $\Psi_0 = 1$  and let  $\Psi = (\Psi_b)_{b \geq 1} = R_0 u$  be the solution of the bulk system with  $\mathfrak{g} = 0$  and source  $u$ , that is

$$\Psi_b = 2q^b F_b, \quad F_t := \sum_{b \geq 0} q^{\max(b, t)} \Psi_b \quad (b \geq 1),$$

the term  $b = 0$  of  $F_t$  being the source  $q^t$ . Then  $t_1 = v^\top R_0 u = \sum_{b \geq 1} q^b \Psi_b$  and, by (23),  $P = R_0 u / (1 - \mathfrak{g}t_1) = B_0 \Psi$ , where  $B_0 = 1/(1 - \mathfrak{g}t_1)$  (Remark 46). So every bulk weight in (24)–(25) is  $B_0 \Psi_b$ , with  $b = 0$  standing for the empty and gap-opening chains. In the variable  $q = x^2$ ,

$$x^{\max(2s, 2b)} = q^{\max(s, b)}, \quad x^{\max(2s+2, 2b)} = q q^{\max(s, b-1)}, \quad x^{\max(2s+1, 2b-1)} = x q^{\max(s, b-1)}, \quad x^{\max(2s+1, 2b+1)} = x q^{\max(s, b)},$$

with  $\max(s, -1) = s$ . At  $y = 1$  the slot  $B_{0z}$  equals  $B_0$ , and (24), (25) combine into

$$\lambda_s^{(1)} = B_0 \sum_{b \geq 0} \Psi_b \left( (1+x) q^{\max(s,b)} + (q+x) q^{\max(s,b-1)} \right), \quad (30)$$

hence  $\langle \lambda^{(1)}, R \rangle = B_0((1+x)X + (q+x)Y)$  with

$$X = \sum_{b \geq 0} \Psi_b (KR)_b, \quad Y = \sum_{b \geq 0} \Psi_b G_b, \quad (KR)_b = \sum_{t \geq 0} q^{\max(b,t)} R_t, \quad G_b = \sum_{s \geq 0} q^{\max(s,b-1)} R_s.$$

All double sums converge absolutely ( $\Psi \in \ell^1$  with  $|\Psi_b| = O(q^{2b} |\Psi|_{\ell^1})$ ,  $R \in \ell^1$ ), so they may be summed in either order.

*Two facts about  $R$ .* From  $\mathcal{T}R = R$ ,  $R_s = 2q^{s+1}(KR)_s$  for every  $s \geq 0$ ; and the normalisation  $\sum_s q^s R_s = \Sigma_1(q_m) = 1$  of Proposition 68 says  $(KR)_0 = G_0 = 1$ . In particular  $R_0 = 2q$ .

*The  $X$  shift identity.* Summing first over  $t$  and splitting off  $b = 0$ ,  $X = 1 + S$  with  $S = \sum_{b \geq 1} \Psi_b (KR)_b$ . Summing first over  $b$  instead,  $X = \sum_t R_t F_t$ . The term  $t = 0$  is  $R_0 F_0 = 2q(1+t_1)$ , since  $F_0 = \sum_b q^b \Psi_b = 1 + t_1$ ; for  $t \geq 1$ ,  $R_t F_t = 2q^{t+1}(KR)_t \cdot \Psi_t / (2q^t) = q \Psi_t (KR)_t$ . Hence  $X = 2q(1+t_1) + qS$ , and eliminating  $S$ ,

$$X = 1 + \frac{X - (1+t_1)R_0}{q}, \quad \text{so} \quad X = \frac{q(1+2t_1)}{1-q}.$$

*The  $Y$  shift identity.* Splitting off  $b = 0$  and using  $G_b = (KR)_{b-1} = R_{b-1}/(2q^b)$  for  $b \geq 1$ ,  $Y = 1 + \sum_{s \geq 0} \Psi_{s+1} R_s / (2q^{s+1})$ . Summing first over  $b$  instead, and using  $q^{\max(s,b-1)} = q^{-1} q^{\max(s+1,b)}$ ,  $Y = \sum_s R_s F_{s+1} / q = \sum_s R_s \Psi_{s+1} / (2q^{s+2})$ . The two expressions give

$$Y = 1 + qY, \quad \text{so} \quad Y = \frac{1}{1-q}.$$

*The closed forms.* With  $q = x^2$ ,  $(1+x)X + (q+x)Y = ((1+x)q(1+2t_1) + q+x)/(1-q)$ , and since  $(1+x)q + q + x = x(1+x)^2$  this is (28). At  $y = q$  only the slot  $s = 0$  of  $E^{(q)}$  changes, from  $B_0$  to  $B_{0z} = 1 + q \mathbf{g}_U S = B_0(1 - t_1 q / (1+q))$  (using  $\mathbf{g}_U S = B_0 - 1 = \mathbf{g}_U t_1 B_0$ ). So  $\Pi_q = B_0((1+x)X + (q+x)Y) + (B_{0z} - B_0)R_0$ , now with  $B_0 = 1/(1 - \mathbf{g}_U t_1)$ , and collecting terms with  $(1+x)^2(1-x+q) = (1+x)(1+q) - q(1-q)$  (both equal  $1+x+x^3+x^4$ ) gives (29).

*Positivity for  $\tau_m \leq 5 \cdot 10^{-3}$ .* By Corollary 111,  $|\mathbf{g}_V t_1| \leq 0.983$ , and  $\mathbf{g}_U < \mathbf{g}_V$ . Hence  $1 - \mathbf{g}_U t_1 \geq 0.017$  for both  $y$ , and  $q|t_1| \leq 0.983(1-q)$ , so  $|t_1| < 0.12$  once  $q \geq 0.9$  (here  $q \geq e^{-0.005}$ ). Since  $(1+x)/(2x) \geq 1$ ,  $1/(2x) > 1/2$  and  $0 < (1-x+q)/(1+q) < 1$ , both brackets are positive, and so are the prefactors.

*Positivity at the poles above that threshold.* This part is a computer-assisted certificate: at  $q_1, \dots, q_{12}$ ,  $\Pi_1$  and  $\Pi_q$  are enclosed in MPFR interval arithmetic and found positive, together with  $1 - \mathbf{g}_U t_1 > 0$  (`code/zeta_probe/rj_certificates/r35rust/src/bin/r54b.rs`). The pole index comes from `code/zeta_probe/rj_certificates/r39pole`, an interval-arithmetic zero count of  $1 - \Sigma_1$  which isolates exactly 13 zeros on  $[0, 0.9988]$  (the right endpoint taken as its binary64 value), each in its own cell. Its tiling of the interval was repaired after a first review found gaps of order  $10^{-77}$  between cells, and the repaired code was rebuilt and re-run from current source by a second reviewer; the trusted base is MPFR and the tool's own code.  $\square$

Only infinitely many poles are needed for Theorems 76 and 81, and the analytic part ( $\tau_m \leq 5 \cdot 10^{-3}$ ) supplies them; the certificate is needed only for the statement “at every travel pole”. The algebraic core is formalised in Lean 4 (§5.10); the eigen-relations of  $R$ , the gate and the analytic estimates behind it enter the formal statements as hypotheses and are not formalised.

**Corollary 73** ((R-J) at the dominant pole).  $\Pi_1(q_1) > 0$  and  $\Pi_q(q_1) > 0$ ; this is the case  $m = 1$  of Theorem 72.

## 5.10 Formalisation of the junction pairing and of the model

The following files in `lean/with_mathlib` are checked by Lean 4 against Mathlib [35]. Each carries `#print axioms` checks on its main statements, and the recorded output lists only the standard axioms; none uses `sorry` or `native_decide`.

- `RJMain.lean` and `RJShift.lean`: the  $X$  and  $Y$  shift identities (`X_shift`, `Y_shift`) and their solutions (`X_closed`, `Y_closed`), from the eigen-recursion of  $R$ , the bulk recursion of  $\Psi$  and the normalisation, including the interchange of the double sums from  $\ell^1$  summability (`summable_swap`).
- `RJClosedForm.lean`: the algebra of (28)–(29) (`closed_form`, `closed_form_yq`) and the positivity step from the gate bound (`gate_bound`, `bracket_pos`, `bracket_yq_pos`).
- `RJIdentities.lean`: auxiliary finite identities for the travel recursion.
- `RJLemmaJ.lean`: the cut criterion at the junction sites used in Appendix D.
- `RJPhi.lean`: the faithfulness of the model, i.e. the affine realisation intertwines the three generators and is injective.
- `RJMetricAll.lean`: `metricAll`, `wordLength` =  $\ell_R + 2c$  for every element, with  $\ell_R, c$  the closed forms `lRTrue`, `cTrue`.

What is *not* formalised: the assembly (M3) of Appendix D; the gate (Appendix B) and every analytic estimate behind it; the identification of the Lean eigen-recursion hypotheses with the operators  $\mathcal{T}$  and  $M_0$  at an actual travel pole; and the interval certificates. The formal statements take these as hypotheses.

## 5.11 Certificate

*Remark 74* (the computational certificate). `code/zeta_probe/assembly_certificate.py` re-derives the statements of this section in two independent arithmetic models and exits 0 with every check passing. Numerical agreement is evidence and is not a proof; the proofs above are what carry the statements. What the certificate adds is an independent arithmetic model and the hypothesis deletions.

*Part 1, exact arithmetic in  $\mathbb{Z}[[q]]/(q^{N+1})$  at the two truncation orders  $N = 28$  and  $N = 40$ , no floating point:* Propositions 37, 38 and 40 and Corollaries 39 and 41, coefficient by coefficient; the bulk block  $b_0$  has coefficients 2, 2, 6, 2, 18, 6, 42, 18, 118, 50, 282, 190, 706, 594 and the travel block the travel-run counts 2, 6, 10, 26, 54, 114, 274, 582, 1298;  $P_{11} = 1 + T_0$ ,  $P_{12} = T_1$ ,  $P_{21} = -S_0$ ,  $P_{22} = S_e = 1 - S_1$ ,  $P_{11}S_e + P_{12}S_0 = 1$ , the Hahn–Exton form (35), and  $S_o = \frac{1-q}{2q}S_0$ .

*Part 2, exact arithmetic in  $\mathbb{Q}[\mathbf{g}][[q]]/(q^{19})$ :* the factorisation of Proposition 45 holds identically in the free parameter  $\mathbf{g}$ , and three hypothesis deletions each break it, with the first surviving order recorded. A rank-two gap term fails at  $q^6$ ; a gap term  $q^{a+2b}$  in place of  $q^{a+b}$  fails at  $q^4$ ; and the asymmetric gapless kernel  $q^{\max(a,b)} + q^{a+2b}$  breaks  $t_0 = b_1$  at  $q^8$ .

*Part 3, two precisions, 40 and 70 significant decimal digits:* at the first ten travel poles,  $q_1 = 0.4494536306$  through  $q_{10} = 0.997756895$ , the dictionary  $b_0 = S_0/S_e$ ,  $t_0 = b_1 = S_1/S_e$ ,  $t_1 = P_{12}/S_e$  holds to relative error  $2.5 \cdot 10^{-26}$  at 40 digits and  $8.8 \cdot 10^{-57}$  at 70, the loss of about 14 digits being the cancellation in the alternating ladders at the highest pole; the two precisions agree on the pole locations to  $1.6 \cdot 10^{-31}$ . The resolvent is evaluated there by the  $O(N)$  two-point solve of Corollary 39, that is by the analytic continuation, the defining Neumann series not converging at

those  $q$ . The Möbius value is cross-checked against an independent dense solve of the truncated system at  $q = 0.5, 0.7, q_1$  and  $y = 1, q$ , agreeing to  $6.2 \cdot 10^{-28}$ . The gate signs  $b_0 > 0$  and  $s < 1$  hold at all ten, with  $b_0 \tau \rightarrow 2$  and  $s \rightarrow \frac{1}{4}$ . (Earlier versions also checked a “lifting increment”  $(1-s)B_V + qb_0$ ; it refers to the  $E$ -sourced Möbius form and not to the model’s bulk sum, and is withdrawn.)

*Part 4, exact rational arithmetic:* the one-sign junction matrix  $[x^{\max(\sigma-1, 2s+1)}]$  has rank 4 over  $\mathbb{Q}$  at  $q = \frac{1}{3}$  on the  $5 \times 5$  block, its interleaved  $3 \times 3$  minor equals  $9/524288$  at  $x = \frac{1}{2}$ , the symmetrised junction matrix (24) has rank 6 on the  $6 \times 6$  block at  $x = \frac{1}{2}$ , and the gap-bridge matrix  $[q^{a+b}]$  has rank 1.

*Part 5, exact integer arithmetic, code/zeta\_probe/tools/sitecost:* the site cost is computed by two independent exact solvers, a min-cost flow and a transportation dynamic program, which agree on all 1 012 664 supply-demand pairs with entries at most 7; on the same range the closed form  $\max(|\alpha|, |\beta|, |\Phi|)$  of Lemma 51 is exact, with 0 misses. Corollary 52 is confirmed on 4 532 157 site configurations, each evaluated by both solvers, namely every crossing count from the minimum to the minimum plus 10, every sign split, both signs of both deposits with  $|d| \leq 24$ , every travel indicator, all four marker data and all eight site types (bulk interior, travel interior in both directions, the near and far junctions for  $k^* > 0$  and  $k^* < 0$ , and the coincident junction at  $k^* = 0$ ): 0 exceptions, with the cost independent of the crossing counts and of the sign splits in every case. Deleting any one of the three pairing costs is witnessed necessary: replacing the sign-flip bounce cost 2 by 0, 1, 3, 4, the pass cost 1 by 0, 2, 3, or the bounce cost 0 by 1, 2 each breaks Corollary 52, with 2544 to 7392 exceptions out of 7524 configurations and the smallest counterexample recorded. The marker closed forms of Corollary 54 hold with 0 exceptions on the full signed range, and the form  $\max(|d_L| - 1, |d_R|)$  fails on 42 of 156 cells at the near junction and on 36 of 156 at its  $k^* < 0$  counterpart, all with  $d_L < 0$ . Corollaries 53 and the shield count are confirmed as recorded in Remark 57.

## 6 The relaxed series $V$

The relaxed series is the travel resolvent dressed by the bulk. The resolvent itself is settled unconditionally; passing from it to  $V$  uses the transfer model (Theorem 66) and the junction pairing (Theorem 72).

**Theorem 75** (the travel resolvent). *The travel resolvent  $G_0^T = \Sigma_0/(1 - \Sigma_1)$  is transcendental over  $\mathbb{Q}(x)$ . This is unconditional: it uses no growth hypothesis, no asymptotic theorem, and no statement about the strand-walk model. Its one numerical input is the contour bound of Lemma 10, of which it consumes only  $|T_2(m\pi)| < 1$ , against a computed worst value 0.027; that bound is a verified enclosure over the parameter region, not a sampled one, and its arithmetic model is stated with the lemma.*

*Proof.*  $\Sigma_0$  and  $1 - \Sigma_1$  are holomorphic on  $|q| < 1$  (Theorem 2, first clause) and  $1 - \Sigma_1 \not\equiv 0$ . The zero set of  $1 - \Sigma_1$  in  $(0, 1)$  is infinite and accumulates exactly at  $q = 1$  (Lemma 12, which consumes from Lemma 10 only  $|T_2| < 1$ , together with Lemma 103). At each such  $q_m$  the numerator does not vanish, by Proposition 25. Hence every  $q_m$  is a pole of  $G_0^T$ , of order the vanishing order of  $1 - \Sigma_1$  there, which is finite. So  $G_0^T$  has infinitely many poles in  $|q| < 1$  accumulating at  $|q| = 1$ ; under  $q = x^2$  these are infinitely many poles in  $|x| < 1$  accumulating at  $|x| = 1$ . An algebraic function over  $\mathbb{Q}(x)$  has only finitely many poles in  $|x| < 1$ , so no algebraic function has such a pole set; hence  $G_0^T$  is transcendental. It is not  $D$ -finite either, in  $q$  or in  $x$ , and  $|q| = 1$  is its natural boundary (Theorems 88 and 92): for  $D$ -finiteness the finiteness of the pole set of an algebraic function is replaced by Lemma 87, which confines the poles in the disc of a solution of a linear differential equation to the finitely many zeros of its leading coefficient.  $\square$

**Theorem 76.** *The relaxed growth series  $V$  is transcendental over  $\mathbb{Q}(x)$ . (Standing: Theorems 66 and 72.)*

*Proof.* By Lemma 67,  $V$  is meromorphic on  $|x| < 1$ , and the local forms below are those of this continuation. By Proposition 68, at a travel pole  $q_m$  the relaxed series has the local form  $V = \mathcal{N}_1(q - q_m)^{-\pi} + (\text{poles of order } < \pi) + H_m$  with  $H_m$  holomorphic,  $\pi \geq 1$ , and  $\mathcal{N}_1 = (2x_m)^{-1} c_\star \Pi_1(q_m)^2$  with  $c_\star \neq 0$ . That proposition consumes two non-vanishing statements, and they are separate:  $S_e(q_m) \neq 0$ , in the quantitative form  $|S_e(q_m)| \geq 0.35\sqrt{\tau_m}$  of Proposition 94, which makes the bulk boundary vectors  $\lambda^{(1)}, \mu^{(1)}$  holomorphic at  $q_m$  and says nothing about vanishing; and  $\Sigma_0(q_m) \neq 0$ , from Proposition 25, which pins the pole order  $\pi \geq p \geq 1$ ; and  $\Pi_1(q_m) \neq 0$  at every travel pole, by Theorem 72.

So the leading Laurent coefficient is non-zero at every travel pole, and  $V$  has a pole at each of them. The travel poles are infinite in number and accumulate at  $q = 1$ , that is at  $|x| = 1$  (Lemma 12); this is the boundary of the disc  $|x| < 1$  of meromorphy, not of the disc of convergence  $|x| < x_1$ , which only the dominant pole bounds. They are locally finite in  $(0, 1)$  (Lemma 103), so an infinite subset of them accumulates at  $q = 1$  and nowhere else. Hence  $V$  has infinitely many poles in  $|q| < 1$  accumulating at  $|q| = 1$ ; under  $q = x^2$  these are infinitely many poles in  $|x| < 1$  accumulating at  $|x| = 1$ . An algebraic function over  $\mathbb{Q}(x)$  has only finitely many poles in  $|x| < 1$ , so an infinite pole set accumulating at the boundary is impossible for one; hence  $V$  is transcendental. No comparison of  $T_2$  against  $\cos W$  is made or needed, and none is available at this strength, the proved bound on  $|T_2|$  being  $0.17\tau^{1/4}$  (Lemma 10), which exceeds  $\sqrt{\tau/2}$  as  $\tau \rightarrow 0$ ; Remark 9 sharpens the size of  $T_2$  and is not used.  $\square$

*Remark 77* (what became of (R)). *What (R) was.* Route B decomposes a relaxed element into two bulk runs, one travel interval of length  $|k|$  and four marker junctions; (R) was that decomposition read as a local form for  $V$  at a travel pole, *together with* a product form for its numerator,  $\mathcal{N}_V = \Sigma_0 \cdot S_0/(1 - S_1) \cdot J$  with  $J$  a junction factor.

*What is now proved.* The  $k$ -sum telescoping (Proposition 37), the bulk dressing (Proposition 38), and the shape of the local form, with a rank-one numerator paired against the null vector of the travel transfer (Proposition 68). These are the whole of (R) except the product form.

*What is refuted.* The product form. The marker junction cost is  $\max(|d_L - 1|, |d_R|)$  (Corollary 54), a max-coupling of rank at least 3 with an explicit non-zero  $3 \times 3$  minor (Proposition 69), so the numerator is not a multiple of the total bulk sum and the step “ $\mathcal{N}_V(q_m) \neq 0$  because  $\Sigma_0(q_m) \neq 0$  and  $S_0(q_m) \neq 0$ ” is not available. Proposition 25 is unaffected and is used, in Theorem 75 and inside Proposition 68; what fails is the deduction that leant on it here.

*What replaces it.* The junction pairing (R-J), that  $\Pi_1$  is non-zero at infinitely many travel poles, which is strictly weaker than (R) and is Theorem 72: at a travel pole  $\Pi_1$  collapses to an explicit function of  $t_1$ , and the gate makes it positive.

*Where it is used.* Only in Theorem 76 and, at  $y = q$ , in Theorem 81. Theorem 75, Propositions 25 and 94, §4 and Appendices A–B do not touch it. All of §5 rests on the transfer model (M), Theorem 66.

*Evidence.* The blocks reproduce their own combinatorial series exactly:  $S_0/(1 - S_1)$  against the bulk-run counts 2, 2, 6, 2, 18, 6, 42, 18, 118, 50, 282, 190, 706, 594 and  $\Sigma_0/(1 - \Sigma_1)$  against the travel-run counts 2, 6, 10, 26, 54, 114, 274, 582, 1298 (`blocks_growth.py`, and `assembly_certificate.py` Part 1 in exact integer arithmetic at two truncation orders).  $V$  itself is computed to 131 exact terms by an independent catalytic transfer, not from the blocks.

**Corollary 78.** *The exponential growth rate of  $v_n$  is  $\beta_2 = 1.4916177871\dots$ , with  $\beta_2^2$  the reciprocal of the smallest travel pole  $q_1$ ; the value  $\frac{3}{2}$  is excluded. It needs only the location of the dominant travel pole: not Remark 9, not Appendix A, and not Appendix B. It does share with Theorem 76 the*

transfer model (M), which is what places  $q_1$  on  $V$  rather than on  $G_0^T$  alone, and at the dominant pole the junction pairing is positive (Corollary 73). The lower bound  $\beta_2$  for  $u_n$  and  $v_n$  is free even of (M), coming from the pure-travel subcount [6]. The same rate for  $u_n$  follows from Proposition 1 (the former hypothesis (T) of Theorem 81, now proved) together with the positivity of  $\Pi_q(q_1)$ ;  $u_n \leq v_n$  gives the upper bound unconditionally.

Whether the number  $\beta_2$  is transcendental is a separate question, open, and with a clean reduction to a value statement for a Hahn–Exton  $q$ -Bessel function; it is treated in Appendix C, which is independent of everything else in this paper.

## 7 The true series $U$

Transcendence of  $U$  needs two things that the analysis of §4 and Appendix B does not supply: that  $U$  has the travel poles at all, and that the residue of  $U$  at each of them is non-zero. The first was the combinatorial hypothesis (T); it is now Proposition 1 and is proved (§7.2). The second was previously the hypothesis (L); its algebraic content is now a theorem (§5.5, Theorem 70), and the residue itself is the junction pairing (R-J) at  $y = q$ , Theorem 72, the same statement that Theorem 76 consumes at  $y = 1$ . This section states (T) in full, with its dependencies, with every place it is used, and with the proof that closes it, and records what the bulk dictionary gives, before turning to the analytic content. A reader should not have to open [8] to see what Theorem 81 rests on.

### 7.1 The bivariate bridge

With  $c(g) \in \mathbb{Z}_{\geq 0}$  the cycle count of §1, set

$$\mathcal{W}(x, y) = \sum_{n, c \geq 0} w_{n, c} x^n y^c, \quad w_{n, c} = \#\{g : \ell_R(g) = n, c(g) = c\},$$

the grading being by *relaxed* length. Then  $V(x) = \mathcal{W}(x, 1)$  and, since  $\ell_T = \ell_R + 2c$  and  $q = x^2$ ,  $U(x) = \mathcal{W}(x, x^2) = \mathcal{W}(x, q)$ : the two series are two evaluations of one bivariate object, at  $y = 1$  and at  $y = q$ , and everything separating them is carried by the marker  $y$ . The second identity is the one place where the metric identity  $\ell_T = \ell_R + 2c$  is used. With  $\ell_R, c$  the site-local closed forms of [6],  $\text{wordLength} = \ell_R + 2c$  holds for every element (Lean, `metricAll` in `RJMetricAll.lean`, resting on `wordLength_eq_lRTrue_add_two_cTrue` and `reaches_of_phiZ` in `PhiLipschitz.lean`), and the word length of the model is the Cayley length in  $W_{\text{univ}}$  (Lean, `RJPhi.lean`); so  $U = \mathcal{W}(x, q)$  unconditionally. That the closed form  $\ell_R$  is also the minimum over relaxed realisations is [6]’s metric formula (its `rem:metric-formula-status`); it bears on reading  $V$  as a relaxed count, and on nothing proved here. The corrected assembly (26) agrees with a breadth-first enumeration of  $W_{\text{univ}}$  as exact series: at  $y = q$  to  $x^{26}$  in all three sectors; at  $y = q^2$  to  $x^{22}$ ; and at symbolic  $y$  in all 816 coefficients with  $\ell_T \leq 31$  ( $5.03 \cdot 10^6$  elements), where a deliberately corrupted model is detected.  $v_0, \dots, v_{19}$  are certified by the same enumeration together with the bound  $\ell_R \geq 2c + 6$ . Earlier versions cited agreement to  $v_{14}$ ; that check (`cycle_weighted_gf.py`) was a sweep of the bridge  $\ell_T = \ell_R + 2c$  and did not assemble the blocks, and the model as then printed fails it.

### 7.2 The former input (T): the travel block carries no marker

(T) is Proposition 1: the denominator  $1 - \Sigma_1$  is  $y$ -free, hence literally the same series in  $\mathcal{W}(x, 1)$  and in  $\mathcal{W}(x, q)$ , so  $U$  has the travel poles  $q_m$  of §2.

*Where it is used.* In the proof of Theorem 81, to transport the pole set of Lemma 12 from  $V$  to  $U$ , and in the same way in Corollary 78 for the growth rate of  $u_n$ . Nothing in §4, Appendix A or Appendix B touches it.

*What it rests on.* Nothing further: it is proved. The localisation of cycles, that a pure-travel element has  $c = 0$ , is Corollary 63, a case of Theorem 62. The metric identity is not needed. Theorem 62 exhibits a relaxed-optimal realisation with no isolated cycle; that realisation is a single open walk, so it is admissible for  $\ell_T$  and gives  $\ell_T \leq \ell_R$ , while  $\ell_T \geq \ell_R$  holds because the relaxed minimum ranges over a superset of the realisations. Hence  $\ell_T = \ell_R$  on pure-travel elements, the two gradings agree termwise, and the travel recursion (2), assembled from pure-travel runs alone, is computed inside a  $c \equiv 0$  sector, so the marker cannot enter it.

In particular the reverse inequality of  $\ell_T = \ell_R + 2c$  – open when earlier versions of this section listed it as a dependency, and since proved in [6] – plays no role here regardless: only the direction that follows from an exhibited realisation is used.

*Evidence.* The pure-travel generating function computed in both gradings to radius 18 is identical coefficient by coefficient, 1, 3, 5, 8, 12, 17, 24, 35, 52, 78, 116, 173, 256, 383, 572, 858, 1276, 1907, 2832, with ratio tending to  $\beta_2$ .

### 7.3 The former input (L): the bulk dictionary and the gate

The marker enters the block assembly through exactly one factor, the gap-bridge kernel  $\mathfrak{g}(q, y) = q/(1 - qy)$  of (M2), summing a maximal run of  $L$  forced gap edges, each of weight  $q$  and each an isolated cycle except the one shielded by the inner block's turn-around. Its two evaluations,  $\mathfrak{g}_V = \mathfrak{g}(q, 1) = \frac{q}{1-q}$  and  $\mathfrak{g}_U = \mathfrak{g}(q, q) = \frac{q}{1-q^2} = \frac{\mathfrak{g}_V}{1+q}$ , are holomorphic and non-zero on  $(0, 1)$ , hence at every travel pole.

The bulk resolvent is semiseparable,  $M_0[b, a] = 2q^b q^{\max(a, b)}$  plus a rank-one gap term of weight  $\mathfrak{g}$ , and the resolvent of  $E$  is a Möbius function of  $\mathfrak{g}$ :

$$B^E(q, y) = \frac{b_0 + \mathfrak{g} \kappa}{1 - \mathfrak{g} t_1}, \quad \kappa = t_0 b_1 - b_0 t_1, \quad (31)$$

with  $b_0, b_1, t_0, t_1$  the  $y$ -free gapless-resolvent sums of Proposition 38, which satisfy the reciprocity  $t_0 = b_1$  (Proposition 40). The factorisation (31) is proved as Proposition 45; its two hypotheses are  $S_e \neq 0$  and  $s \neq 1$ , which the paper was carrying already (Remark 47). Write  $s = \mathfrak{g}_V t_1$ , so that  $q t_1 = s(1 - q)$ . What the assembly consumes is the *gate*

$$b_0 > 0 \quad \text{and} \quad s := \frac{q}{1-q} t_1 < 1, \quad t_1 = \frac{P_{12}}{S_e}, \quad (32)$$

on the off-diagonal cocycle entry  $P_{12}$ , together with the dictionary identifying the entries of (31) with the blocks of (1),

$$S_e = 1 - S_1, \quad S_o = \frac{1-q}{2q} S_0, \quad b_0 = \frac{S_0}{1 - S_1} = \frac{2q}{1 - q} \cdot \frac{S_o}{S_e}, \quad (33)$$

$S_o$  being identified in Lemma 101 and the two resolvent lines in Corollary 41. The remaining entry  $t_1 = P_{12}/S_e$  is an identity between independently defined objects (Corollary 39), not a definition, so all four lines of (33) now stand on the same footing.

*What is proved.* The factorisation (31) and the dictionary (33) are Theorem 70. Three ingredients previously listed as verified rather than proved are now proved: the reciprocity  $t_0 = b_1$  (Proposition 40), the entry  $t_1 = P_{12}/S_e$  (Corollary 39), and the identification of the resolvent quantities  $b_0, b_1, t_0$  with the blocks (Proposition 38). That the bulk transfer is  $M_0$  with the rank-one

gap term and source  $u$  is part of (M), now Theorem 66. On the numerical side, the kernel  $K(a, b) = q^{\max(a, b)}$  on active half-sizes reproduces  $S_0/(1 - S_1) = 2q + 2q^2 + 6q^3 + 2q^4 + 18q^5 + 6q^6 + 42q^7 + \dots$  (`blocks_growth.py`), and the whole dictionary is re-derived in exact integer arithmetic and at two precisions in `assembly_certificate.py`.

*Where it is used.* In Theorem 72, through  $t_1$  and  $1 - \mathbf{gt}_1$ ; and inside Corollary 111, where the half  $b_0 > 0$  is proved from a sign chain running through the closed form in (33). The other half of the gate,  $|s| < 1$ , uses neither: it is Theorem 109 with Lemma 110, and it is unconditional.

## 7.4 The gate block in closed form

Both columns of the cocycle satisfy the scalar three-term recursion  $y_{n+1} = (1 + q^3 - 2(1 - q)q^{2n+2})y_n - q^3y_{n-1}$ ; setting  $x = q^n$  turns it into the linear  $q$ -difference equation

$$Y(qx) + q^3Y(x/q) = (1 + q^3 - 2(1 - q)q^2x^2)Y(x), \quad (34)$$

whose regular solution is, in closed form, a Hahn–Exton  $q$ -Bessel function [23, 19, 30],

$$Y_3(x) = \sum_{k \geq 0} d_k x^{2k+3}, \quad d_k = \frac{(-2)^k (1 - q)^k q^{k^2+3k}}{(q^2; q^2)_k (q^5; q^2)_k} = J_{3/2}^{(3)} \text{ normalised}, \quad (35)$$

and  $P_{12} = \frac{2q^3}{1-q^3} Y_3(1)$ . The denominator  $S_e$  is the block of (1), controlled by Lemma 110. It remains to bound the numerator residual against its confluent limit; this is where the two-Bessel identity enters.

## 7.5 The amplitude atom, formally verified

As  $q \rightarrow 1$ , equation (34) degenerates to  $\tau Y'' - (2\tau/x)Y' + 2Y = 0$ , solved by  $x^{3/2} J_{3/2}(Wx)$ . The  $q$ -corrected connection coefficient across the turning point  $X = \nu = \frac{3}{2}$  is governed by the following identity, whose point is that the confluence is *entire*; it passes *through* the turning point, bypassing the Liouville–Green singularity where asymptotic methods stall.

**Lemma 79** (two-Bessel confluence; amplitude normalisation). *The regular solution of (34) has the confluent form*

$$Y_3(x) = \gamma_{\text{cl}} x^{3/2} \left[ J_{3/2}(X) + \frac{\tau X}{2} J_{5/2}(X) \right] + O(\tau^2), \quad X = wx, \quad \gamma_{\text{cl}} = \frac{3\sqrt{\pi}}{4} \left( \frac{w}{2} \right)^{-3/2},$$

and its single-solution envelope satisfies  $A/A_{\text{cl}} = 1 + \tau h(X) + O(\tau^2)$  with the explicit rational

$$h(X) = \frac{1 + 3/(2X^2)}{1 + 1/X^2} \in [1, \tfrac{3}{2}] \quad (X > 0).$$

Consequently  $\gamma/\gamma_{\text{cl}} = 1 + O(\tau)$  with explicit constant  $\frac{3}{2}$ .

*Proof.* From the  $q$ -Pochhammer confluence  $d_k = \tau^{-k} D_k (1 - k\tau + O(\tau^2))$  with  $D_k = (-1)^k / [2^k k! (\frac{5}{2})_k]$ , the leading sum is  $\gamma_{\text{cl}} x^{3/2} J_{3/2}(X)$  by the Bessel representation of  ${}_0F_1$ , and the  $-k\tau$  correction contributes  $\frac{\tau X}{2} \gamma_{\text{cl}} x^{3/2} J_{5/2}(X)$  via  ${}_0F_1(\cdot; \frac{7}{2}; \cdot)$ . Both series are entire in  $X$ . The envelope of  $J_{3/2} + \frac{\tau X}{2} J_{5/2}$  follows from  $A^2 = M_{3/2}^2 + (\frac{\tau X}{2})^2 M_{5/2}^2 + \tau X (J_{3/2} J_{5/2} + Y_{3/2} Y_{5/2})$  and the closed half-integer cross-term identity

$$J_{3/2}(X) J_{5/2}(X) + Y_{3/2}(X) Y_{5/2}(X) = \frac{4}{\pi X^2} \left( 1 + \frac{3}{2X^2} \right),$$

which with  $M_{3/2}^2 = \frac{2}{\pi X} (1 + \frac{1}{X^2})$  gives  $h(X) = (1 + \frac{3}{2X^2}) / (1 + \frac{1}{X^2})$ ; this is monotone decreasing from  $h(0^+) = \frac{3}{2}$  to  $h(\infty) = 1$ , so  $1 \leq h \leq \frac{3}{2}$ .  $\square$

*Remark 80* (machine verification). The cross-term identity and the bound  $1 \leq h(X) \leq \frac{3}{2}$  have been formalised and checked in Lean 4 against Mathlib (file `AtomN.lean`); the proof terms depend only on the standard axioms `propext`, `Classical.choice`, `Quot.sound`, with no `sorry`. The identity is proved by a single `linear_combination` off  $\sin^2 + \cos^2 = 1$ , and the bound by `nlinarith`.

**Theorem 81.** *The growth series  $U$  of  $W_{\text{univ}}$  is transcendental over  $\mathbb{Q}(x)$ . (Standing: Theorems 66 and 72.)*

*The input (T) that earlier versions of this theorem carried is gone: Proposition 1 is proved, via Theorem 62 and Corollary 63, and the proof does not use the metric identity.*

*The hypothesis (L) that earlier versions of this theorem carried is gone: its content, the factorisation (31) with the dictionary (33) and the reciprocity  $t_0 = b_1$ , is Theorem 70. The residue is the square of  $\Pi_q(q_m)$ , which is Theorem 72.*

*Independently of (M), (T) and (R-J), the half  $|s| < 1$  of the gate (32) holds at every travel pole. The numerator amplitude estimate ( $\star$ ) (at every travel pole  $q_m$ ,  $|P_{12}(q_m)| w_m^3 < 2$ ) is Theorem 109 (Appendix B):  $|P_{12}| \leq 0.619 \tau^{3/2}$  for  $\tau \leq 5 \cdot 10^{-3}$  and  $|P_{12}| w^3 \leq 0.75$  at the six poles above that threshold. The passage to  $|s| < 1$  is Corollary 111:  $|s| \leq 0.983$  on the uniform range and  $|s| \leq 0.296$  at the six exceptional poles, with the denominator lower bound  $|S_e| \geq 0.63\sqrt{\tau}$  supplied by Lemma 110. The remaining half  $b_0 > 0$  is proved in the same corollary from a sign chain through the closed form for  $b_0$ , which is Corollary 41. The leading-order picture of Remark 16 (the phase-matched zero giving  $|P_{12}|/\tau^{3/2} \rightarrow 1/(4\sqrt{2})$ , a fourfold margin below the threshold  $1/\sqrt{2}$ ) is consistent with, and explained by, the second-difference identity of Proposition 102.*

*Proof.* By Lemma 67 at  $y = x^2$ ,  $U$  is meromorphic on  $|x| < 1$ , and the local forms below are those of this continuation. By (T),  $U$  inherits the travel poles. By Proposition 68 at  $y = q$ , at each travel pole  $U$  has the local form  $\mathcal{N}_q(q - q_m)^{-\pi} + (\text{lower order}) + H_m$  with  $\mathcal{N}_q = (2x_m)^{-1} c_\star \Pi_q(q_m)^2$  and  $c_\star \neq 0$ ; the boundary vectors are holomorphic there because the gate (32) holds at every travel pole. Indeed, on the uniform range  $\tau \leq 5 \cdot 10^{-3}$ ,  $|P_{12}| \leq 0.619 \tau^{3/2}$  (Theorem 109) and  $|S_e| \geq 0.63\sqrt{\tau}$  (Lemma 110) give  $|s| = \frac{q}{1-q} |P_{12}|/|S_e| \leq \frac{q}{e\tau-1} \cdot \frac{0.619}{0.63} \leq 0.983 < 1$ , and  $b_0 > 0$  follows from the sign chain of the same lemmas through the closed form of Corollary 41; the six exceptional poles are checked directly (Corollary 111). By Theorem 72 the leading coefficient  $\mathcal{N}_q$  is non-zero at every  $q_m$ , so  $U$  has an uncanceled pole at each of them; these accumulate at  $q = 1$  (Lemmas 12 and 103), so  $U$  has infinitely many poles in the unit disc accumulating at its boundary and is transcendental over  $\mathbb{Q}(x)$ , as in Theorem 76. Transcendence does not consume the exceptional-pole enumeration of Theorem 109: the uniform range  $\tau \leq 5 \cdot 10^{-3}$  alone carries infinitely many poles (Lemma 12 produces them at  $\tau \downarrow 0$ ); the enumeration only upgrades “at infinitely many travel poles” to “at every travel pole” in the gate. For orientation: the computed values over  $m = 5, \dots, 71$  are  $|P_{12}|/\tau^{3/2} \rightarrow 1/(4\sqrt{2}) = 0.1768$  and  $s \rightarrow \frac{1}{4}$ , well inside every bound above.  $\square$

*Remark 82* (the assembly is machine-checked). The composition just performed (the gate arithmetic, the two halves of (32), the passage to the pole set, and the step from infinitely many poles to non-algebraicity) is verified in Lean 4 (`UAssembly.lean`, seven theorems, no `sorry`, axioms `propext`, `Classical.choice`, `Quot.sound`), each analytic input entering as an explicit hypothesis: Theorem 109, Lemma 110, Corollary 111, Lemma 12, the reduction (L) (the Lean hypothesis `hred`), and the finiteness of an algebraic function’s pole set in the disc. The file formalises *no* analysis: no  $q$ -series, no steepest descent, no numerics; its content is exactly that those hypotheses *compose*, so its hypothesis list is an audit of what Theorem 81 rests on, not a verification of any of them. In particular `hred`, the hypothesis “the gate implies a non-zero residue”, is assumed there and not derived; its content is Theorem 72, whose identities and positivity step are checked in `RJClosedForm.lean` and `RJMain.lean` with the eigen-relations and the gate as hypotheses (§5.10).

The quantitative step is isolated as `gate_ratio_lt_one`, whose hypothesis  $K < c$  (numerator constant strictly below denominator constant) is what the gate needs and what a numerator bound alone cannot supply: with  $K = 0.619$  from Theorem 109 and the constant 0.35 of Proposition 94 it is *false*, which is precisely why Lemma 110’s 0.63 is required.

*Remark 83* (the standing of Theorem 81). The unconditional content of this section is the gate: at every travel pole,  $|P_{12}| \leq 0.619 \tau^{3/2}$  (Theorem 109),  $|S_e| \geq 0.63\sqrt{\tau}$  (Lemma 110) and hence  $|s| \leq 0.983 < 1$  (Corollary 111), together with the infinitude and accumulation of the travel poles (Lemma 12) and the fact that these facts compose to non-algebraicity (Remark 82). The factorisation (31), the reciprocity  $t_0 = b_1$ , the dictionary (33) and the half  $b_0 > 0$  of the gate are Theorem 70; the junction pairing is Theorem 72.

(T) is now proved. Proposition 1 follows from Corollary 63: a pure-travel element has no gap edge, so by Theorem 62 some relaxed-optimal realisation is a single open walk, which bounds  $\ell_T \leq \ell_R$ ; the reverse inequality is immediate. The metric identity  $\ell_T = \ell_R + 2c$  is not used.

(R-J) is Theorem 72, and (M) is Theorem 66. The standing of Theorem 81 is therefore that of its inputs: the gate (Appendix B, which contains verified interval-arithmetic enclosures); the metric identity and the faithfulness of the model (Lean); the assembly (M3) (a hand proof, Appendix D, with two independent adversarial reviews and exact series agreement to  $x^{26}$ ); and, for the statement at every travel pole rather than infinitely many, the interval certificate at the first twelve poles. The assembly is not formalised.

*Remark 84* (what the turning-point atom does and does not give). Lemma 79 discharges the amplitude *through the turning point*  $X = \frac{3}{2}$ , which is the obstruction to any Liouville–Green treatment. It does *not* supply the amplitude at the *pole*  $X = w$  to the  $O(\tau^2)$ -relative accuracy the numerator gate requires. There the two-Bessel form carries a relative defect whose  $O(1)$  part is exactly  $\frac{1}{36}$ ; i.e. the classical turning-point-to-pole connection factor  $\frac{36}{35}$ , which is itself derivable and, once applied, matches  $Y_3(1/q)$  to  $O(\tau)$ -relative; and whose *residual*  $O(\tau)$  part (the amplitude drift from the turning point to the pole) is not supplied by this route. Because  $P_{12} = \frac{2q^3}{3(1-q^3)}Y_3(1/q) - \frac{2}{3}S_e$  is an  $O(\sqrt{\tau})$  cancellation, that residual is amplified to the size of  $P_{12}$  itself. This residual is closed in Appendix B, not by resumming the amplitude at all, but by the second-difference identity of Proposition 102:  $P_{12}$  is the symmetric second difference of the  $q$ -cosine at its own zero, so the gate needs only  $\max|\psi''| \leq 3.5w$  on a window of width  $\tau$ , which Theorem 109 proves; Lemma 110 and Corollary 111 then close the remaining halves of the gate ( $|S_e| \geq 0.63\sqrt{\tau}$  and  $b_0 > 0$ ). The two off-diagonal cocycle entries obey the Abel relation  $x_N Y_N - X_N y_N = -1$  (the discrete  $q$ -Wronskian, conserved because  $\det M_n \equiv 1$ ), whence

$$P_{12} = \frac{X_N S_e - 1}{x_N} \quad \text{exactly,}$$

$x_N, X_N$  being the leading components of the same three-term recursion. Hence  $P_{12}$  is fixed by the *same* family of Hahn–Exton blocks that controls  $S_e$ : one analytic class; that of Remark 9; not two. In this precise sense  $U$  sits on  $V$ ’s footing. What  $U$  additionally requires is the pole asymptotics of  $x_N, X_N$ ; again Hahn–Exton, again governed by the exact  $q$ -Wronskian of Olde Daalhuis [41],  $W_q(J_\nu, Y_\nu) = q^{\nu(\nu-1)}(1-q^2)/\Gamma_{q^2}(\frac{1}{2})^2$  for the numerically satisfactory pair, whose  $q \rightarrow 1$  expansion is explicit via Moak’s  $q$ -Stirling formula [36] (e.g.  $\Gamma_{q^2}(\frac{1}{2}) = \sqrt{\pi}(1 - \frac{3}{8}\tau + O(\tau^2))$ ). The block-asymptotic assembly is carried out *elementarily* in Remark 16: the numerator amplitude has a derived closed-form expansion with exact rational constants, and the phase-matching of the travel-pole equation against the amplitude’s zero (both collapsing to the same sinh-product family) derives the gate value  $1/(4\sqrt{2})$ .  $U$ ’s analytic inputs thereby coincide with  $V$ ’s: the further pieces (( $\star$ ) and the closure of the gate) are proved in Appendix B. (The orthogonal arithmetic route of §9 would settle  $U$  without any of this analysis, but is itself open.)

## 8 Beyond algebraicity: $D$ -finiteness, functional equations and the natural boundary

Theorems 76 and 81 exclude algebraic equations. The pole set that proves them excludes much more, because it is infinite and lies inside the disc  $|x| < 1$  on which  $U$  and  $V$  are meromorphic (Lemma 67), well beyond their radius of convergence  $x_1 = 0.6704\dots$ . This section records what it excludes. Every statement below about  $U$  or  $V$  has the standing of Theorems 76 and 81 (Theorems 66 and 72); every statement about the travel resolvent  $G_0^T$  alone is unconditional, as Theorem 75 is.

### 8.1 Poles of $U$ on the negative axis

**Proposition 85.** *At every travel pole  $q_m$ ,  $U$  has a pole at  $x = -x_m$  as well as at  $x = x_m$ , both of order  $\pi$ , the order of the pole of  $(I - \mathcal{T})^{-1}$  at  $q_m$ . At  $x = -x_m$  the junction pairing is negative:*

$$\Pi_q|_{x=-x_m} = \frac{2q_mB_0(1-x_m)}{1+x_m} \left( t_1 \frac{1+x_m+q_m}{1+q_m} - \frac{1}{2x_m} \right) < 0, \quad B_0 = \frac{1}{1-\mathfrak{g}_U t_1}.$$

(Standing: Theorems 66 and 72; at the six poles with  $\tau_m > 5 \cdot 10^{-3}$ , the enumeration standing of Theorem 109.)

*Proof.* The travel resolvent  $(I - \mathcal{T})^{-1}$ , the bulk vector  $\Psi$ ,  $t_1$  and  $B_0$  depend on  $x$  only through  $q = x^2$ ;  $x$  itself enters  $\mathcal{W}(x, q)$  only through the monomials of (40)–(41) and the prefactor  $x^{-1}$ . The proofs of Proposition 68 and of (29) use only  $q = x^2$ , never the sign of  $x$  (the reductions  $x^{\max(2s+1, 2b\pm 1)} = x q^{\max(\cdot, \cdot)}$  and the final polynomial identities hold for either root). Near  $x = -x_m$  one has  $q - q_m = (x + x_m)(x - x_m)$  with the second factor non-zero, so  $U = \mathcal{N}_q(-2x_m)^{-\pi}(x + x_m)^{-\pi} + (\text{lower order})$  with  $\mathcal{N}_q = c_\star(-2x_m)^{-1}\Pi_q|_{x=-x_m}^2$ , and (29) at  $x = -x_m$  is the displayed formula. The prefactor is positive:  $0 < x_m < 1$ , and  $1 - \mathfrak{g}_U t_1 > 0$  because  $|\mathfrak{g}_U t_1| < |\mathfrak{g}_V t_1| = |s| < 1$  (Corollary 111). For the bracket write  $t_1 = s(1-q)/q$  and use  $(1+x+q)/(1+q) = 1+x/(1+x^2) \leq \frac{3}{2}$ :

$$2x_m |t_1| \frac{1+x_m+q_m}{1+q_m} \leq 3|s| \frac{1-q_m}{x_m}.$$

For  $\tau_m \leq 5 \cdot 10^{-3}$ ,  $|s| \leq 0.983$  and  $(1-q_m)/x_m \leq \tau_m e^{\tau_m/2} \leq 0.00502$ , so the right side is below 0.015. At the six poles with  $\tau_m > 5 \cdot 10^{-3}$ ,  $|s| \leq 0.296$  (Corollary 111) and  $(1-q_m)/x_m \leq (1-q_1)/x_1 = 0.8212$ , since  $q \mapsto (1-q)/\sqrt{q}$  decreases; the right side is at most 0.730. In both cases  $|t_1|(1+x_m+q_m)/(1+q_m) < 1/(2x_m)$ , the bracket is negative,  $\mathcal{N}_q \neq 0$ , and  $-x_m$  is a pole of order exactly  $\pi$ .  $\square$

At the dominant pole the bracket is  $-0.2155$ , and it tends to  $-\frac{1}{2}$  along the ladder. Only the uniform range is needed for infinitely many poles at  $-x_m$ ; the six-pole list upgrades this to every travel pole.

*Remark 86* ( $V$  at  $-x_m$ : verified, not proved). For  $V$  the same substitution in (28) gives

$$(1 - \mathfrak{g}_V t_1) \Pi_1|_{x=-x_m} = \frac{2q_m(1-x_m)}{1-q_m} \left( t_1 - \frac{1-x_m}{2x_m} \right),$$

and here the gate does not decide the sign. With  $t_1 = s(e^\tau - 1)$ ,  $s \rightarrow \frac{1}{4}$ , and  $(1-x_m)/(2x_m) = (e^{\tau/2} - 1)/2 = \frac{\tau}{4} + O(\tau^2)$ , the two terms cancel at leading order and the bracket is  $O(\tau^2)$ ; its sign is decided by the  $O(\tau)$  correction to  $s$ , which is not proved. Evaluated at 90 digits at the first twelve travel poles, the bracket is positive, from 0.117 at  $m = 1$  to  $2.9 \cdot 10^{-7}$  at  $m = 12$ , and close to  $\tau_m^2/8$ . So  $V$  has a pole at  $-x_m$  for  $m \leq 12$ ; for all  $m$  this is open. Nothing below uses it: every statement about  $V$  uses only the poles  $x_m$ .

## 8.2 $D$ -finiteness

**Lemma 87.** *Let  $\Omega \subset \mathbb{C}$  be a domain, let  $a_0, \dots, a_r, b$  be holomorphic on  $\Omega$  with  $a_r \neq 0$ , and let  $f$  be meromorphic on  $\Omega$  with  $a_r f^{(r)} + \dots + a_1 f' + a_0 f = b$  off the poles of  $f$ . Then every pole of  $f$  in  $\Omega$  is a zero of  $a_r$ .*

*Proof.* Let  $z_0$  be a pole of  $f$  of order  $k \geq 1$ . Then  $f^{(j)}$  has a pole of order exactly  $k + j$  at  $z_0$ , so for  $j < r$  the term  $a_j f^{(j)}$  has a pole of order at most  $k + r - 1$ , and  $b$  is holomorphic. If  $a_r(z_0) \neq 0$ , the term  $a_r f^{(r)}$  has a pole of order exactly  $k + r$ , which nothing else in the equation cancels. Hence  $a_r(z_0) = 0$ .  $\square$

**Theorem 88.**  *$U$  and  $V$  are not  $D$ -finite: neither satisfies a linear differential equation  $\sum_{j=0}^r a_j(x) f^{(j)}(x) = b(x)$  with polynomial coefficients and  $a_r \neq 0$ , nor even one whose coefficients  $a_j$  and  $b$  are holomorphic on a neighbourhood of the closed disc  $|x| \leq 1$ . Equivalently,  $(u_n)$  and  $(v_n)$  are not  $P$ -recursive. The same holds for the travel resolvent  $G_0^T$ , as a series in  $q$  and as a series in  $x$ , and for  $G_0^T$  it is unconditional. (Standing for  $U$  and  $V$ : Theorems 66 and 72.)*

*Proof.* Let  $f$  be one of the series and suppose it satisfies such an equation. By Lemma 67 (for  $G_0^T$ , Theorem 2, first clause)  $f$  continues to a meromorphic function on the open unit disc  $D$ . The expression  $\sum_j a_j f^{(j)} - b$  is then meromorphic on  $D$  and vanishes near 0, so by the identity theorem it vanishes on  $D$  off the poles of  $f$ . By Lemma 87 every pole of  $f$  in  $D$  is a zero of  $a_r$ . Since  $a_r \neq 0$  is holomorphic on a neighbourhood of the compact closed disc, its zeros there are isolated and hence finite in number. But  $f$  has infinitely many poles in  $D$ :  $V$  at every  $x_m$  (Theorem 76),  $U$  at every  $\pm x_m$  (Theorem 81 and Proposition 85), and  $G_0^T$  at every  $q_m$  in the variable  $q$  and at every  $\pm x_m$  in the variable  $x$  (Theorem 75). This is a contradiction.  $D$ -finiteness of a power series is equivalent to  $P$ -recursiveness of its coefficients [54, §6.4].  $\square$

The pole argument does not reach differential algebraicity:  $1/\cos(\pi/(1-x))$  has poles at  $x = 1 - 2/(2n+1)$ ,  $n \geq 1$ , infinitely many in  $|x| < 1$  and accumulating at  $x = 1$ , and satisfies an algebraic differential equation. Whether  $U$  and  $V$  are  $D$ -algebraic is open; we expect not.

**Corollary 89.** *The bivariate series  $\mathcal{W}(x, y) \in \mathbb{Z}[y][[x]] \subset \mathbb{Z}[[x, y]]$  is not  $D$ -finite in  $(x, y)$ . (Standing: Theorems 66 and 72.)*

*Proof.*  $D$ -finiteness is preserved under substitution of algebraic power series without constant term [32]. If  $\mathcal{W}$  were  $D$ -finite, the substitution  $y = x^2$  would make  $\mathcal{W}(x, x^2) = U$   $D$ -finite (Corollary 139), contrary to Theorem 88.  $\square$

## 8.3 Linear $q$ -difference and Mahler equations

**Proposition 90.** *Let  $f$  be  $U$  or  $V$ , as series in  $x$ , or  $G_0^T$ , as a series in  $q$ , and let  $a_0, \dots, a_r, b$  be polynomials with  $a_0 a_r \neq 0$ .*

(i) *If  $Q \in \mathbb{C}^\times$  is not a root of unity, then  $\sum_{i=0}^r a_i(x) f(Q^i x) \neq b(x)$ .*

(ii) *If  $k \geq 2$  is an integer, then  $\sum_{i=0}^r a_i(x) f(x^{k^i}) \neq b(x)$ .*

*For  $G_0^T$  both statements are unconditional; for  $U$  and  $V$  the standing is that of Theorems 76 and 81. When  $Q$  is a root of unity the case is open: the pole set can be invariant under  $x \mapsto Qx$ , as that of  $U$  is under  $x \mapsto -x$  (Proposition 85), and the argument below gives nothing.*

*Proof.* Let  $D$  be the open unit disc,  $P$  the pole set of  $f$  in  $D$ , and  $\rho > 0$  the radius of convergence of  $f$ ;  $P$  lies in  $\rho \leq |x| < 1$ , has finitely many points in each disc  $|x| \leq R < 1$ , and is infinite (proof of Theorem 88). An equation as in (i) or (ii) holds as an identity of power series near 0.

(i) If  $|Q| > 1$ , replacing  $x$  by  $Q^{-r}x$  and  $i$  by  $r - i$  gives an equation of the same form with  $Q^{-1}$  in place of  $Q$  and new leading and trailing coefficients  $a_r(Q^{-r}x)$ ,  $a_0(Q^{-r}x)$ ; so we may assume  $|Q| \leq 1$ . Then every  $f(Q^i x)$  is meromorphic on  $D$ , and the equation holds on  $D$  by the identity theorem. Let  $E$  be the finite zero set of  $a_0$  in  $D$ . At  $p \in P \setminus E$  the term  $a_0 f$  is singular and  $b$  is not, so some  $f(Q^i x)$  with  $1 \leq i \leq r$  is singular at  $p$ , that is  $Q^i p \in P$ . Thus every  $p \in P \setminus E$  has a successor  $Q^i p \in P$ ,  $1 \leq i \leq r$ , and every point of  $P$  is the successor of at most  $r$  points. Follow successors from any  $p \in P$ ; along the chain the exponent of  $Q$  increases strictly.

If  $|Q| < 1$  the moduli decrease strictly, so the chain stays in the finite set  $P \cap \{|x| \leq |p|\}$  and ends, necessarily at a point of  $E$ . A chain of  $n$  steps ends at modulus at most  $|Q|^n |p| < |Q|^n$ , and that modulus is at least  $\rho$ , so  $n < N := \log \rho / \log |Q|$ . Hence every point of  $P$  is reached backwards from  $E$  in fewer than  $N$  steps, and  $|P| \leq |E| \sum_{n < N} r^n$  is finite.

If  $|Q| = 1$  the chain stays on the circle  $|x| = |p|$ , which meets  $P$  in a finite set, and its points are pairwise distinct because  $Q$  is not a root of unity; so it ends in  $E$ . Hence  $P$  lies on the finitely many circles  $|x| = |e|$ ,  $e \in E$ , each meeting  $P$  in finitely many points, and  $P$  is finite. Both cases contradict the infinitude of  $P$ .

(ii) Here every  $f(x^{k^i})$  is meromorphic on  $D$  and the equation holds on  $D$ . Work on the positive axis: let  $P_+ = P \cap (0, 1)$  and let  $E_+$  be the finite zero set of  $a_0$  in  $(0, 1)$ . As in (i), every  $p \in P_+ \setminus E_+$  has a successor  $p^{k^i} \in P_+$  with  $1 \leq i \leq r$ ; it is smaller than  $p$ , so every chain ends in  $E_+$ , and reading a chain backwards,  $p = e^{k^{-n}}$  for some  $e \in E_+$  and some integer  $n \geq 0$ . Hence, for  $0 < R < 1$  and  $\ell = \log(1/R)$ ,

$$\#(P_+ \cap (0, R]) \leq \sum_{e \in E_+} \#\{n \geq 0 : k^n \leq \log(1/e)/\ell\} \leq |E_+| \left(1 + \log_k \frac{L}{\ell}\right), \quad L = \max_{e \in E_+} \log(1/e),$$

which grows like  $\log(1/\ell)$  as  $R \uparrow 1$ . On the other hand, by Lemma 12 there is a travel pole with  $w \in (m\pi, (m+1)\pi)$  for every  $m \geq 4$ , at  $x = e^{-1/w^2} < e^{-1/((m+1)\pi)^2}$ , equivalently  $q = e^{-2/w^2} < e^{-2/((m+1)\pi)^2}$ , and it lies in  $P_+$  (Theorems 76, 81, 75). Distinct windows give distinct poles, so  $P_+ \cap (0, R]$  has at least  $\lfloor c/(\pi\sqrt{\ell}) \rfloor - 4$  points, with  $c = 1$  in the variable  $x$  and  $c = \sqrt{2}$  in  $q$ . This grows like  $\ell^{-1/2}$ , faster than  $\log(1/\ell)$ , a contradiction for  $R$  close to 1.  $\square$

The count in (ii) is made on the real axis on purpose. It uses only the travel poles that are proved to exist, and it needs no statement about where on the circle the poles of  $f$  accumulate; compare Remark 93. Mahler equations are those of [39]. The catalytic equation of §2 is a  $q$ -difference equation in the catalytic variable, not in  $x$ , and is not affected by (i).

## 8.4 The natural boundary

We use the following rationality theorem.

**Theorem 91** (Pólya–Bertrandias). *Let  $\Omega \subset \mathbb{C}$  be a simply connected domain containing 0 and stable under complex conjugation, and let  $\varphi$  be the Riemann map of the unit disc onto  $\Omega$  with  $\varphi(0) = 0$ ,  $\varphi'(0) > 0$ . If  $\varphi'(0) > 1$  and  $f \in \mathbb{Z}[[x]]$  is the Taylor series at 0 of a function meromorphic on  $\Omega$ , then  $f$  is a rational function.*

This is Pólya’s theorem [46], which replaced the disc of Borel’s rationality theorem by a domain measured by transfinite diameter; Bertrandias [5] extended it to  $p$ -adic and adelic domains, whence the name. Bost and Chambert-Loir prove a generalisation to curves over number fields [9, Thm. 7.9], from which they recover the Pólya–Bertrandias criterion [9, Example 7.10], and Cantor’s rationality theorem [11, Thm. 5.2.2] is a capacity-theoretic generalisation. Meromorphy suffices, but the strict inequality  $\varphi'(0) > 1$  is needed: for  $\Omega$  the unit disc,  $\varphi'(0) = 1$ , and  $\sum_n x^{2^n}$  is not rational.

**Theorem 92.** (a) *The travel resolvent  $G_0^T = \Sigma_0/(1-\Sigma_1) \in \mathbb{Z}[[q]]$  has  $|q| = 1$  as a natural boundary. This is unconditional.*

(b)  *$U$  and  $V$  have  $|x| = 1$  as a natural boundary. (Standing: Theorems 66 and 72.)*

*More precisely, in each case there is no point  $\zeta$  of the unit circle and no disc  $B$  about  $\zeta$  such that the function continues meromorphically to  $D \cup B$ ,  $D$  the open unit disc.*

*Proof.* Let  $f$  be  $G_0^T$  (in  $q$ ),  $U$  or  $V$  (in  $x$ ). In each case  $f$  has integer coefficients (the travel-run counts,  $u_n, v_n$ ), is meromorphic on  $D$  (Lemma 67), and is not rational, having infinitely many poles in  $D$  (Theorems 75, 76, 81). Suppose  $f$  continues meromorphically to  $D \cup B$  with  $B$  the disc of radius  $\delta > 0$  about  $\zeta$ ,  $|\zeta| = 1$ . Shrinking  $\delta$ , we may assume  $\delta < |\operatorname{Im} \zeta|$  when  $\zeta \notin \mathbb{R}$ , so that  $B$  and its conjugate  $\bar{B}$  are disjoint; when  $\zeta = \pm 1$ ,  $\bar{B} = B$ . Because the coefficients are real,  $\overline{f(\bar{x})} = f(x)$  near 0, hence on  $D$ , and  $x \mapsto \overline{f(\bar{x})}$  continues  $f$  meromorphically to  $\bar{B}$ , agreeing with  $f$  on  $\bar{B} \cap D$ . So  $f$  is meromorphic and single-valued on

$$\Omega = D \cup B \cup \bar{B},$$

which is stable under conjugation and simply connected: it is the disc  $D$  with one or two disjoint discs attached, each meeting  $D$  in a convex set. Let  $\varphi$  be its Riemann map,  $\varphi(0) = 0$ ,  $\varphi'(0) > 0$ . By uniqueness of the Riemann map,  $\overline{\varphi(\bar{z})} = \varphi(z)$ , so  $\varphi$  has real Taylor coefficients (Schwarz reflection). The map  $\psi = \varphi^{-1}|_D$  sends  $D$  into  $D$  with  $\psi(0) = 0$  and is not onto, since  $\Omega \neq D$ ; by the Schwarz lemma  $0 < \psi'(0) < 1$ , that is  $\varphi'(0) > 1$ . Theorem 91 now makes  $f$  rational, a contradiction.  $\square$

A  $D$ -finite series continues analytically along every path avoiding the finitely many zeros of the leading coefficient of its equation, so Theorem 92 gives a second proof of the polynomial-coefficient case of Theorem 88. The first proof is elementary and also covers coefficients holomorphic near the closed disc.

*Remark 93* (where on the circle the poles accumulate). Theorem 92 does not locate the singularities on the circle. The poles proved above accumulate only at  $x = \pm 1$  (at  $q = 1$  for  $G_0^T$ ), yet every point of the circle is singular. Numerically the pole set is much larger. The function  $1 - \Sigma_1 = \cos(Z; q^2)$  has 64 zeros in  $|q| < 0.94$  by the argument principle: the travel poles  $q_1, q_2$  and 62 non-real zeros in 31 conjugate pairs, the first at  $q = -0.4075 \pm 0.5449i$ . At each non-real zero  $\Pi_q$  was found non-zero at 40 digits, so these are candidate poles of  $G_0^T$  and of  $U$  off the real axis. Plotted, the zeros approach roots of unity along straight lines. We conjecture that the poles are dense on the circle; a route is a  $q$ -cosine resummation at  $q = \zeta e^{-t}$ ,  $\zeta$  a root of unity. The natural boundary does not depend on this.

## 9 An orthogonal arithmetic route

For each prime  $p$ ,  $U \bmod p \in \mathbb{F}_p[[q]]$  is algebraic over  $\mathbb{F}_p(q)$  iff its coefficient sequence is  $p$ -automatic (Christol [13, 14]), iff its  $p$ -kernel is finite. We find the  $p$ -kernel to be *maximally* non-automatic: through the accessible depth its size tracks the full  $p$ -ary tree with *no* deficit. (An apparent deficit of 1 at level 3; 1, 4, 13, 39 vs. the full 1, 4, 13, 40 at  $p = 3$ ; is a *comparison-length artifact*: a level- $k$  decimation subsequence retains only  $\sim N/p^k$  terms, so at a short prefix two distinct decimations can coincide; the deficit disappears once the comparison length reaches 9.) Two instances are machine-checked in Lean 4; both proofs use `native_decide`, so besides the three standard axioms they depend on the compiler-trust axiom `_native.native_decide.ax_1_1` of the declaration concerned. For the true series itself, the 3-kernel of  $(u_n \bmod 3)$  through level 3 is the full ternary tree, deficit 0: all  $1 + 3 + 9 + 27 = 40$  decimation words of length  $\leq 3$  are pairwise

distinct: at comparison lengths 9 and 10 alike; on the 271-term prefix  $n \leq 270$  (`UKernel.lean`, from the validated `tools/u_modp_rust` output, whose  $u_n$  are self-checked against the 43 known terms and byte-identical to the earlier validated  $n \leq 180$  run on the overlap). For the sibling block  $\Sigma_1$ , transcendental over  $\mathbb{Q}(q)$  by Corollary 3, the same level-3 full tree is certified in `SigmaKernel.lean` (built from the  $A/C$  recursion over  $\mathbb{F}_3$ ), with the short-prefix deficit-1 confirmed spurious. The Berlekamp–Massey [34] linear complexity of  $(u_n \bmod 3)$  is  $\approx N/2$ . This is the strongest finite-data evidence for non-automaticity, and hence for transcendence over  $\mathbb{Q}(x)$ , independent of all of §§4–7. A proof for  $U$  would require a non-automaticity criterion valid for *dense*-support series (the coefficients carry no lacunary gaps), which the standard gap/Mahler methods do not provide.

## 10 Conclusion

The arithmetic of A396406 separates cleanly. The Pólya–Carlson dichotomy for the catalytic building blocks is unconditional (Theorem 2), and so is the lower bound  $\beta_2$  for the growth rate; the matching upper bound, and with it the identification of  $\beta_2$  as the rate, consumes the transfer model (M), Theorem 66, and nothing else (Corollary 78). For the two denominators  $\Sigma_1, S_1$  the exclusion of the rational alternative needs no growth hypothesis: it follows from the infinitude of their level sets in  $(0, 1)$ , hence from the same Lemma 10 that the rest of the paper consumes (Corollary 3). For the two numerators it rests on Hypothesis 5, verified out to degree 1156 and unproved, except for the joint statement that  $\Sigma_0$  and  $S_0$  are not both rational, which the pole route supplies. **The travel resolvent  $\Sigma_0/(1 - \Sigma_1)$  is transcendental over  $\mathbb{Q}(x)$ , unconditionally** (Theorem 75), and it needs no growth hypothesis. Its non-cancellation inputs are two and are separate. The denominator input is Proposition 94, proved in Appendix A by an elementary chain with explicit numerical constants: an annulus representation, an Euler bound, a Poisson theta envelope, and a  $q$ -Wronskian invariant. No asymptotic theorem is cited on that path. The numerator input is Proposition 25, the identity  $\Sigma_0(q_m)S_0(q_m) = 2q_m/(1 - q_m)$ , which follows from the closed forms  $\Sigma_0 = \frac{2q}{Z}\mathfrak{s}(Z)$ ,  $S_0 = \frac{2q}{z_0}\mathfrak{s}(z_0)$  and the  $q$ -Pythagorean identity at a zero of the  $q$ -cosine; it is exact, it holds at every pole with no exceptional set, and it consumes no asymptotics. Remark 9 gives the denominator conclusion by a different and shorter route (the saddles  $z_{\pm}^* = \pm iW/2$  with all constants derived explicitly, the two saddles supplying the global cancellation that no partial-summation method can), but that route is a sketch only (Remark 11), and no theorem uses it. Both  $V$  and  $U$  take their pole set from Lemma 12, which consumes only the rectangle bound  $|T_2| \rightarrow 0$  of Lemma 10. The block assembly is developed in §5, on the observation that the travel and the bulk block are the same object, the resolvent of  $q^{\max(a,b)}$  against a geometric weight: this proves the travel telescoping, the bulk dressing, the reciprocity  $t_0 = b_1$  with its cocycle form  $\det P = 1$ , the identity  $t_1 = P_{12}/S_e$ , the Möbius factorisation (31), and the shape of the local form at a travel pole, whose leading numerator is the square of the junction pairing  $\langle \lambda, R \rangle$ . It also refutes the product form of the numerator: the marker junction has rank at least 3 (Proposition 69); at a travel pole the pairing nevertheless collapses to an exact closed form in  $t_1$ , positive at every pole (Theorem 72).  **$V$  is transcendental over  $\mathbb{Q}(x)$**  (Theorem 76). The true series  $U$  stands on the same *analytic* footing: its further analytic inputs are proved in Appendix B: the gate block is the symmetric second difference of the  $q$ -cosine at its own zero (Proposition 102), the two-interval uniform certificate of Theorem 109 gives  $|P_{12}| \leq 0.619\tau^{3/2} < \tau^{3/2}/\sqrt{2}$  for  $\tau \leq 5 \cdot 10^{-3}$  with the six poles above that threshold checked directly, and the exact identity  $S_e = \frac{qZ}{2}\mathfrak{s}(Z) - P_{12}$  turns the same certificate into the denominator bound that closes the gate ( $|s| \leq 0.983$ ; Corollary 111), consistent with the phase-matched leading values  $|P_{12}|/\tau^{3/2} \rightarrow 1/(4\sqrt{2})$  and  $s \rightarrow \frac{1}{4}$ .  **$U$  is transcendental over  $\mathbb{Q}(x)$**  (Theorem 81). The input (T) that earlier versions of this theorem carried is gone: Proposition 1 is now proved (§7.2),

via Theorem 62 and Corollary 63, and does not depend on the metric identity. (R-J) is Theorem 72, and the hypothesis (L) of earlier versions is Theorem 70. Neither series is  $D$ -finite, and  $|x| = 1$  is a natural boundary of both (Theorems 88 and 92, by the Pólya–Bertrandias theorem); for the travel resolvent the natural boundary  $|q| = 1$  is unconditional. Two questions remain open: whether the poles are dense on the circle, and whether  $U$  and  $V$  are  $D$ -algebraic. Every statement in §5 beyond the operator algebra rests on the transfer model (M), now Theorem 66, whose assembly step is a hand proof not yet formalised. No input separates  $U$  from  $V$ . Two finite ranges are computer-assisted: the six gate poles above  $\tau = 5 \cdot 10^{-3}$  and the positivity of the pairing at the first twelve poles; neither is needed for transcendence, only for the statements “at every travel pole”. The series in question is the growth series of the universal group  $W_{\text{univ}}$ ; for a particular rational triangle it agrees with that of  $W_T$  only to the effective universality depth of [6]. The turning-point amplitude atom on the  $U$ -side is the elementary rational bound of Lemma 79, formally verified in Lean 4. The independent mod- $p$  route of §9 would settle  $U$  by a completely different mechanism, if a dense-support non-automaticity criterion can be found. Whether the number  $\beta_2$  is transcendental is untouched by all of this and remains open; Appendix C reduces it to a value statement for a Hahn–Exton  $q$ -Bessel function and shows the obstruction to be the arithmetic ratio  $\rho = \frac{1}{2}$ .

## A An explicit lower bound for $S_e$ at the travel poles

Throughout  $z_0 = \sqrt{2(1-q)}$ ,  $Z = z_0/\sqrt{q}$ ,  $w = \sqrt{2/\tau}$ ,  $f(z) = \cos(z; q^2) = C(z^2)$ ,  $C(y) = \sum_{j \geq 0} (-1)^j q^{j^2+j} y^j / (q; q)_{2j}$ , and  $s(z) = \sin(z; q^2)$ ; by Proposition 17,  $S_e = f(z_0)$  and the travel poles are the solutions of  $f(Z) = 0$ .

**Proposition 94.** *At every travel pole  $q_m$  with  $\tau_m \leq 1.29 \cdot 10^{-4}$  one has  $|S_e(q_m)| \geq 0.35\sqrt{\tau_m}$ . The finitely many (Lemma 103) poles with  $\tau_m > 1.29 \cdot 10^{-4}$  (the dominant pole  $q_1 = 0.449453 \dots$  ( $|S_e|/\sqrt{\tau_1} = 0.581$ ; `star_gate_certificate.py`, Corollary 111) and the thirty-nine tabulated below it,  $\tau_{41} = 1.2354 \cdot 10^{-4}$  being the first below the threshold) satisfy the same bound by direct high-precision computation.*

The proof occupies the rest of the appendix: an annulus representation turns all derivative bounds into cancellation-free integrals, and the pole condition supplies the value-level input.

**Lemma 95.** *For  $0 < r < 1$ ,  $C(y) = \frac{1}{2\pi} \int_0^{2\pi} \Theta(y/x^2)(x; q)_\infty^{-1} d\theta$  with  $x = re^{i\theta}$  and  $\Theta(t) = \sum_{j \in \mathbb{Z}} (-1)^j q^{j^2+j} t^j = (q^2; q^2)_\infty (q^2 t; q^2)_\infty (1/t; q^2)_\infty$ .*

*Proof.* Euler’s identity gives  $1/(q; q)_{2j} = [x^{2j}](x; q)_\infty^{-1}$ ; extract the coefficient by Cauchy’s formula on  $|x| = r$ . For  $j \leq -1$  the integrand  $x^{-2j}(x; q)_\infty^{-1}$  is analytic in  $|x| < 1$  (the zeros of  $(x; q)_\infty$  lie at  $x = q^{-k}$ ,  $|q^{-k}| \geq 1$ ), so the negative-index terms vanish and the sum completes to the bilateral one, which is the triple product. Equivalently this is the  $p = q^2$  case of the  $q$ -Borel– $q$ -Laplace inversion of [58] and [37, Lemma 1], the  $q^2$ -Borel transform of  $C$  being the even part of  $(x; q)_\infty^{-1}$ .  $\square$

**Lemma 96.** *For  $0 < r \leq \frac{1}{2}$ :  $(x; q)_\infty^{-1} \leq \exp(w_r \cos \theta + K_r)$  in modulus, with  $w_r = r/(1-q)$  and  $K_r = r^2/(2(1-r)(1-q^2))$ .*

*Proof.*  $-\log|1-u| = \operatorname{Re} u + \operatorname{Re} \sum_{n \geq 2} u^n/n \leq \operatorname{Re} u + \frac{1}{2}|u|^2/(1-|u|)$  for  $|u| < 1$ ; summing over  $u = xq^j$  gives  $\sum_j \operatorname{Re}(xq^j) = w_r \cos \theta$  and  $\sum_j r^2 q^{2j}/(2(1-rq^j)) \leq r^2/(2(1-r)(1-q^2)) = K_r$ .  $\square$

**Lemma 97.** *Let  $p = q^2$ ,  $t = \rho e^{i\varphi}$ ,  $a = (\log \rho - \tau) + i\tilde{\varphi}$  with  $\tilde{\varphi} \in (-\pi, \pi]$  representing  $\varphi + \pi$ . For  $n \in \{0, 1, 2\}$ ,*

$$|\Theta^{(n)}(t)| \leq \sqrt{\pi/\tau} e^{(\log \rho - \tau)^2/(4\tau)} \rho^{-n} (|a|/(2\tau) + \tau^{-1/2})^n G(\tilde{\varphi}), \quad G(\tilde{\varphi}) = \sum_{k \in \mathbb{Z}} e^{-\pi^2(k - \tilde{\varphi}/2\pi)^2/\tau}.$$

*Proof.*  $\Theta(t) = \theta_p(-pt)$  with  $\theta_p(y) = \sum_n p^{n(n-1)/2} y^n$ . Poisson summation gives the *exact* identity

$$S(a) := \sum_{j \in \mathbb{Z}} e^{-\tau j^2 + aj} = \sqrt{\pi/\tau} \sum_{k \in \mathbb{Z}} e^{a_k^2/(4\tau)}, \quad a_k := a - 2\pi i k. \quad (36)$$

Since  $\operatorname{Re}(a_k^2) = (\log \rho - \tau)^2 - (\tilde{\varphi} - 2\pi k)^2$ , the imaginary part of  $a$  contributes *decay*, and  $\sum_k e^{\operatorname{Re}(a_k^2)/(4\tau)} = e^{(\log \rho - \tau)^2/(4\tau)} G(\tilde{\varphi})$ ; no cancellation is involved, the term  $-(\tilde{\varphi} - 2\pi k)^2$  being precisely what constitutes  $G$ . Differentiating (36) in  $a$ , the  $j$ - and  $j(j-1)$ -weighted sums are  $S'$  and  $S'' - S'$ , and  $\partial_a e^{a_k^2/(4\tau)} = (a_k/2\tau) e^{a_k^2/(4\tau)}$ , so the  $k$ -th image carries the weight

$$W_1(a_k) = \frac{|a_k|}{2\tau}, \quad W_2(a_k) = \frac{|a_k|^2}{4\tau^2} + \frac{1}{2\tau} + \frac{|a_k|}{2\tau}.$$

At the radius  $\rho = e^\tau$  of Lemma 98 one has  $a_k = i(\tilde{\varphi} - 2\pi k)$  exactly, so  $|a_k| = |\tilde{\varphi} - 2\pi k|$ . To pull the weight out of the  $k$ -sum, note that for  $k \neq 0$  the excess  $|a_k| - |a_0|$  is paid for by the Gaussian factor, which is smaller by  $e^{-((\tilde{\varphi} - 2\pi k)^2 - \tilde{\varphi}^2)/(4\tau)}$ ; the aggregate excess is maximised at the nearest image and is at most  $\max_{\delta > 0} 2\delta e^{-\pi\delta/\tau} = 2\tau/(\pi e) \leq 0.2342\tau$ , which is below  $\tau^{-1/2}$  for  $\tau \leq 0.0198$ . Hence  $\sum_k W_n(a_k) e^{\operatorname{Re}(a_k^2)/(4\tau)} \leq (|a|/(2\tau) + \tau^{-1/2})^n \sum_k e^{\operatorname{Re}(a_k^2)/(4\tau)}$ , which is the stated bound.  $\square$

**Lemma 98.** For  $\tau \leq 0.0198$  and every  $\xi \in [qZ, Z]$ , with  $r^2 = \xi^2 e^{-\tau}$ :  $M_2 := \max_{[qZ, Z]} |f''| \leq 7w^4$ .

*Proof.*  $f''(z) = 2C'(z^2) + 4z^2 C''(z^2)$ , and differentiating Lemma 95 under the integral,  $|C^{(n)}(y)| \leq (2\pi r^{2n})^{-1} \int_0^{2\pi} |\Theta^{(n)}(t)| (x; q)_\infty^{-1} d\theta$  at  $\rho = y/r^2 = e^\tau$ , which zeroes  $(\log \rho - \tau)^2$ . The key step is the exact unfolding: as  $\theta$  runs over  $[0, 2\pi)$  the variable  $u = \pi - 2\theta$  runs over two periods, consecutive branches carrying opposite signs of  $\sin(u/2)$ , so

$$\frac{1}{2\pi} \int_0^{2\pi} e^{w_r \cos \theta} G(\pi - 2\theta) d\theta = \frac{1}{2\pi} \int_{\mathbb{R}} \cosh(w_r \sin \frac{u}{2}) e^{-u^2/(4\tau)} du,$$

an identity; the derivative weights unfold with it, becoming polynomials in  $u$ , and  $|\sin(u/2)| \leq |u|/2$ . The even weight gives an exact Gaussian moment,  $\int v^2 e^{-(v-s)^2/(4\tau)} dv = \sqrt{4\pi\tau} (2\tau + s^2)$  with  $s = \tau w_r \leq \sqrt{2\tau} e^{\tau/2}$ , and  $\sqrt{\pi/\tau} \cdot \sqrt{4\pi\tau} = 2\pi$  cancels the prefactors; the odd weight does *not*, giving instead  $\int |v| e^{-(v-s)^2/(4\tau)} dv = \sqrt{4\pi\tau} s \operatorname{erf}(s/(2\sqrt{\tau})) + 4\tau e^{-s^2/(4\tau)}$ , and the error function and the exponential are kept, not discarded. Writing  $s = \tau w_r$ , the two assembled factors are

$$Q_1 = \tau^{-1/2} + \frac{s \operatorname{erf}(s/(2\sqrt{\tau})) + 2\sqrt{\tau/\pi} e^{-s^2/(4\tau)}}{2\tau}, \quad Q_2 = \frac{2\tau + s^2}{4\tau^2} + \tau^{-3/2} [\text{same bracket}] + \frac{1}{\tau},$$

the prefactor  $\sqrt{\pi/\tau}$  of Lemma 97 cancelling against  $\sqrt{4\pi\tau}$  in every term. On the whole range,  $\sup \sqrt{\tau} Q_1 = 1.8273$  and  $\sup \tau Q_2 = 3.6596$ , both attained at  $\tau = 0.0198$ ,  $\xi = Z$ . With the weights of Lemma 97 this assembles, for  $\tau \leq 0.0198$  (so  $r \leq 0.1981 < \frac{1}{2}$ ,  $K_r \leq 0.631$ ,  $e^{\tau w_r^2/4} \leq e^{0.506}$ ), to

$$|C'(y)| \leq e^{K_r - \tau + \tau w_r^2/4} (1.83 \tau^{-1/2}) r^{-2}, \quad |C''(y)| \leq e^{K_r - 2\tau + \tau w_r^2/4} (3.670/\tau) r^{-4}.$$

With  $4\xi^2/r^4 \leq (2/\tau) e^{3\tau} (1 + 2\tau)$  the total  $M_2 \tau^2$  is at most 25.04 at  $\tau = 0.0198$ , decreasing to 20.2 at  $\tau = 1.29 \cdot 10^{-4}$ , in both cases below the target  $7w^4 \tau^2 = 28$ . Those two numbers come from pairing the largest  $K_r$  and the largest  $\tau w_r^2/4$ , both attained at  $\xi = Z$ , with the largest  $r^{-2}$ , attained at  $\xi = qZ$ ; taking instead the supremum over  $\xi \in [qZ, Z]$  of the assembled bound at a single  $\xi$  gives 23.33 at  $\tau = 0.0198$ , 20.06 at  $\tau = 1.29 \cdot 10^{-4}$  and 19.84 as  $\tau \rightarrow 0$ , so the true margin against 28 is 16.7% rather than the 10.6% the factored form reports. Every constant in this proof, and the two  $\xi$ -suprema, are regenerated by `annulus_certificate.py` at two precisions; see Remark 99.  $\square$

*Remark 99* (what the certificate checks, and what it found). `annulus_certificate.py` recomputes, at 30 and again at 60 digits with zero disagreement: the window geometry ( $\sup r = 0.19802$ ,  $\sup K_r = 0.62963$ ,  $\sup \tau w_r^2/4 = 0.50497$  against the quoted  $\frac{1}{2}$ , 0.631, 0.506); the excess constant  $2/(\pi e) = 0.234199$  of Lemma 97; Lemma 96 against the true  $|(x; q)_\infty^{-1}|$  on a  $(\tau, \xi, \theta)$  grid; the unfolding identity of this proof against numerical quadrature; the two moment suprema above; the assembled  $M_2\tau^2$  on a 400-point  $\tau$  grid with the  $\xi$ -supremum taken by search rather than at an endpoint; and the chain of Proposition 94 below. It also checks that the bound dominates the true  $\max |f''|$  where both are computable, the ratio  $\max |f''|/(\text{bound})$  coming out between 0.023 and 0.045 at  $\tau = 0.0198, 0.0127, 5 \cdot 10^{-3}, 10^{-3}$ , consistent with Remark 100. Two constants of an earlier version of this proof did not survive it and are corrected above:  $\sup \sqrt{\tau} Q_1 = 1.82733$  is not bounded by 1.825, and  $2/(\pi e) = 0.234199$  is not bounded by 0.234. Both corrections move  $M_2\tau^2$  by less than 0.01 and neither affects any conclusion. The arithmetic model is mpmath `mpf` at the two stated precisions, not interval arithmetic.

*Remark 100* (how lossy Lemma 98 is, and why one sample cannot say). The ratio  $M_2/w^4$  does not converge:  $f$  is the  $q$ -cosine, so  $f''(z) \approx -(W/z_0)^2 \cos(zW/z_0)$  with  $(W/z_0)^2 = e^{-\tau}/(\tau(1-q)) \sim w^4/4$ , while the window  $[qZ, Z]$  spans a phase of only  $(1-q)Z/\tau \sim \sqrt{2\tau}$  radians. As  $\tau$  varies the window slides through the cosine and  $M_2/w^4$  sweeps  $[0, \frac{1}{4}]$ ; a single value of  $\tau$  therefore reports an arbitrary point of that sweep and says nothing about the maximum. The maximum is attained where the window covers an extremum of the phase, i.e. along the extreme-phase points  $w = m\pi$ , and there  $M_2/w^4 \uparrow \frac{1}{4}$ : measured 0.2499993 at  $w = 120\pi$ , against 0.22599 at the top of the range  $\tau = 0.0198$  and 0.02337 at  $\tau = 0.0163$ . Over the 165 values of  $\tau$  scanned,  $M_2 < \frac{1}{4}w^4$  (`m2_certificate.py`, mpmath at two precisions, agreeing to ten digits at every point; a scan, not a supremum proof). Lemma 98 is therefore lossy by a factor of about 28 at the top of the range, and its target  $7w^4$  clears the measured values by about 31.

*Proof of Proposition 94.* At a pole  $f(Z) = 0$ , so the Koornwinder–Swarttouw rules give the exact facts  $f(qZ) = qZ s(Z)$  and  $s(qZ) = s(Z)$ ; the mean value theorem on  $[qZ, Z]$  then pins  $f'(\zeta) = -q s(Z)/(1-q)$  at some  $\zeta \in (qZ, Z)$ , and a second mean value on  $[z_0, Z] \subset [qZ, Z]$  gives

$$|S_e| = (Z - z_0) |f'(\tilde{\zeta})| \geq \frac{q(1 - \sqrt{q})}{1 - q} Z |s(Z)| - (1 - \sqrt{q})(1 - q)Z^2 M_2.$$

The sine factor is bounded below by an invariant of Casoratian ( $q$ -Wronskian) type for the Hahn–Exton pair; the closed-form  $q$ -Wronskian is due to Swarttouw [56] (eq. (4.3.8)); see also [41, eq. (4.21)]. Concretely, with  $c_n = \cos(Zq^n; q^2)$ ,  $s_n = \sin(Zq^n; q^2)$  and  $F_n = c_n^2 - Zq^{n+1}c_n s_n + q s_n^2$ , the rules give the exact telescoping  $F_n - F_{n+1} = -Zq^{n+1}(1-q)c_{n+1}s_{n+1}$  with  $F_n \rightarrow 1$ ; the form has smallest eigenvalue  $\lambda_n \geq q - Z/2$  (by  $\sqrt{a^2 + b^2} \leq a + b$ ),  $\sum_{n \geq 0} Zq^{n+1}(1-q) = qZ$ , and the discrete Gronwall gives  $F_0 \geq e^{-0.61Z} \geq 1 - 0.61Z$  for  $Z \leq \min(0.2, q/3)$ ; at the pole  $F_0 = q s(Z)^2$ , so  $|s(Z)| \geq \sqrt{1 - 0.61Z}$ . (The Gronwall sum is  $qZ/(2q - Z)$ , whose supremum over the range is  $0.5568Z$ , so 0.61 has room;  $Z \leq \min(0.2, q/3)$  holds throughout,  $Z = 0.19999$  at  $\tau = 0.0198$  and  $Z = 0.016063$  at  $\tau = 1.29 \cdot 10^{-4}$ .) Inserting Lemma 98, the error fraction is  $28\sqrt{2}(1.02)\sqrt{\tau} \leq 0.46$  for  $\tau \leq 1.29 \cdot 10^{-4}$ , a range containing every pole with  $m \geq 41$ , whence  $|S_e(q_m)| \geq (1 - 0.02 - 0.46)\sqrt{\tau_m}/2 \geq 0.35\sqrt{\tau_m}$  there. Evaluated rather than rounded, the same chain gives  $0.3856\sqrt{\tau_m}$  at the threshold and more below it, the exact error fraction being 0.4520 against the quoted 0.46 (`annulus_certificate.py`). The tabulated poles are verified numerically at 60+ digits (`qtrig_verify.py`, mpmath, no interval arithmetic:  $|S_e|/\sqrt{\tau} \in [0.698, 0.708]$  as asserted by the script; the tabulated set is poles 2, ..., 60 in the indexing of §2 (the dominant pole is not among them, its value 0.581 coming from `star_gate_certificate.py`) and the table omits poles 53 and 60 (rows 52, 59), whose  $\tau < 7.7 \cdot 10^{-5}$  lies inside the proved range), and the

two ranges overlap on every tabulated pole with  $m \geq 41$ ; only poles  $2, \dots, 40$  (table rows  $1, \dots, 39$ ) are load-bearing. The polynomial identities underlying the telescoping and the pole reduction are machine-checked (`GramCasoratian.lean`: `F_shift`, `F_drift`, `pole_qsine`, `sigma1_term`; each audits inside the standard three, the first two on `propext`, `Quot.sound` alone; no `sorry`).  $\square$

On the intermediate window  $1.29 \cdot 10^{-4} < \tau_m \leq 5 \cdot 10^{-3}$  the bound also follows, with margin, from Lemma 110 ( $0.63 > 0.35$ ), independently of any enumeration of the poles there.

## B Proof of the amplitude estimate: the gate holds at every travel pole

This appendix proves the estimate  $(\star)$  of Theorem 81 (at every travel pole,  $|P_{12}(q_m)| w_m^3 < 2$ ,  $w_m = \sqrt{2/\tau_m}$ ; quantitatively  $|P_{12}| \leq \frac{3.5\sqrt{2}}{8} \tau^{3/2} < 0.619 \tau^{3/2}$  for  $\tau \leq 5 \cdot 10^{-3}$ , and  $|P_{12}| w^3 \leq 0.75$  at the six poles above that threshold) and then closes the full gate (32): the same certificate yields the denominator bound  $|S_e| \geq 0.63\sqrt{\tau}$  (Lemma 110), hence  $s < 1$  at every travel pole, and, granting the transfer model (M) for the closed form of  $b_0$  (Corollary 41), also  $b_0 > 0$  there (Corollary 111). Nothing in this appendix depends on (M) except the statement just named as depending on it. Throughout,  $h = \tau/2$ ,  $\sigma^2 = \tau/2$ ,  $\psi(t) = \cos(Ze^t; q^2)$ ; by Proposition 102,  $P_{12} = -\frac{1}{2}[\psi(h) + \psi(-h)]$  and  $\psi(0) = 0$ , so

$$|P_{12}| \leq \frac{h^2}{2} \max_{|t| \leq h} |\psi''(t)|. \quad (37)$$

### The gate block is a second difference at a zero

The following exact identity is not needed for Theorem 76, but it is the sharpest available description of the object the  $U$ -gate must bound, and it makes the size  $\tau^{3/2}$  structural rather than numerical.

**Lemma 101** (the gate block in closed form). *Identically in  $0 < q < 1$ , with  $c(u) = \cos(u; q^2)$  and  $\mathfrak{s}(u) = q^{-1/2} \sin(q^{1/2}u; q^2)$ :*

$$P_{12} = \frac{qZ}{2} \mathfrak{s}(Z) - c(z_0), \quad S_o := \sum_{j \geq 0} \frac{(-1)^j [2(1-q)]^j q^{j^2+2j} (1-q)}{(q; q)_{2j+1}} = \frac{z_0}{2} \mathfrak{s}(z_0),$$

$S_o$  being the odd bulk block of the closed form  $b_0 = \frac{2q}{1-q} S_o/S_e$  of Corollary 41.

*Proof.* From (35),  $P_{12} = \frac{2q^3}{1-q^3} \sum_k d_k$  with  $d_k = (-2)^k (1-q)^k q^{k^2+3k} / [(q^2; q^2)_k (q^5; q^2)_k]$ ; since  $(q; q)_{2k+3} = (q; q^2)_{k+2} (q^2; q^2)_{k+1}$  and  $(q; q^2)_{k+2} = (1-q)(1-q^3)(q^5; q^2)_k$ , this is

$$P_{12} = \sum_{k \geq 1} (-1)^{k-1} \frac{[2(1-q)]^k q^{k^2+k+1} (1-q^{2k})}{(q; q)_{2k+1}}. \quad (38)$$

Since  $q^{1/2}Z = z_0$  and  $z_0^2 = 2(1-q)$ , the  $j$ -th summand of  $\frac{qZ}{2} \mathfrak{s}(Z) = \frac{z_0}{2} \sin(z_0; q^2)$  is  $(-1)^j (1-q)[2(1-q)]^j q^{j^2+j} / (q; q)_{2j+1}$ , and that of  $c(z_0)$  is  $(-1)^j [2(1-q)]^j q^{j^2+j} / (q; q)_{2j}$ . Their difference carries the factor  $(1-q) - (1-q^{2j+1}) = q^{2j+1} - q$ : the  $j = 0$  terms cancel, and what remains is (38) term by term. The  $S_o$  identity is the same computation one half-step down: the  $j$ -th summand of  $\frac{z_0}{2} \mathfrak{s}(z_0)$  is  $(-1)^j (1-q)[2(1-q)]^j q^{j^2+2j} / (q; q)_{2j+1}$ , which is  $S_o$ 's summand verbatim. All three matchings are exponent bookkeeping of the kind machine-checked in `QTrigIdentities.lean`; the identities are verified off-pole (hence as identities) and at the six exceptional poles to  $10^{-100}$  (`star_gate_certificate.py`); the collapse identities behind Proposition 102 are further verified at the 58 tabulated poles (`halfstep_verify.py`).  $\square$

**Proposition 102.** *At every travel pole, with  $Z = z_0/\sqrt{q}$  the corresponding zero of  $\cos(\cdot; q^2)$ ,*

$$P_{12} = -\frac{1}{2}[\cos(z_0; q^2) + \cos(z_0/q; q^2)].$$

*Since  $z_0 = Ze^{-\tau/2}$  and  $z_0/q = Ze^{+\tau/2}$ , and since  $\cos(Z; q^2) = 0$ , the right-hand side is exactly the symmetric second difference of  $\cos(\cdot; q^2)$  at its own zero, taken in the variable  $\log u$  with step  $\tau/2$ .*

*Proof.* Write  $c(u) = \cos(u; q^2)$  and  $\mathfrak{s}(u) = q^{-1/2} \sin(q^{1/2}u; q^2)$ . The shift rules and the pole invariant are Lemma 24:  $c(qu) = c(u) + q^{3/2}u \mathfrak{s}(\sqrt{q}u)$ ,  $\mathfrak{s}(qu) = \mathfrak{s}(u) - uc(\sqrt{q}u)$ , and  $q^{3/2}\mathfrak{s}(z_0)\mathfrak{s}(Z) = 1$  at every travel pole. Taking  $u = Z/\sqrt{q}$  in the first shift rule gives  $c(z_0) - c(z_0/q) = qZ \mathfrak{s}(Z)$ . Taking  $u = Z$  in the Hahn–Exton  $q$ -Wronskian  $\mathfrak{s}(u)c(qu) - c(u)\mathfrak{s}(qu) = u$  and in the first shift rule, and using  $c(Z) = 0$  twice, gives  $\mathfrak{s}(Z)c(qZ) = Z$  and  $c(qZ) = q^{3/2}Z \mathfrak{s}(z_0)$ , whence again the pole invariant and  $c(qZ) = Z/\mathfrak{s}(Z)$ . Substituting the closed form  $P_{12} = \frac{qZ}{2}\mathfrak{s}(Z) - c(z_0)$  of Lemma 101 into  $c(z_0) + c(z_0/q) = 2c(z_0) - qZ \mathfrak{s}(Z)$  yields the boxed identity.  $\square$

The two algebraic steps of the proof are machine-checked (`SelowerSkeleton.lean`): `shift_at_pole` is the instantiation of the half-step rule at  $u = z_0/q$ , written so that the index bookkeeping is explicit, and `P12_second_difference` is the elimination; both are stated in doubled form over an arbitrary commutative ring and depend only on `propext`. The identity is *verified* at all 58 tabulated poles to a worst relative deviation of  $2 \cdot 10^{-44}$  (`halfstep_verify.py`). The consequence, (37) above, is that the gate is a bound on a second derivative on a window of width  $\tau$ , not an asymptotic evaluation.

**Lemma 103** (The travel poles are locally finite). *For every  $\tau_\star > 0$  there are only finitely many travel poles with  $\tau_m > \tau_\star$ .*

*Proof.* By Proposition 17 the travel poles are the zeros in  $q \in (0, 1)$  of  $g(q) := \cos(Z; q^2) = C(Z^2)$  with  $Z^2 = 2(1 - q)/q$ , i.e. of

$$g(q) = \sum_{j \geq 0} (-1)^j \frac{2^j q^{j^2} (1 - q)^j}{(q; q)_{2j}}.$$

First,  $g$  has no zero near  $q = 0$ . For  $0 < q \leq \frac{1}{10}$  one has  $(q; q)_{2j} \geq \prod_{i \geq 1} (1 - q^i) \geq 1 - \frac{q}{1-q} \geq \frac{8}{9}$  and  $(1 - q)^j \leq 1$ , so  $|g(q) - 1| \leq \frac{9}{8} \sum_{j \geq 1} 2^j q^{j^2} \leq \frac{9}{8} (2q + 4q^4/(1 - q)) \leq 0.24 < 1$ , whence  $g(q) > 0$  there.

Second,  $g$  is real-analytic on  $(0, 1)$ : the series converges locally uniformly, since on  $q \leq 1 - \delta$  the  $j$ -th term is bounded by  $2^j q^{j^2}/(q; q)_\infty$  and  $q^{j^2}$  dominates  $2^j$ . Hence the zeros of  $g$  are isolated in  $(0, 1)$ , and any zero set with an accumulation point in the *open* interval would force  $g \equiv 0$ , contradicting  $g(\frac{1}{10}) > 0$ .

Now fix  $\tau_\star > 0$ . The poles with  $\tau_m > \tau_\star$  lie in  $q \in [\frac{1}{10}, e^{-\tau_\star}]$ , a compact subinterval of  $(0, 1)$ . An analytic function on a domain whose zeros are isolated has only finitely many zeros on any compact subset; hence there are finitely many such poles.  $\square$

*Remark 104.* Only local finiteness is proved here; that the poles *do* accumulate at  $q = 1$  (equivalently  $\tau_m \downarrow 0$ ) is Lemma 12. The present lemma says merely that they cannot accumulate anywhere inside  $(0, 1)$ , so that any threshold  $\tau_\star$  leaves a finite set to be checked directly. It is used four times: to make the splice in Proposition 94 well posed, to license the phrase “for all large  $m$ ” in Theorem 76, to make the exceptional set of Theorem 109 finite, and in Lemma 12’s accumulation clause (its proof also shows  $g > 0$  on  $(0, \frac{1}{10}]$ , so every travel pole has  $q_m > \frac{1}{10}$ ).

**Lemma 105.** *For  $0 < \rho < 1$ , with  $\theta \sim N(\pi/2, \tau/2)$ :  $\mathbb{E}[(\rho e^{i\theta}; q)_\infty^{-1}] = \sum_{M \geq 0} q^{M^2/4} (i\rho)^M / (q; q)_M =: \Phi(\rho)$ , and  $\operatorname{Re} \Phi(\rho) = \cos(\rho/\sqrt{q}; q^2)$ . The sum is the  $q$ -Airy function of Kajiwara–Masuda–Noumi–Ohta–Yamada [27] at base  $\sqrt{q}$ , and the representation is Zhang’s  $q$ -Laplace transform of the second kind at level  $\frac{1}{2}$  [58]; only the probabilistic packaging is ours.*

*Proof.* Euler's identity  $(x; q)_\infty^{-1} = \sum_M x^M / (q; q)_M$  integrated termwise (absolute uniform convergence on  $|x| = \rho < 1$ ), with  $\mathbb{E}[e^{iM\theta}] = i^M q^{M^2/4}$ , splitting  $M$  by parity and matching exponents  $((2j+1)^2 = 4(j^2 + j) + 1$ ; Lean `parity_exponent`) identifies  $\text{Re } \Phi$  with the cosine series of Proposition 17.  $\square$

**Lemma 106.** Write  $A(\xi) = (\rho e^{i(\pi/2+\xi)}; q)_\infty^{-1}$ . Then  $\log A(\xi) = \sum_{k \geq 1} (i\rho)^k e^{ik\xi} / (k(1-q^k))$ ; the cumulants  $\Lambda_n = i^n \sum_k k^{n-1} (i\rho)^k / (1-q^k)$  satisfy  $|\Lambda_n| \leq (n-1)! \rho / ((1-q)(1-\rho)^n)$ ; Gaussian integration by parts gives, at radius  $\rho(t) = z_0 e^t$ ,

$$\psi'(t) = \frac{2}{\tau} \text{Im } \mathbb{E}[\xi A], \quad \psi''(t) = \frac{4}{\tau^2} \text{Re } \mathbb{E}[(\frac{\tau}{2} - \xi^2) A];$$

and for complex  $a, b$  with  $\text{Re}(1 - b\sigma^2) > 0$ ,  $\mathbb{E}[\xi^m e^{a\xi + b\xi^2/2}] = \beta^{-1/2} e^{a^2\sigma^2/(2\beta)} M_m$ ,  $\beta = 1 - b\sigma^2$ ,  $M_0 = 1$ ,  $M_1 = \mu = a\sigma^2/\beta$ ,  $M_m = \mu M_{m-1} + (m-1)(\sigma^2/\beta) M_{m-2}$ .

*Proof.*  $-\log(1-u) = \sum u^m/m$  summed over  $u = \rho e^{i\theta} q^j$ ;  $1 - q^k \geq 1 - q$  and a binomial estimate;  $\rho \partial_\rho A = -i \partial_\theta A$  with  $\mathbb{E}[f'] = \mathbb{E}[f\xi/\sigma^2]$  once and twice; completion of the square and analytic continuation in  $(a, b)$ . The exponent bookkeeping is machine-checked (`QTrigIdentities.lean`).  $\square$

**Lemma 107.** At every travel pole with  $\tau \leq 5 \cdot 10^{-3}$  and every  $t \in [-\tau, h]$ :  $|\psi'(t)| \leq 1.10 w$ .

**Lemma 108.** At every travel pole with  $\tau \leq 5 \cdot 10^{-3}$ , writing  $\mathfrak{s}(u) = q^{-1/2} \sin(q^{1/2}u; q^2)$ :  $\cos(qZ; q^2) = Z/\mathfrak{s}(Z)$ , and  $|\mathfrak{s}(Z)| \geq Z/(\tau \max_{-\tau \leq t \leq 0} |\psi'(t)|) \geq 0.90$ . Moreover, for  $t \in [-\tau, 0]$ ,  $|\Phi(\rho(t))| \geq q^{3/4}(|\mathfrak{s}(Z)| - 0.49\tau)$ .

*Proof of Lemmas 107 and 108.* The identity is the pole reduction underlying Proposition 102 (the Hahn–Exton  $q$ -Wronskian at the zero, cited there). The bound in Lemma 108 is the mean value theorem on  $[-\tau, 0]$ , where  $qZ = Ze^{-\tau}$  exactly and all radii satisfy  $\rho(t) \leq z_0$ . For the last bound:  $\text{Im } \Phi(\rho) = q^{3/4} \mathfrak{s}(\rho/\sqrt{q})$  (the odd half of the parity split in Lemma 105), and the drift of  $\mathfrak{s}(\rho(t)/\sqrt{q})$  over  $t \in [-\tau, 0]$  is the even/odd  $G_4$  term of the certificate,  $\leq 0.49\tau$  (`budget.py`, `G4_coef`). For Lemma 107: the principal term is the  $m = 1$  tilted moment of Lemma 106 at  $a = \Lambda_1$ ,  $b = \Lambda_2$ , at most  $e^{c_1\sqrt{\tau}}|\beta|^{-1/2}L_1(\text{Re } \beta)^{-1} \cdot w \leq 1.088w$ , where  $L_1$  is the interval constant bounding  $|\Lambda_1|/w$  (Lemma 106,  $n = 1$ ) and  $c_1 \leq 0.177$  bounds the exponent residual per  $\sqrt{\tau}$  (the two leading halves of the exponent cancel exactly; every residual carries a geometric tail); the tail (linear orders 3..10 with  $\xi$ -weighted moments, the product convolution, and the degree- $\geq 11$  remainder) is at most 0.131 absolute, i.e.  $\leq 0.0066w$  at the top endpoint  $w = 20$ . All constants are produced by the committed certificate `code/zeta_probe/budget.py` (function `uniform_stack`), which works in 30-digit floating point with the interval constants realised as endpoint evaluations. That evaluation is *re-certified in directed-rounding interval arithmetic* by `budget_iv.py`: the same stack is evaluated with  $\tau$  entering as an *interval*, over 160 cells covering  $[1.2 \cdot 10^{-4}, 5 \cdot 10^{-3}]$  and 160 covering  $[10^{-8}, 1.2 \cdot 10^{-4}]$  (contiguous, with both outer endpoints clamped exactly), giving enclosures valid for *every*  $\tau$  in each cell; so on those ranges neither the floating-point arithmetic nor the monotonicity of any atom is assumed. The certified maxima are 3.349 and 2.650, against the budget 3.5. On the remaining range  $(0, 10^{-8}]$ , which no finite set of cells can cover, the power-counting argument below (every term a positive power of  $\tau$ ) is what proves the bound. The companion `star_gate_certificate.py` works in the floating-point model at 45–260 digits with explicit truncation control.  $\square$

**Theorem 109** (the amplitude estimate  $(\star)$ ). At every travel pole,  $|P_{12}(q_m)| w_m^3 < 2$ . Quantitatively: for  $\tau \leq 5 \cdot 10^{-3}$ ,  $\max_{|t| \leq h} |\psi''(t)| \leq 3.5 w$ , so (37) gives  $|P_{12}| \leq \frac{3.5\sqrt{2}}{8} \tau^{3/2} < 0.619 \tau^{3/2} < \tau^{3/2}/\sqrt{2}$ ; and the six poles with  $\tau > 5 \cdot 10^{-3}$ ,

$$\tau_1 = 0.79972, \quad \tau_2 = 9.049 \cdot 10^{-2}, \quad \tau_3 = 3.248 \cdot 10^{-2}, \quad \tau_4 = 1.656 \cdot 10^{-2}, \quad \tau_5 = 1.001 \cdot 10^{-2}, \quad \tau_6 = 6.701 \cdot 10^{-3},$$

satisfy  $|P_{12}|w^3 = 0.745, 0.530, 0.510, 0.505, 0.503, 0.502$  by direct evaluation at 120-digit precision with explicit truncation control (`star_gate_certificate.py`; the underlying identities are independently verified by `halfstep_verify.py` at the 58 tabulated poles, 47–54 digits on the five it shares with this list: the dominant pole is covered by `star_gate_certificate.py` alone). The first is the dominant pole  $q_1 = 1/\beta_2^2$  of Corollary 78. Enumeration of the exceptional set: every travel pole has  $q_m > \frac{1}{10}$  and the set is finite (Lemma 103); the count of six is established by a derivative-majorant subdivision for  $\tau \geq 0.03$  (three poles; in the arithmetic model disclosed above) and needs no enumeration for  $\tau \leq 5 \cdot 10^{-3}$ , where the uniform certificate covers every pole; on the window  $5 \cdot 10^{-3} < \tau < 0.03$  the three listed poles are located by a  $\Delta w \approx 5 \cdot 10^{-4}$  scan (they sit on the  $(k + \frac{1}{2})\pi$  ladder to three decimals, and  $|\cos(Z; q^2)|$  swings to 1.00 between them with minimum 0.019 outside the root cells). The scan is the weakest step of the enumeration and is stated as such; transcendence in Theorem 81 does not consume it.

*Proof.* By Lemma 106,  $\psi'' = \frac{4}{\tau^2} \operatorname{Re}[-e^{\Lambda_0} G_Q(\Lambda_2 \sigma^4 / \beta + \mu^2) + e^{\Lambda_0} \mathbb{E}[(\frac{\tau}{2} - \xi^2) e^Q (T - 1)]]$ , three terms, with  $Q = \Lambda_1 \xi + \Lambda_2 \xi^2 / 2$ ,  $G_Q = \mathbb{E}[e^Q] = \beta^{-1/2} e^{\Lambda_1^2 \sigma^2 / (2\beta)}$ ,  $\beta = 1 - \Lambda_2 \sigma^2$ ,  $\mu = \Lambda_1 \sigma^2 / \beta$  (Lemma 106 at  $m = 0, 1$ ), and  $T = e^\Omega$ ,  $\Omega = \sum_{n \geq 3} \Lambda_n \xi^n / n!$ , the exponential of the higher cumulants. The  $\mu^2$  term's phase is pinned by the pole:  $|\cos \arg \Phi(\rho(t))| = |\psi(t)| / |\Phi(\rho(t))| \leq \max |\psi'| |t| / |\Phi|$  with  $\psi(0) = 0$  (Lemma 107) and  $|\Phi| \geq q^{3/4} (|\mathfrak{s}(Z)| - 0.49\tau)$  (Lemma 108); its argument's sine obeys  $|\sin 2 \arg \mu| \leq 13.3 \tau^{3/2}$ , the two  $O(1/w)$  tilts of  $\Lambda_1$  and  $\beta$  cancelling to  $O(\tau)$  because they are the same Lambert data shifted by one index. Every remaining quantity is bounded by an interval constant in scaled variables, each resting on a one-line monotonicity ( $(1 - e^{-\tau})/\tau$  decreasing;  $e^{c\tau}$ ,  $\rho^2 = 2(1 - q)e^\tau$  increasing;  $w, q$  decreasing), with tails via  $1 - q^k \geq (1 - q)kq^{k-1}$  and the signed cancellations via the alternating-series bound  $\tau^2/6 - \tau^3/12 \leq 1 + q - 2(1 - q)/\tau \leq \tau^2/6$ . The degree- $\geq 11$  remainder is handled on two intervals: on  $[1.2 \cdot 10^{-4}, 5 \cdot 10^{-3}]$  by an exponential majorant whose profile  $e^{0.097/\sqrt{\tau} \tau^{4.5}}$  is increasing exactly there (threshold  $1.16 \cdot 10^{-4}$ ), total  $\leq 3.03w$ ; on  $(0, 1.2 \cdot 10^{-4}]$  with no exponential device (linear remainder geometric with ratio  $\frac{1}{2}$  on the cutoff region; all products by  $|e^\Omega - 1 - \Omega| \leq \frac{1}{2} |\Omega|^2 e^{|\Omega|}$  with  $|\Omega| \leq 0.45$  on  $|\xi| \leq t_0 \kappa^{-1/3}$ ; the intermediate zone suppressed by the Gaussian; here  $t_0 = 1 - \rho$  and  $\kappa = \rho/(1 - q)$ ), total  $\leq 2.10w$ . Both are endpoint evaluations of quantities proved monotone; on  $[10^{-8}, 5 \cdot 10^{-3}]$  the same conclusion is obtained without any monotonicity assumption by the interval-arithmetic certification `budget_iv.py` described above, and the monotonicity argument is what carries the remaining range  $(0, 10^{-8}]$ . The assembly is the committed script `budget.py`.  $\square$

**Lemma 110** (the denominator at the pole). *At every travel pole, exactly:  $S_e = \frac{qZ}{2} \mathfrak{s}(Z) - P_{12}$ . Consequently, for  $\tau \leq 5 \cdot 10^{-3}$ ,*

$$|S_e| \geq 0.90 \cdot \frac{qZ}{2} - |P_{12}| \geq \left[ \frac{0.90}{\sqrt{2}} \sqrt{qv} - 0.619\tau \right] \sqrt{\tau} \geq 0.63 \sqrt{\tau}, \quad v = \frac{1-q}{\tau},$$

and  $S_e$  has the sign of  $\mathfrak{s}(Z)$ .

*Proof.* The identity is Lemma 101 rearranged, since  $S_e = \cos(z_0; q^2)$  (Proposition 17). For the bound:  $|\mathfrak{s}(Z)| \geq 0.90$  (Lemma 108),  $\frac{qZ}{2} = \sqrt{q(1 - q)/2} = \sqrt{\tau/2} \sqrt{qv}$ ,  $|P_{12}| \leq 0.619 \tau^{3/2}$  (Theorem 109), and  $\sqrt{qv} \geq \sqrt{(1 - \tau)(1 - \tau/2)} \geq 1 - \frac{4}{5}\tau$  for  $\tau \leq \frac{1}{2}$  (since  $(1 - \frac{4}{5}\tau)^2 \leq (1 - \tau)(1 - \frac{\tau}{2})$  there); at  $\tau = 5 \cdot 10^{-3}$  the bracket is  $\geq 0.630$  (exact endpoint value 0.6309) and it increases as  $\tau$  decreases. The sign statement holds because  $0.90 \cdot \frac{qZ}{2} > |P_{12}|$  with the same margin.  $\square$

**Corollary 111** (the gate closes). *At every travel pole the gate (32) holds; with Theorem 72 this gives  $\Pi_1(q_m), \Pi_q(q_m) > 0$  for  $\tau_m \leq 5 \cdot 10^{-3}$  (the enumeration standing of the six-pole list is that of Theorem 109). The half  $|s| < 1$  is proved here outright; the half  $b_0 > 0$  consumes the closed form*

$b_0 = \frac{2q}{1-q} S_o/S_e$ , which is Corollary 41 and therefore rests on the transfer model (M), Theorem 66. Quantitatively, for  $\tau \leq 5 \cdot 10^{-3}$ :

$$|s| = \frac{q}{1-q} \frac{|P_{12}|}{|S_e|} \leq \frac{\tau}{e^\tau - 1} \cdot \frac{0.619}{0.63} \leq 0.983 < 1,$$

and  $b_0 > 0$ : with the closed form  $b_0 = \frac{2q}{1-q} S_o/S_e$  of Corollary 41 and  $S_o = \frac{z_0}{2} \mathfrak{s}(z_0)$  (Lemma 101), the pole invariant  $q^{3/2} \mathfrak{s}(z_0) \mathfrak{s}(Z) = 1$  (Proposition 102’s proof) gives  $\text{sign } S_o = \text{sign } \mathfrak{s}(Z) = \text{sign } S_e$  (Lemma 110), so  $S_o/S_e > 0$ . At the six poles with  $\tau > 5 \cdot 10^{-3}$  the gate quantities are evaluated directly (*star\_gate\_certificate.py*):  $|s| = 0.296, 0.256, 0.252, 0.251, 0.251, 0.250$  and  $b_0\tau = 2.174, 2.001, 2.000, 2.000, 2.000, 2.000$ ; moreover  $|S_e|/\sqrt{\tau} \geq 0.58$  at all six, so the bound  $0.35\sqrt{\tau}$  of Proposition 94 holds at the dominant pole, as its statement records.

## Machine verification of the $q$ -trigonometric layer

Every series identity of the  $q$ -trigonometric sections and of this appendix is proved by matching the  $j$ -th summand; the matching is exponent bookkeeping, which is where such arguments fail in practice. That bookkeeping, and the Casoratian constancy behind the half-step invariant, are machine-checked in Lean 4 (*QTrigIdentities.lean*; the last four rows of the table below live in *SelowerSkeleton.lean*); convergence and the interchange of summation are analytic and are not formalised.

| Paper step                                   | Lean declaration                   | axioms               |
|----------------------------------------------|------------------------------------|----------------------|
| half-step rule for $c$ , exponent match      | <code>cshift_exponent</code>       | <code>propext</code> |
| half-step rule for $s$ , exponent match      | <code>sshift_exponent</code>       | <code>propext</code> |
| parity split of $\Phi$                       | <code>parity_exponent</code>       | <code>propext</code> |
| functional equation of $F$                   | <code>qairy_exponent</code>        | <code>propext</code> |
| $P_{12}$ $k$ -sum split, low branch          | <code>P12_split_lo</code>          | <code>propext</code> |
| $P_{12}$ $k$ -sum split, high branch         | <code>P12_split_hi</code>          | <code>propext</code> |
| Casoratian constancy                         | <code>casoratian_re_const</code>   | standard             |
| constancy $\Rightarrow$ every rung           | <code>casoratian_re_eq_zero</code> | standard             |
| shift rule at the pole                       | <code>shift_at_pole</code>         | <code>propext</code> |
| $P_{12}$ as a second difference              | <code>P12_second_difference</code> | <code>propext</code> |
| scaling $\tau^2 w = \sqrt{2} \tau^{3/2}$     | <code>tau_sq_mul_w_sq</code>       | standard             |
| gate threshold $2/w^3 = \tau^{3/2}/\sqrt{2}$ | <code>gate_threshold</code>        | standard             |

The algebraic skeleton of Proposition 94 is likewise machine-checked (*SelowerSkeleton.lean*): `pole_facts` derives the two exact facts  $s(qZ) = s(Z)$  and  $c(qZ) = qZ s(Z)$  from the two shift rules together with  $c(Z) = 0$ ; `F_at_pole` collapses the invariant to  $q s(Z)^2$  at a pole; `form_lower_bound` is the quadratic-form estimate behind  $\lambda_n \geq q - Z/2$ ; and `prod_one_sub_ge` is the Weierstrass bound  $\prod(1 - \delta_i) \geq 1 - \sum \delta_i$  used in the discrete Gronwall step. The analytic inputs (the annulus representation, the Poisson envelope and the mean value theorem) are hypotheses there, not formalised.

“standard” means `propext`, `Classical.choice`, `Quot.sound`. Every file named in this paper lives in the Mathlib-based package of the accompanying development; none of its 62 files contains `sorry`, each ends with a `#print axioms` audit, and every declaration cited anywhere in this paper appears in one of those audits. (The audits are selective in the other direction: a file may prove auxiliary lemmas that its audit block does not list.) Some declarations report “does not depend on any axioms”, which is the legitimate output for a goal closed by `decide` or `rfl` (for instance `OverDetermination.linear_order_19_ok`) and is also, character for character, what an audit

prints for a declaration that failed to elaborate; the audits quoted here were therefore read from a cold elaboration of each file one at a time (`lake env lean`, exit status checked), not from a warm `lake build`, which does not replay the output for an up-to-date library. Three declarations cited in this paper are proved by `native_decide` and `carry`, besides the standard axioms, the compiler-trust axiom named at their point of use (Remark 6; §9). The realness of the ladder coefficient is what makes the added term in `casoratian_re_const` purely imaginary, and is carried as an explicit hypothesis.

## The boundary of the formalisation: the analytic layer

The combinatorial layer of this paper is formalised (§5.6), the analytic layer is not, and the obstruction is one and the same for every statement in it: Mathlib carries no  $q$ -Pochhammer symbol  $(a; q)_n$  or  $(a; q)_\infty$ , no product form of a theta function, no Bessel or  $q$ -Bessel function, no  $q$ -difference equations, and no steepest-descent, stationary-phase or van der Corput machinery. It does carry the two-variable Jacobi theta series with its summability, analyticity and modular functional equation, and the Mellin transform; it does not carry the Jacobi triple product, which is the identity that connects the two. The list below records, for each analytic statement of this paper, the object whose absence blocks a formalisation. The statements are at PROVED, not at VERIFIED, and this table is the reason.

| Statement            | missing prerequisite                                                  |
|----------------------|-----------------------------------------------------------------------|
| Lemma 96             | $(x; q)_\infty$ and its logarithm                                     |
| Lemma 101            | $(q; q)_n$ ; the $q$ -cosine and $q$ -sine                            |
| Proposition 102      | $(q; q)_n$ ; the $q$ -cosine                                          |
| Lemma 110            | $(q; q)_\infty$ ; the $q$ -sine                                       |
| Lemma 95             | the Jacobi triple product; $(q^2; q^2)_\infty$                        |
| Lemma 103            | analyticity of a $q$ -series and its zero set                         |
| Lemma 12             | the same, plus a limit at $q \rightarrow 1$                           |
| Lemma 13             | the same                                                              |
| Proposition 94       | $(q; q)_\infty$ ; quantitative $q$ -trigonometric bounds              |
| Lemma 97             | a Gaussian majorant for a theta series                                |
| Lemma 98             | second derivatives of a $q$ -series                                   |
| Lemma 107            | derivative bounds for the $q$ -sine                                   |
| Lemma 108            | the $q$ -sine and $q$ -cosine at a zero                               |
| Theorem 109          | a quantitative Taylor bound on a $q$ -phase                           |
| Lemma 105            | the $q$ -Airy function; a Gaussian expectation                        |
| Lemma 106            | cumulant bounds; Gaussian integration by parts                        |
| Lemma 7              | $\log \Gamma$ with an explicit error term; $\log(q; q)_\infty$        |
| Lemma 10             | Mellin–Barnes contour shifting; a verified contour-integral enclosure |
| Lemma 79, confluence | Bessel $J_\nu$ at half-integer order                                  |

The first five rows would be reached by a  $q$ -Pochhammer layer alone, together with the Jacobi triple product; the next four would be reached by that layer together with Mathlib’s existing complex analysis; the remainder need asymptotic machinery that is a larger absence than the  $q$ -series one. The half of Lemma 79 that is an explicit rational inequality *is* verified (`AtomN.lean`); the confluence half is the row above (Remark 80).

## C The number $\beta_2$ : value transcendence

This appendix concerns the *number*  $\beta_2 = 1/\sqrt{q^*}$  of Corollary 78, not the function  $V$ . Nothing in §§3–10 or in Appendices A–B depends on any statement made here; the material is recorded because

it locates the obstruction to value transcendence precisely, at the arithmetic ratio  $\rho = \frac{1}{2}$ , one notch beyond the proven  $\rho \geq 1$  theory.

*Remark 112* (the number  $\beta_2$ : a value-transcendence question). Whether the number  $\beta_2$  is transcendental is open, but the question has a clean reduction. Since  $\beta_2 = 1/\sqrt{q^*}$  with  $\Sigma_1^T(q^*) = 1$ , the number  $\beta_2$  is algebraic iff  $q^*$  is; if  $q^*$  were algebraic, the series  $\Sigma_1^T$  would take the *rational* value 1 at an algebraic interior point of its disc. For the Jacobi-theta class this is impossible: by Nesterenko's theorem on Eisenstein values [38],  $\sum_n \alpha^{n^2}$  is transcendental at *every* algebraic  $\alpha$  with  $0 < |\alpha| < 1$  [18]. The blocks here are theta-like; their coefficients ride the squares  $q^{(j+1)^2}$ , which is the source of the natural boundary of Theorem 2; but their Euler-product denominators  $(q; q)_{2k+1}$  spread the support densely, and no analogue of [38] is known off the modular class.

Concretely the reduction localizes on a single confluent basic hypergeometric function. Writing  $\Sigma_1^T(q) = 1 - S(q)$ ,

$$S(q) = \sum_{k \geq 0} \frac{(-2(1-q))^k q^{k^2}}{(q; q)_{2k}} = {}_1\phi_1(0; q; q^2, 2q(1-q)),$$

a  $q$ -Bessel-class function (using  $(q; q)_{2k} = (q; q^2)_k (q^2; q^2)_k$ );  $\beta_2$  is algebraic iff the least positive zero  $q^* = 0.449453630\dots$  of  $S$  is. Certified exclusions at 430 digits (residual  $10^{-442}$ ; PSLQ with post-verification of every candidate): neither  $q^*$  nor  $\beta_2$  satisfies an integer polynomial of degree  $\leq 8$  with height  $10^{20}$ , degree  $\leq 12$  with height  $10^{18}$ , or degree  $\leq 16$  with height  $10^{14}$ ; no bilinear relation ties  $q^*$  to the fold point  $q_c$  of the zero curve (height  $10^{12}$ ). The continued fraction of  $q^*$  (383 certified partial quotients, two-precision guard) is Gauss–Kuzmin generic: digit frequencies match, Khinchin mean 2.63 (2.685), Lévy exponent 3.21 (3.276), largest quotient  $a_{170} = 8250$  (probability  $\approx 6\%$  over 383 draws, consistent with chance). The number offers no arithmetic foothold of any tested kind. We record what the natural tools give here, since it locates the difficulty precisely. First,  $S$  is *neither*  $q$ -holonomic *nor* a Mahler function in  $q$  (verified to  $O(q^{48})$ ): the diagonal specialization  $z = 2q(1-q)$  ties the argument to the base and destroys the single-variable  $q$ -difference structure, so the methods of Mahler–Nishioka [39] and of holonomy do not apply. Second, two elementary refinements fail *decisively*, not for want of effort: an ultrametric argument is void; the  $v$ -adic and archimedean sums of  $S(\alpha)$  are unrelated numbers, so  $|S(\alpha)|_v = 1$  does not bound the real value; and a Liouville/height bound is beaten by the  $q$ -Pochhammer denominators, whose growth over the conjugates of modulus  $> 1$  is cubic ( $\sim K^3$ ) against the merely quadratic ( $\sim K^2$ ) tail, Kronecker-sharply (equality would need Mahler measure 1, i.e. a root of unity). A more promising reformulation uses two variables. Set  $G(q, z) = {}_1\phi_1(0; q; q^2, z) = \sum_k (-1)^k q^{k(k-1)} z^k / (q; q)_{2k}$ , so  $S(q) = G(q, 2q(1-q))$ ; unlike  $S$ , the function  $G$  is  $q$ -holonomic in  $z$ , satisfying the second-order  $q$ -difference equation

$$q G(q, z) - (q + 1 - qz) G(q, q^2 z) + G(q, q^4 z) = 0.$$

One checks that  $z^* := 2q^*(1 - q^*) = 0.4948901\dots$  is the *smallest positive zero* of  $G(q^*, \cdot)$ . Hence  $\beta_2$  algebraic would force an *algebraic zero of a  $q$ -Bessel-class function at an algebraic base*; and the classical analogue is false: the zeros of the Bessel function  $J_\nu$  are transcendental (Siegel [50]). A  $q$ -analogue of Siegel's theorem for the zeros of  ${}_1\phi_1$  would therefore settle  $\beta_2$ . The right framework is arithmetic. Writing the equation as a system  $Y(q^2 z) = A(z)Y(z)$  with  $Y = (G(z), G(q^2 z))^T$ , the companion matrix

$$A(z) = \begin{pmatrix} 0 & 1 \\ -q & q + 1 - qz \end{pmatrix}$$

is *polynomial* in  $z$  with  $\mathbb{Z}[q]$  entries and  $\det A = q$ , so the iterated matrices carry no denominators and the Casoratian obeys  $W(q^2 z) = qW(z)$ , i.e.  $W(z) = cz^{1/2}$ : the two solutions form a  $q$ -Bessel pair with Frobenius exponents  $0, \frac{1}{2}$ . The second solution is explicit,  $\tilde{G}(z) = z^{1/2}H(z)$  with

$H(z) = \sum_k (-1)^k q^{k^2} (1 - q) z^k / (q; q)_{2k+1}$ , and the pair satisfies the exact  $q$ -Wronskian identity

$$q G(z) H(q^2 z) - G(q^2 z) H(z) = q - 1,$$

so at the zero  $z^*$  one has the clean relation  $G(q^2 z^*) H(z^*) = 1 - q$ . Moreover  $G(q, \cdot)$  is *entire of order zero*; a  $q$ - $E$ -function, not a  $q$ - $G$ -function. Hence  $\beta_2$  transcendental would follow from a  $q$ -analogue of the Siegel–Shidlovskii theorem for this  $q$ - $E$ -function: such a theorem gives  $G(q, \alpha)$  transcendental at every algebraic  $\alpha \neq 0$  off the singular locus, whereas  $G(q^*, z^*) = 0$  is algebraic; forcing  $q^*$ , and with it  $\beta_2$ , transcendental (and by the shared  $q$ -connection core, likewise  $U$ ). All the structural hypotheses hold here (polynomial companion matrix, constant determinant, entire order-zero solution); the sole open point is whether the available  $q$ -Siegel–Shidlovskii results (André [2]; Di Vizio–Hardouin [15]; and the Siegel-method theory of [1]) extend to a  $1\phi_1$  that is *irregular* at  $z = \infty$ ; a precise, expert-checkable question, not an absence of theory. (Note the contrast with the *function-level* results of this paper, which say nothing about individual interior values.)

For *irrationality* (weaker than transcendence) the classical architecture transports verbatim.  $G$  and  $H$  are Hahn–Exton (third Jackson)  $q$ -Bessel functions:  $G(q, z) \propto J_{-1/2}(\sqrt{z}/q; q^2)$  and  $H(q, z) \propto J_{+1/2}(\sqrt{z}/q; q^2)$  (verified; the zeros correspond exactly), i.e. the  $q$ -analogues of  $\cos$  and  $\sin$ ; and  $z^*$  is the  $q$ -analogue of  $(\pi/2)^2$ . Niven’s irrationality proof for  $\pi$  runs on the Lommel-polynomial ladder  $J_{m+1/2}(r) = R_{m,1/2}(r) J_{1/2}(r) - R_{m-1,3/2}(r) J_{-1/2}(r)$  evaluated at the zero of  $J_{-1/2} = \sqrt{2/\pi r} \cos$ ; the exact  $q$ -analogue of every ingredient exists (Koelink–Swarttouw [29]): the ladder  $J_{\nu+1}(x; q^2) = ((1 - q^{2\nu})/x + x) J_\nu - J_{\nu-1}$ , the  $q$ -Lommel connection  $J_{\nu+m} = R_{m,\nu} J_\nu - R_{m-1,\nu+1} J_{\nu-1}$  with  $x^m R_{m,\nu} \in \mathbb{Z}[q, x^2]$ , and the second-kind functions carrying  $q^{m^2/2}$  prefactors; all verified numerically for our pair. Moreover, since a rational zero would make  $G(q^*, z^*) = 0$  a *rational value* at rational base and argument,  $\beta_2$  is irrational as soon as this Hahn–Exton function omits the value 0 at rational points: a plain *value-irrationality* statement, strictly weaker than every transcendence target above. What blocks even this is now a single quantified invariant: the *arithmetic ratio*  $\rho$  of the series ( $b$ -adic decay exponent of the terms over  $b$ -adic growth exponent of the cleared denominators,  $q = s/t$  in lowest terms). Every proven  $q$ -irrationality theorem (Tschakaloff,  $q$ -exponential, the Bézivin–Bundschuh–Väänänen classes) lives at  $\rho \geq 1$ , where the Pochhammer denominator itself supplies the decay; our  $G$  has an independent  $q^{k^2}$  numerator against the double Pochhammer  $(q; q)_{2k}$ , giving  $\rho = \frac{1}{2}$ . All three classical constructions fail by exactly this margin, measured: the  $z$ -Padé remainder  $q^{3.5n^2}$  against height  $q$ -degree  $\approx 9n^2$ ; the  $z$ -direction continued fraction (convergents in  $\mathbb{Z}[q, z]$ , Pincherle limit the  $H$ -ratio) converges only geometrically against  $t^{n^2}$  heights; and the  $\nu$ -ladder at the zero has  $J_{m+1/2}(x_0) = R_{m,1/2}(x_0) J_{1/2}(x_0) \sim C x_0^m$ ; geometric, because the  $q$ -factorial is geometric where Niven’s  $m!$  is super-exponential; against  $t^{m^2}$ . The  $\nu$ -ladder is in fact fully costed: the  $q$ -Lommel coefficients satisfy  $\deg_q p_m = m^2$  *exactly* (immediate from the recurrence, verified symbolically), so their denominators at  $q = s/t$  are the pure power  $t^{m^2}$ : carrying *no* cyclotomic structure, so the lcm-compression that powers the little- $q$ -Legendre proofs has nothing to act on; while by Poincaré–Perron [45, 44] the  $m$ -recurrence, whose coefficient tends to  $x_0 + 1/x_0$  with *distinct* characteristic roots  $x_0, 1/x_0$ , admits *only geometric* solutions: no theta-type  $q^{m^2}$ -decaying solution exists in the  $\nu$ -direction for either kind (the second kind’s  $q^{\nu^2/2}$  prefactor is exactly cancelled by the  $(q^{-\nu+1}; q)_\infty$  blow-up of its shifted argument). Hence the cleared  $\nu$ -ladder forms grow like  $t^{m^2} x_0^{-m} \rightarrow \infty$  for *every* rational  $q = s/t$ , including  $s = 1$ . The  $b \leftrightarrow z$  transformation  $1\phi_1(0; q; q^2, z) = \frac{(z; q^2)_\infty}{(q; q^2)_\infty} 1\phi_1(0; z; q^2, q)$  redistributes but conserves  $\rho$ . The structural summary: the theta-type decay lives only in the  $z$ -direction, where the equation is *irregular* at  $z = \infty$  (Newton polygon slopes  $\{0, 1\}$ ) and every integral structure carries the double-Pochhammer weight  $\rho = \frac{1}{2}$ ; the  $\nu$ -direction is *regular* (Poincaré class) with clean  $\mathbb{Z}[q, X]$  arithmetic but only geometric decay.

Decay and arithmetic live in different directions and never meet; the same irregularity that obstructs  $q$ -Siegel–Shidlovskii creates the arithmetic deficit. The precise open door remains a  $q$ -irrationality criterion at  $\rho = \frac{1}{2}$ ; one notch beyond the proven  $\rho \geq 1$  theory.

*Remark 113* (the  $\beta_2$  family).  $q^*$  is the first member of an infinite family. On  $(0, 1)$  the parabola  $w(q) = 2q(1 - q)$  falls like  $1 - q$  while every zero branch falls like  $(1 - q)^2$  (the confluent limit): the branches are  $z_j(q) = 2q(1 - q)\lambda_j(q)$  with  $\lambda_j$  the eigenvalue branches of Theorem 19, and  $w = z_j$  means  $\lambda_j = 1$ . So  $w$  overtakes each branch at least once, distinct branches at distinct points, and  $S$  has infinitely many zeros  $q_1^* < q_2^* < \dots \rightarrow 1$ , which are the travel poles  $q_j$  (Proposition 22). That  $w$  overtakes each branch *exactly* once, so that the  $j$ -th zero is the crossing of the  $j$ -th branch,  $w(q_j^*) = z_j(q_j^*)$ , is Proposition 29(i), at the standing of Hypothesis 28 (verified at  $j = 2$  to 43 digits: exactly one zero of  $G(q_2^*, \cdot)$  lies below  $w$ ). The accumulation obeys the classical cosine-zero law

$$1 - q_j^* \sim \frac{8}{\pi^2(2j - 1)^2}, \quad \text{measured } (1 - q_j^*) \frac{(2j - 1)^2 \pi^2}{8} = 0.679, 0.961, 0.986, 0.993, 0.9956, 0.9970, 0.9979, 0.998$$

The second member is  $q_2^* = 0.913486638731047141696\dots$ , i.e.  $\beta_2^{(2)} := 1/\sqrt{q_2^*} = 1.046282352687417855\dots$ ; neither satisfies an integer polynomial of degree  $\leq 8$  with height  $10^{10}$ , and no bilinear relation ties  $q_1^*$  to  $q_2^*$  (PSLQ, 100 digits). The family interpolates between the open head ( $j = 1$ , the wild  $\beta_2$ ) and the classical tail (the  $\pi^2$ -law): the roots become “provably transcendental in the limit” while the first, the one of combinatorial interest, remains untouched; the confluent boundary of the step-ratio mechanism made visible inside a single object. (That the higher roots are genuine singularities of  $U$  itself is the non-vanishing  $\Pi_q(q_m) \neq 0$ , which is Theorem 72.)

Three refinements (all computations at 115 digits, 30 cosine and 10 sine members). First, the sine side has its own family,  $H(q_j^s, 2q(1 - q)) = 0$  with  $1 - q_j^s \sim 2/(j\pi)^2$  (law ratios  $0.913, \dots, 0.9991$  for  $j \leq 10$ ), and the two families *interlace*:  $q_j^* < q_j^s < q_{j+1}^*$  (verified  $j \leq 9$ ); its head is  $q_1^s = 0.815032640290\dots$ . Since  $H(q, 2q(1 - q)) = (1 - q)\Sigma_0(q)/(2q)$ , the sine family is exactly the zero set of  $\Sigma_0$  in  $(0, 1)$ , that is, of  $q \mapsto J_{1/2}^{(3)}(z_0(q); q^2)$  with  $z_0 = \sqrt{2(1 - q)}$  (Lemma 23 and the normalisation after Proposition 17). Half of the interlacing, at least one sine member in each  $(q_j^*, q_{j+1}^*)$ , is Proposition 30(b), at the standing of Hypothesis 28; that there is exactly one is verified only. Second, the accumulation law has an exact-looking rational second order, identical for both families:

$$1 - q_j^* = \frac{2}{\mu_j} \left( 1 - \frac{8}{9\mu_j} + \frac{B}{\mu_j^2} + O(\mu_j^{-3}) \right), \quad \mu_j = \left( \frac{(2j-1)\pi}{2} \right)^2 \text{ resp. } (j\pi)^2,$$

with  $A = -8/9$  matched to 12 digits (fit residual  $3 \cdot 10^{-18}$  out of sample) and  $B$  derived below. The constant  $A$  is derived, and the derivation separates two regimes that must not be conflated. Writing  $\varepsilon = 1 - q$ , the logarithm of the ratio of the  $k$ -th term of  $G(1 - \varepsilon, \varepsilon^2 y)$  to  $(-1)^k y^k / (2k)!$  equals  $\frac{k}{2}\varepsilon + c_2(k)\varepsilon^2 + O(\varepsilon^3)$  with

$$c_2(k) = -\frac{k^3}{9} - \frac{k^2}{12} + \frac{23k}{72};$$

to first order  $G$  is  $\cos\sqrt{y}e^{\varepsilon/2}$ , so at *fixed*  $j$  the zero branch obeys  $z_j = \varepsilon^2 \mu_j (1 - \frac{1}{2}\varepsilon + O(\varepsilon^2))$ : the fixed- $j$  confluent correction is  $-\frac{1}{2}$  (measured  $-0.5004$  at  $\varepsilon = 0.002$ , converging linearly). The crossing, however, sits in the *uniform* regime  $\mu\varepsilon \approx 2$ , where the  $k^3$ -term acts as the operator  $-\frac{\varepsilon^2}{9}D^3$ ,  $D = (u/2)d/du$  on  $\cos u$ ,  $u = \sqrt{X}$ ; at the zeros  $D^3 \cos u = \frac{u^3}{8} \sin u + O(u^2)$ , while even powers of  $D$  are cos-weighted and drop out (parity protects the order). With  $m = \varepsilon\mu$  the crossing equation  $2(1 - \varepsilon)e^{\varepsilon/2} = \varepsilon\mu(1 - \varepsilon^2\mu/36)$  becomes  $m = 2 - \varepsilon + \varepsilon m^2/36 + O(\varepsilon^2)$ , whence  $m = 2 - \frac{8}{9}\varepsilon$ :

$$A = -1 + \frac{1}{9} = -\frac{8}{9},$$

identically for the sine family (its  $\varepsilon^2$ -coefficient carries the same leading  $-k^3/9$ ). The two regimes carry different constants ( $-\frac{1}{2}$  versus the crossing's  $-\frac{8}{9}$ ). One more order of the same calculus derives  $B$  exactly. The ingredients: the exact prefactor  $2(1-\varepsilon)e^{\varepsilon/2} = 2 - \varepsilon - \frac{3}{4}\varepsilon^2 - \dots$ ; the cubic coefficient  $c_3(k) = -\frac{k^3}{9} - \frac{k^2}{12} + \frac{17k}{72}$  (the  $m^3$ -terms cancel in its defining sum, so  $c_3$  shares the leading  $-k^3/9$ ); the quartic  $c_4(k) = \frac{k^5}{450} + O(k^4)$  together with  $\frac{1}{2}c_2(D)^2 = \frac{D^6}{162} + \frac{D^5}{108} + \dots$ ; and one subtlety: at order  $\varepsilon^6 u^9$  the cube  $\frac{1}{6}(c_2(D)\varepsilon^2)^3$  contributes  $+\frac{u^9}{2239488} \sin u$  through the leading term of  $D^9$ , cancelling the  $\delta^3/6$  of the cosine expansion *identically*; truncating the exponential before the cube gives a spurious  $m^5$ -term. With that cancellation the crossing equation closes as  $m = 2 - \frac{8}{9}\varepsilon - \frac{1891}{8100}\varepsilon^2 + O(\varepsilon^3)$ , and inversion gives

$$B = 2m_2 + \frac{64}{81} = \frac{1309}{4050} = 0.3232098765432\dots,$$

confirmed by a five-order free fit to the computed roots:  $B = 0.3232098765429\dots$  agrees to 13 digits (with  $A = -8/9$  simultaneously reproduced to 17), and the  $(A, B)$ -constrained fit has out-of-sample residuals below  $4 \cdot 10^{-16}$ . The accumulation law of the  $\beta_2$  family therefore begins

$$1 - q_j^* = \frac{2}{\mu_j} \left( 1 - \frac{8}{9\mu_j} + \frac{1309}{4050\mu_j^2} + O(\mu_j^{-3}) \right),$$

with both corrections rational and derived. Three closing observations. (a) *Universality*: the sine family obeys the *same* law; a constrained fit to 22 sine members gives  $B_{\text{sin}} \cdot 4050 = 1309.000000$  to the fit precision  $2 \cdot 10^{-8}$ ; so through second order the two interlaced families share one accumulation series, with only  $\mu_j$  distinguishing them. (b) Every ingredient of the crossing calculus is rational (the  $c_n(k)$  are rational polynomials;  $D^n \cos u$  has odd  $u$ -powers on sin and even on cos, an invariant of  $D = (u/2)d/du$  preserved by the one-line induction; the  $\delta$ -elimination therefore stays in odd powers and the  $m$ -equation in rational coefficients), so we conjecture the expansion is rational to all orders. The third constant resists identification:  $C = -0.4745357072870\dots$  is stable to  $10^{-14}$  across model orders and members through  $j = 55$ , excludes every denominator up to  $\sim 10^6$  (PSLQ), and the fitted higher coefficients grow ( $F \approx 4.3$ ), marking the series as asymptotic-divergent; which caps the extractable precision and leaves  $C$ 's (conjecturally rational, large-denominator) value to the third-order calculus. (c) At  $j = 1$  the same series becomes a rational-coefficient expansion in  $4/\pi^2$  evaluated at the wild head: divergent, hence a resummation question. If a Borel-type summation of the rational series could be shown to reproduce  $q^*$ , the arithmetic of  $\pi^2$  would enter the problem for the first time through the family; we record this as a direction and have tested its carrying capacity twice. A Borel–Padé resummation of the seven known coefficients reproduces the four accessible samples (both family heads and both second members) to residuals  $7.9 \cdot 10^{-3}$ ,  $1.7 \cdot 10^{-6}$ ,  $4.6 \cdot 10^{-9}$ ,  $7.4 \cdot 10^{-11}$  at  $\mu = 2.47, 9.87, 22.2, 39.5$ . An attempted transseries extraction from these residuals then *refuted its own premise*: the log-slopes between consecutive samples are 1.14, 0.48, 0.24, matching the truncation signature  $7/\mu = 1.17, 0.47, 0.24$  exactly (three points), not an exponential  $e^{-\beta\mu}$ ; the residuals are the  $x^7$ -tail of the divergent series, not a nonperturbative signal. At the head this is the standard resurgence degeneracy: seven coefficients place the evaluation at optimal truncation, where truncation floor and first transseries term coincide in scale ( $e^{-t_0\mu_1} \approx 0.01$ , with  $t_0 \approx 0.75 \pm 1.70i$  the Padé–Borel singularities), so the head residual  $-0.008(4)$  *cannot* be attributed to a nonperturbative sector on present data. That cost has now been paid: the crossing calculus has been automated in exact rational arithmetic (weight-graded formal Newton for the zero shift; parity and residual controls exact), producing the law's coefficients through order thirty; *all rational*, e.g.  $A_3 = -1695089/3572100$  (whose denominator  $2^2 3^6 5^2 7^2$  explains the failed PSLQ) and  $A_4 = 10199641/535815000$ . The coefficients are computed by running the calculus over  $\mathbb{F}_p$  and recovering each  $A_n$  by rational reconstruction from twenty 62-bit primes; the values are independently

confirmed by extracting  $A_n$  from the accumulation law applied to the roots themselves, computed to 105 digits.

The Borel geometry is only partly determined by these coefficients. The signs of  $A_n$  follow a three-positive, three-negative pattern from  $n \approx 12$ , indicating a conjugate pair of singularities near  $60^\circ$ ; the pattern is not exact over the full range. This period-6 oscillation dominates the naive growth diagnostics, and must be removed before the Gevrey order can be read off. Setting

$$G_n := \frac{1}{6}(\log |A_n| - \log |A_{n-6}|),$$

which is phase-matched and therefore free of the oscillation, the model  $|A_n| \sim C^n (n!)^s$  gives  $G_n \sim \log C + s \log n$ , so  $s$  is the slope of  $G_n$  against  $\log n$ . Least squares yields  $s = 0.844$  on  $16 \leq n \leq 30$ ,  $0.897$  on  $23 \leq n \leq 30$  and  $1.033$  on  $26 \leq n \leq 30$ : the estimate rises towards 1 as the fitting window advances, and the data are consistent with Gevrey-1. (The unnormalised ratio  $\log |A_n|/(n \log n)$ , which equals  $s + (\log C - s)/\log n$ , is  $0.313$  at  $n = 14$  and  $0.411$  at  $n = 30$ ; its corrections are of order  $1/\log n$  and it is therefore useless for extrapolation at these lengths.)

For the singularity modulus three independent estimates are available and agree only in order of magnitude: the phase-matched growth constant gives  $|t_0| = 1/C = 3.59\text{--}3.98$  over  $24 \leq n \leq 30$ ; a three-term recurrence fit, appropriate to a conjugate pair, gives  $|t_0| = 3.285, 3.394, 3.493, 3.558$  at  $n_0 = 22, 25, 28, 30$ , extrapolating in  $1/n_0$  to  $4.30$ , with  $\arg t_0 = 60.8^\circ, 61.7^\circ, 62.4^\circ, 62.3^\circ$ ; and the Padé–Borel approximants give  $|t| = 2.61, 2.56, 2.96, 3.09, 3.33$  for  $[6/6]$  through  $[14/14]$ . Thus  $|t_0|$  lies between 2.9 and 4.3 and is not pinned. Gevrey-1 growth, and with it the  $n!$ -Borel framework, is supported but not established; the location of the singularities is not established; and the Borel representation below remains a conjecture whose analytic premise is open. One structural constraint is available: by the weight lemma, only Bernoulli numbers of index  $\lesssim n$  reach  $A_n$  (a  $B_j$  occurring in  $c_m$  sits at  $k$ -degree  $\deg c_m - j$  and reaches  $A_n$  only if  $j \leq 2n - m + 1$ ), which excludes the Gevrey-2 growth that  $B_{2n}$  would otherwise force, but does not pin the class. Where testable the resummation succeeds: at the sine head the Padé–Borel ladder converges  $1.7 \cdot 10^{-6} \rightarrow 3.7 \cdot 10^{-11}$  (any transseries prefactor  $\lesssim 10^{-6}$  unless phase-suppressed), while at the cosine head the plain ladder reaches its approximation wall at  $\sim 10^{-3}$  with residuals oscillating about zero. We therefore pose the *Borel representation conjecture*:  $1 - q_j^* = (2/\mu_j) \mathcal{S}(1/\mu_j)$  for both families and all  $j$ , where  $\mathcal{S}$  is the Borel–Laplace sum of the explicit rational series  $\sum A_n x^n$ ; in particular

$$q^* = 1 - \frac{8}{\pi^2} \mathcal{S}\left(\frac{4}{\pi^2}\right),$$

the first exact (conjectural) bridge between  $q^*$  and  $\pi^2$ -arithmetic. Numerically the head representation is confirmed to  $\sim 10^{-4}$  by conformal-map Borel resummation (the residual crosses zero smoothly with no plateau above the estimator scatter), and the *same* resummation machinery reproduces the three lower- $x$  heads to  $10^{-13}$ – $10^{-26}$ ; the coefficients are now computed through  $A_{22}$ , still rational (e.g.  $A_{21}, A_{22}$  have denominators  $\sim 10^{93}$ ), consistent with the rationality of the expansion at every order.

Two honest boundaries frame the bridge. *Analytically*, its proof is *not* a cite-the-theorems exercise: it reduces to a single hard lemma; a  $q$ -uniform, turning-point (Olver/Airy-grade) factorial remainder bound  $|A_n| \leq C \sigma^n n!$  for the Hahn–Exton crossing as  $q \rightarrow 1$ ; which has no citable form in the literature and is plausibly the program’s recurring degenerate-saddle obstruction in Borel-summability costume. *Arithmetically*, the Borel transform  $b(t) = \sum (A_n/n!)t^n$  carries a sharp, exceptionless prime law: for every odd prime  $p \leq 43$ ,  $p \mid \text{den}(b_n)$  exactly when  $2n + 1 \geq p$ , with  $p$ -adic valuation exactly 2 at the onset  $n = (p - 1)/2$ , and thereafter non-vanishing with valuation growing linearly *on average* (least-squares slopes  $2.38, 1.17, 0.73, \dots$  for  $p = 3, 5, 7$ ; the

valuation is non-monotone locally, e.g.  $v_5$  dips  $6 \rightarrow 5$  and  $v_{17}$  dips  $2 \rightarrow 1$ ). The clean-before-onset half is a theorem for all  $p$ . (i) The polynomials  $c_n(k)$  built from the calculus are  $p$ -integral for  $n \leq p-2$  (von Staudt–Clausen [25] at the Faulhaber layer). (ii)  $A_n$  depends only on the calculus data through order  $2n$ ; this is now proved, in two steps. First, by series reversion  $A_n$  is a function of  $m_1, \dots, m_n$  alone (the order- $n$  inverse coefficient uses only the first  $n$  forward coefficients; verified by perturbing  $m_{>n}$ ). Second,  $m_i$  depends only on  $c_1, \dots, c_{2i}$ : in the turning-point grading, where  $k \sim \varepsilon^{-1/2}$  assigns  $\varepsilon^a u^b$  the weight  $2a - b$  (physical order = weight/2) and every operation is additive on weights, the polynomial  $c_m$  enters at minimum weight  $2m - \deg_k c_m$ , attained by its top-degree term. Here  $\deg_k c_m = m + [m \text{ even}]$ : the degree- $m$  coefficient of the  $\varepsilon^m$  log-ratio  $\ell_m(i)$  equals  $(-1)^m [x^m] \log((e^x - 1)/x)$ , which vanishes for every odd  $m \geq 3$  because  $\log((e^x - 1)/x) = \frac{x}{2} + \log(2 \sinh(x/2)/x)$  and the last term is *even* in  $x$ ; the Faulhaber sum then raises the  $k$ -degree by one. Hence  $c_m$  has minimum weight  $m - [m \text{ even}]$  and first reaches physical order  $i$  only when  $m \leq 2i$ , so  $m_i$  (and therefore  $A_n$ ) sees no  $c_m$  with  $m > 2n$ . (iii) The assembly introduces no prime  $p$  at output order  $n < p$ : the Bernoulli denominators of the Faulhaber sums acquire  $p$  only when  $(p-1) \mid 2m$ , i.e.  $2m \geq p-1$ , and all factorial and division indices stay below  $p$ . For  $n < (p-1)/2$  one has  $2n \leq p-3$ , so every ingredient feeding  $A_n$  is  $p$ -integral and  $p \nmid \text{den}(A_n)$ .

The matching lower direction; that  $p$  *does* appear at  $n = (p-1)/2$ , with valuation exactly 2; reduces to a unit. At  $n = (p-1)/2$  the sole prime- $p$  source is  $c_{p-1}$  (all lower  $c$  are  $p$ -integral), and it enters  $A_n$  *linearly*, since  $c_{p-1}^2$  has weight  $2(p-2) > 2n$ . Only the top three coefficients of  $c_{p-1}$  (degrees  $\geq p-2$ ) survive the weight cutoff; of the two that carry  $p^2$ ; the  $k^1$  and  $k^p$  monomials (von Staudt–Clausen at the Faulhaber layer,  $\deg_k c_{p-1} = p$ ); only  $k^p$  reaches  $A_n$ , as  $k^1$  sits at weight  $2p-3 > 2n+1$ . The coefficient of that top monomial in  $A_{(p-1)/2}$  is a *unit*,  $\partial A_{(p-1)/2} / \partial c_{p-1}[k^p] = \pm 1$ , and this now has a leading-symbol proof. The monomial perturbs the confluent expansion of  $G$  by  $\varepsilon^{p-1} k^p$ ; the turning-point map (the  $D^p$ -action,  $D = (u/2) d/du$ ) sends  $k^p$  to  $a_p u^p \sin u + b_p u^{p-1} \cos u$  with  $a_p = (-1)^{n-1} 2^{-p}$  the leading sine coefficient and  $b_p = (-1)^n n(2n+1) 2^{-p}$  the leading cosine coefficient ( $p = 2n+1$ ; both closed forms follow from the recursion  $(A, B) \mapsto (\frac{u}{2}(A' - B), \frac{u}{2}(B' + A))$ ). At a simple zero  $u_*$  of  $\cos$  the cosine term vanishes and every sine power below  $u^p$  contributes only at higher order in  $\varepsilon$ , so the zero shift is  $\delta = \varepsilon^{p-1} a_p u_*^p$  (the factor  $\sin u_* = \pm 1$  cancels against  $G' = -\sin$ ). With  $m = \varepsilon u^2$  and the leading balance  $u_*^2 = 2/\varepsilon$  (from  $\varepsilon u^2 = 2(1 - \varepsilon)e^{\varepsilon/2} \rightarrow 2$ ),

$$\delta m = 2\varepsilon u_* \delta = 2a_p \varepsilon^p u_*^{p+1} = a_p 2^{n+2} \varepsilon^n, \quad \text{so} \quad \frac{\partial m_n}{\partial c_{p-1}[k^p]} = a_p 2^{n+2} = \pm 2^{1-n},$$

and with the reversion Jacobian  $\partial A_n / \partial m_n = 2^{n-1}$  (from  $\varepsilon \sim 2/\mu$ ) the product is  $\pm 1$ . Only the top sine power  $u^p$  reaches order  $\varepsilon^n$ , so the magnitude is exact; the sign is  $(-1)^n$  (verified for all  $n \leq 10$ , independently of  $p$ ). Since  $c_{p-1}[k^p]$  has  $p$ -adic valuation  $-2$  while the other reaching monomials ( $k^{p-1}, k^{p-2}$ ) have valuation  $\geq -1$  against  $p$ -integral multipliers,  $v_p(\text{den } A_{(p-1)/2}) = 2$  exactly. Thus the onset law;  $p \mid \text{den}(A_n) \iff 2n+1 \geq p$ , with valuation 2 at  $n = (p-1)/2$ ; is *proved for all odd*  $p$ . The average-linear (Legendre/factorial) growth; not the logarithmic growth an  $\text{lcm}(1, \dots, 2n)$  or  $\zeta$ -type denominator would give; makes  $\text{den}(A_n)$  super-geometric ( $\log \text{den}(A_n)/n$  rises  $5.4 \rightarrow 10.2$  over  $n \leq 20$ , unbounded). Consequently  $b(t)$  is *not* a  $G$ -function (super-geometric denominators alone suffice, independent of holonomy), so  $\sum A_n x^n$  is not of André arithmetic-Gevrey type [2] and lies outside the  $G$ -function,  $E$ -function, and  $E$ -operator (Fischler–Rivoal [20]) classes entirely. The closest existing frame, the  $E$ -operator theory, has exactly one blocking gap; the hypothesis that  $b$  is a  $G$ -function; which the prime law refutes rather than supplies. The missing tool is therefore genuinely new, not an incremental hypothesis-removal: a value theory for Borel–Laplace sums whose Borel transform is non-holonomic with factorial denominators, evaluated at a point  $(4/\pi^2)$  transcendental over  $\mathbb{Q}$ . This constant is a canonical first target for such a theory. Third, the family

sum converges to

$$T := \sum_{j \geq 1} (1 - q_j^*) = 0.7357490877157094 \dots (\pm 3 \cdot 10^{-17}, \text{ tail summed via the fitted law and Hurwitz zeta}),$$

for which no relation with  $\{1, \pi^{-2}, \pi^{-4}, \log 2, \gamma, \pi^2/6\}$  exists at height  $10^4$  within the stated uncertainty (and  $T \neq 2/e$ ): the classical tail is  $\pi^2$ -rational piece by piece, but the wild head shifts the sum off every tested constant; a sum rule, if one exists, must come from global structure not yet visible.

The failure pattern above is not an artifact of particular constructions; it is forced by exact arithmetic facts, which we record as a theorem-level ledger.

**Proposition 114** (Arithmetic no-go ledger at  $\rho = \frac{1}{2}$ ). *Let  $q = s/t \in (0, 1)$  with  $\gcd(s, t) = 1$ , and let  $g_k = (-1)^k q^{k(k-1)} / (q; q)_{2k}$  be the Taylor coefficients of  $G(q, \cdot)$ . Then:*

- (i) (exact denominators) *in lowest terms  $g_k = (-1)^k s^{k^2-k} t^{k^2+2k} / \prod_{i=1}^{2k} (t^i - s^i)$ ; in particular  $\text{den}(g_k) = \prod_{i=1}^{2k} (t^i - s^i) = t^{k(2k+1)(1+o(1))}$  exactly, while  $|g_k| = (s/t)^{k^2(1+o(1))}$ , so the arithmetic ratio is  $\rho = \log(t/s)/(2 \log t) \leq \frac{1}{2}$ , with equality iff  $s = 1$ .*
- (ii) ( $\nu$ -ladder arithmetic) *the  $q$ -Lommel polynomials of Koelink–Swarttouw [29] satisfy  $\deg_q p_m = m^2$  exactly, with leading  $q$ -coefficient  $(-1)^m$ ; hence at  $q = s/t$  their denominators are the pure power  $t^{m^2}$ , carrying no cyclotomic structure.*
- (iii) ( $\nu$ -direction regularity) *the ladder recurrence  $u_{m+1} = ((1 - q^{2m+1})/x_0 + x_0)u_m - u_{m-1}$  has coefficients converging exponentially to  $x_0 + 1/x_0$  with distinct characteristic roots  $x_0^{\pm 1}$ ; by Poincaré–Perron [45, 44] every nonzero solution satisfies  $|u_m|^{1/m} \rightarrow x_0^{\pm 1}$ . No theta-type ( $q^{cm^2}$ -decaying) solution exists in the  $\nu$ -direction, for either kind. Consequently the cleared  $\nu$ -ladder forms are  $\gg t^{m^2} x_0^{-m} \rightarrow \infty$  for every rational  $q$ , including  $s = 1$ .*
- (iv) (cyclotomic floor) *any linear form built over the coefficients  $g_0, \dots, g_M$  has decay at best  $q^{M^2(1+o(1))}$ , while its cyclotomic support is  $\{\Phi_d(t, s) : d \leq 2M\}$ ; even if every  $\Phi_d$  appeared at most once; the idealized best case of any lcm/acceleration argument; the denominator exceeds  $t^{\sum_{d \leq 2M} \varphi(d)} = t^{(12/\pi^2 + o(1))M^2}$ , and  $12/\pi^2 = 1.2159 \dots > 1$ . Hence no cyclotomic-compression (Rhin–Viola-type [49]) refinement of the Padé, continued-fraction, or ladder families can produce integer forms tending to 0. Moreover the true solved Padé denominators contain growing non-cyclotomic irreducible factors (at  $n = 3$ , one of degree 26 with coefficients up to 4; no integer  $d$  has  $\varphi(d) = 26$ ), which no lattice-of-cyclotomes argument addresses.*

*Proof.* (i)  $q^{k(k-1)} = s^{k(k-1)} / t^{k(k-1)}$  and  $1/(q; q)_{2k} = t^{k(2k+1)} / \prod_{i \leq 2k} (t^i - s^i)$ ; the numerator  $s^{k^2-k} t^{k^2+2k}$  is coprime to each factor since  $\gcd(st, t^i - s^i) = 1$ . (ii) Induction:  $p_{m+1} = (X + 1 - q^{2m+1})p_m - Xp_{m-1}$ , and the top  $q$ -degree term of  $p_{m+1}$  is  $-q^{2m+1} \cdot (\text{top of } p_m)$ , of degree  $m^2 + 2m + 1 = (m+1)^2$ , which cannot cancel against  $Xp_{m-1}$  (degree  $(m-1)^2$ ). (iii) The perturbation  $(1 - q^{2m+1})/x_0 - 1/x_0$  decays geometrically, so the Poincaré–Perron theorem [45, 44] applies with the limit equation  $u_{m+1} = (x_0 + 1/x_0)u_m - u_{m-1}$ , whose roots  $x_0 \neq 1/x_0$  are simple. (iv) A form using  $g_0, \dots, g_M$  vanishes to order at most  $M$  at  $z = 0$ , so its value at a fixed  $z_0$  inside the disc is  $\gg |g_{M+1} z_0^{M+1}| = q^{M^2(1+o(1))}$  in the truncation-based families; its denominators involve  $(t^i - s^i)$  for  $i \leq 2M$ , i.e.  $\Phi_d(t, s)$  for  $d \leq 2M$ , and  $\sum_{d \leq N} \varphi(d) = \frac{3}{\pi^2} N^2 + O(N \log N)$  gives the floor with  $N = 2M$ . The degree-26 factor is exhibited by the exact  $n = 3$  computation (`code/zeta_probe/beta2_qniven_ladder.py`).  $\square$

*Remark 115 (scope).* Proposition 114 closes every construction whose coefficients live in the natural  $\mathbb{Z}[q^{\pm 1}, z]$ -lattices of the  $q$ -difference module; Taylor Padé and Hermite–Padé, the argument-direction continued fraction, the  $\nu$ -ladders of both kinds; together with all cyclotomic-compression refinements of them. It does *not* exclude a fundamentally different family of forms; producing one is exactly what a  $\rho = \frac{1}{2}$  criterion would mean. The ledger thereby locates  $\beta_2$ 's arithmetic at a precise frontier of  $q$ -Diophantine approximation.

The remaining freedom can be narrowed further: the finite places of  $\mathbb{Q}$  are provably useless here. The series  $S(q) = \sum_k g_k z^{*k}$ ,  $z^* = 2q(1 - q)$ , has rational terms, so besides its real sum it possesses  $s$ -adic and  $t$ -adic *avatars*; the same series summed in  $\mathbb{Q}_s$  and  $\mathbb{Q}_t$ , where it also converges (the valuations of the terms tend to  $+\infty$ ).

**Lemma 116** (adelic units). *Let  $q = s/t \in (0, 1)$ ,  $\gcd(s, t) = 1$ , and  $z^* = 2q(1 - q)$ . For every  $k \geq 1$  and every prime  $p$ : if  $p \mid t$  with  $e = v_p(t)$ , then*

$$v_p(g_k z^{*k}) = k^2 e + k[p = 2] \geq k^2,$$

*(exact, verified symbolically), and if  $p \mid s$  then  $v_p(g_k z^{*k}) \geq k^2$ . Consequently the series converges in  $\mathbb{Q}_p$  for every  $p \mid st$ , and the finite-place avatars satisfy  $S_p \equiv 1 \pmod{p^{v_p(st)}}$ ; in particular  $S_t \equiv 1 \pmod{t}$  and  $S_s \equiv 1 \pmod{s}$ : they are units, hence nonzero; regardless of whether the real avatar vanishes.*

*Proof.* Each factor  $t^i - s^i$  of  $\text{den}(g_k)$  is a  $p$ -unit for every  $p \mid st$  (it is  $\equiv -s^i$  or  $\equiv t^i \pmod{p}$ ), so by Proposition 114(i),  $v_p(g_k) = (k^2 + 2k)e$  for  $p \mid t$  and  $v_p(g_k) = (k^2 - k)v_p(s)$  for  $p \mid s$ . From  $z^* = 2s(t - s)/t^2$ : for  $p \mid t$ ,  $v_p(t - s) = 0$  and  $v_p(z^*) = [p = 2] - 2e$ , giving  $v_p(g_k z^{*k}) = (k^2 + 2k)e + k([p = 2] - 2e) = k^2 e + k[p = 2]$ ; for  $p \mid s$ ,  $v_p(z^*) = v_p(2s) \geq v_p(s) \geq 1$ , giving  $v_p(g_k z^{*k}) \geq (k^2 - k) + k = k^2$ . All are  $\geq 1$  for  $k \geq 1$ , so each avatar is 1 plus terms of positive valuation.  $\square$

*Remark 117* (exact adelic balance, and the shape of any future criterion). Lemma 116 has two consequences. First, it extends the no-go ledger from the module's coefficient lattices to the *values*: no divisibility or smallness can be sourced from the finite places, since the finite-place avatars are units; a hypothetical rationality of  $\beta_2$  would be a maximally place-asymmetric event, the real avatar vanishing while every finite avatar equals a unit  $\equiv 1$ . Second, it makes the  $\rho = \frac{1}{2}$  balance exact in the sense of the product formula: the term  $g_k z^{*k}$  carries archimedean decay  $(s/t)^{k^2(1+o(1))}$ ,  $s$ -adic and  $t$ -adic numerator content  $s^{k^2} t^{k^2}$ , and denominator  $t^{2k^2(1+o(1))}$ ; the books balance with *zero margin* at the  $k^2$  scale, so any successful construction must be lossless at all three places simultaneously and win at the geometric scale. Every mechanism surveyed above loses a factor  $t^{ck^2}$  somewhere; a  $\rho = \frac{1}{2}$  criterion is therefore forced to be a purely *archimedean cancellation* phenomenon. This suggests the precise principle that would settle  $\beta_2$ , which we record as a conjecture: for the irregular rank-two  $q$ -difference module with polynomial companion matrix and  $\det A = q$ , the vanishing locus is *adelically rigid*: the entire function  $G_v(q, \cdot)$  cannot vanish at a rational point  $z_0$  at which all finite-place avatars are units (as at  $z_0 = z^*$ , by Lemma 116) at one place  $v$  of  $\mathbb{Q}$  while the avatars at all other convergent places are units. The finite-place half of this rigidity is exactly Lemma 116; its archimedean half implies  $\beta_2 \notin \mathbb{Q}$ . This is a  $q$ -analogue, for irregular  $q$ -difference modules, of the place-uniformity enjoyed by  $E$ - and  $G$ -function values, where the exceptional sets of Siegel–Shidlovskii–André theory are independent of the place.

Further analysis of the archimedean obstruction yields several exact structural facts and a faithful reduction. All are verified numerically to 40–60 digits (`code/zeta_probe/beta2_*.py`).

**Proposition 118** (structure of the zero curve). Write  $Q = q^2$  and let  $0 < z_1(q) < z_2(q) < \dots$  be the positive zeros of  $G(q, \cdot)$ .

- (i) (base change)  $G(q, z) = \frac{(q^2; q^2)_\infty}{\sqrt{q}(q; q^2)_\infty} z^{1/4} J_{-1/2}^{(3)}(\sqrt{z}/q; q^2)$ , where  $J_\nu^{(3)}$  is the Hahn–Exton (third Jackson)  $q$ -Bessel function; thus  $G$  is, up to an explicit prefactor, the Hahn–Exton  $q$ -cosine, and its zeros are those of a classical, well-studied object.
- (ii) (zero lattice)  $z_k(q) Q^{k-1} \rightarrow 1$  as  $k \rightarrow \infty$ , with theta-suppressed dressing  $\delta_k := z_{k+1} Q^k - 1 = O(c^k Q^{k^2})$  (numerically  $\delta_0 = -0.5051$ ,  $\delta_1 = -1.17 \cdot 10^{-2}$ ,  $\delta_2 = -1.03 \cdot 10^{-5}, \dots$ );  $z^* = z_1(q^*)$  is the maximally dressed member, displaced from the bare lattice point  $Q^0 = 1$  by an  $O(1)$  correction; it is not a torsion/lattice point of the connection curve.
- (iii) (power sums are rational) the symmetric functions  $\sigma_p(q) := \sum_{n \geq 1} z_n(q)^{-p}$  lie in  $\mathbb{Q}(q)$  for every  $p \geq 1$  (Newton’s identities on the  $\mathbb{Q}(q)$ -coefficients  $g_k$ ); e.g.  $\sigma_1 = 1/((1-q)(1-q^2))$ . Consequently  $z^*(q) = \lim_{p \rightarrow \infty} \sigma_p(q)/\sigma_{p+1}(q)$  with geometric rate  $(z_1/z_2)^p$ , giving an all-orders  $\mathbb{Q}(q)$ -approximation of the zero curve.

Two of the no-go’s ingredients also sharpen. First, the cause of  $\rho = \frac{1}{2}$  is exactly the double Pochhammer:  $\rho = \log(t/s)/(2 \log t)$ , and the factor 2 in the denominator is the  $2k$  in  $(q; q)_{2k}$ ; a single Pochhammer would give  $\rho > 1$  (whence Liouville) when  $t > s^2$ . The intractability is intrinsic to the base- $q^2$  Hahn–Exton structure, not to the choice of construction. Second, the finite-place primes are *disjoint* from  $st$ :  $\prod_{i \leq 2N} (t^i - s^i)$  is coprime to  $st$  (only the prime 2, via lifting-the-exponent when  $s, t$  are both odd, contributes an  $O(N)$  term), so the arithmetic of every partial-sum denominator is supported off  $\{s, t\}$ ; orthogonal to the unit residues of Lemma 116. The Hankel/determinantal escape is likewise closed exactly:  $\det(g_{i+j})_{0 \leq i, j \leq n}$  has denominator of cubic archimedean size ( $\log |\text{den}| \sim c n^3$ ), one power worse than the quadratic remainder.

Finally, the Nesterenko route is blocked more precisely than by non-holonomicity alone. Nesterenko’s method needs a finitely generated  $\delta$ -closed ( $\delta = q d/dq$ )  $\mathbb{C}(q)$ -algebra containing  $S$ , i.e. that  $S$  be *differentially algebraic*; non-holonomicity does not suffice (the quasimodular  $E_2$  is non-holonomic yet  $\delta$ -algebraic). One computes  $\delta S = \sum_k a_k (k^2 - \frac{kq}{1-q} + L_k)$  with  $L_k = \sum_{i=1}^{2k} i q^i / (1 - q^i) \rightarrow (1 - E_2)/24$ :  $S$  carries *every* finite Eisenstein truncation  $L_k$ , not merely the limit, which is precisely why it escapes the modular differential ring. Whether  $S$  is differentially algebraic is itself open, but the exclusion is now proof-grade over a wide window:  $S$  satisfies no algebraic differential equation in the  $\delta$ -jets at any of the profiles (order, degree,  $\deg_q$ ) = (2, 12, 3), (3, 8, 3), (4, 5, 6), (5, 4, 8), (6, 3, 15), (8, 2, 35), (12, 1, 120); exact linear algebra on 2000 coefficients modulo a prime, every matrix of full column rank; since a primitive integer relation survives reduction, full rank modulo one prime excludes a relation over  $\mathbb{Q}$  at that profile outright. (The pipeline is validated on  $E_2$ , for which it recovers the  $q$ -Chazy equation  $\delta^3 E_2 = E_2 \delta^2 E_2 - \frac{3}{2} (\delta E_2)^2$  at its exact profile.) The same exclusion holds verbatim for the zero branch  $z_1(q)$ . The Nesterenko route is blocked *modulo the differential transcendence of  $S$* , which is itself open; and, with no functional equation in  $q$  available for  $S$  (no shift,  $q$ -difference, or Mahler equation exists), no current method can even attack it.

*Remark 119* (reduction to a theta non-resonance). All routes converge on the same faithful reformulation:  $\beta_2$  is irrational iff  $G(q^*, z^*) \neq 0$  is forced by the arithmetic, and (Ramis–Sauloy–Zhang [48]) this is a *theta non-resonance* statement; the connection ratio  $R_q(z) = [P_{21} y_{\text{irr}}]/[P_{11} y_{\text{reg}}]$ , an explicit  $\theta_{q^2}$ -quotient, must satisfy  $R_q(2q(1-q)) \neq -1$  at every algebraic  $q$ . The evidence points toward truth: in the confluent limit  $q \rightarrow 1$  this is exactly Lindemann’s theorem that  $\cos$  has no algebraic zero. But the reduction is not yet shown *easier*:  $R_q(z) = -1$  is a functional-equation-consistent value that a nonconstant theta-quotient must attain somewhere, and on the diagonal  $z = 2q(1-q)$

the theta-arguments appear algebraically dependent, so no current independence theorem (Bertrand; Nesterenko [38]; Schneider–Lang [31]) forbids it. The archimedean obstruction is thus the exact mirror of the proven finite-place balance: units ( $\equiv 1$ ) at every  $p \mid st$ , and a putative  $-1$  at  $\infty$ , with neither forbidden. The required connection-entry computation is carried out below.

**Proposition 120** (the connection entries). *Write  $Q = q^2$ ,  $\theta_b(x) = \sum_{n \in \mathbb{Z}} b^{n(n-1)/2} x^n$ . The equation for  $G$  has, at  $z = \infty$ , the formal solution basis  $f_1(z) = \theta_Q(-z) \hat{g}(z)$  and  $f_2(z) = h(z)/\theta_Q(-z/q)$ , where  $h(z) = \sum_k c_k z^{-k}$  is convergent (entire in  $1/z$ ,  $c_k = O(Q^{k^2/2})$ ) while  $\hat{g}(z) = \sum_k b_k z^{-k}$  is  $q$ -Gevrey divergent, with the closed form*

$$b_k = \frac{q^{k-k(k-1)/2}}{(q; q)_k}$$

(equivalent to the recursion  $b_k = [-(q+1)Q b_{k-1} - Q^{3-k} b_{k-2}]/(q(Q^k - 1))$ ; verified symbolically for  $k \leq 8$ ). By Euler's theorem  $\sum_j x^j/(q; q)_j = 1/(x; q)_\infty$ , the  $q$ -Borel transform is therefore explicit:

$$\hat{B}(\xi) = \sum_{k \geq 0} b_k q^{k(k+1)/2} \xi^{-k} = \sum_{k \geq 0} \frac{(q^2/\xi)^k}{(q; q)_k} = \frac{1}{(q^2/\xi; q)_\infty},$$

meromorphic on  $\mathbb{C}^*$  with poles exactly on  $\xi \in q^{2+\mathbb{N}_0}$ , and satisfying  $\hat{B}(\xi) + \frac{q+1}{\xi} \hat{B}(q\xi) + (\xi^{-2} - 1) \hat{B}(q^2\xi) = 0$  identically (direct computation from the product form; in particular  $\hat{B}(-1) = (q+1) \hat{B}(-q)$ ). The theta-kernel  $q$ -Laplace

$$g^{[\lambda]}(z) = \frac{\sum_{m \in \mathbb{Z}} q^{m(m-1)/2} (\lambda/z)^m \hat{B}(\lambda q^m)}{\theta_q(\lambda/z)}, \quad f_1^{[\lambda]} = \theta_Q(-z) g^{[\lambda]}(z),$$

is then singular only for  $\lambda q^{\mathbb{Z}}$  meeting the pole set, i.e. for the single class  $[\lambda] \in q^{\mathbb{Z}}$  of  $E_q = \mathbb{C}^*/q^{\mathbb{Z}}$  (the positive-real lattice class, near which the zeros of  $G$  lie); every other class is regular. For  $\lambda \notin q^{\mathbb{Z}}$  the bilateral sum converges absolutely on compacts of  $\mathbb{C}^* \setminus (-\lambda q^{\mathbb{Z}})$  and  $Lf_1^{[\lambda]} = 0$  exactly: the theta shifts  $\theta_q(q^{-2}x) = q^{-3}x^2\theta_q(x)$  and  $\theta_q(q^{-4}x) = q^{-10}x^4\theta_q(x)$  turn  $f_1^{[\lambda]}(Qz)$  and  $f_1^{[\lambda]}(Q^2z)$  into the same kernel sum with  $\hat{B}(\lambda q^m)$  replaced by  $\hat{B}(\lambda q^{m+2})$ ,  $\hat{B}(\lambda q^{m+4})$ ; collecting the coefficient of  $q^{m(m-1)/2} (\lambda/z)^m / \theta_q(\lambda/z)$  in  $Lf_1^{[\lambda]}/\theta_Q(-z)$  then yields  $q[\hat{B}(\xi) + \frac{q+1}{\xi} \hat{B}(q\xi) + (\xi^{-2} - 1) \hat{B}(q^2\xi)]$  at  $\xi = \lambda q^m$ , identically zero. (Each step was verified numerically to 40 digits; the kernel normalization was validated on convergent input to 34 digits, and  $Lf_1^{[\lambda]} \approx 10^{-61}$  numerically.) Writing  $G = C_1 f_1^{[\lambda]} + C_2^{[\lambda]} f_2$ , numerically to 59–60 digits:

- (i) since  $\det A = q$ , every Casoratian  $\text{Cas}(u, v)(z) := u(z)v(Qz) - u(Qz)v(z)$  of solutions satisfies  $\text{Cas}(Qz) = q \text{Cas}(z)$  (immediate from  $u(Q^2z) = a u(Qz) - q u(z)$ ), so  $C_1 = \text{Cas}(G, f_2)/\text{Cas}(f_1^{[\lambda]}, f_2)$  and  $C_2^{[\lambda]} = \text{Cas}(f_1^{[\lambda]}, G)/\text{Cas}(f_1^{[\lambda]}, f_2)$  are  $Q$ -elliptic; proven, and verified to  $10^{-60}$ ;
- (ii)  $C_1$  is a constant, independent of both  $z$  and the resummation direction  $\lambda$ : this is proven in Theorem 121 below (the limit argument applies along every orbit class  $\neq [1]_Q$ , and ellipticity extends the value); the slope-1 component of  $G$  is Stokes-invariant, and all  $q$ -Stokes ambiguity is carried by  $C_2^{[\lambda]}$ , nonconstant elliptic with  $\lambda^2$  in its zero divisor;
- (iii) (dressing-product formula, verified to 59 digits)

$$C_1 = \frac{1}{(Q; Q)_\infty \prod_{k \geq 1} z_k Q^{k-1}} :$$

the Stokes-invariant connection constant is the regularized product of the zero dressings;

(iv) at  $q = q^*$  the entire apparatus evaluates:  $C_1(q^*) = 2.69898382616391375\dots$ , and the zero condition  $G(q^*, z^*) = 0$  takes the pinned form

$$C_2^{[\lambda]}(z^*) = -C_1 \frac{f_1^{[\lambda]}(z^*)}{f_2(z^*)} = 0.011592994120624266\dots \quad (\lambda = \tfrac{3}{2}),$$

verified to 30 digits. This is the explicit instance of the theta non-resonance: every object in it is now constructed.

The constant  $C_1$  is in fact *modular*, and this upgrades the dressing-product formula into an exact evaluation of the zero product.

**Theorem 121** (the connection constant, and the zero product of the Hahn–Exton  $q$ -cosine). *For  $0 < q < 1$ ,*

$$C_1 = \frac{1}{(q; q)_\infty}, \quad \text{and consequently} \quad \prod_{k \geq 1} z_k(q) Q^{k-1} = (q; q^2)_\infty,$$

where  $z_1 < z_2 < \dots$  are the positive zeros of  $G(q, \cdot)$ . Likewise, for the second solution's underlying entire function  $H$  with zeros  $w_1 < w_2 < \dots$ ,  $\prod_{k \geq 1} w_k(q) q Q^{k-1} = (q^3; q^2)_\infty$ . (Verified at three bases including  $q^*$ , to 41, 23, 26 digits for  $C_1$  and 58 digits for the product.)

*Proof.* (1) *An exact identity.* From  $(q; q)_{2k} = (q; q)_\infty / (q^{2k+1}; q)_\infty$  and Euler's expansion  $(x; q)_\infty = \sum_{j \geq 0} (-1)^j q^{j(j-1)/2} x^j / (q; q)_j$  at  $x = q^{2k+1}$ , together with  $k(k-1) + 2kj = (k+j)(k+j-1) - j(j-1)$  and the substitution  $m = k + j$ , absolute convergence of the double sum permits the rearrangement

$$(q; q)_\infty G(z) = \sum_{m \geq 0} (-1)^m q^{m(m-1)/2} z^m P_m(1/z), \quad P_m(w) := \sum_{j=0}^m c_j w^j, \quad c_j = \frac{q^{j-j(j-1)/2}}{(q; q)_j}$$

(the  $c_j$  coincide with the  $b_j$  of Proposition 120; identity checked symbolically through  $z^8$  and numerically to 41 digits).

(2) *The constant.* Fix  $u < 0$  and put  $z = uQ^{-N}$ . The moduli  $q^{m(m-1)/2}|z|^m$  peak at  $m_0 = N + O(1)$  and decay like  $q^{(m-m_0)^2 + O(1)}$  away from it. For  $1 \leq j \leq m \leq 2N$ ,  $2Nj - j(j-1)/2 \geq Nj$ , so  $|c_j z^{-j}| \leq Cq^{Nj}|u|^{-j}$  and  $|P_m(1/z) - 1| \leq C'q^N$  uniformly; the terms with  $m > 2N$  contribute a relative  $O(q^{N^2})$ . The negative- $m$  part of  $\theta_Q(-z)$  is  $O(|z|^{-1})$ , negligible against the peak. Hence  $(q; q)_\infty G(z) / \theta_Q(-z) \rightarrow 1$  along  $z = uQ^{-N}$ , for each fixed  $u < 0$ .

(3) *Geometric zero dressing.* On the circle  $|z - Q^{1-k}| = \varepsilon q^k Q^{1-k}$  the identity of step (1) and the bounds of step (2) give  $|(q; q)_\infty G - \theta_Q(-z)| \leq Cq^k |\theta_Q(-z)|_{\text{peak}}$ , while  $\theta_Q(-z)$  has a simple zero at  $Q^{1-k}$  with derivative bounded below there; Rouché gives  $z_k = Q^{1-k}(1 + O(q^k))$ , i.e.  $\log(z_k Q^{k-1}) = O(q^k)$ , absolutely summable.

(4) *The product.*  $G(z) = \prod_k (1 - z/z_k)$  (order 0, genus 0,  $G(0) = 1$ ) and  $\theta_Q(-z) = (Q; Q)_\infty (z; Q)_\infty (Q/z; Q)_\infty$  (Jacobi). Along  $z = -R$ ,  $R \rightarrow \infty$ ,

$$\frac{G(-R)}{\theta_Q(R)} = \frac{1}{(Q; Q)_\infty (-Q/R; Q)_\infty} \prod_{k \geq 1} \frac{1 + R/z_k}{1 + RQ^{k-1}},$$

and for  $x, y > 0$  the function  $R \mapsto \log \frac{1+Rx}{1+Ry}$  is monotone from 0 to  $\log(x/y)$ , so each factor is dominated by  $|\log(z_k Q^{k-1})|$ , summable by step (3). Dominated convergence gives  $G(-R)/\theta_Q(R) \rightarrow 1/[(Q; Q)_\infty \prod_k z_k Q^{k-1}]$  as  $R \rightarrow \infty$ ; evaluating along the subsequence  $R = |u|Q^{-N}$  by step (2), the limit equals  $1/(q; q)_\infty$ , whence  $\prod_k z_k Q^{k-1} = (q; q)_\infty / (Q; Q)_\infty = (q; q^2)_\infty$  by Euler's  $(q; q)_\infty = (q; q^2)_\infty (q^2; q^2)_\infty$ . Finally, in the basis of Proposition 120: along the negative axis (a regular

class,  $[-1] \notin q^{\mathbb{Z}}$  one has  $g^{[\lambda]}(z) \rightarrow 1$  and  $f_2(z)/\theta_Q(-z) = O(1/(\theta_Q(R)\theta_Q(R/q))) \rightarrow 0$ , so the limit just computed is the entry  $C_1$ ; hence  $C_1 = 1/(q; q)_{\infty}$ . The  $H$ -statement is identical with comparison function  $\theta_Q(-qz)$  and ratio  $(1-q)/(q; q)_{\infty}$ . Finally, steps (1)–(2) apply verbatim along  $z = uQ^{-N}$  for any  $u \in \mathbb{C}^*$  whose orbit avoids the zero class  $[1]_Q$  of  $\theta_Q(-\cdot)$ , using the standard theta lower bound  $|\theta_Q(-z)| \geq c([u]) \cdot (\text{peak term})$  off that class; along each such orbit  $C_1(z)$  is constant (ellipticity) and  $f_2/f_1 \rightarrow 0$ ,  $g^{[\lambda]} \rightarrow 1$ , so  $C_1([u]) = 1/(q; q)_{\infty}$  on every class  $\neq [1]_Q$ , and by meromorphy  $C_1 \equiv 1/(q; q)_{\infty}$ : for every regular  $\lambda$ . In particular  $C_1$  is  $z$ - and  $\lambda$ -independent.  $\square$

*Remark 122* (Diophantine consequences at  $z^*$ ). At  $q = q^*$  the identity splits off the candidate rational zero:

$$\prod_{k \geq 2} z_k(q^*) Q^{k-1} = \frac{(q; q^2)_{\infty}}{z^*} = 0.988299884628746 \dots \quad (\text{verified to 30 digits}).$$

Were  $\beta_2$  rational, the *modular* value  $(q; q^2)_{\infty}$  (transcendental at algebraic  $q$  by Nesterenko's theorem applied to the Euler products) would equal the rational  $z^*$  times the tail dressing product  $\prod_{k \geq 2} z_k Q^{k-1}$ . Thus the entire modular sector of the problem is now explicit; the Stokes-invariant connection constant is  $1/(q; q)_{\infty}$ , the zero product is  $(q; q^2)_{\infty}$ ; and *all* remaining unknown arithmetic is provably confined to the tail dressing product, equivalently to the  $q$ -Stokes cocycle  $C_2^{[\lambda]}$ : the non-modularity of the  $\beta_2$  problem *is* its  $q$ -Stokes data. A transcendence statement for that tail (an algebraic independence of the higher-zero dressings from the modular values, or a rigidity theorem for the  $q$ -Stokes cocycle) would settle  $\beta_2$ ; no current theorem provides it.

The cocycle itself is now explicit. Writing  $g = g^{[\lambda]}$  for the resummed series and  $h$  for  $f_2$ 's convergent one,

$$C_2^{[\lambda]}(z) = \theta_Q(-z/q) \frac{g(z) G(Qz) + z^{-1} g(Qz) G(z)}{-(z/q) g(z) h(Qz) + z^{-1} g(Qz) h(z)}$$

This is an identity: it is the ratio of the Casoratian evaluations  $\text{Cas}(f_1, G) = \theta_Q(-z)[g(z)G(Qz) + z^{-1}g(Qz)G(z)]$  and  $\text{Cas}(f_1, f_2) = \frac{\theta_Q(-z)}{\theta_Q(-z/q)}[-(z/q)g(z)h(Qz) + z^{-1}g(Qz)h(z)]$ , both instances of  $\theta_Q(Qw) = w^{-1}\theta_Q(w)$  (and verified numerically to  $10^{-61}$ ); with divisor now fully described: an *explicit* zero lattice  $z \in q^{-1}Q^{\mathbb{Z}}$  from the theta factor (the computed zero at  $z = 1/q$ , exact to  $10^{-63}$ ), a direction-dependent zero at  $z = \lambda^2$  arising from the exact *resummation resonance*  $g^{[\lambda]}(\lambda^2)G(Q\lambda^2) + \lambda^{-2}g^{[\lambda]}(Q\lambda^2)G(\lambda^2) = 0$  (verified to  $10^{-62}$  at  $\lambda \in \{\frac{3}{2}, 2, \frac{13}{5}\}$ ; a proof for general regular  $\lambda$  is open), and poles at the zeros of the  $h$ -bracket. Substituting into the pinned zero condition at  $z^*$  and cancelling the common theta factor reduces the entire  $\beta_2$  question to *one* relation between the two modular values  $\theta_Q(-z^*)$ ,  $(q; q)_{\infty}$  and five values of the non-modular pair  $(g^{[\lambda]}, h)$  on the orbit of  $z^*$ . The remaining input is a transcendence/independence statement for those five values; supplying it is one of three research programs (a non-modular extension of Nesterenko's method via the deformation of this connection, a  $\rho < \frac{1}{2}$ -regime irrationality criterion, or adelic rigidity of the vanishing locus); none is currently available.

The first of those programs now has its foundation: the deformation of the module in  $q$  closes finitely.

**Proposition 123** (the deformation module). *Let  $L = q - a\sigma + \sigma^2$  with  $a = q + 1 - qz$ ,  $\sigma f(z) = f(Qz)$ , and let  $\theta = z\partial_z$ ,  $\delta = q\partial_q$  (acting on all  $q$ -dependence, including the base  $Q = q^2$ , so that  $\delta\sigma = \sigma\delta + 2\sigma\theta$ ). Then*

$$L(\theta G) = -qz \sigma G, \quad L(\delta G) = -qG + q(1 + 3z) \sigma G + 4q \theta G - 2a \sigma \theta G$$

(two commutator computations; verified to 51 digits). Consequently the  $\mathbb{Q}(q, z)\langle\sigma\rangle$ -module generated by  $\{G, \theta G, \delta G\}$  has finite rank  $\leq 6$ , with explicit  $\sigma$ -companion relations  $\sigma^2 G = a\sigma G - qG$ ,  $\sigma^2 \theta G = a\sigma \theta G - q\theta G - qz\sigma G$ , and  $\sigma^2 \delta G = a\sigma \delta G - q\delta G - qG + q(1 + 3z)\sigma G + 4q\theta G - 2a\sigma \theta G$ .

This finiteness is of the first jet only: the module is not closed under a second application of  $\theta$  or  $\delta$ . Both  $\theta^2 G$  and  $\delta^2 G$  escape  $\text{span}_{\mathbb{Q}(q, z)}\{G, \sigma G, \theta G, \sigma \theta G, \delta G, \sigma \delta G\}$  (verified by exact modular linear algebra at  $q = \frac{1}{3}$ ,  $z$ -series to 70 terms, coefficient degrees  $\leq 6$ : the augmented rank strictly exceeds the plain rank). The two derivations are moreover linked by the exact identity

$$\delta G = (\theta^2 - \theta)G + C, \quad C = \sum_{k \geq 0} (-1)^k \left( \sum_{i=1}^{2k} \frac{iq^i}{1-q^i} \right) \frac{q^{k(k-1)} z^k}{(q; q)_{2k}}$$

( $= z^2 G_{zz} + C$ ; immediate from  $g_k$ 's logarithmic  $q$ -derivative), whose weight  $C$  carries the truncated Eisenstein sums  $\sum_{i \leq 2k} iq^i/(1 - q^i)$  term by term; the same off-modular data that obstructs the differential closure. In particular, differentiating the pinned relation of Proposition 120 in  $q$  does not close: the second  $q$ -derivative leaves the module, consistent with Proposition 131. The value of the rank-6 structure is at the function level; it is the module whose difference Galois group is computed in Theorem 128; not as a Nesterenko precondition in the descent variable, which Proposition 131 shows does not exist.

**Corollary 124** (equation of the zero curve). *Implicit differentiation of  $G(q, z_1(q)) = 0$  gives*

$$z_1'(q) = - \frac{z_1(\delta G)(q, z_1)}{q(\theta G)(q, z_1)},$$

with  $\theta G, \delta G$  the first-jet generators of Proposition 123 (formula checked against a finite-difference derivative to 20 digits at  $q = 0.3$ ). The right-hand side is not a closed vector field; iterating  $d/dq$  leaves the rank-6 module (Proposition 123); so this is one exact relation, not an autonomous system. The number  $q^*$  is the crossing of the zero curve with the diagonal:  $z_1(q^*) = 2q^*(1 - q^*)$  (residual  $< 10^{-27}$ ).

**Lemma 125** (irreducibility of the  $\sigma$ -module). *For  $0 < q < 1$  the rank-2  $\sigma$ -module of  $G$  (in the sense of [47]) is irreducible over  $\mathbb{C}(z)$ : the Riccati equation*

$$u(z)u(Qz) - a(z)u(z) + q = 0, \quad a = q + 1 - qz,$$

has no solution  $u \in \mathbb{C}(z)$ .

*Proof.* Suppose  $u = P/R$  with  $P, R \in \mathbb{C}[z]$  coprime, of degrees  $p, r$ , leading coefficients  $\alpha, \beta$ . Clearing denominators,

$$F(z) := P(Qz)P(z) - a(z)P(z)R(Qz) + qR(z)R(Qz) \equiv 0.$$

At  $z = 0$ :  $F(0) = P(0)^2 - (q + 1)P(0)R(0) + qR(0)^2 = 0$ ; if  $R(0) = 0$  then  $P(0) = 0$ , contradicting coprimality, and symmetrically; so  $P(0)R(0) \neq 0$  and  $u(0) \in \{1, q\}$ . *Top degree*: the three terms have degrees  $2p, p + r + 1, 2r$ . If  $p = r$ , the top degree  $2p + 1$  is attained by the middle term alone, with coefficient  $q\alpha\beta Q^r \neq 0$ ; if  $|p - r| \geq 2$ , the top degree is attained by the first or third term alone, with coefficient  $\alpha^2 Q^p$  or  $q\beta^2 Q^r$ . Both are impossible, so  $|p - r| = 1$ . *Root pairing*: let  $\zeta \neq 0$  be a root of  $R$  of multiplicity  $m$ , so  $P(\zeta) \neq 0$ . At  $z = \zeta/Q$  the three terms of  $F$  have orders  $k, k + m + [a(\zeta/Q) = 0], m_0 + m$ , where  $k = \text{ord}_{\zeta/Q} P$  and  $m_0 = \text{ord}_{\zeta/Q} R$ ; since  $F \equiv 0$  the minimum order cannot be attained by a single term, forcing  $k \geq m_0 + m \geq m$ . Symmetrically, a root  $\xi \neq 0$  of  $P$  of multiplicity  $k$  forces  $\text{ord}_{Q\xi} R \geq k$  (coprimality gives  $\text{ord}_{Q\xi} P = 0$ , so the first term has order  $k$  at  $\xi$ , the third  $\text{ord}_{Q\xi} R$ , and the middle at least  $k + \text{ord}_{Q\xi} R$ ; again no single minimum). The two maps  $\zeta \mapsto \zeta/Q, \xi \mapsto Q\xi$  are mutually inverse on the nonzero root sets, and chains close at length two by coprimality. Summing multiplicities,  $p \leq r \leq p$ , so  $p = r$ ; contradicting  $|p - r| = 1$ .  $\square$

**Proposition 126** (the symmetric square). *Products  $w = y\tilde{y}$  of solutions of  $L$  satisfy the third-order equation*

$$w(Q^3z) = \frac{A_1(A_0A_1 - q)}{A_0} w(Q^2z) + q(q - A_0A_1) w(Qz) + \frac{q^3A_1}{A_0} w(z),$$

with  $A_0 = a(z)$ ,  $A_1 = a(Qz)$ , obtained by eliminating the mixed product  $m = y\tilde{y}(Q\cdot) + \tilde{y}y(Q\cdot)$  through  $m(Qz) = 2A_0w(Qz) - qm(z)$ . Its characteristic roots at  $z = 0$  are  $\{1, q, q^2\}$ ; verified on the solutions  $G^2$ ,  $z^{1/2}GH$ ,  $zH^2$  to 41–43 digits.

**Lemma 127** (divisor orientation). *Let  $r \in \mathbb{C}(z)^\times$  satisfy the rational-quotient relation of Proposition 126,  $r_0r_1r_2 = c_2r_0r_1 + c_1r_0 + c_0$  with  $r_k = r(Q^kz)$ ,  $c_2 = A_1(A_0A_1 - q)/A_0$ ,  $c_1 = q(q - A_0A_1)$ ,  $c_0 = q^3A_1/A_0$ . On every  $Q$ -orbit in  $\mathbb{C}^*$  disjoint from the singular orbit  $z_aQ^\mathbb{Z}$  ( $z_a = (q + 1)/q$ , the zero of  $a$ ), the maximal-modulus zero-or-pole of  $r$  cannot be a pole, and the minimal-modulus zero-or-pole cannot be a zero.*

*Proof.* Off  $z_aQ^\mathbb{Z}$  the  $c_i$  are finite and  $c_0 \neq 0$ . If the maximal-modulus zero-or-pole  $\zeta$  on such an orbit were a pole, evaluate at  $z = Q^{-2}\zeta$ : there  $r_0, r_1$  are finite and nonzero (all higher-modulus orbit points carry no zero or pole),  $r_2$  has a pole, so the left side has a pole while the right side is finite. If the minimal-modulus zero-or-pole  $\zeta'$  were a zero, rewrite the relation for  $s = 1/r$  as  $c_0s_0s_1s_2 + c_1s_1s_2 + c_2s_2 = 1$  and evaluate at  $z = \zeta'$ :  $s_0$  has a pole,  $s_1, s_2$  are finite and nonzero,  $c_0(\zeta') \neq 0$ , so the left side has a pole while the right side is 1.  $\square$

**Theorem 128** (the difference Galois group contains  $\mathrm{SL}_2$ ). *For every  $0 < q < 1$ , the difference Galois group over  $\mathbb{C}(z)$  of the module of  $G$  contains  $\mathrm{SL}_2$ .*

*Proof.* For  $N \geq 1$  suppose  $\mathrm{Sym}^N M$  has a rank-1 submodule, i.e. there is a nonzero solution  $w$  of the  $N$ -th symmetric-power equation with  $r := w(Q\cdot)/w \in \mathbb{C}(z)$ . The products  $f_1^{[\lambda]N-k} f_2^k$  ( $0 \leq k \leq N$ ) form a solution basis (their Casoratian is a power of  $\mathrm{Cas}(f_1, f_2) \neq 0$ ; checked numerically for  $N = 2$ ), so  $w = \sum_k c_k f_1^{N-k} f_2^k$  with constants  $c_k$ , and the case split is  $\lambda$ -robust since Stokes jumps preserve  $k_0 := \min\{k : c_k \neq 0\}$ .

*Case  $k_0 < N$ .* Along a ray in a regular class,  $f_2/f_1$  decays super-polynomially, so the rational function  $r$  has the Poincaré asymptotic expansion of the quotient of the dominant component:

$$r \sim (-1/z)^{N-k_0} (-z/q)^{k_0} \left( \frac{\hat{g}(Qz)}{\hat{g}(z)} \right)^{N-k_0} \left( \frac{h(Qz)}{h(z)} \right)^{k_0}.$$

The order- $j$  coefficient  $d_j$  of  $\hat{g}(Qz)/\hat{g}(z)$  satisfies  $d_j \sim -b_j$  with  $b_j = q^{(3j-j^2)/2}/(q; q)_j$  (verified:  $d_j/(-b_j) = 0.99986, 0.99998, 0.999998$  at  $j = 10, 12, 14$ ,  $q = \frac{1}{3}$ , converging to 1). Since this factor enters  $r$  to the power  $N - k_0 \geq 1$  against rational, hence geometrically bounded, prefactors, the coefficients of  $r$  inherit the growth  $b_j \sim q^{-j^2/2}$ , which is super-geometric, whereas a rational function's expansion at  $\infty$  has geometrically bounded coefficients. Since matching asymptotic expansions are unique, no rational  $r$  exists.

*Case  $k_0 = N$ .* Then  $r = (-z/q)^N \rho^N$  with  $\rho := h(Q\cdot)/h$  meromorphic on  $\mathbb{C}^*$ ;  $\rho^N \in \mathbb{C}(z)$  and single-valuedness force  $\rho \in \mathbb{C}(z)$  (an  $N$ -th root of a rational function is single-valued on  $\mathbb{C}^*$  only if every divisor multiplicity, including at 0 and  $\infty$ , is divisible by  $N$ ). Then  $f_2(Qz)/f_2(z) = -(z/q)\rho(z)$  is rational, giving a rational solution of the rank-2 Riccati equation; contradicting Lemma 125.

Hence no  $\mathrm{Sym}^N M$  has a rank-1 submodule.  $N = 1$  is irreducibility;  $N = 2$  excludes the imprimitive (dihedral) case; and a projectively finite irreducible subgroup  $(A_4, S_4, A_5)$  fixes a line in some  $\mathrm{Sym}^N$  with  $N \leq 12$  (the classical binary polyhedral invariants), which is likewise excluded. By the classification of algebraic subgroups of  $\mathrm{GL}_2$ , the group contains  $\mathrm{SL}_2$ .  $\square$

**Proposition 129** (no  $\sigma\delta$ -integrability). *For every  $0 < q < 1$  there is no  $B \in \text{gl}_2(\mathbb{C}(z))$  and no  $c \in \mathbb{C}(z)$  with  $\theta(A) = \sigma(B)A - AB + cA$ , where  $A = \begin{pmatrix} 0 & 1 \\ q & a \end{pmatrix}$  and  $\theta = z\partial_z$ .*

*Proof.* Writing  $B = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}$ , the entries give  $\gamma = -q\beta_1$ ,  $\delta = \alpha_1 + a\beta_1 - c$ , and eliminating  $\alpha$  (the central terms  $c$  cancel identically; verified symbolically) leaves the necessary condition

$$qa\beta_3 = a_1(aa_1 - q)\beta_2 - a(aa_1 - q)\beta_1 + qa_1\beta + qz(a_1 + q^2a),$$

$\beta_k = \beta(Q^k z)$ ,  $a = a(z)$ ,  $a_1 = a(Qz)$ . A polynomial part of degree  $m \geq 0$  contributes top coefficient  $\propto \beta_m(1 - q^{2+2m}) \neq 0$  at degree  $m + 3$ , above the inhomogeneity: so  $\beta$  has no polynomial part. If  $\beta$  has a pole off the orbit  $z_a Q^{\mathbb{Z}}$ , the maximal-modulus pole appears in exactly one term (the slot  $z = \zeta Q^{-3}$ , coefficient  $qa \neq 0$  there), and the minimal-modulus pole likewise (slot  $z = \zeta'$ , coefficient  $qa_1$ ): both impossible; so the poles lie in the window  $\{z_a Q^{-1}, \dots, z_a Q^3\}$  ( $a(z_a Q^k) = (q+1)(1 - Q^k)$  vanishes only at  $k = 0$ ). Sweeping the slots  $z = z_a Q^{-4}, \dots, z_a Q^{-1}$  kills the orders at  $z_a Q^{-1}, \dots, z_a Q^2$  one by one (each pole appears in a single term with nonvanishing coefficient), and the order at  $z_a Q^3$  is killed at the slot  $z = z_a Q$ ; coefficient  $a_1(aa_1 - q)$  with  $(aa_1 - q)(z_a Q) = (q+1)^2(1 - q^2)(1 - q^4) - q$ ; or, where that vanishes (one root  $q \approx 0.78027$ ), at the slot  $z = z_a Q^2$ ; coefficient  $a(aa_1 - q)$  with  $(aa_1 - q)(z_a Q^2) = (q+1)^2(1 - q^4)(1 - q^6) - q$  (one root  $q \approx 0.86753$ ); the two polynomials are coprime, so for every  $q \in (0, 1)$  at least one slot applies. Hence  $\beta \equiv 0$ , contradicting  $z(a_1 + q^2a) = z[(q+1)(1 + q^2) - 2q^3z] \neq 0$ . (Cross-check: at  $q = \frac{1}{3}$  the exact linear system for poles of order  $\leq 3$  in the window has  $\text{rank}(A) = 15$ ,  $\text{rank}[A|b] = 16$ .)  $\square$

**Corollary 130** (differential transcendence of  $G$ ). *For every  $0 < q < 1$  the function  $G(q, \cdot)$  satisfies no algebraic differential equation over  $\mathbb{C}(z)$ : the Hahn–Exton  $q$ -cosine is differentially transcendental.*

*Proof, via parametrized difference-Galois theory.* We apply the parametrized  $(\sigma, \theta)$ -difference Galois theory with derivation  $\theta = z\partial_z$ , over the difference-Galois formalism of [47] and in the form of [16, 3]. The determinant equation  $\sigma(y) = \det(A)y = qy$  has the solution  $y = z^{1/2}$  (indeed  $\sigma(z^{1/2}) = (Qz)^{1/2} = qz^{1/2}$ , as  $Q = q^2$ ), which is algebraic over  $\mathbb{C}(z)$ ; hence the parametrized difference Galois group of  $\sigma(y) = \det(A)y$  lies in  $\mathbb{C}^\times$ . This is exactly the hypothesis of Dreyfus–Hardouin–Roques [16, Prop. 2.9], which for a rank-2 system whose difference Galois group contains  $\text{SL}_2$  (here Theorem 128) gives the dichotomy: the parametrized group  $\text{Gal}^\theta$  either is conjugate into  $\text{GL}_2(\mathbb{C})$  (all solutions differentially algebraic) or contains  $\text{SL}_2(\tilde{\mathbb{C}})$  (all nonzero solutions differentially transcendental), and the *first* alternative holds if and only if there exists  $B \in \text{gl}_2(\mathbb{C}(z))$  with  $\theta(A) = \sigma(B)A - AB$  (their integrability condition (7);  $\theta(\det A) = 0$  since  $\det A = q$  is constant in  $z$ , so no trace term appears). By Proposition 129 no such  $B$  exists; indeed none exists even allowing a scalar twist  $cA$ . Therefore the second alternative holds:  $\text{Gal}^\theta \supseteq \text{SL}_2(\tilde{\mathbb{C}})$ , and every nonzero solution, in particular  $G(q, \cdot)$ , is differentially transcendental over  $\mathbb{C}(z)$ .  $\square$

The transcendence of the *number*  $q^*$  is a value-level statement, and the natural route (Nesterenko’s multiplicity method fed by Corollary 130) meets a precise structural obstruction, which we isolate.

**Proposition 131** (where the differential closure fails in the descent variable). *Write  $\delta = q\partial_q$ ,  $\theta = z\partial_z$ , and let  $u_j(q) := G(q, Q^j z^*(q))$  denote the orbit values along the diagonal  $z^*(q) = 2q(1 - q)$ .*

- (i) (orbit collapse) *The  $\sigma$ -orbit is not an obstruction to finiteness: the  $z$ -recurrence evaluated on the diagonal has coefficients  $a(Q^j z^*) = q + 1 - qQ^j z^* \in \mathbb{Q}(q)$ , so  $u_{j+2} = a(Q^j z^*)u_{j+1} - qu_j$  and every orbit value lies in  $\text{span}_{\mathbb{Q}(q)}\{u_0, u_1\}$  (verified for  $j \leq 3$  to 32 digits).*

(ii) (the  $E_2$ -coupled jet structure) *The  $q$ -derivative of  $G$  obeys the exact identity*

$$\delta G = (\theta^2 - \theta)G + \varphi(q)G - T, \quad \varphi(q) = \sum_{i \geq 1} \frac{iq^i}{1 - q^i}, \quad T = \sum_{k \geq 0} (-1)^k \left( \sum_{i > 2k} \frac{iq^i}{1 - q^i} \right) \frac{q^{k(k-1)} z^k}{(q; q)_{2k}}$$

(verified to 41 digits), where  $\varphi$  is the quasimodular Eisenstein weight ( $24\varphi = 1 - E_2$  up to normalization) and the correction  $T$  has coefficients smaller than  $G$ 's by the factor  $\sum_{i > 2k} iq^i / (1 - q^i) = O(kq^{2k})$ . Expanding these weights as Lambert series and interchanging the (absolutely convergent) sums exhibits  $T$  as an Eisenstein-weighted trace of the module over the forward  $\sigma$ -orbit:

$$T(q, z) = \sum_{d \geq 1} \frac{2q^d}{1 - q^d} (\theta G)(q, q^{2d}z) + \sum_{d \geq 1} \frac{q^d}{(1 - q^d)^2} G(q, q^{2d}z)$$

(verified to 41 digits): the entire non-modular content of  $\delta G$  is an orbit convolution of  $(G, \theta G)$  against  $E_2$ -type weights. On the diagonal the orbit values collapse by (i) into four generators  $u_0, u_1, \theta$ -analogues  $v_0, v_1$  (the  $v$ 's obey the inhomogeneous transported recurrence  $v_{j+2} = a_j v_{j+1} - qv_j - qQ^j z^* u_{j+1}$ ), but the collapse coefficients accumulate into transfer-product series  $A, B, C, D \in \mathbb{Z}[[q]]$  (integrality is immediate: transfer data and Lambert weights lie in  $\mathbb{Z}[[q]]$ ), which are not of any classical type: they satisfy no relation over  $\text{span}\{1, \varphi, \sum \sigma_3(n)q^n\}$  with rational-function coefficients of degree  $\leq 8$ , and no linear  $q$ -ODE of order  $\leq 3$  and degree  $\leq 6$  (exact linear algebra on 45 coefficients). The identity relocates, and does not remove, the obstruction. Since  $\delta$  and  $\theta$  commute, each application of  $\delta$  raises the  $z$ -jet order by two:  $\delta^n S$  involves  $\theta^{2n} G|_{z=z^*}$  with unit leading coefficient.

(iii) (the actual obstruction) *The  $z$ -jets do not close:  $\theta^2 G$  and  $\delta^2 G$  escape the rank-6 module of Proposition 123, and by Corollary 130 the tower  $\{\theta^m G\}_m$  satisfies no algebraic relation over  $\mathbb{C}(z)$ . A finitely generated  $\delta$ -closed  $\mathbb{C}(q)$ -algebra containing  $S$  would force an algebraic relation over  $\mathbb{C}(q)$  among the diagonal restrictions  $\{\theta^m G(q, z^*(q))\}_{m \leq M}$  for some finite  $M$ ; a statement equivalent to  $S$  being differentially algebraic in  $q$ , for which there is no supporting structure and proof-grade negative evidence across a wide window: no algebraic differential equation at profiles up to  $(2, 12, 3)$ ,  $(3, 8, 3)$ ,  $(4, 5, 6)$ ,  $(5, 4, 8)$ ,  $(6, 3, 15)$ ,  $(8, 2, 35)$ ,  $(12, 1, 120)$  in (order, degree,  $\deg_q$ ), by full column rank on 2000 coefficients modulo a prime (`beta2_diffalg_search.py`, `beta2_ade_sweep.py`;  $E_2/q$ -Chazy control recovered).*

Thus the Nesterenko route closes if and only if  $S$  is differentially algebraic in  $q$ ; the obstruction is carried by the  $z$ -jet tower; whose function-level independence is exactly Corollary 130; and not by the  $\sigma$ -orbit, which collapses to rank two.

*Remark 132* (the value-level descent as a joint multiplicity estimate). Nesterenko's method derives transcendence of a value from three ingredients: a finite ring closed under the relevant operator, a *multiplicity estimate* bounding  $\text{ord } P(\text{generators}) \leq c(\deg P)^\kappa$ , and routine analytic size/Siegel-lemma constructions. Theorem 128 and Proposition 129 furnish precisely the nondegeneracy a multiplicity estimate consumes (no invariant subvariety; no first integral). But Proposition 131 shows the first ingredient fails in the descent variable  $q$ : the finiteness of the  $\beta_2$  module is intrinsically *joint* in  $(\sigma_z, \delta_q)$ , and  $\sigma_z$  is a non-infinitesimal  $q$ -lattice shift, outside the Pfaffian/differential multiplicity theories of Nesterenko and Binyamini. Thus the transcendence of  $q^* = \beta_2^{-2}$  reduces to a *joint  $q$ -difference-differential multiplicity estimate* for the rank- $\leq 6$  module at  $(q^*, z^*)$ : a bound  $\text{ord}_{(q^*, z^*)} P(G, \theta G, \delta G) \leq cN^\kappa$  ( $N = \deg P$ ) in the local ring, uniform along the  $\sigma$ -orbit. Such an estimate for finite  $(\sigma, \delta)$ -modules is not available in the literature; it is the sole non-routine input remaining, and the precise open problem on which the (ir)rationality of  $\beta_2$  now turns.

*Remark 133* (the obstruction inside  $q$ -Siegel–Shidlovskii theory). The same wall appears, quantified, in the value-transcendence theory built on Siegel’s method. The approach of Amou–Matala-aho–Väänänen [1] (and the Matala-aho-school Baker-type refinements) proves non-vanishing of values of Poincaré-type  $q$ -difference solutions at algebraic points under two hypotheses: the system is Poincaré-type at 0 (met here:  $A(0)$  is invertible,  $\det A = q$ ), and a *Siegel margin*  $\gamma < \Gamma$  holds, where  $\gamma$  is the ratio of the logarithmic height of the cleared denominators to that of the analytic decay. For the Hahn–Exton class that ratio is exactly  $\gamma = \rho = \log(t/s)/(2 \log t) \leq \frac{1}{2}$  of Proposition 114: the double Pochhammer  $(q; q)_{2k}$  is the factor 2; so the margin *fails*, and the theory does not apply. Thus Conjecture Z (transcendence of the Hahn–Exton zeros at algebraic base, which by  $z^* = z_1(q^*)$  implies  $\beta_2$  transcendental) reduces to extending  $q$ -Siegel–Shidlovskii past the margin for a class that is additionally irregular at  $\infty$ . The connection theory of this paper supplies the irregular- $\infty$  data explicitly; the margin itself is the value-theoretic face of the multiplicity estimate of Remark 132, and is equal in strength to  $\beta_2$ -transcendence, not weaker. A detailed program is recorded in the repository ([paper/journal/qSS\\_program.md](#)).

One further mechanism deserves record, because it isolates which half of the Siegel scheme is actually blocked. Under the rationality hypothesis all tail power sums  $\tau_p = \sum_{k \geq 2} z_k^{-p} = \sigma_p - z^{*-p}$  are rational, and they are the moments of a positive measure with infinite support (unit atoms at  $1/z_k$ ); hence every Hankel determinant  $\det[\tau_{i+j+2}]_{i,j \leq n}$  is *strictly positive for free*; no auxiliary nonvanishing argument is needed; and, being a positive rational, is bounded below by the reciprocal of its denominator ( $\leq t^{8n^3/3+O(n^2)}$ ). Analytically these determinants decay like  $Q^{n^3/3(1+o(1))}$  (atoms at  $\approx Q^{k-1}$ ; verified: strictly positive with super-quadratic exponent growth at  $q^*$ ). The comparison  $2 \log(t/s) > 8 \log t$  fails for every rational  $q$ : the positivity route conserves the ledger at a factor 4. The instructive point is that *nonvanishing was never the bottleneck* of any construction in this ledger; positivity supplies it gratis here; the entire difficulty of the problem resides in the size-versus-height comparison. Nor is there an Apéry-type denominator miracle hiding in that comparison: the *measured* reduced denominators of the partial sums  $S_N(s/t)$  equal 96–100% of the naive clearing  $\prod_{i \leq 2N} (t^i - s^i)$  in logarithmic size, with the ratio tending to 1 as  $N$  grows (exact rational arithmetic,  $N \leq 12$ , at  $q = \frac{1}{2}$  and  $\frac{2}{5}$ ); the cancellation is an  $O(N)$  boundary effect in an  $O(N^2)$  exponent. The arithmetic of the ledger is exactly its worst-case costing.

*Remark 134* (backward-orbit normal form of the zero condition). The zero condition admits a compression into a single explicit rational sequence. If  $G(q, z_1) = 0$ , the three-term recurrence along the backward  $\sigma$ -orbit,  $u_{-m} = (a(Q^{-m} z_1) u_{-m+1} - u_{-m+2})/q$  with  $u_n = G(q, Q^n z_1)$ , loses one solution constant:  $u_{-m} = u_1 r_m(q)$  with  $r_0 = 0$ ,  $r_1 = -1/q$ , and  $r_m \in \mathbb{Q}(q, z_1)$  computable by the same recurrence. The asymptotics of Theorem 121 then give the exact evaluation

$$\lim_{m \rightarrow \infty} r_m(q) Q^{m(m+1)/2} (-z_1)^{-m} = \frac{\theta_Q(-z_1)}{(q; q)_\infty G(q, Q z_1)},$$

with geometric rate  $Q^m$  and  $q$ -Gevrey correction series  $\sum_j b_j z_1^{-j} Q^{mj}$  (verified to 17 digits at  $q^*$  by  $m = 24$ ). Under the hypothesis  $q^* = s/t$  the normalized terms are rationals of height  $(st)^{m^2+O(m)}$  approximating the limit to  $O(Q^m)$ ; accelerated forms that cancel  $J$  correction terms gain decay  $Q^{mJ}$  but pay the quadratic heights of the  $b_j$ , and the arithmetic ratio stays  $\leq \frac{1}{2}$ ; the balance of Proposition 114 once more. The normal form is recorded as the sharpest single-sequence statement of what any future method must exploit: the entire rationality hypothesis is equivalent to the arithmetic of this one explicit sequence. The  $\theta$ -jet admits the same normal form one level up: along the backward orbit,

$$\frac{(\theta G)(q, Q^{-m} z_1)}{G(q, Q^{-m} z_1)} - m \longrightarrow -\frac{z_1 \theta'_Q(-z_1)}{\theta_Q(-z_1)} \quad (\text{rate } Q^m; \text{ verified to 10 digits at } q = 0.3),$$

an exact theta log-derivative: each backward step increments  $\theta \log \theta_Q(-z)$  by exactly one (quasi-periodicity), and the  $\hat{g}$ -correction vanishes in the limit.

The *forward* orbit admits the complementary normal form, in the language of rational dynamics. The Riccati relation gives the explicit continued fraction  $q/(a(z) - q/(a(Qz) - \dots))$ , which by Pincherle's theorem [33] converges to the minimal-solution ratio  $q H(Qz)/H(z)$ ; the zero condition  $G(q, z_1) = 0$  forces the *dominant* ratios  $w_n := G(q, Q^{n+1}z_1)/G(q, Q^n z_1)$  to obey  $w_1 = a(z_1)$  exactly and thereafter the rational map

$$w_{n+1} = a(Q^n z_1) - \frac{q}{w_n},$$

so that under the hypothesis  $q, z_1 \in \mathbb{Q}$  the entire forward orbit is an explicit  $\mathbb{Q}$ -orbit converging to the attracting fixed point 1 with  $w_n - 1 = \Theta(Q^n)$  (verified to 25 digits at  $q^*$ ). Its heights compound quadratically ( $t^{O(n^2)}$ ), so the Liouville comparison  $Q^n$  vs.  $t^{-n^2}$  is again consistent: both normal forms; backward theta-growth and forward rational dynamics; conserve the balance of Proposition 114, and together they exhaust the one-orbit reformulations of the zero condition.

Combining the two orbits with the  $q$ -Wronskian makes the relation-count of the rationality hypothesis explicit. Under  $G(q, z_1) = 0$  the unknown transcendental data reduces to three numbers ( $g_1 := G(q, Qz_1)$ ,  $H(q, z_1)$ , and the backward limit  $\rho_\infty$ ), bound by exactly two relations:  $g_1 H(q, z_1) = 1 - q$  (the  $q$ -Wronskian at the zero) and  $\rho_\infty g_1 = \theta_Q(-z_1)/(q; q)_\infty$  (the backward normal form). Eliminating  $g_1$ ,

$$\frac{\rho_\infty}{H(q, z_1)} = \frac{\theta_Q(-z_1)}{(1 - q)(q; q)_\infty}$$

(verified to 32 digits at  $q^*$ ), with modular right-hand side. Three unknowns, two relations: the rationality hypothesis is consistent at this level, and a proof requires precisely one further independent relation; the multiplicity statement of Remark 132, in yet another equivalent form.

*Remark 135* (truncation roots, and the exact deficit). A scheme not covered by the lattice ledger prices the wall in its simplest form. Clearing the truncation of  $S$  gives integer polynomials  $P_N = (q; q)_{2N} \sum_{k \leq N} (-2)^k q^{k^2} (1 - q)^k / (q; q)_{2k} \in \mathbb{Z}[q]$  of degree  $2N^2 + N$ , whose heights are only  $e^{O(N)}$  (the pentagonal sparsity of  $q$ -Pochhammer coefficients; measured  $\log_2 H = 2.6, 4.6, \dots, 12.1$  for  $N = 2, \dots, 7$ ), and whose root nearest  $q^*$  satisfies  $|q_N - q^*| \asymp q^{*(N+1)^2}$  (verified  $N \leq 7$ ). If  $q^* = s/t$  then  $|P_N(q^*)| \geq t^{-(2N^2+N)}$ , while analytically  $|P_N(q^*)| = e^{O(N)} q^{*(N+1)^2}$ ; contradiction would require  $q^* < t^{-2}$ , impossible for  $q^* = s/t \geq 1/t$ . The deficit is exactly the factor 2 in the clearing degree; the double Pochhammer once more, now in  $s$ -independent form. The same computation shows that an equivalent representation of the root condition with *single*-Pochhammer denominators (clearing degree  $\sim N^2/2$ ) would yield  $q^* \neq s/t$  for all  $s < \sqrt{t}$ . No such representation exists within the Rogers–Ramanujan/Bailey framework [21]: those transformations require the argument to be a  $q$ -power, and the argument here is the catalytic point  $z = 2q(1 - q)$ . In this precise sense the obstruction to  $\beta_2$ 's irrationality is the non- $q$ -power (catalytic) argument itself.

The catalytic locus is moreover *seat-independent*. Writing  $S(q) = {}_1\phi_1(0; q; Q, z^*)$  ( $Q = q^2$ ), the parameter–argument transformation  ${}_1\phi_1(0; b; Q, x) = \frac{(x; Q)_\infty}{(b; Q)_\infty} {}_1\phi_1(0; x; Q, b)$  gives the equivalent zero condition

$$T(q) := {}_1\phi_1(0; 2q(1 - q); q^2, q) = \sum_{k \geq 0} \frac{(-1)^k q^{k^2}}{(q^2; q^2)_k (2q(1 - q); q^2)_k} = 0$$

(verified to 51 digits; the prefactor cannot vanish since  $z^* < 1$ ), in which the argument is the  $q$ -power  $q$  and the catalytic value sits in the *parameter*; equivalently,  $q^*$  is the point where the Hahn–Exton  $q$ -Bessel of  $q$ -dependent order  $\nu(q) = \log_{q^2}(2q(1 - q)) - 1$  vanishes at the fixed argument  $q^{-1/2}$ . The clearing degree of  $T$ 's truncations is again  $\sim 2N^2$ : the deficit is invariant under the swap, and in

either seat it is the non-special value  $2q(1 - q)$  that lies outside every Rogers–Ramanujan/Bailey lattice. The well-poised generalization does not help: a WP-Bailey specialization producing a single-Pochhammer form would require  $2q(1 - q)$  to be a  $q$ -power (the only special-value type such summations admit), whereas at  $q = q^*$  one has  $q^* < 2q^*(1 - q^*) < 1$ , strictly between consecutive  $q$ -powers (the two loci where  $2q(1 - q)$  is a  $q$ -power,  $q = \frac{1}{2}$  and  $q = \frac{2}{3}$ , both miss  $q^*$ ). The catalytic argument is thus  $q$ -power-free in every seat, which is the transformation-theoretic content of the  $\rho = \frac{1}{2}$  wall.

**Proposition 136** (effective non-rationality of  $q^*$ ). *The least positive root  $q^*$  of  $S$  admits no rational representation  $s/t$  with  $t \leq 10^{1044}$ . Consequently, if  $\beta_2 = u/v$  in lowest terms then  $u > 10^{522}$ .*

*Proof.* All steps are certified interval arithmetic (outward rounding; `mpmath.iv`; script `beta2_effective_irrational` with the series tail bounded by  $(q; q)_{2k} \geq (1 - 2q)/(1 - q) > 0$  for  $q < \frac{1}{2}$  and term ratios  $< \frac{1}{2}$  beyond the truncation. (i)  $S > 0$  on  $[0, 0.40]$  (400 certified chunks). (ii)  $S' \leq -1$  on  $[0.40, 0.46]$  (600 certified chunks;  $S'$  evaluated term-by-term with the logarithmic-derivative weights), so  $S$  has at most one zero in  $[0.40, 0.46]$ . (iii) with  $c$  the 2090-digit truncation of  $q^*$ ,  $a = c - 2 \cdot 10^{-2090}$  and  $b = c + 2 \cdot 10^{-2090}$ , certified evaluation gives  $S(a) > 0 > S(b)$  (working precision 2130 digits, series tail enclosed in  $\pm 10^{-2125}$ ); by (i)–(ii),  $q^* \in [a, b]$ . (iv) the interval  $[a, b]$  contains the rational  $s_0/t_0$  (Stern–Brocot construction) with  $9.5 \cdot 10^{1044} < t_0 < 10^{1045}$ , verified by exact rational comparison, and certified  $S(s_0/t_0) \neq 0$ . If  $m/n \in [a, b]$  with  $n \leq 10^{1044}$  and  $m/n \neq s_0/t_0$ , then  $|m/n - s_0/t_0| \geq 1/(nt_0) > 10^{-2089} > b - a = 4 \cdot 10^{-2090}$ , impossible; so  $m/n = s_0/t_0$ , whence  $n = t_0 > 10^{1044}$ , a contradiction. Thus no rational with denominator  $\leq 10^{1044}$  lies in  $[a, b]$  at all, in particular none equals  $q^*$ . For  $\beta_2 = u/v$  in lowest terms,  $q^* = v^2/u^2$  in lowest terms, so  $u^2 > 10^{1044}$ .  $\square$

## D Proof of the block assembly (M3')

**Status.** This appendix proves the assembly step (M3) of Theorem 66 in the corrected form (26). It is a *hand proof*. Two independent adversarial reviews reconstructed it and found it correct; the second also compared an independent tuple enumerator against the model, with no mismatch. The proof is *not* formalised. The Lean development formalises the objects it is about: the site costs, the cut predicate and  $\ell_R^{\text{true}}, c^{\text{true}}$  (`Realisation.lean`, `CorrectedSpan.lean`), the junction-cut characterisation (`RJLemmaJ.lean`), and the metric identity  $|g| = \ell_R^{\text{true}}(g) + 2c^{\text{true}}(g)$  (`metricAll`). The proof is supported by the computational checks recorded in §7.1: with  $y$  symbolic, all 816 coefficients with  $\ell_R + 2c \leq 31$  agree with exhaustive enumeration in all three sectors ( $5.03 \times 10^6$  elements), and a deliberately corrupted model is detected; at  $y = q$  the model agrees to  $x^{26}$  and at  $y = q^2$  to  $x^{22}$ . The checks are evidence, not part of the proof.

In this appendix we write  $k, \varepsilon, \delta$  for the marker data  $k^*, \varepsilon^*, \delta^*$  of the main text, and `Elt`, `SameElt` for the element type of the Lean development and its equality of the four normal-form fields [6, Theorem “Normal form and lamp model”, `thm:normal-form`].

### D.1 Setting and statement

**Elements.** Edges and sites are indexed by  $\mathbb{Z}$ . Site  $s$  lies between edge  $s - 1$  and edge  $s$ . For  $k \in \mathbb{Z}$ , the *travel indicator* is

$$f_j^{(k)} = \begin{cases} +1, & 0 \leq j < k, \\ -1, & k \leq j < 0, \\ 0, & \text{otherwise.} \end{cases}$$

Let  $\mathcal{G}$  be the set of tuples  $(k, \varepsilon, \delta, d)$  such that

- $k \in \mathbb{Z}$ ,  $\varepsilon \in \{\pm 1\}$  and  $\delta \in \{0, 1\}$ ;
- $d : \mathbb{Z} \rightarrow \mathbb{Z}$  has finite support;
- $d_j \equiv f_j^{(k)} \pmod{2}$  for all  $j$ .

**Lemma 137** (classes). *The map  $g \mapsto (g.\text{kstar}, g.\text{eps}, g.\text{delta}, g.\text{d})$  induces a bijection from SameElt-classes of  $\text{Elt}$  onto  $\mathcal{G}$ . The functions  $\ell_R^{\text{true}}$  and  $c^{\text{true}}$  of `CorrectedSpan.lean` are constant on classes, so they are functions on  $\mathcal{G}$ .*

*Proof.* SameElt is by definition equality of the four fields, so the induced map is well defined and injective. The fields of an  $\text{Elt}$  satisfy the conditions defining  $\mathcal{G}$ : `hpar` gives the parity condition, and `hsupp` gives finite support.

For surjectivity, take  $(k, \varepsilon, \delta, d) \in \mathcal{G}$ . Put  $\text{supp} = \text{supp } d \cup \{j : f_j^{(k)} \neq 0\}$ . This set is finite, and it gives an  $\text{Elt}$  with the prescribed fields.

The only field not determined by the class is `supp`. It enters  $\ell_R^{\text{true}}$  and  $c^{\text{true}}$  only through  $\text{occTrue} = \text{supp} \cap \{j : d_j \neq 0 \vee f_j \neq 0\}$ . By `hsupp`, this intersection equals  $\{j : d_j \neq 0 \vee f_j \neq 0\}$ , which is independent of `supp`. All the other ingredients are functions of  $(k, \varepsilon, \delta, d)$ : `mu`, `siteCost`, `cut` and `ShieldFires`.  $\square$

From now on we write  $g = (k, \varepsilon, \delta, d) \in \mathcal{G}$  and  $f = f^{(k)}$ .

**Local data.** These are transcribed from `Realisation.lean`. Write  $[\cdot]$  for the Iverson bracket.

$$\begin{aligned} \alpha_s &= d_{s-1} - [s = 0] + \varepsilon[s = k][\delta = 0], & \beta_s &= d_s - \varepsilon[s = k][\delta = 1], \\ \Phi_s &= f_{s-1} + [s = 0] - [s = k][\delta = 0], & \text{site}(s) &= \max(|\alpha_s|, |\beta_s|), \\ \mu_j &= \begin{cases} 2, & d_j = f_j = 0, \\ \max(|d_j|, |f_j|), & \text{otherwise,} \end{cases} & \text{cut}(s) &\iff \alpha_s = \beta_s = \Phi_s = 0. \end{aligned}$$

The term  $-[s = 0]$  is the virtual arrival `vArr`. It is the “ $\pm 1$  correction” at the junction.

**Corrected span.** These are transcribed from `CorrectedSpan.lean`. Let  $\text{occ} = \{j : d_j \neq 0 \vee f_j \neq 0\}$ . Then

$$\begin{aligned} A &= \begin{cases} \min(0, \min \text{occ}), & \text{occ} \neq \emptyset, \\ 0, & \text{otherwise,} \end{cases} & B &= \begin{cases} \max(-1, \max \text{occ}), & \text{occ} \neq \emptyset, \\ -1, & \text{otherwise,} \end{cases} \\ \ell_R^{\text{true}}(g) &= \sum_{j=A}^B \mu_j + \sum_{s=A}^{B+1} \text{site}(s), & c^{\text{true}}(g) &= \#\{s : A < s < B + 1, \text{cut}(s)\} + [\text{SF}(g) \wedge \text{cut}(0)], \end{aligned}$$

where the *boundary shield* is

$$\text{SF}(g) \iff k = 0 \wedge \delta = 0 \wedge (\forall j < 0 : d_j = 0) \wedge (\exists j \geq 0 : d_j \neq 0).$$

**Series.** Put  $q = x^2$  and

$$\mathfrak{g} = \sum_{L \geq 1} q^L y^{L-1}.$$

The bulk vector  $P = (P_b)_{b \geq 1}$  is the solution of

$$P_b = 2q^b \left( q^b + \sum_{a \geq 1} K_{\mathfrak{g}}(a, b) P_a \right), \quad K_{\mathfrak{g}}(a, b) = q^{\max(a, b)} + q^{a+b} \mathfrak{g}, \quad (39)$$

which exists and is unique by Lemma 140. From it we form

$$S = \sum_{b \geq 1} q^b P_b, \quad B_0 = 1 + \mathfrak{g}S, \quad B_{0z} = 1 + y \mathfrak{g}S,$$

and, for  $s \geq 0$ ,

$$\mu_s^{(y)} = B_0 x^{2s+1} + \sum_{b \geq 1} \frac{P_b}{2} (x^{\max(2b-1, 2s+1)} + x^{\max(2b+1, 2s+1)}), \quad (40)$$

$$E_s^{(y)} = B_0^{[s]} x^{2s} + B_0 x^{2s+2} + \sum_{b \geq 1} P_b (x^{\max(2s, 2b)} + x^{\max(2s+2, 2b)}), \quad B_0^{[0]} = B_{0z}, \quad B_0^{[s]} = B_0 \quad (s \geq 1), \quad (41)$$

$$\lambda_s^{(y)} = E_s^{(y)} + 2\mu_s^{(y)}. \quad (42)$$

Let  $D_e = \text{diag}(e_s)_{s \geq 0}$  with  $e_s = 2q^{1+s}$ , let  $K(s, t) = q^{\max(s, t)}$ , and let  $\mathcal{T} = D_e K$ . Define

$$\mathcal{W}_+ = x^{-1} \langle \mu^{(y)}, (I - \mathcal{T})^{-1} D_e \lambda^{(y)} \rangle, \quad \mathcal{W}_- = x^{-1} \langle \tfrac{1}{2} E^{(y)}, (I - \mathcal{T})^{-1} D_e \lambda^{(y)} \rangle,$$

and let  $\mathcal{W}_0$  be the branch sum (49) of §D.8, a double sum over the left and right chain kinds (summable, not finite).

**Theorem 138** ((M3')). *The following identity holds in  $\mathbb{Z}[y][[x]]$ :*

$$\sum_{[g] \in \text{Elt/SameElt}} x^{\ell_R^{\text{true}}(g)} y^{c^{\text{true}}(g)} = \sum_{g \in \mathcal{G}} x^{\ell_R^{\text{true}}(g)} y^{c^{\text{true}}(g)} = \mathcal{W}_+ + \mathcal{W}_- + \mathcal{W}_0.$$

*The three terms are the contributions of the sectors  $k > 0$ ,  $k < 0$  and  $k = 0$ . Moreover*

$$\mathcal{W}_+ + \mathcal{W}_- = \frac{1}{2x} \langle \lambda^{(y)}, (I - \mathcal{T})^{-1} D_e \lambda^{(y)} \rangle,$$

*and each of  $\mathcal{W}_+$ ,  $\mathcal{W}_-$ ,  $\mathcal{W}_0$  lies in  $\mathbb{Z}[y][[x]]$ .*

**Corollary 139.** *By metricAll (Lean:  $|g| = \ell_R^{\text{true}}(g) + 2c^{\text{true}}(g)$  for every  $\text{Elt}$ , all of which are reachable), the specialisation  $y = x^2$  of Theorem 138 is the growth series  $\sum_{[g]} x^{|g|}$  of the group modelled by  $\text{Elt}$ . This group is  $W_{\text{univ}}$ . For a particular triangle group  $W_T$ , the growth series agrees with this one only up to the depth at which  $\rho_T$  stops being faithful on  $W_{\text{univ}}$  [6, Theorem “Effective universality”, thm:effective-universality]. For the (1, 2) triangle the first deviation is at depth 33 [6, thm:universality-sharp].*

The first equality of Theorem 138 is Lemma 137. The rest of this appendix proves the second.

## D.2 Convergence and integrality

We use the  $x$ -adic topology on  $\mathbb{Z}[y][[x]]$ . A family of series is *summable* if only finitely many of its members have a nonzero coefficient of any given  $x^n$ . We write  $\text{ord}$  for the  $x$ -adic order.

**Lemma 140.** (a) For each  $n$ , only finitely many  $g \in \mathcal{G}$  have  $\ell_R^{\text{true}}(g) = n$ , and for these  $c^{\text{true}}(g) \leq n + 1$ . So the left side of Theorem 138 is a well-defined element of  $\mathbb{Z}_{\geq 0}[y][[x]]$ .

(b) Equation (39) has a unique solution with  $P_b \in \mathbb{Z}[y][[x]]$  and  $\text{ord } P_b \geq 4b$ . Every  $P_b$  is divisible by 2.

(c)  $(I - \mathcal{T})^{-1} = \sum_{n \geq 0} \mathcal{T}^n$  converges entrywise, and every pairing in the definitions of  $\mathcal{W}_{\pm}$  is a summable sum.

(d)  $\mu^{(y)}$ ,  $E^{(y)}$ ,  $\lambda^{(y)}$ ,  $\mathcal{W}_+$ ,  $\mathcal{W}_-$  and  $\mathcal{W}_0$  all have coefficients in  $\mathbb{Z}[y]$ .

*Proof.* (a) Every edge of  $[A, B]$  has  $\mu_j \geq 1$ , so  $B - A + 1 \leq n$ . Moreover  $|d_j| \leq \mu_j \leq n$  and  $|k| \leq B - A + 1$ . This leaves finitely many choices of  $(k, \varepsilon, \delta, d)$ . Also  $c^{\text{true}} \leq \#\{s : A < s < B + 1\} + 1 = (B - A + 1) + 1 \leq n + 1$ .

(b) Let  $\Psi(P)_b = 2q^b(q^b + \sum_a K_{\mathfrak{g}}(a, b)P_a)$ . The entry  $K_{\mathfrak{g}}(a, b)$  has order  $\geq 2 \max(a, b) \geq 2a$ , so  $\sum_a$  is summable whenever  $\text{ord } P_a \geq 0$ . If two inputs satisfy  $\text{ord}(P_a - P'_a) \geq m$  for all  $a$ , then  $\text{ord}(\Psi(P)_b - \Psi(P')_b) \geq 2b + 2 + m \geq m + 4$ . So  $\Psi$  is a contraction. Starting from  $P = 0$ , the iterates therefore converge to the unique fixed point. Every iterate has  $\text{ord } P_b \geq 4b$  and carries the factor 2. The coefficients lie in  $\mathbb{Z}[y]$ , since the coefficient of  $x^n$  in  $\mathfrak{g}$  is 0 or  $y^{n/2-1}$ .

(c)  $\mathcal{T}(s, t) = 2x^{2s+2+2 \max(s, t)}$  has order  $\geq 2s + 2t + 2$ . So  $(\mathcal{T}^n)(s, t)$  has order  $\geq 2n + 2t$ , the Neumann series converges, and for fixed  $s$  the sums over  $t$  are summable. By (b),  $\text{ord } \mu_s^{(y)}, \text{ord } E_s^{(y)} \geq 2s$ , so the outer pairings are summable as well.

(d)  $\mu_s^{(y)}$  is integral because  $P_b/2$  is integral, by (b). For  $\mathcal{W}_{\pm}$ , factor  $D_e = 2D'$  with  $D' = \text{diag}(q^{1+s})$ . The push-through identity gives

$$(I - D_e K)^{-1} D_e = D_e (I - K D_e)^{-1} = 2D' (I - K D_e)^{-1}.$$

Hence

$$\mathcal{W}_- = x^{-1} \langle \frac{1}{2} E^{(y)}, 2D' (I - K D_e)^{-1} \lambda^{(y)} \rangle = \langle E^{(y)}, (x^{-1} D') (I - K D_e)^{-1} \lambda^{(y)} \rangle,$$

and  $x^{-1} D' = \text{diag}(x^{2s+1})$  is integral. **So  $\mathcal{W}_-$  is integral because  $D_e$  carries the factor 2.** The  $\frac{1}{2}$  on  $E^{(y)}$  is cancelled by that factor, and  $E^{(y)}$  itself is not divisible by 2: its  $s = 0$  entry has constant term 1. The same factorisation shows  $\mathcal{W}_+ \in \mathbb{Z}[y][[x]]$ , with the factor 2 to spare.  $\mathcal{W}_0$  is a summable double sum of products  $x^j w_L w_R$  with weights from 1,  $\mathfrak{g}S$ ,  $y\mathfrak{g}S$  and  $P_b/2$ , as (49) shows.  $\square$

All rearrangements below are of families of monomials with nonnegative integer coefficients, and each family is summable by Lemma 140(a). So they are legitimate in  $\mathbb{Z}[y][[x]]$ .

## D.3 Decomposition of an element: chains and the clamp analysis

**Chains.** A *chain* is a finitely supported  $c : \mathbb{Z}_{\geq 1} \rightarrow 2\mathbb{Z}$ . Its *length* is  $N(c) = \max\{i : c_i \neq 0\}$ , with  $N(0) = 0$ . There are three kinds of chain:

- *empty* (kind E):  $N = 0$ ;
- *gap-opening* (kind Z):  $N \geq 1$  and  $c_1 = 0$ ;
- *deposit-opening* (kind  $N_{\sigma, b}$ ):  $c_1 = 2\sigma b$  with  $\sigma = \pm 1$  and  $b \geq 1$ .

Given  $g = (k, \varepsilon, \delta, d)$ , put  $k_- = \min(0, k)$  and  $k_+ = \max(0, k)$ . The *left chain* and the *right chain* of  $g$  are

$$c_i^L = d_{k_- - i}, \quad c_i^R = d_{k_+ - 1 + i} \quad (i \geq 1).$$

Both consist of even deposits, since  $f = 0$  outside  $[k_-, k_+)$ . The *travel deposits* are  $d_j$  for  $k_- \leq j < k_+$ . There are  $|k|$  of them, all odd. Write  $\ell = N(c^L)$  and  $r = N(c^R)$ .

**Lemma 141** (bijection). *The map*

$$g \mapsto (k, \varepsilon, \delta, (d_j)_{k_- \leq j < k_+}, c^L, c^R)$$

*is a bijection from  $\mathcal{G}$  onto the set of tuples consisting of  $k \in \mathbb{Z}$ ,  $\varepsilon$ ,  $\delta$ , a word of  $|k|$  odd integers, and two chains.*

*Proof.* The three index sets  $\{j < k_-\}$ ,  $[k_-, k_+)$  and  $\{j \geq k_+\}$  partition  $\mathbb{Z}$ , and the parity condition says exactly that the deposits are odd on the middle set and even off it.  $\square$

**Lemma 142** (span; clamp analysis). *For every  $g \in \mathcal{G}$ ,*

$$A = k_- - \ell, \quad B = k_+ - 1 + r.$$

*Hence:*

(i) *The edges of  $[A, B]$  are the  $\ell$  left-chain edges, the  $|k|$  travel edges and the  $r$  right-chain edges.*

(ii) *The site range  $[A, B + 1]$  consists of the following sites:*

- *the junction sites  $k_-$  and  $k_+$ , which coincide when  $k = 0$ ;*
- *the  $|k| - 1$  travel-interior sites  $k_- < s < k_+$ ;*
- *the chain-interior sites:  $A < s < k_-$  on the left and  $k_+ < s < B + 1$  on the right;*
- *the outer sites  $A$  and  $B + 1$ , but only when  $\ell > 0$ , resp.  $r > 0$ .*

(iii) *The junction site  $k_-$  lies in the open interval  $(A, B + 1)$  if and only if  $\ell > 0$  or  $k < 0$ , i.e.  $k_- < k_+$ . Symmetrically,  $k_+$  lies in it if and only if  $r > 0$  or  $k > 0$ . The exception is  $k = 0$ : there site 0 lies in the open interval if and only if  $\ell > 0$  and  $r > 0$ .*

*Proof.*  $\text{occ}$  consists of the travel range  $[k_-, k_+)$ , the edges  $k_- - i$  with  $c_i^L \neq 0$ , and the edges  $k_+ - 1 + i$  with  $c_i^R \neq 0$ .

*Case  $k > 0$ .* Here  $0 \in \text{occ}$ . So  $\min \text{occ} = -\ell \leq 0$  (it is 0 when  $\ell = 0$ ), and the clamp  $\min(0, \cdot)$  is inactive. Likewise  $\max \text{occ} = k - 1 + r \geq 0 > -1$ , and the clamp  $\max(-1, \cdot)$  is inactive.

*Case  $k < 0$ .* The argument is symmetric, using  $-1 \in \text{occ}$ .

*Case  $k = 0$ .* Now  $\text{occ} = \{-i : c_i^L \neq 0\} \cup \{i - 1 : c_i^R \neq 0\}$  and the clamps are active:

- If  $\ell = 0$ , then  $\text{occ} \subseteq [0, \infty)$  or  $\text{occ} = \emptyset$ , and either way  $A = 0 = -\ell$ .
- If  $r = 0$ , then  $B = -1 = r - 1$ .
- If  $\ell > 0$ , resp.  $r > 0$ , the unclamped values  $-\ell$ , resp.  $r - 1$ , already lie on the correct side of the clamp.

So the formula for  $A$  and  $B$  holds in all cases.

Items (i) and (ii) now follow by listing  $[A, B]$  and  $[A, B + 1]$ . When  $\ell = 0$ , the outer site  $A$  coincides with the junction site  $k_-$ , and similarly on the right.

For (iii),  $A < k_-$  if and only if  $\ell > 0$ , and  $k_- < B + 1 = k_+ + r$  if and only if  $k_- < k_+$  or  $r > 0$ . The symmetric statement for  $k_+$  is proved the same way.  $\square$

So the clamps in  $\mathbf{ATrue}$  and  $\mathbf{BTrue}$  change nothing when  $k \neq 0$ . When  $k = 0$  they do exactly two things. First, they keep site 0 in the site range even when both chains are empty. Second, they put site 0 *outside* the open interval unless both chains are nonempty. That second effect is what SF partly undoes (§D.8).

**Lemma 143** (locality). *Let  $s$  be a site of  $[A, B + 1]$  other than a junction site. Then  $\alpha_s = d_{s-1}$ ,  $\beta_s = d_s$  and  $\text{site}(s) = \max(|d_{s-1}|, |d_s|)$ . Moreover:*

- (a) *A travel-interior site has odd cost  $\max(|d_{s-1}|, |d_s|)$ , and it is never a cut, because  $\Phi_s = f_{s-1} = \pm 1$ .*
- (b) *A chain-interior site is a cut if and only if  $d_{s-1} = d_s = 0$ . Every chain-interior site lies in  $(A, B + 1)$ .*
- (c) *An outer site has cost  $|c_N|$ , where  $c$  is the corresponding chain and  $N$  its length. It is not a cut, and it does not lie in  $(A, B + 1)$ .*

In particular  $\ell_R^{\text{true}}$  splits as a sum of the following parts:

- left-chain edges and left chain-interior and outer sites;
- travel edges and travel-interior sites;
- right-chain edges and right chain-interior and outer sites;
- the junction sites.

The cut count  $c^{\text{true}}$  splits the same way, except that the junction sites are governed by Lemma 142(iii) and by SF.

*Proof.* For  $s \notin \{0, k\}$  all the brackets vanish, so  $\alpha_s = d_{s-1}$ ,  $\beta_s = d_s$  and  $\Phi_s = f_{s-1}$ . For the cut criterion see also `RJLemmaJ.bulk_cut`.

For (a), if  $k_- < s < k_+$  then  $s - 1 \in [k_-, k_+)$ , so  $d_{s-1}$  and  $d_s$  are odd and  $f_{s-1} = \pm 1$ .

For (b), if  $s$  is chain-interior then  $s - 1$  and  $s$  are chain edges with  $f = 0$ . Such  $s$  lie strictly between  $A$  and  $B + 1$  by Lemma 142.

For (c), at the outer site  $B + 1$  (when  $r > 0$ ) we have  $\alpha = d_B = c_r^R \neq 0$  and  $\beta = 0$ . The site  $A$  is symmetric.

The cost and cut conditions of every non-junction site involve only the deposits of one block, which gives the splitting.  $\square$

## D.4 Chain generating functions

For a chain  $c$  of length  $N \geq 1$ , define its *weight*  $\omega(c)$  as  $x^{\ell(c)} y^{\gamma(c)}$ , where

$$\ell(c) = \underbrace{\sum_{i=1}^N (|c_i| + 2[c_i = 0])}_{\text{edges}} + \underbrace{\sum_{i=2}^N \max(|c_{i-1}|, |c_i|)}_{\text{interior sites}} + \underbrace{|c_N|}_{\text{outer site}},$$

$$\gamma(c) = \#\{2 \leq i \leq N : c_{i-1} = c_i = 0\}.$$

The empty chain has weight 1. By Lemmas 142 and 143, the left and right chains of  $g$  contribute exactly  $\omega(c^L)$  and  $\omega(c^R)$  to  $x^{\ell_R^{\text{true}}} y^{c^{\text{true}}}$ . Everything else comes from the junction sites and the travel block. Note that the junction site is *not* part of  $\omega$ .

**Lemma 144** (bulk). *For  $b \geq 1$ ,*

$$\sum_{c: |c_1|=2b} \omega(c) = P_b, \quad \sum_{c: c_1=2\sigma b} \omega(c) = \tfrac{1}{2}P_b \quad (\sigma = \pm 1), \quad \sum_{c \text{ of kind } Z} \omega(c) = \mathfrak{g}S.$$

*Consequently*

$$\sum_{c \text{ of kind } E \text{ or } Z} \omega(c) = B_0.$$

*Proof.* Let  $Q_b = \sum_{|c_1|=2b} \omega(c)$ . We split according to what follows edge 1.

- $N = 1$ . The weight is  $x^{2b}$  for the edge times  $x^{2b}$  for the outer site, i.e.  $q^{2b}$ , for each of the two signs.
- *Next deposit*  $c_2 = \pm 2a \neq 0$ . Site 2 costs  $x^{2\max(a,b)}$ .
- *A gap run*  $c_2 = \dots = c_{L+1} = 0$  with  $L \geq 1$ , followed by  $c_{L+2} = \pm 2a$ . Site 2 costs  $x^{2b}$ . The  $L$  gap edges give  $q^L$ . The  $L - 1$  sites strictly inside the gap cost 1 and are cuts, giving  $y^{L-1}$ . The site before  $c_{L+2}$  costs  $x^{2a}$ .

The tail  $(c_{i+1})_{i \geq 1}$  in the second case, and  $(c_{L+1+i})_{i \geq 1}$  in the third, is again a chain with first entry of magnitude  $2a$ . Its interior sites, outer site and cuts are exactly those counted in its own  $\omega$ . So

$$Q_b = 2q^b \left( q^b + \sum_a (q^{\max(a,b)} + q^b \mathfrak{g} q^a) Q_a \right),$$

which is (39). We have  $\text{ord } Q_b \geq 4b$ , so  $Q = P$  by the uniqueness in Lemma 140(b).

The negation  $c \mapsto -c$  preserves  $\omega$ , which gives the signed statement.

A chain of kind Z is a gap run of length  $L \geq 1$ , followed by a chain with  $|c_1| = 2a$ . Its weight is  $q^L y^{L-1} \cdot x^{2a} \cdot \omega(\text{tail})$ . Here the sites strictly inside the gap are the  $L - 1$  cuts, and  $x^{2a}$  is the site between the gap and the deposit. Summing gives  $\mathfrak{g} \sum_a q^a P_a = \mathfrak{g}S$ .  $\square$

## D.5 Travel block

**Lemma 145** (travel). *Let  $k \neq 0$ . Fix everything except the signs of the travel deposits, and write  $|d_j| = 2s_j + 1$ . The travel edges and travel-interior sites contribute*

$$\prod_j x^{2s_j+1} \prod_{\text{interior}} x^{2\max(s_j, s_{j+1})+1} = x^{-1} \prod_j (x^{2s_j+2}) \prod_{\text{interior}} x^{2\max(s_j, s_{j+1})}.$$

*This does not depend on the signs.*

*Proof.* The run has  $|k|$  edges and  $|k| - 1$  interior sites. Moving one  $x$  from each interior site to the edge that follows it leaves one extra  $x$ , which is where the factor  $x^{-1}$  comes from. Sign independence follows from Lemma 143(a).  $\square$

Summing over the travel words with the two end magnitudes fixed, and counting the  $2^{|k|}$  sign choices as the factor 2 in each  $e_s$ , gives

$$\sum_{|k| \geq 1} \sum_{\text{words}} (\dots) = x^{-1} \sum_{n \geq 0} \langle v, (D_e K)^n D_e w \rangle = x^{-1} \langle v, (I - \mathcal{T})^{-1} D_e w \rangle. \quad (43)$$

This formula needs a condition on the ends: the end weights  $v_{s_{\text{near}}}$  and  $w_{s_{\text{far}}}$  must be the *sign-averaged* end weights. That is, the ends together must contribute the factor  $\frac{1}{2} \cdot 2$  per end sign, which the next lemma makes precise. Here  $n = |k| - 1$ , and  $K$  is symmetric, so the reading direction does not matter.

**Lemma 146** (sign decoupling). *Let  $k \neq 0$ . Let  $\sigma_n$  and  $\sigma_f$  be the signs of the travel deposits adjacent to the near junction (site 0) and the far junction (site  $k$ ). Write the contribution of the two junction sites, together with the adjacent chains, as  $N(\sigma_n) \cdot F(\sigma_f, \varepsilon, \delta)$ . Here  $N$  is summed over the chain at site 0 and  $F$  over the chain at site  $k$ . Suppose that*

$$\Lambda := \sum_{\varepsilon, \delta} F(\sigma, \varepsilon, \delta) \quad \text{does not depend on } \sigma. \quad (44)$$

*Then the sum of  $N \cdot F$  over  $\varepsilon, \delta$  and the end signs equals  $2 \cdot \bar{N} \cdot \Lambda$  when  $|k| = 1$  (so  $\sigma_n = \sigma_f$ ), and  $4 \cdot \bar{N} \cdot \Lambda$  when  $|k| \geq 2$ . Here  $\bar{N} = \frac{1}{2} \sum_{\sigma} N(\sigma)$ . In both cases this is what (43) produces with  $v = \bar{N}$  and  $w = \Lambda$ .*

*Proof.* If  $|k| \geq 2$  the two signs are independent, and  $\sum_{\sigma_n, \sigma_f, \varepsilon, \delta} NF = (\sum_{\sigma} N)(\sum_{\sigma_f} \Lambda) = 2\bar{N} \cdot 2\Lambda$ .

If  $|k| = 1$  there is one travel edge and one sign  $\sigma$ , so  $\sum_{\sigma, \varepsilon, \delta} N(\sigma)F(\sigma, \varepsilon, \delta) = \sum_{\sigma} N(\sigma)\Lambda = 2\bar{N}\Lambda$ , using (44).

In both cases these are the edge factors 2 carried by  $D_e$  at the end edges.  $\square$

*Remark 147* (decoupling holds only after the sum over  $\varepsilon$ ). Hypothesis (44) is essential when  $|k| = 1$ , and it fails before the sum over  $\varepsilon$ .

Take  $k = -1$ , so that  $d_{-1} = \sigma(2s + 1)$  is both the near and the far travel deposit. Then  $N(\sigma)$  involves  $|d_{-1} - 1|$ , which is  $2s$  or  $2s + 2$  according to  $\sigma$ . For fixed  $\varepsilon$  and  $\delta = 1$ , the far site involves  $|d_{-1} - \varepsilon|$ , which is also  $2s$  or  $2s + 2$  according to  $\sigma\varepsilon$ . So for fixed  $\varepsilon$  the sum  $\sum_{\sigma} N(\sigma)F(\sigma, \varepsilon, 1)$  pairs the cheap near cost with the cheap or the expensive far cost depending on  $\varepsilon$ . It is *not* equal to  $\bar{N} \sum_{\sigma} F(\sigma, \varepsilon, 1)$ .

Only after summing over  $\varepsilon$  does the far weight lose its  $\sigma$ -dependence, because the substitution  $\varepsilon \mapsto -\varepsilon$  exchanges the two values. Then the product factorises.

For  $k = +1$ , the near weight depends only on  $|d_0|$  (see (45)), so there the factorisation is harmless. The model nevertheless uses the  $\varepsilon$ -summed far vector  $\lambda^{(y)}$  in both sectors, and that is legitimate only because of (44).

Condition (44) is verified in (46) and (48) below. In both, the  $\sigma$ -independence comes from the involution  $\varepsilon \mapsto -\varepsilon$ , combined with negating the far chain where necessary. Negating the far chain preserves  $\omega$  and all cut conditions.

## D.6 Sector $k > 0$

The near junction is site 0. There  $\alpha_0 = d_{-1} - 1$ ,  $\beta_0 = d_0$  and  $\Phi_0 = f_{-1} + 1 = 1$ . So **site 0 is never a cut** (RJLemmaJ.J\_site0\_pos). Its chain is  $c^L$ , with  $u := c_1^L = d_{-1}$ . Let  $d_0 = \sigma(2s + 1)$ . The cost is  $\max(|u - 1|, 2s + 1)$ , which does not depend on  $\sigma$ . Summing over  $c^L$  with Lemma 144:

- $u = 0$  (kinds E, Z): the cost is  $2s + 1$  and the weight is  $B_0$ ;
- $u = +2b$ : the cost is  $\max(2b - 1, 2s + 1)$  and the weight is  $\frac{1}{2}P_b$ ;
- $u = -2b$ : the cost is  $\max(2b + 1, 2s + 1)$  and the weight is  $\frac{1}{2}P_b$ .

Hence

$$N(\sigma) = \bar{N} = \mu_s^{(y)}. \quad (45)$$

No  $y$  occurs here: site 0 is never a cut, and in any case it lies in  $(A, B + 1)$  only when  $\ell > 0$ .

The far junction is site  $k$ . Put  $d_{k-1} = \sigma(2s + 1)$  and  $v := c_1^R = d_k$ . Then

$$\alpha_k = \sigma(2s + 1) + \varepsilon[\delta = 0], \quad \beta_k = v - \varepsilon[\delta = 1], \quad \Phi_k = f_{k-1} - [\delta = 0] = [\delta = 1].$$

By Lemma 142(iii), site  $k$  lies in  $(A, B + 1)$  if and only if  $r > 0$ .

$\delta = 1$ . Here  $\Phi_k = 1$ , so there is no cut. The cost is  $\max(2s + 1, |v - \varepsilon|)$ .

- $v = 0$ : the cost is  $2s + 1$  for both  $\varepsilon$ , contributing  $2B_0x^{2s+1}$ .
- $v = \pm 2b$ : as  $\varepsilon$  runs over  $\pm 1$ ,  $|v - \varepsilon|$  runs over  $\{2b - 1, 2b + 1\}$ . Each sign of  $v$  has weight  $\frac{1}{2}P_b$ , so the contribution is  $P_b(x^{\max(2b-1, 2s+1)} + x^{\max(2b+1, 2s+1)})$ .

The total is  $2\mu_s^{(y)}$ .

$\delta = 0$ . Here  $\Phi_k = 0$  and  $\beta_k = v$ . As  $\varepsilon$  runs over  $\pm 1$ ,  $|\alpha_k|$  takes the value  $2s + 2$  (for  $\varepsilon = \sigma$ ) and  $2s$  (for  $\varepsilon = -\sigma$ ).

- $\varepsilon = \sigma$ : the cost is  $\max(2s + 2, |v|)$  and there is no cut. The contribution is  $B_0x^{2s+2} + \sum_b P_b x^{\max(2s+2, 2b)}$ .
- $\varepsilon = -\sigma$ : the cost is  $\max(2s, |v|)$ . Site  $k$  is a cut if and only if  $s = 0$  and  $v = 0$  (RJLemmaJ.J\_sitek\_pos:  $\delta = 0$ ,  $d_{k-1} = -\varepsilon$ ,  $d_k = 0$ ). The cut is *counted* if and only if moreover  $r > 0$ , i.e. the right chain has kind Z; for kind E, site  $k = B + 1$  is the outer end. So for  $v = 0$  the weight is  $1 + y^{[s=0]} \mathbf{g}S = B_0^{[s]}$ , and the contribution is  $B_0^{[s]}x^{2s} + \sum_b P_b x^{\max(2s, 2b)}$ .

The total is  $E_s^{(y)}$ .

Therefore

$$\sum_{\varepsilon, \delta} F(\sigma, \varepsilon, \delta) = E_s^{(y)} + 2\mu_s^{(y)} = \lambda_s^{(y)} \quad \text{for both } \sigma, \quad (46)$$

which verifies (44).

Combining Lemmas 143, 144, 145 and 146 with (43):

$$\sum_{g: k > 0} x^{\ell_R^{\text{true}}} y^{c^{\text{true}}} = x^{-1} \langle \mu^{(y)}, (I - \mathcal{T})^{-1} D_e \lambda^{(y)} \rangle = \mathcal{W}_+.$$

There is no shield term, because SF requires  $k = 0$ .

## D.7 Sector $k < 0$ , and the factor $\frac{1}{2}$ on $E$

The near junction is site 0. Its chain is now  $c^R$ , with  $v = c_1^R = d_0$ , and the adjacent travel deposit is  $d_{-1} = \sigma(2s + 1)$ . Then

$$\alpha_0 = \sigma(2s + 1) - 1, \quad \beta_0 = v, \quad \Phi_0 = f_{-1} + 1 = 0.$$

So  $|\alpha_0| = 2s$  for  $\sigma = +1$ , and  $|\alpha_0| = 2s + 2$  for  $\sigma = -1$ . Site 0 is a cut if and only if  $d_{-1} = 1$  and  $d_0 = 0$  (RJLemmaJ.J\_site0\_neg). It lies in  $(A, B + 1)$  if and only if  $r > 0$ , by Lemma 142(iii). So:

- $\sigma = +1$ : the contribution is  $B_0^{[s]}x^{2s} + \sum_b P_b x^{\max(2s, 2b)}$ . The  $y$  appears exactly for  $s = 0$  with  $c^R$  of kind Z.
- $\sigma = -1$ : the contribution is  $B_0x^{2s+2} + \sum_b P_b x^{\max(2s+2, 2b)}$ , with no cut.

Thus  $N(+1) + N(-1) = E_s^{(y)}$ , and

$$\overline{N} = \frac{1}{2} E_s^{(y)}. \quad (47)$$

**The  $E/2$  argument.** Here, unlike the case  $k > 0$ , the near weight depends on  $\sigma$ . The sum over  $\sigma$  of the near end produces the whole of  $E_s^{(y)}$ . But the edge carrying  $\sigma$  is a travel edge, and its two signs are already counted by the factor 2 in  $e_s$ . Pairing  $E^{(y)}$  with  $D_e$  would therefore count that sign twice. The correct near vector is the sign average,  $\frac{1}{2}E^{(y)}$ . This is exactly what Lemma 146 says. Its integrality was recorded in Lemma 140(d).

The far junction is site  $k$ . Its chain is  $c^L$ , with  $u = c_1^L = d_{k-1}$ , and  $d_k = \sigma(2s+1)$ . Then

$$\alpha_k = u + \varepsilon[\delta = 0], \quad \beta_k = \sigma(2s+1) - \varepsilon[\delta = 1], \quad \Phi_k = f_{k-1} - [\delta = 0] = -[\delta = 0].$$

By Lemma 142(iii), site  $k$  lies in  $(A, B+1)$  if and only if  $\ell > 0$ .

$\delta = 0$ . Here  $\Phi_k = -1$ , so there is no cut. The cost is  $\max(|u + \varepsilon|, 2s+1)$ . By the same computation as for  $\delta = 1$  in §D.6, with the chains mirrored, the sum over  $\varepsilon$  and  $c^L$  is  $2\mu_s^{(y)}$ .

$\delta = 1$ . Here  $\Phi_k = 0$  and  $\alpha_k = u$ , while  $|\beta_k|$  is  $2s$  for  $\varepsilon = \sigma$  and  $2s+2$  for  $\varepsilon = -\sigma$ . Site  $k$  is a cut if and only if  $u = 0$ ,  $s = 0$  and  $\varepsilon = \sigma$  (RJLemmaJ.J\_sitek\_neg). The cut is counted if and only if  $\ell > 0$ , i.e. the left chain has kind Z. The sum over  $\varepsilon$  and  $c^L$  is  $E_s^{(y)}$ , with the slot  $B_{0z}$  at  $s = 0$ .

Hence

$$\sum_{\varepsilon, \delta} F(\sigma, \varepsilon, \delta) = \lambda_s^{(y)} \quad \text{for both } \sigma. \quad (48)$$

By Lemma 146, which is now needed in earnest at  $k = -1$  (Remark 147),

$$\sum_{g: k < 0} x^{\ell_R^{\text{true}}} y^{c^{\text{true}}} = x^{-1} \langle \frac{1}{2}E^{(y)}, (I - \mathcal{T})^{-1} D_e \lambda^{(y)} \rangle = \mathcal{W}_-.$$

Finally,  $\mu^{(y)} + \frac{1}{2}E^{(y)} = \frac{1}{2}\lambda^{(y)}$  by (42), which gives  $\mathcal{W}_+ + \mathcal{W}_- = (2x)^{-1} \langle \lambda^{(y)}, (I - \mathcal{T})^{-1} D_e \lambda^{(y)} \rangle$ .

## D.8 Sector $k = 0$ : the complete case analysis

When  $k = 0$  there is no travel, every  $d_j$  is even, and site 0 is the only junction site. Put  $u = c_1^L = d_{-1}$  and  $v = c_1^R = d_0$ . Then

$$\alpha_0 = u - 1 + \varepsilon[\delta = 0], \quad \beta_0 = v - \varepsilon[\delta = 1], \quad \Phi_0 = 1 - [\delta = 0] = [\delta = 1].$$

The term  $-1$  is **vArr**. By Lemmas 142, 143 and 144,

$$\mathcal{W}_0 = \sum_{\varepsilon = \pm 1} \sum_{\delta \in \{0,1\}} \sum_{(u, w_L)} \sum_{(v, w_R)} x^{j(\varepsilon, \delta, u, v)} w_L w_R y^\chi, \quad j = \max(|u - 1 + \varepsilon[\delta = 0]|, |v - \varepsilon[\delta = 1]|), \quad (49)$$

where:

- $(u, w_L)$  ranges over the left kinds:  $E : (0, 1)$ ,  $Z : (0, \mathfrak{g}S)$  and  $N_{\pm, b} : (\pm 2b, \frac{1}{2}P_b)$  for  $b \geq 1$ ;
- $(v, w_R)$  ranges over the same list for the right chain;
- $\chi \in \{0, 1\}$  records whether site 0 is counted in  $c^{\text{true}}$ .

Only  $w_R$  of kind Z can carry the extra  $y$ :  $y^\chi w_R$  is  $y\mathfrak{g}S$  when  $\chi = 1$ . The chain interiors are accounted for inside  $w_L$  and  $w_R$ .

**Site-0 cost, by branch.**

| $\delta$ | $\varepsilon$ | $j(\varepsilon, \delta, u, v)$ |
|----------|---------------|--------------------------------|
| 0        | +1            | $\max( u ,  v )$               |
| 0        | -1            | $\max( u - 2 ,  v )$           |
| 1        | +1            | $\max( u - 1 ,  v - 1 )$       |
| 1        | -1            | $\max( u - 1 ,  v + 1 )$       |

For example,  $u = v = 0$  gives 0, 2, 1 and 1 in the four rows. These are the identity and three elements of length  $\leq 2$ .

**When is site 0 a cut?**  $\text{cut}(0)$  holds if and only if all three of the following hold:

- $\delta = 0$ , since  $\Phi_0 = [\delta = 1]$ ;
- $v = 0$ ;
- $u = 1 - \varepsilon$ , that is,  $u = 0$  if  $\varepsilon = +1$  and  $u = 2$  if  $\varepsilon = -1$ .

**When is site 0 counted?** Site 0 is counted if and only if  $\text{cut}(0)$  holds and one of the following holds:

- (I)  $0 \in (A, B + 1)$ , which by Lemma 142(iii) means  $\ell > 0$  and  $r > 0$ ;
- (S)  $\text{SF}(g)$  holds. This means  $\delta = 0$ ,  $\ell = 0$  (the left chain has kind **E**; recall  $d_j = 0$  for all  $j < 0$  says exactly that), and  $r > 0$ .

Since  $v = 0$  and  $r > 0$  force the right chain to have kind **Z**, we get  $\chi = 1$  only if  $\delta = 0$  and  $c^R$  has kind **Z**. The full table follows. Rows are the left kind, and the **N** kinds are split by  $u$ .

| $\delta$ | $\varepsilon$ | left kind ( $u$ )                   | right kind | $\text{cut}(0)$      | (I)                   | (S)                      | $\chi$   |
|----------|---------------|-------------------------------------|------------|----------------------|-----------------------|--------------------------|----------|
| 0        | +1            | <b>E</b> ( $u = 0$ )                | <b>Z</b>   | yes                  | no ( $\ell = 0$ )     | <b>yes</b>               | <b>1</b> |
| 0        | +1            | <b>Z</b> ( $u = 0$ )                | <b>Z</b>   | yes                  | yes                   | no                       | <b>1</b> |
| 0        | +1            | <b>N</b> ( $u \neq 0$ )             | <b>Z</b>   | no ( $u \neq 0$ )    | yes                   | no                       | 0        |
| 0        | -1            | <b>E</b> ( $u = 0$ )                | <b>Z</b>   | no ( $u \neq 2$ )    | no                    | yes                      | 0        |
| 0        | -1            | <b>Z</b> ( $u = 0$ )                | <b>Z</b>   | no ( $u \neq 2$ )    | yes                   | no                       | 0        |
| 0        | -1            | <b>N</b> <sub>+,1</sub> ( $u = 2$ ) | <b>Z</b>   | yes                  | yes                   | no                       | <b>1</b> |
| 0        | -1            | <b>N</b> , $u \neq 2$               | <b>Z</b>   | no                   | yes                   | no                       | 0        |
| 0        | $\pm 1$       | <b>E</b>                            | <b>N</b>   | no ( $v \neq 0$ )    | no                    | yes                      | 0        |
| 0        | $\pm 1$       | <b>Z</b> , <b>N</b>                 | <b>N</b>   | no ( $v \neq 0$ )    | yes                   | no                       | 0        |
| 0        | $\pm 1$       | <b>E</b>                            | <b>E</b>   | $[\varepsilon = +1]$ | no ( $\ell = r = 0$ ) | no ( $r = 0$ )           | 0        |
| 0        | $\pm 1$       | <b>Z</b> , <b>N</b>                 | <b>E</b>   | possible             | no ( $r = 0$ )        | no ( $\ell > 0, r = 0$ ) | 0        |
| 1        | $\pm 1$       | any                                 | any        | no ( $\Phi_0 = 1$ )  | –                     | no ( $\delta = 1$ )      | 0        |

The table exhausts all branches. There are  $2 \times 2$  marker choices, the left kinds **E**, **Z**, **N** <sub>$\pm, b$</sub> , and the right kinds **E**, **Z**, **N** <sub>$\pm, b$</sub> . Every marker choice with  $\delta = 0$  appears in the first eleven rows, split by right kind, and  $\delta = 1$  is the last row. The rows marked (S)=yes with  $\chi = 0$  are the cases in which the shield *fires but site 0 is not a cut*. They contribute nothing, which is why **cTrue** conjoins **SF** with  $\text{cut}(0)$ . The first row is the only branch in which the shield changes the count. There site 0 is a genuine cut, but it lies at the clamped left end  $A = 0$  of the span and so falls outside the open interval.

**Summary.** The right  $\mathbf{Z}$  weight is  $y\mathbf{g}S$  exactly in the following cases:

- $\delta = 0$ ,  $\varepsilon = +1$  and the left chain has kind  $\mathbf{E}$  or  $\mathbf{Z}$ ;
- $\delta = 0$ ,  $\varepsilon = -1$  and  $u = +2$ .

In every other case it is  $\mathbf{g}S$ . This is branch for branch the rule used by the bivariate enumeration check of §7.1. This proves

$$\sum_{g: k=0} x^{\ell_R^{\text{true}}} y^{c^{\text{true}}} = \mathcal{W}_0.$$

Integrality is clear from (49):  $P_b/2 \in \mathbb{Z}[y][[x]]$ , and the sum is summable because  $\text{ord } P_b \geq 4b$ .

*Remark 148* (the regrouping of  $\mathcal{W}_0$ ). Let  $\Psi = (\Psi_b)_{b \geq 1}$  be the ungapped bulk vector, the solution of (39) with  $\mathbf{g} = 0$ , which depends on  $q$  alone. By the Sherman–Morrison formula (23),  $P_b = B_0 \Psi_b$ , and  $\mathbf{g}S = B_0 - 1$ . So every weight in (49) is affine in  $B_0$ :  $w \in \{1, B_0 - 1, \frac{1}{2}B_0 \Psi_b\}$ , and  $y^\chi w_R = y(B_0 - 1)$  when  $\chi = 1$ . Each term  $x^j w_L w_R y^\chi$  is therefore a polynomial of degree at most 2 in  $B_0$ , and collecting powers gives

$$\mathcal{W}_0 = A_0 + B_0 A_1 + B_0^2 A_2,$$

with  $A_0, A_1, A_2$  absolutely convergent sums of monomials in  $x$  and  $y$  times at most two entries of  $\Psi$ . The  $A_i$  contain no travel resolvent. As functions of  $q$  they are singular only where  $\Psi$  is, i.e. where  $I - M_0$  is not invertible ( $S_e = 0$ , Corollary 44), and  $B_0 = 1/(1 - \mathbf{g}t_1)$  only where  $1 - \mathbf{g}t_1 = 0$ . Neither occurs at a travel pole (Proposition 94, Corollary 111), so  $\mathcal{W}_0$  has no travel pole.

## D.9 Conclusion of the proof

The sectors  $k > 0$ ,  $k < 0$  and  $k = 0$  partition  $\mathcal{G}$ . Lemma 141 identifies each sector with the product set of its blocks, and Lemma 143 shows that  $x^{\ell_R^{\text{true}}} y^{c^{\text{true}}}$  is the product of the block weights and the junction-site factors. The three sector computations above then give Theorem 138. The rearrangements are justified by Lemma 140.  $\square$

**What is and is not established here.** The following are established:

- the identity of Theorem 138, as a hand proof;
- integrality;
- $x$ -adic convergence.

The following are not established here:

- any statement about a specific rational triangle beyond the universality depth (Corollary 139);
- a Lean formalisation of the assembly.

The inputs taken from Lean are the definitions of `siteCost`, `cut`, `ATrue`, `BTrue`, `lRTrue`, `cTrue` and `ShieldFires`, the junction lemmas of `RJLemmaJ.lean`, and `metricAll` (for the corollary only).

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